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CENSUS OF INDIA, 1901.

VOLUME VI.

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The Lower Provinces of Bengal and their Feudatories.

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PART I.

THE REPORT.

BY

E. A. GAIT, F.S.S.

OF THE INDIAN CIVIL SERVICE,

HONORARY ANTHROPOLOGICAL SECRETARY TO THE ASIATIC SOCIETY OF BENGAL AND SUPERINTENDENT OF CENSUS OPERATIONS, BENGAL.

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INTRODUCTION.

DATE OF CENSUS.

THE fourth regular Census of Bengal was taken with that of the whole of India on the 1st March 1901. A full account of the procedure adopted in connection with the taking of the Census and the compilation of the results has been given in a separate report, but it may be interesting to note briefly a few of the more important facts connected with the operations.

ARRANGEMENTS FOR THE ENUMERATION.

Owing to various circumstances, of which the dearth of literate persons is one of the chief, a Census is a matter of special difficulty in India, and especially so in Bengal, where the population far exceeds that dealt with by a single Census Superintendent in any other country in the world, while the absence of a regular detailed survey necessitates very elaborate preliminary arrangements, in order to obtain a complete list of villages, and to portion them out amongst the different grades of Census Officers in such a way that there may be no cases of omission or overlapping. Special care has to be taken to obtain a full enumeration of the boat population, on the network of rivers in East Bengal, and of the wild tribes of the Chota Nagpur Plateau and other remote tracts.

Owing to the general illiteracy of the population, the European method of giving to the head of each family a form or schedule on which to record the desired particulars for his household is impracticable in India, and the Census was taken, as on previous occasions, by persons specially appointed for the purpose, called Enumerators, each of whom dealt with a specified number of houses, usually about 40. But even so, the men available were seldom well qualified for their duties. They were, of course, literate, but their general education was usually of a very low standard, and in order to obtain correct entries in the different columns of the Census schedules, it was necessary to drill them very carefully. With this object in view a regular scale of Census Officers was appointed. The persons who were entrusted with the training of the Enumerators and the examination of the work done by them were called Supervisors and, on the average, there was one Supervisor to every 12 Enumerators. The Supervisor was usually a non-official, and although he was selected with special reference to his educational fitness, and was of a much better stamp than the Enumerators, it was still necessary to supplement his written instructions by oral teaching, and to keep a careful watch over his work. For this purpose every district was parcelled out into charges, each under a Charge Superintendent, who was in almost all cases an official. The Charge Superintendents were, in their turn, subordinate to the District Magistrate and his Subdivisional Officers, who were assisted in the task of general supervision by such gazetted officers as had not been detailed to take immediate charge of the work in particular charges. Excluding Calcutta and the Native States, there were, in round numbers, 1,500 Charge Superintendents, 28,000 Supervisors and 384,000 Enumerators.

The first direct step towards the taking of the Census was the numbering of the houses. A house was defined as the residence of a commensal family, and each such house was given a separate number. When all had been numbered, a statement showing the number of houses and of each grade of Census Officers was compiled and sent to the Provincial Superintendent, who used it to correct the rough indent for forms which had been previously sent to the Press.

House numbering.

The next step was the preparation of the preliminary record, *i.e.*, the entry in the enumeration-schedules of the necessary particulars regarding all persons ordinarily resident in each house. The information to be recorded included:—name, religion, sex, age, civil condition, caste, occupation, parent-tongue, birthplace, literacy or illiteracy, language in which literate, and certain infirmities. In order to obtain an accurate return, it was necessary that the Enumerators should be very carefully trained beforehand. This training was carried out during November and December. The Charge Superintendents were first thoroughly taught, either at head-quarters or by officers of the higher grades deputed for the purpose; the Supervisors were next instructed by the Charge Superintendents, and the Enumerators by the Supervisors. Classes were held at which the rules were explained and schedules were filled in experimentally. The mistakes made in preparing these test schedules were corrected and explained on the spot.

The Preliminary Record.

Everything possible was done to give the Enumerators a thorough knowledge of their duties, but even so, experience on previous occasions had shown that numerous errors would still remain. The correction of these mistakes in the enumeration-schedules would have made them very unsightly. To avoid this, the original entries were made on plain paper, and the actual Census record was not prepared until the rough draft had been shown to, and corrected by, the Supervisors. The testing, however, was by no means confined to the Supervisors. Every single officer who could be spared was given a share in the operations, and for two days all Government offices were closed in order to set the whole staff free for the inspection of the work.

The preliminary record was prepared in the Mufassal between the 20th January and the 10th February. In towns it was begun and finished ten days later. The period remaining before the 1st March was utilised in checking the entries as described above.

The actual Census was taken between 7 P.M. and midnight on the 1st March. Each Enumerator visited in turn every house in his beat, and brought the record up to date by striking out the entries relating to persons no longer present, and entering the necessary particulars for all new-comers. In a few remote tracts this final revision was commenced a few hours earlier, so that it might be completed by nightfall. In the Tributary States of Chota Nagpur and in the greater part of Sikkim there was no final revision; the preliminary record was there treated as the actual Census. Special arrangements, which need not be detailed here, were made for the enumeration of travellers by rail, road, steamer and boat, for wood-cutters in the Sundarbans and other forests, and for ports, large fairs, tea-gardens, &c.

The Actual Census.

THE PROVISIONAL RESULTS.

On the morning after the Census, the Enumerators repaired to a place previously fixed by their Supervisors, and prepared an abstract showing the number of houses, and of persons, male and female, in their blocks. These abstracts, after being checked by a second Enumerator, were posted by the Supervisor in a summary for his Circle. The Circle summaries were checked and posted in a Charge summary, which was sent to head-quarters, where the provisional totals for the district were compiled. The first district to communicate its totals was Balasore (on the second day after the Census), which was followed closely by Puri, Darbhanga, and other districts. The figures for the whole Province had been received and reported to Government by the 10th March, or within nine days of the Census. The total population, as shown by these provisional figures, differed by only 2,982, or less than ·004 per cent. from the result arrived at after detailed tabulation. The greatest degree of accuracy was attained in Puri, Backergunge and the 24-Parganas, where the preliminary totals differed from the final figures by only 2, 5 and 24, respectively.

PREPARATION OF FINAL TABLES.

At previous enumerations, the information contained in the schedules was extracted on "abstraction sheets" (one for each final Table), which were divided by rules into spaces corresponding to the headings of the Table concerned. A separate sheet was used for each Enumerator's book of schedules, and a tick was made in the appropriate column corresponding to each entry therein. When the whole book had been abstracted the ticks were counted. The figures thus obtained were added up for the Police Circle (this was called tabulation), and the figures for the latter were compiled into a total for the district.

THE SLIP SYSTEM.

On the present occasion this method was abandoned, under the orders of the Census Commissioner for India, in favour of what is known as the slip, or card, system. A separate slip containing all the prescribed details was prepared for each person enumerated, and these slips were then sorted for all the final Tables in turn. Each sorter was supplied with a set of pigeon-holes, which were labelled to indicate their contents. For instance, when sorting by caste, one pigeon-hole would be labelled "Bráhman," another "Káyasth," and so on; all the slips on which Bráhman was shown as the caste were placed in the pigeon-hole labelled "Bráhman," and all those for Káyasths into the hole labelled "Káyasth." When the sorting for a Table had been completed, the slips in each hole were counted, and the result was noted on a form called the "sorter's ticket." The figures in the sorter's tickets were then posted in "tabulation registers," and added up to form the district total. This method of working out the results of a Census was invented by Herr Von Mayr in connection with the Bavarian Census of 1872, and has since been adopted by almost all civilised countries, including many of the colonies. It has many obvious advantages. It is much less complicated than the old method; the work is more easily tested; and by putting together and sorting at one time the slips for a large number of persons, the operation previously known as tabulation was entirely dispensed with.

In order to reduce the amount of writing to be done, slips of different colours were used for the different religions, and symbols were printed on

them to indicate sex and civil condition. The symbols used in Bengal were as follows:—

	Unmarried.	Married.	Widowed.
Male ...			
Female ...			

The selection of the right slip thus obviated the necessity of making any entry for religion, sex or civil condition. The labour of copying was still further reduced by the judicious use of abbreviations.

Preparation of slips.

In 1881, when the Tables were far less elaborate, the results were compiled in three central offices, while in 1891 the work was carried out at the head-quarters of each district. On the present occasion the slips were usually written up in the districts, and they were then sent for sorting to six central offices. A certain amount of slip-copying, as it was called, was also carried out in the central offices, partly in order to give the officers in charge time to organize their establishments before the more difficult part of the work—sorting—began, and partly because in some cases there were special reasons why the slips could not conveniently be copied locally. This arrangement worked admirably, and within ten weeks of the Census almost the whole of the 78 million slips had been prepared. When the work was in full swing the number of men employed in the slip-copying offices exceeded five thousand.

Sorting the slips.

The sorting took longer. There were in all eighteen Tables to be prepared; and some of them, such as those connected with castes and occupations, were very complicated, and involved much correspondence with district officers and a great deal of local enquiry. The work, however, was practically finished by the end of September.

Compilation.

Then followed compilation and revision, which, for the more elaborate Tables, proved to be by far the most tedious part of the work. The Caste Tables in their original form were ready about the end of January 1902, and the Occupation Tables a month later. But the detailed examination of the statistics in the head-office and the checking of all entries which appeared doubtful, either because they differed widely from the results of the last Census, or because of discrepancies in the figures for males and females, which were purposely kept separate throughout the earlier stages of the work, or for any other reason, took up much time. The last of the Tables was thus not finally passed until after the end of May.

The Report.

Until the revision of the Tables was well advanced it was impossible to devote much time to the writing of the report, and on the 1st of July more than half of it still remained to be written. The Census Commissioner for India was anxious that it should be completed at the earliest possible moment, and with great efforts the actual writing of the report was completed by the 7th September, and the printing three weeks later, or more than six months earlier than in 1891. It was impossible, however, in the stress of work to devote as much time as one

could have wished to the careful revision of the proofs, or to the improvement of the style and arrangement of what was unavoidably a very hurried draft.

The report has run to much greater length than I had intended, but in a great Province like Bengal it is impossible to deal fully with the statistics within the limits which are sufficient elsewhere. The population of Bengal is about a quarter of that of the whole of India, but the results of the Census of the other three-quarters are treated of in no less than twenty-two different Census Reports. In the Assam Report, for example, 20 pages are devoted to the variations in the population of that Province, which comprises about 6 million persons. At the same rate, about 250 pages would be required for the 78 millions of Bengal, but the actual length of the corresponding chapter of my report is only 88 pages. Nor is it merely a question of population. The local conditions in different parts of Bengal vary very greatly, and in respect of caste, language, religion, social customs and material condition, Bengal, Bihar, Orissa and Chota Nagpur differ from each other as much as do any two adjoining Provinces in the Empire. In the Himalayan district of Darjeeling and the Sikkim State, again, the population is drawn mainly from Nepal and Tibet, and the report would not be complete if the tribes there met with and the languages which they speak were left unnoticed. In order to curtail the report as much as possible, a great deal of interesting descriptive material which had been collected has been omitted. Several of the general discussions not immediately connected with the statistics, such as those regarding the popular religious beliefs, the origin and nature of caste, and caste precedence, have been inserted in accordance with the request of the Census Commissioner for India. But even here I have tried not to be too diffuse, and I have not by any means made full use of my material. The subject of caste precedence alone, if the numerous memorials received from the representatives of the different castes had been at all fully discussed, would have taken up more space than the whole Caste Chapter now does; and, in the same way, the notes on Hindu Godlings might easily have occupied more pages than the whole of the Chapter on Religion as it now stands.

After all a Census Report is, in the main, a work of reference, and completeness is more important than brevity, especially in India, where there is no body of professional statisticians ready and eager to pounce on the raw material provided for them at the Census, and to make the required deductions. Unless the Census Superintendent himself analyses the figures and points to the conclusions to be drawn from them, they are in danger of being left unnoticed altogether.

Cost of the Census.

The total expenditure of all kinds on the present Census of Bengal has been less than Rs. 3,90,000, or under Rs. 5 per 1,000 of the population, compared with Rs. 7,00,000, or rather more than Rs. 9·7 per 1,000 in 1891. This large reduction in cost is due, to a great extent, to the introduction of the slip system, but considerable savings have also been effected in other directions, by improved record-room arrangements, the employment of men on lower pay and strict economy at all points.

Acknowledgments.

In conclusion I have to express my gratitude to the District Magistrates and other officers of the general Administration for their cordial co-operation at all stages of the operations. Fully occupied, as they already were, with their ordinary duties, the Census must have been a heavy additional burden, and cannot well have

been regarded otherwise than as an unmitigated nuisance. But in spite of this, with scarcely an exception, they gave it their ungrudging attention, and it was owing to their efforts that a very high standard of accuracy in the actual enumeration was attained. Nor did their labours cease when the Census was taken. In most districts the slips were copied locally, and the arrangements for, and supervision of, this operation took up much of their time. In the course of sorting the slips and compiling the final tables, numerous local references, both official and demi-official, were unavoidably necessary, and the correspondence and enquiries in connection with these references must have formed an irksome addition to the ordinary office work, but in no case did a request for information fail to receive prompt and courteous attention. There were also innumerable special reports on religion, caste, marriage customs and the like, many of which involved a great deal of local enquiry and research. The general information contained in this volume is based mainly on the information thus obtained; and, as already stated, a great deal of most interesting material has been received which I have not yet been able to utilize.

I am under still greater obligations to the officers named in the margin, who were associated with me in the compilation of the results, and of whose industry and devotion it is impossible to speak too highly. Mr. Howard at Dacca dealt with a population of about 15½ millions, and managed his huge office with entire success. The work was done very quickly and methodically, and a high standard of accuracy was attained. The same may be said of the work done by Babus Monmohan Roy and Jamini Mohan Das, whose management of the offices at Berhampur and Cuttack was all that could be desired. Mr. Moberly, at Patna, had an office nearly as large as that at Dacca and a very indifferent class of clerks. In spite of this disadvantage he laboured unceasingly, and was making good progress when his health gave way and he was obliged to go on leave. His place was taken by Babu Monmohan Roy, who had closed his office at Berhampur, and, in spite of failing health, brought the operations at Patna to a successful conclusion. Babu Srinath Chakravarti, my Personal Assistant, has worked assiduously throughout, and has managed the head-office very satisfactorily. He has a good head for figures, and has been of very great assistance to me.

Mr. Moberly.
„ Howard.
Babu Jamini Mohan Das.
„ Monmohan Roy.
Mr. Manmatha Nath Ghosh.

Lastly, I have to acknowledge my indebtedness to Mr. Chalmers, Superintendent of Government Printing, Bengal, for the close personal attention which he has always given to Census matters, and for the great help which he has given me at all stages of the work. The form of slip used for working out the results, which has been described above, was devised by him, after many different devices had been tried and rejected, and it was found to answer our requirements in all respects. It was adopted, not only by me for use in Bengal, but also by the Census Superintendents of Assam and several other Provinces. The arrangement of the form in which the different Tables should be printed was another matter in respect of which I am under special obligations to Mr. Chalmers.

MAP
OF
BENGAL

SHEWING THE DENSITY OF POPULATION PER SQUARE MILE OF EACH DISTRICT AND STATE ACCORDING TO THE CENSUS OF 1901.

Scale of Miles.

REFERENCES.

Province or State Boundary .....

District .....

1. Angul, Govt. Estate ..... 114 persons per sq. mile.
2. Pal Lahara, Native State 69 ..... "
3. Talcher ..... 151 ..... "
4. Athmallik ..... 56 ..... "
5. Athgar ..... 331 ..... "
6. Narsinghpur ..... 199 ..... "
7. Baramba ..... 235 ..... "
8. Tigiria ..... 492 ..... "
9. Hindol ..... 152 ..... "
10. Nayagar ..... 252 ..... "
11. Khandpara ..... 284 ..... "
12. Ranpur ..... 227 ..... "
13. Nilgiri ..... 239 ..... "
14. Seraikela ..... 222 ..... "
15. Kharsawan ..... 365 ..... "

Native States ..... N. S.

REFERENCES.

Under 200 persons per sq. mile .....

200 persons per sq. mile and under 400 .....

400 ..... do. ..... do. ..... 500 .....

500 ..... do. ..... do. ..... 600 .....

600 ..... do. ..... do. ..... 655 .....

772 ..... do. ..... do. ..... 800 .....

800 ..... do. ..... do. ..... and over .....

NOTE.—The figures under each district name give the actual number of inhabitants per sq. mile.

Reg. No. 618, Census, Bengal.—Apr. 08.—1,000.

Photo. S. I. O., Calcutta.

REPORT

ON THE

CENSUS OF BENGAL, 1901.

Chapter I.

DISTRIBUTION OF THE POPULATION.

General Description of Bengal.

Area, population, and boundaries.

The territory ruled by the Lieutenant-Governor of Bengal covers an area of 190,000 square miles and has a population of more than 78 millions. Its extent is thus half as great again as that of the United Kingdom, and its population is nearly double. It includes Bengal Proper with a population of 41 millions, or about the same as that of the whole Madras Presidency including its feudatories; Bihar with 23 millions, or rather less than that of the Bombay Presidency with the Native States attached to it; Orissa with four millions, or rather more than the population of Upper Burma; Chota Nagpur with five millions, or rather less than that of Lower Burma; and, lastly, the Native States, viz., Kuch Bihar, Hill Tippera, Sikkim and the twenty-six Tributary States of Orissa and Chota Nagpur. The aggregate population of these States approaches four millions, and is nearly half that of the Central India Agency. This great Province is bounded on the north by the Himalayas; on the east by Assam and the range which divides Assam from Burma; on the south by Madras and the Bay of Bengal; and on the west by the Central Provinces and the United Provinces of Agra and Oudh. It contains tracts of greatly varying physical features, including the alluvial plains of the Ganges and the Brahmaputra and the deltas of these great rivers, the crystalline plateaux of Chota Nagpur and the Tributary States and the hills stretching from their south-eastern extremity northwards to the Ganges at Rájmahál, the narrow strip of alluvium comprising the Orissa Commissionership, and, lastly, a small tract of the Sub-Himalaya, the Sikkim State and its ceded area, which forms the greater part of the modern district of Darjeeling. The most distinctive feature of the Province is its network of rivers—the Ganges and Brahmaputra with their affluents and distributaries. These rivers are of use in many ways. They furnish an admirable and cheap means of transport; they contain an inexhaustible supply of fish, and they bring down vast quantities of fertilising silt which they distribute over the surface of the delta.

Changes in elevation.

2. In Chota Nagpur and Orissa there is no evidence of any change in late geological times, but the rest of the Province has undergone great vicissitudes. It is supposed that there was formerly a continuous chain of hills connecting the Rájmahál range with the remains of the Peninsula system still in existence in Assam, and that the subsidence of this area was due to the same disturbances that resulted in the raising of the Himalayas. There are reasons for supposing that the Indus and Ganges were once connected, and it is thought that before these changes the waters of the Ganges and its tributaries found their way to the sea down the valley of the Indus. The Gangetic plain appears to have been formed from the silt of the great Himalayan rivers. There are no marine deposits on the southern face of the Himalayas, nor have any been brought to light by borings at Allahabad, Calcutta, and elsewhere. The Calcutta bore hole revealed ancient land surfaces at a depth of 30 and again at 382 feet, and it is

known that the soil of the Sundarbans has sunk considerably in quite recent times. It is thus probable that a great part of the Bay of Bengal was once dry land, and that the gradual raising of the surface of the country by the action of the rivers has been discounted by successive subsidences, which have been accompanied by upheavals elsewhere.* It is believed that the formation of the depression along which the Padma now flows was connected with the elevation of the Tippera Hills, and that the raising of the Madhupur jungle was contemporaneous with the sinking of the country occupied by the Sylhet jhils.†

CHANGES IN RIVER SYSTEM—THE GANGES.

3. The above changes, though interesting in themselves, belong to prehistoric times. They are thus of less practical importance than the great alterations that have been taking place during the last three or four hundred years in the river system of the Province. From the dawn of history until probably some time in the 16th century the Bhágirathi formed the main channel of the Ganges. In the eyes of the Hindus this river, and not the Padma, is still the sacred stream, and on its banks were situated the great capitals of Gaur (Lakhnauti), Pandua, Rájmahál, Nabadvip, and Sátgaon. Its earliest bed, known as the Saraswati, left the modern Hooghly at Sátgaon and pursued a more westerly course to a point near the place where the Damodar now joins the Hooghly. Large vessels sailed up this river in the 16th century. Its silting up led to the establishment of the port and town of Hooghly by the Portuguese in 1637. The Damodar, it is said, formerly joined the Bhágirathi at Sátgaon, and it is only since the middle of the 18th century that it left this course and burst into the old channel of the Bhágirathi. Local traditions have preserved no record of the supplanting of the Bhágirathi by the Padma as main channel of the Ganges, and it is probable that it was effected very gradually.‡ The whole country below Rájmahál and Murshidabad was formerly part of the true Ganges delta, where the river was split up into various channels, all of which were busy depositing silt and so raising their beds and blocking up their mouths. The process doubtless proceeded most rapidly in the Bhágirathi, which was then the main channel, and in time the river was obliged to seek another course by which to discharge the bulk of its accumulated water. In this way the Ichámati,§ the Jalangi, the Mátábhángá, the Kumár or Nabagangá, and the Gorai probably each in turn became the main outlet of the Ganges. The river tended ever eastwards, cutting right across the old drainage channels of the country, until at last it was met and stopped by the Brahmaputra.‖

THE BRAHMAPUTRA.

4. The Brahmaputra flowed round the foot of the Garo Hills east of the Madhupur jungle, and after discharging its silt into the Sylhet jhils, united with the Megna. This is the course shown on the maps of Rennell's survey of 1785, and it was not till the beginning of this century that, having raised its bed and lost its velocity, it was no longer able to hold its own against the Megna, and being forced to seek another outlet for its banked-up waters, it suddenly broke westwards and joined the Ganges near Goalundo.¶ It now competes with the Ganges in depositing its detritus in the eastern part of the Delta; and as the

* There is evidence to show that the elevation of the Himalayas, if not still in progress, has only recently ceased.

† Such alterations of level have occurred in historic times, *e.g.*, the submergence of two thousand square miles in the Rann of Cutch in 1819. The Assam earthquake of 1897 also resulted in some small changes of elevation.

‡ The Padma is mentioned in the *Ain-i-Akbári*. It is also shown as a wide river in the map published in 'Da Asia,' by De Barros, who died in 1570. This map, however, is not very reliable.

§ The Ichámati may be an older river which was cut in halves by the Padma, as there is a stream of the same name in Pabna on the north bank of the Padma. In Dhrubananda Misra's Káyastha Kárika which purports to contain Ballála Sena's rules for the Káyasths, the country of the Bangaja Káyasths, is said to be bounded on the east by the (old) Brahmaputra, on the west by the Madhumati, and *on the north by the Ichámati*. The name Mátábhánga also survives north of the Padma as the name of a thana in the Kuch Bihar State.

‖ Mr. F. G. Shillingford has suggested that the Kosi formerly flowed into the Brahmaputra, and that on its junction with the Ganges the united mass of water opened up the passage now called the Padma, and the old channel of the Bhágirathi was then left comparatively dry.

¶ These and other changes in the great rivers of the Gangetic plain are well described and explained by Fergusson in his paper on "Some recent changes in the Delta of the Ganges"—*Journal of the Geological Society*, Vol. XVIII, page 321. Reasons are there given for thinking that at a still earlier period the Brahmaputra had flowed, as it now does, west of the Madhupur jungle, and that its diversion to the east was due to the elevation of that tract and the simultaneous subsidence of the land now covered by the Sylhet jhils.

quantity of silt brought down by it is estimated to be about twice as great as that borne by the Ganges, the coast line is thus being rapidly thrust forward.

THE KARÁTOYÁ AND MAHÁNANDÁ.

5. Along the Northern Frontier of Bengal numerous rivers debouch from the Himalayas. These now discharge their waters by various channels into the Ganges or the Brahmaputra, but it appears that formerly, when these great rivers were still 150 miles apart, the Himalayan streams united to form a great independent river which found its own way to the sea. The elevated tract known as the Bárind formed an obstacle which could not be pierced so easily as the more recent alluvium around it, and the outlet of the Himalayan streams was thus diverted to one side or the other. Sometimes when the trend of the rivers was eastwards they flowed down the channel of the Karátoyá, the memory of which is preserved in the *Puránas*, though in some parts the traces of its course have now well nigh disappeared. It bore a high character for sanctity, and its mermaid goddess, whose image has been found among the ruins of Mahásthán, was widely worshipped. Even now the old course of the river at Mahásthán is still a favourite place of pilgrimage. It is mentioned in the *Jogini Tantra* as the western boundary of the ancient kingdom of Kámarupa, and it was along its right bank that Bakhtyár Khilji marched on his ill-fated invasion of Tibet. In the narrative of that expedition, it is described as being three times the width of the Ganges. It was no doubt the great river crossed by Hiuen Tsiang on his way to Kámarupa and by Husain Sháh on his invasion of the same country.* It is shown in Van den Broucke's map (*cir* 1660) as flowing into the Ganges. Its most recent bed, which is still in existence, joins the Atrai some thirty miles east of Pabna, and the latter flows into the Jamuna,† as the present course of the Brahmaputra is called, about the same distance above the junction of that river and the Padma.‡ But in an alluvial country the course of the rivers, especially when they have a rapid current, is constantly changing, and it thus often happened that the Tista and its neighbours worked their way westwards and found an outlet on the other side of the Bárind down the channel of the Mahánandá.

Though less famous than its rival, the Karátoyá, the Mahánandá appears to have had a greater influence on the recent ethnic distribution of the people. The Rájbansis are the main element in the population east of its course, while to the west they are scarcely found at all. East of this river Muhammadanism is the main religion, while to the west Hinduism is most prevalent. It is also a linguistic boundary, Hindi being spoken to the west of it and Bengali to the east. Prior to the Muhammadan occupation it was the dividing line between the Rárh and the Bárendra country. Unlike the Karátoyá, the Mahánandá is still a considerable stream.

South of the Padma there is no trace of any river bearing the name either of the Karátoyá or of the Mahánandá; but remembering that the former was called Bhángmáti by the historian of Bakhtyár Khilji, it may perhaps be identified with the Mátábhángá, which flows through Chuadanga almost due south of Pabna. The latter river had formerly an outlet towards the east of the Delta; but owing to the gradual silting up of this tract, it subsequently left its bed and turning west occupied in turn parts of the channels of the Kumár, Ichámáti, and Churni rivers, and eventually rejoined the Bhágirathi not far from Chakdaha. It has been suggested that the Haringhátá was the original estuary of the Karátoyá and its affluents, and it is possible that the Bhairab was the ancient channel of the Mahánandá. Its tortuous course can still be traced on both sides of the Jalangi and the Mátábhángá, and it is only near the Padma, almost opposite the point where the Mahánandá flows into it, that all upward traces of this old river disappear.

* Possibly the Sankosh and even the Manás joined it. There are the remains of a river called Manás in Rangpur and Bogra, and this may possibly indicate the former course of the river of that name which now flows into the Brahmaputra above Goalpara in Assam. The Pauránic name of the Kosi was Kausikí, and it is interesting to note that a Naiad of this name was worshipped on the bank of the Karátoyá.

† The name probably indicates that the Brahmaputra has here occupied the bed of a river called the Jamuna. There is still an affluent of the Atrai of this name, and in this country of constant fluvial changes, its present insignificance is no argument against its having once been a large and important river.

‡ The Tista, which is really the modern Karátoyá, though deprived of many of its affluents, flowed south-west instead of south-east, at the time of Rennell's survey, and joining the Atrai in Dinajpur fell into the Padma. It broke eastwards in the destructive floods of 1787.

EARLY HISTORY OF THE PROVINCE.

6. The history of the changes in these rivers* is of importance in connection with the ethnic distribution of the people. For a correct appreciation of the distribution of the population by race and religion and of the caste distinctions that now exist, it is also necessary to know something of the history of the Province. A brief outline of the more important points is therefore given below.

BIHAR.

7. The Province of Bihar† is known to us from very early times. The ancient kingdom of Magadha comprised the country now included in the districts of Patna, Gaya, and Shahabad. Its capital was at Rájagriha, some thirty miles north-east of Gaya. North of the Ganges was Videha or Mithila, which included the modern districts of Darbhanga, Saran, Champaran, and North Muzaffarpur; the south of the latter district constituted the small kingdom of Vaisáli. To the east lay Anga, including Monghyr, Bhagalpur, and Purnea, as far as the Mahánandá river. There are constant references to these countries in the *Mahábhárata*. Magadha is even mentioned under the name of Kikota in the *Rig Veda*. It was in Magadha that Buddha developed his religion and that Mahávirá founded the cognate creed of the Jains. Soon after Buddha's death a Sudra, named Nanda, wrested the throne from the Kshattriyas and founded a new dynasty. He made his capital at the confluence of the Sone and the Ganges near the modern Patna. Chandra Gupta, a contemporary of Alexander the Great, overthrew this family and founded the Maurya dynasty. He successfully resisted Seleucus, one of Alexander's generals, and it was at his court that Megasthenes compiled his great work on India. His grandson, Asoka, established a hegemony over the whole of Northern India, including a great part of Bengal and Orissa. He was the great protagonist of Buddhism and sent his missionaries to every known country. In the fourth century the Gupta dynasty rose to power. Their capital also was at Patna, and their supremacy was acknowledged by the kings of Bengal and Kámarupa. They were Hindus by religion. In Hiuen Tsiang's time North Bihar was divided into Vriti to the north and Vaisali to the south, both countries stretching westwards to the Mahánandá. South of the Ganges were Hiranya Parvana (Monghyr) and Champa (South Bhagalpur, the Sonthal Parganas, and Birbhum). The rulers of both these kingdoms were probably Khetauris of Mál origin. In the ninth century the Buddhist dynasty founded by Gopála included Magadha in its dominions, and eventually fixed its capital at Odantapuri. The last of this line was defeated in 1197 A.D. by Bakhtyár Khilji, whose soldiers destroyed Odantapuri and massacred the Buddhist monks assembled there.

BENGAL PROPER.

9. Very little is known of Bengal Proper until the rise of the Pála dynasty. At the time of the *Mahábhárata*, Northern and Eastern Bengal formed with Assam the powerful kingdom of Prágjyotisha, or Kámarupa as it was subsequently called, and its ruler, Bhagadatta, was one of the great chiefs who fought in the battle of Kurukshettra. This kingdom stretched westwards as far as the Karátoyá river. It was ruled by a succession of princes of Mongoloid stock, and was still flourishing when visited by Hiuen Tsiang in the seventh century. South-east of Prágjyotisha, between the Karátoyá and the Mahánandá, lay Pundra or Paundravardhana, the country of the Pods, which, according to Cunningham, has given its name to the modern Pabna; its capital may have been at Mahásthán on the right bank of the old Karátoyá river. This kingdom was in existence in the third century (B.C.) and Asoka's brother found shelter there in the guise of a Buddhist monk. It was still flourishing when Hiuen Tsiang travelled in India, and it is mentioned as a place of pilgrimage in the 11th century.‡

* As pointed out in the Ain-i-Akbari the distribution of the Muhammadan Sirkars in Bengal depended on the courses of the Padma, Bhágirathi and Megna. The boundaries of the older Hindu divisions of the country were also determined with references to these rivers.

† Excluding Malda and the Sonthal Parganas. Malda was always treated as a part of Bengal prior to the British occupation, while the Sonthal Parganas belongs more properly to Chota Nagpur.

‡ *Ballála Charitra*, by Ananda Bhatta. Ballála Sena's wife went there, accompanied by a Brahman priest, to worship a phallic emblem, but the priest who received the offerings seems to have been a Buddhist.

10. East of the Bhágirathi and south of Pundra lay Vanga (called Samatata by Hiuen Tsiang), which has given its name to the modern Province of Bengal. Its people are described in the *Raghuvansa* as living in boats, and they are clearly the ancestors of the Chandáls, who at the present day inhabit this part of the country. On the west of the Bhágirathi lay Karna Suvarna (Burdwan, Bankura, Murshidabad, and Hooghly), whose king, Sashánka or Narendra, the last of the Guptas, was a fanatical worshipper of Siva, and invaded Magadha and cut down the sacred *bodhi* tree early in the seventh century.* Lastly, there was the kingdom of Támralipta or Suhma comprising what now constitutes the districts of Midnapore and Howrah. The rulers of this country seem to have been Kaibarttas.

THE PÁLA AND SENA KINGS.

11. During the ninth century the Pála dynasty rose to power in the country formerly known as Pundra and Anga. Like the kings of Pundra, they were Buddhists, but they were tolerant towards Hinduism. They gradually extended their power westwards, and absorbed the greater part of Magadha. They were driven from Bengal Proper about the middle of the eleventh century by Sámanta Sena, who, starting from Navadvip, gradually established his sway over Eastern and Southern Bengal. By degrees the whole of Bengal Proper as far east as the Karátoyá and the Brahmaputra came under Sena rule.

12. The Senas were Hindus and during their rule Buddhism was actively discouraged. The best remembered king of this dynasty is Ballála Sena, who reorganised the caste system and introduced Kulinism amongst the Brahmans, Vaidyas, and Káyasths. To him is attributed the division of Bengal into four parts, viz., Rárh, west of the Bhágirathi, corresponding roughly to Karna Suvarna; Barendra† between the Mahánandá and the Karátoyá, corresponding to Pundra; Bágri (Bágdi) or South Bengal, and Banga or Eastern Bengal. The last King, Lakshmana Sena, was still ruling at Gaur at the time of Bakhtyár Khilji's invasion at the end of the 12th century. He fled to Bikrampur in the Dacca district, where his descendants exercised a precarious sovereignty for another 120 years.‡ Many Brahmans fled with him, and this explains how Bikrampur has become one of the great strongholds of Brahmanism in Bengal.

ORISSA.

13. The earliest name for Orissa was Kalinga, a country stretching from the mouth of the Ganges to that of the Krishna. Its capital about half way down the coast was on the site of the modern city of Kalingapatam. Later the term Kalinga was applied only to the delta of the Godaveri: that of the Mahánadi became known as Utkala or Odra. In the caves at Khandagiri are to be seen the earliest memorials of Buddhistic life, the latest of which date from about the first century of our era. The *Mádala Panjiká*, or palm-leaf records of the temple of Jagannath, speak of various invasions from the north and of a subsequent invasion, about 320 A.D., of Yavanas from over the sea. These invaders, who, whatever their race, were staunch Buddhists, ruled for 150 years. They were expelled from Orissa by Yayáti Kesari, the founder of the Lion dynasty, which ruled until 1132 A.D. These kings were worshippers of Siva, and their progeniter is reputed to have imported 10,000 Brahmans from Oudh. It was they who built the temples at Bhuvanesvar, and during their rule Buddhism gradually gave way to Sivaism. The Lion kings were succeeded by the Chola or Gangetic line, who ruled till the middle of the 16th century, when the Bráhman apostate, Kálá Páhár, conquered the country for the Pathan kings of Bengal. Under the Gangetic kings Vaishnavism became the State religion of Orissa. The chief monuments of their rule are the great Temple of Jagannath at Puri and the Sun Temple at Kanárak.

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* The capital of this country has been identified by Colonel Waddell with a suburb of Burdwan and by Mr. Beveridge with Rangamati in the Murshidabad district. The latter view is the one which seems most likely to receive general acceptance.

† The name survives in the Bárind or elevated tract of old alluvium which extends over a considerable part of the modern districts of Rajshahi, Dinajpur, Malda, and Bogra.

‡ Ghiyasuddin, son of Firuz Shah, made conquests in Eastern Bengal and established himself at Sonargaon under the name of Bahadur Shah, where he struck coins in 1311. Thirty years previously Moghisuddin had been Governor at Sonargaon, but he became insubordinate and was overthrown by an imperial army assisted by the troops of the "Zamindar" Dhinwsj Rai, doubtless Dhinaj Madhab who is supposed to be a great grandson of Lakshmana Sena.

MUHAMMADAN PERIOD—CONQUEST OF BENGAL AND BIHAR.

14. Bakhtyár Khilji, a Pathan General of Muhammad Ghori, conquered Bihar in 1197. Two years later he advanced to Bengal and took possession of Gaur and Nabadvip without a struggle. He unsuccessfully invaded Tibet, and in his retreat lost the greater part of his army at the hands of the Meches east of the Karátoyá. The greater part of Bengal gradually came under the control of the Muhammadan Governors who ruled at Gaur or Lakhnauti until 1338, when Muhammad Tughlak declared himself independent.

RULERS OF BIHAR.

15. Eight years before this date South Bihar had been separated from Bengal and annexed to Delhi. North Bihar apparently belonged to Bengal for some time longer, as the Bengal King Háji Ilyás is reputed to be the founder of Hájipur. In 1397 the whole of Bihar became part of the kingdom of Jaunpur, but a century later it was again taken possession of by the Emperors of Delhi, who continued to hold it except for a short time when the Bengal king Husain Shah and his son, Nasrat Shah, obtained temporary possession of the country north of the Ganges. Under the Moghals the capital of the country was the town of Bihár in the south of the Patna district, and from this town the whole province took its name.

INDEPENDENT KINGS OF BENGAL.

16. From 1338 till 1576 Bengal was ruled by various lines of independent kings, mostly of Pathan origin. In the latter year Akbar defeated Daud, who was then king, and Bengal was annexed to the Moghal Empire, to which it continued to belong until it passed into the possession of the East India Company. The capital was usually at Gaur or the neighbouring towns of Pandua and Rajmahal until 1608, when it was moved to Dacca. About a century later Murshid Kuli Khan made Murshidabad his head-quarters, and so it remained until the end of Moslem rule.

NORTH AND SOUTH-EAST BENGAL.

17. In North Bengal the Khen dynasty ruled until the end of the 15th century, when it was overthrown by Husain Shah, but the country was not permanently held. Biswa Singh, the progenitor of the Koch kings, founded a new dynasty, whose rule extended from the Karátoyá as far as Central Assam, and it was not until 1661 that the country as far as Goalpara was permanently acquired by Mir Jumla. Previous to the 17th century the Chittagong Division and Noakhali were usually in the hands of the Tiparas or of the Maghs, and it was only after the transfer of the capital to Dacca that this tract was gradually annexed.

ORISSA.

18. Orissa (including Midnapore), which had been wrested from the Hindu kings by Kálá Páhár, the General of Suláiman, King of Bengal, in 1567, remained in the possession of the Afghans until 1592, when Mán Singh annexed it. It was placed under separate Governors, but Midnapore and Balasore were subsequently transferred to Bengal. In 1752 Alivardi Khan ceded the Province to the Mahrattas, in whose possession it remained until its conquest by the British in 1803.

CHOTA NAGPUR AND ORISSA TRIBUTARY STATES.

19. Chota Nagpur,* including the Tributary States of Chota Nagpur and Orissa, is called Jhárkand in the Akbarnamah. The country was ruled by chiefs of various aboriginal tribes, the Cheros being predominant in Palamau, the Mundas in Ranchi, and the Bhuiyás and Gonds in the Orissa States. The south of Chota Nagpur Proper was annexed by Akbar and Palamau by Sháh Jahán. Tho remoter chiefs appear to have remained independent until their subjugation by the Mahrattas towards the end of the 18th century.

CHARACTER OF MUHAMMADAN RULE IN BENGAL.

20. During Muhammadan rule the authority of the Central Government varied with the character of the King or Governor for the time being. If he was energetic and masterful, the whole Province accepted his authority, but if he was weak and indolent, the local rulers became practically independent. Chief among these were the Bárah Bhuiyás of Eastern and Southern Bengal, of whom Rája Pratápáditya of Jessore and Isá Khán of Khizrpur, who is

* Chota Nagpur is a corruption of Chutia Nagpur. The name is derived from Chutia, near Ranchi the residence of the old Munda Rájas.

mentioned by Ralph Fitch, are the best known. The Rájas of Vishnupur in Bankura, or Mallabhum as it was then called, and Burdwan were also practically independent so long as they paid the revenue assessed on their estates.*

Soil.

21. The soil of the Ganges plain, consists partly of the older alluvium or *bhángar*, a yellowish clay with frequent deposits of *kankar*, and partly of the newer alluvium, which varies from sand and sandy clay in the upper course of the rivers to a fine silt consolidating into clay in the delta. Before the delta is reached the newer alluvium forms the low land or strath (*khádar*) through which the rivers flow, which is flooded in the rains and which has evidently been cut out from the *bhángar* or older alluvium. In the delta, on the other hand, where the rivers have grown torpid, the silt is deposited in their beds and on their banks, which are thus gradually raised above the level of the surrounding country, until at last the river breaks through to the adjacent low land and repeats the process. In the delta therefore the newer alluvium is found above and not below the level of the surrounding country, and great marshes or *bils* are often found within the enclosures formed by the high banks of rivers. In some parts a black loam is met with. This is an accumulation of decayed vegetable matter deposited in the marshes just referred to.

East of the Bhágirathi the newer alluvium is everywhere prevalent except in the Bárind in North Bengal and the Madhupur jungle in the south of Mymensingh. West of the Bhágirathi and in Bihar the *bhángar* or older alluvium predominates, except near the course of the great rivers. On the Chota Nagpur Plateau there are extensive areas of rock, laterite, and gravel which are unfit for cultivation and, except in the valleys, the patches of fertile ground are small and infrequent.

Climate, Seasons and rainfall.

22. The climate of Bengal is distinctly tropical and the mean yearly temperature ranges from 80° at Cuttack to 74° in Chota Nagpur. The mean temperature during the cold weather months is about 64° and during the hot weather 83°. The highest temperature recorded in Calcutta in 1901 was 108·2° on the 12th June, and the lowest 50·1° on the 20th and 25th January. The atmosphere, especially in the east and south-east, is extraordinarily humid. The rainfall is heaviest in Eastern Bengal and in the Himalayan Terai, where it often exceeds 100″, and lowest in the southern districts of Bihár, where the average is only about 41″. The early part of the year is usually almost rainless, but about the beginning of February atmospheric disturbances begin to occur which are generally accompanied by heavy showers of rain and occasionally hail. The rainfall gradually increases until June, when the local sea breezes give way to the steadier winds of the south-west monsoon which supplies the province with the greater part of its annual rainfall.† The monsoon current fades away in September and its departure is signalized by showers which sometimes continue into November. There is often a little rain again about the end of December. This is the ordinary course of events, which is also most beneficial to the crops, but it frequently happens that the total rainfall is below the average, or that its distribution is abnormal. In such cases much harm may be caused to the crops, resulting in extreme cases in total failure. The *ághani* or cold weather rice requires copious showers in May and a punctual commencement of the monsoon, but the sufficiency of the rainfall in September is the chief factor in producing a good outturn. For the early rice, Indian corn and millets, *i.e.*, the *bhadoi* crops, the pre-monsoon showers are of most importance, while the *rabi* or spring rice, wheat, barley and pulses depend on the showers that follow the monsoon and the Christmas rain.

Agriculture.

23. The inhabitants of Bengal are for the most part agriculturists. The choice of crop is determined chiefly by the elevation of the land and the climatic conditions. The character of the soil is of less importance, but, speaking generally, rice does

* In the Ain the revenue of Bengal as it stood in 1582, excluding numerous abwábs or special imposts, was fixed at Rs. 1,06,85,944. This was levied from the raiyats in specie as the equivalent of the fourth share of the gross produce. In 1765, when the East India Company acquired the *diwáni*, the *net* amount of all revenue collected by authority in Bengal was Rs. 2,56,24,223.

† The bulk of the rain comes from the Bay of Bengal, but parts of Bihar, Orissa and Chota Nagpur get a portion of their supply from westerly winds from Bombay.

best on clay, wheat on a rich loam, and barley on a more sandy soil, while maize will thrive on shallow rocky soils which would support very few other crops. Taking the Province as a whole, more than sixty per cent. of the entire cultivated area is under rice, and in 1891 the estimated outturn amounted to about sixteen and-a-half million tons of cleaned rice. The proportion of rice cultivation is greatest in the deltaic districts, where the humidity of the climate and the swampy condition of the country are peculiarly favourable to its growth. Jute is the only other important crop which will thrive under these conditions; the area on which it is grown has more than doubled during the last twenty years, and in 1900 it is estimated that the gross outturn aggregated 1,200,000 tons. Of this, about half was exported, the estimated value being rather more than 7 millions sterling; the rest was consumed in the Province, as will be explained further on. This rapid extension of jute cultivation is one of the great factors in the growing prosperity of Eastern Bengal.

24. During the cold weather oilseeds of various kinds are extensively cultivated in most districts, usually as a second crop on land that has already yielded a harvest of early rice. The average outturn of all kinds of oilseeds during the last seven years amounts to about six and a half million tons. Of this, about a third of a million tons, valued at two and a half millions sterling, was exported. In the dryer districts of Bihar, in addition to rice and oilseeds, wheat, barley, and maize are largely cultivated, and also opium and indigo. The last-mentioned staple grows well in Bengal also, especially on the *chars*, but it is not a favourite crop with the cultivators and, owing to this and to falling prices, the area planted with it is declining rapidly. In Bihar, too, the reduction in price has caused a decrease of about a quarter in the area under indigo as compared with 1893, and has led the planters to seek for some other crop to replace or supplement it. The exports of indigo in 1900 were valued at a little more than a million sterling, or barely a third of the estimated value of the exports in 1844. The revenue from the cultivation of opium, which is a Government monopoly, amounted to 274 lakhs of rupees in 1899, against nearly 549 lakhs in 1881. The cultivation of the sugarcane is common throughout Bengal, but the amount grown (estimated at 9,000,000 tons in 1900) is not sufficient to meet the local demand, and large quantities of beet-root sugar are imported. In Eastern Bengal sugar is manufactured from the juice of the date-palm: no figures are available as to the quantity of sugar thus obtained, but it is believed that the total outturn, though still considerable, has fallen off a good deal during the last 20 years. Tobacco is grown for home consumption in every district and in Rangpur, Jalpaiguri, Kuch Bihar and Darbhanga considerable quantities are produced for export. Much of it is taken to Nepal and some finds its way to Burma, where it is made up into cheroots.

Estimated Outturn in 1900.

	Tons.
Wheat	... 572,600
Barley	... 440,200
Maize	... 673,300
Opium	... 2,220
Indigo	... 2,400

25. The silkworm is reared in Western Bengal and in the districts where the Presidency and Rajshahi Commissionerships meet. The industry was threatened with extinction, owing to diseases amongst the worms, but the subject has been carefully investigated and there now seems a fair prospect of the further spread of these diseases being prevented. The exports of raw silk in 1899 were valued at nearly 68 lakhs of rupees as compared with nearly 49 lakhs in 1891.

26. Tea is grown chiefly in the Duars of Jalpaiguri and in the Darjeeling district, and also, though to a smaller extent, in Chittagong, Ranchi, and Hazaribagh. The tea-gardens of Jalpaiguri attract crowds of coolies from the barren uplands of Chota Nagpur and the Sonthal Parganas. In Darjeeling most of the labour force is recruited from Nepal. The estimated outturn of tea in Bengal in 1900 was 46½ million pounds, valued at £976,187, against 27¼ million pounds, valued at £800,000, in 1891. This rapid expansion of tea cultivation has now received a check, owing to a serious fall in prices, and the immediate future of the industry is far from reassuring.

PRICES OF FOOD-GRAINS.

27. The prices of food-grains have risen everywhere during the past decade. At the Patna mart in 1900 rice sold at 14·99, wheat at 14·13, and gram at 17·70 seers per rupee, as compared with 18·23, 17·30, and 23·33 seers respectively in 1891. At the Dacca mart only 13·15 seers of rice could be obtained for a rupee in 1900,

against 16·63 seers ten years earlier, of wheat only 10·46 against 13·92 seers, and of gram only 12·19 against 15·38 seers.

This general rise of prices has enriched the cultivating classes, but it has been the reverse of beneficial to the rest of the population. So far, however, as the labouring classes are concerned, the greater cost of living has, to a great extent, been met by a general rise in wages. In Bihar wages show an advance of barely 3 per cent. but in all other parts of the Province they have risen to the extent of at least 10 per cent. The increase is most marked in the case of skilled labourers. There has also been a movement of the labouring classes from Bihar where wages are low to Bengal Proper, where they are high.

PASTURE.

28. Buffaloes are bred in Purnea and Rangpur, sheep in Bihar, and goats throughout the Province. In Mymensingh numerous herds of cattle and buffaloes are grazed in the *háors* or basins which are filled with water in the rains, but dry up in the cold weather. Large quantities of cheese (known in the market as Dacca cheese) are made and exported even as far as Turkey. But on the whole Bengal is not a pastoral country, and as a rule very little stock is bred except such as is required locally. The amount of grazing ground is generally very limited, and it is all that the raiyats can do to provide fodder for the cattle kept by them for ploughing and for milk.

MANUFACTURES.

29. Dacca and Santipur were formerly famous for their fine muslins, and early in the century the value of muslins exported to Europe, and especially to France, was very great. From Dacca alone the exports in 1817 were valued at a hundred and fifty-two lakhs of rupees. Ordinary cotton goods were also exported in large quantities, and as early as 1755 efforts were being made to induce weavers to settle in the neighbourhood of Calcutta. The introduction of machinery in Europe, however, not only killed the Indian export trade, but has flooded the country with cheap piece-goods, and has so seriously crippled the indigenous manufacture that many persons belonging to the weaver castes have been driven to abandon the loom for the plough.

The manufacture of silk also is decadent, and the quantity exported in 1899 was valued at only ten lakhs of rupees, against twenty-five lakhs in 1882.

On the other hand, the number of jute mills is rapidly increasing, and in 1900 there were 38 mills with nearly 14,000 looms and employing more than 100,000 labourers compared with 21 mills with 5,000 looms in 1881. Nearly half the raw jute produced in Bengal is now consumed in these mills, and the value of gunny-bags, rope, and other goods exported in 1900 was over four millions sterling, against only one million twenty years previously. Other large industries are also springing up, such as paper mills, iron foundries and pottery and machinery works, and for miles above Calcutta the banks of the Hooghly present a scene of industrial activity which bids fair in time to rival that of the largest manufacturing towns in Europe. These mills are at present chiefly under European supervision and supported by European capital. It may be hoped that the natives of the country will follow the lead thus given them, and in the meantime a lucrative employment is opened out to the ever-growing class of landless labourers in Bihar.

MINES—COAL.

30. The chief mining industry is coal. The principal coalfields at present known are those of Karharbári or Giridih, Raniganj, Jheria and Karanpura. They are estimated to contain 1,500,000,000 tons of coal. The Karharbári field lies in the valley of the Barákar and those of Raniganj and Jheria in that of the Damodar; these have been made accessible by rail and are rapidly being developed, but the extensive Karanpura coalfields at the head of the Damodar valley are not yet worked owing to their distance from the existing lines of railway. There are also smaller fields still practically unworked at Ramgarh, Daltonganj, and Talcher. All the above coals are somewhat bituminous with a rather high percentage of ash. In the Darjeeling district near the Nepal frontier there is a narrow field of anthracite coal, but it is doubtful if it could be successfully extracted on a large scale.

The first mine was opened in 1820, but it is only in recent years, since the establishment of through railway communication that the production of coal has

advanced by leaps and bounds. There were 46 mines in 1872 and 73 in 1881, while in 1900 the number had risen to 280. The total output in 1872 was less than a third of a million tons; in 1881 it was still less than a million; in 1891 it was less than a million and three quarters, but by 1900 it had risen to close on five million tons, valued at nearly 10 millions sterling. The exports of coal in the latter year amounted to more than half-a-million tons, or more than four times the quantity exported five years earlier.

Other Minerals.

31. A certain amount of iron ore is worked, chiefly at Barákar near Asansol, where there are iron works at which pig-iron pipes and various kinds of castings are turned out. The total quantity of iron ore extracted in 1900 was 57,000 tons, or nearly three times the quantity obtained ten years previously. This industry, however, is still in its infancy; its ultimate success depends in a great measure on the coking qualities of the Bengal coal. The coke hitherto made at Barákar is of inferior quality, but it has not yet been manufactured according to modern methods. The coal of Giridih and Jheria, on the other hand, is said to furnish an excellent, hard coke.

Mica is found in various parts of Bihar and Chota Nagpur, and there are numerous mines, chiefly in the Hazaribagh district. The total output in 1900 was 429 tons, valued at £28,000, or nearly five times the quantity obtained ten years previously. Soap-stone is found in Manbhum, and is made into cups, images, and the like, but the industry is small and decadent. The saltpetre of Indian commerce is obtained mainly from the Patna Division and Monghyr. It occurs as a natural efflorescence on the surface of the ground, and its manufacture affords employment to thousands of the caste (Nunia) to which it gives its name. The quantity recorded as having been produced in the Province in 1900 is estimated at 160,000 hundredweight, valued at £80,000, or rather less than the outturn in 1891. Fuller's earth is obtained in the Bhagalpur Division; granite in Gaya; lime-stone, sand-stone, and laterite in parts of Bihar, West Bengal, and Orissa; and slate in Monghyr, but the quantity of these minerals hitherto extracted is small, and accurate statistics are not available. Brick, fire and pottery clays are found in various parts, the finest being that of Raniganj, which is used in the pottery works of Messrs. Burn & Co. for the manufacture of stoneware, glazed drain-pipes, bricks, tiles, etc.

Railways.

32. In 1872 there were less than 900 miles of railway in the whole of Bengal, viz., 176 miles on the Eastern Bengal State Railway (Eastern Section 149 miles and Southern Section 27 miles) and 731 miles on the East Indian Railway, including (1) the main line to Chausa 418 miles, (2) the loop line 250 miles, and (3) branches 53 miles. During the next nine years, 407 miles were constructed, the chief items being 230 miles on the northern section of the Eastern Bengal Railway and the first 70 miles of the Tirhut State Railway; the Darjeeling-Himalayan Railway was also commenced. Between 1881 and 1891 progress was much more rapid. More than 350 miles were added to the Eastern Bengal Railway system, including 158 miles on the Bihar section and 86 miles on the Dacca section. An addition of 253 miles was made to the Tirhut State Railway, and the first 140 miles of the Bengal-Nagpur Railway were constructed. The Bengal Central Railway (125 miles) was commenced and finished, and also that portion of the Bengal and North-Western Railway (112 miles) which runs through this Province. The total length of the railway lines completed during the decade was 1,051 miles. In the decennium that has just passed no less than 1,614 miles of railway have been opened. The largest addition consists of 524 miles on the Bengal-Nagpur Railway, which has brought Midnapore, Singhbhum, Manbhum, and the Orissa districts into direct railway communication with Calcutta. The East Indian Railway has added 320 miles to its length, chiefly on the Gaya-Mogulserai (107 miles) and South Bihar (79 miles) branches. The other important additions include the Tirhut State Railway (203 miles), the Assam-Bengal Railway (159 miles), and the Bengal-Duars Railway (102 miles). An interesting feature of the decade was the construction of light railways between Howrah and Amta, Howrah and Sheakhala, and Ranaghat and Krishnagar. It seems probable that these lines are the precursors of many others linking up important trade centres with the existing railway systems.

These improved communications not only open out new markets for produce, and thereby raise prices and increase the profits of the cultivators, but they also facilitate migration, and thus tend to relieve the pressure of population in the more thickly-populated tracts. Districts along the line of march to the great places of pilgrimage, such as Puri and Gaya, are no longer so liable as they were formerly to the importation of epidemic disease, but on the other hand disease can be carried to a much greater distance than was formerly the case.*

CANALS.

33. The construction of canals as a means of irrigation dates from 1868, when the Sone, Orissa, and Midnapore canal schemes were commenced. The Sone Canals draw their water from the Sone river and run through the west of Patna and Gaya and the northern and central part of Shahabad. The main canals were completed by 1881; three quarters of the branch canals and distributaries were ready by the same date, and the remainder by 1891. The Orissa Canals draw their water from the Mahánadi, Bráhmini, and Baiturni rivers, and serve the deltaic country between the railway line and the coast. Three-fifths of the whole system had been completed by 1881. The Midnapore Canals are dependent on the Cossye river. There are canals in Saran and Champaran also, but these are on a comparatively small scale. These projects cost in all more than six crores of rupees. The Sone Canals were originally estimated to be capable of irrigating a million acres, the Orissa Canals about half a million, and the Midnapore Canals about one-eighth of a million acres. The area actually paying water-rate, however, has in no case reached even half these estimates, except in 1896, when the Sone Canals carried water to 555,126 acres. From the experience then gained it would seem that this is about the maximum area that they are capable of irrigating in a year of severe drought.

The benefit to the country resulting from the construction of these canals cannot be gauged by the direct financial results, which are disappointing, especially in Orissa, where the receipts seldom cover the actual working expenses.† They have, however, added greatly to the annual yield of the land watered by them; in some parts a great rise in rents has taken place, and cultivation has been extended to areas which were previously uncultivable waste; and in a famine year the direct saving to Government is enormous. It is estimated that in 1896 the Sone Canals added at least 230,000 tons to the stock of food-grains in Bihar and saved to Government at least a third of their capital outlay. It has recently been decided to construct a similar, but less costly system of canals for the protection of the cultivated area in the submontane tracts of North Bihar, where the numerous streams and the rapid fall of the country make irrigation comparatively easy.

34. Though primarily constructed with a view to irrigation, the larger canals are also of use as a means of transport, the total length available for navigation being nearly 500 miles. There are also several canals which were constructed solely for navigation purposes, and a considerable expenditure is incurred annually in keeping the Bhágirathi and other rivers in the Nadia district open for boat traffic. During the

	Miles.
Hijli Tidal Canal ...	29
Calcutta and Eastern Canals	47
Orissa Coast Canal ...	102½
Total ...	178½

* It is impossible to illustrate the effect of railways on the population by any graphic method owing to their two-fold operation. There can be no doubt as to the great benefit that a tract derives from the construction of a railway through it, but it does not follow that this benefit will be reflected by an immediate growth of the population. Very frequently the tendency is the other way. The cultivator benefits by the rise in prices, but there is no wasteland available for new settlers, while on the other hand the landless labourer is enabled to move more easily and for shorter periods to places where there is a greater demand for his services. Thus the construction of the railway through Orissa has stimulated the emigration of Oriyas to Bengal, without attracting to Orissa any new cultivators, or in fact any one at all, except a few railway officials. Again, the variations in the population are not necessarily dependent on the railway. There has been a loss of population along the new line of railway from Patna to Gaya and through the south-east of Midnapore. On the other hand the Chandpur subdivision of Tippera which has been tapped by the Assam-Bengal Railway shows an increase of 30 per cent. since 1891. In none of these cases can the variation be attributed to improved communications. The growth of the Chandpur subdivision was equally rapid in the two previous decades when there was no railway, while the decrease in Gaya and the part of Midnapore through which the railway runs is due to fever and plague more than to emigration.

† In his recent report on the settlement of Orissa, Mr. Maddox reckoned that the increase of land revenue due to the canals was between a quarter and half a lakh. He estimated that a further sum of a third of a lakh is saved yearly on account of remissions which would be required from time to time if the canals and embankments were not in existence. On the other hand, he calculated that the people of the Orissa Division benefit by the canals to the extent of 40 lakhs of rupees a year.

last few years operations have been in progress for improving the *bil* route between the Madhumati and Kumár rivers, with a view to providing a channel that will eventually be suitable for the passage of large steamers throughout the year.

In addition to canals for providing water artificially, numerous embankments are maintained by Government as a protection against floods. Of these, the best known are the Orissa embankments with a total length of nearly seven hundred miles, the embankments of the Bhágirathi and Damodar, and those on the seacoast of Midnapore.

ROADS.

35. There are nearly 40,000 miles of roads in Bengal. In the province as a whole, this gives about one mile of road to every four miles of area, but the circumstances of different localities vary a great deal. In the Patna Division there is on the average one mile of road for every two miles of area, and the ratio is also high in the Presidency, Bhagalpur, and Rajshahi Divisions. In the Dacca and Chittagong Divisions, on the other hand, where the network of waterways dispenses to a great extent with the necessity for roads, and in the sparsely populated tracts of Chota Nagpur, the mileage is less than half the provincial average. Nearly 7,000 miles of road were constructed during the decade preceding the present census, chiefly as relief works during the famine of 1896. The total length of metalled roads is 4,606 miles, or nearly 10 per cent. more than in 1891.

FAMINES.

36. The people of Bengal are mainly agricultural and the outturn of the crops grown by them is dependent on the amount and distribution of the rainfall. It follows that any serious failure in the latter must have a very disastrous effect on the community as a whole. In former times when railways did not exist and roads were few, the privations of the people in famine years were terrible. It was estimated at the time that in the famine of 1769 about a third of the inhabitants of Purnea died of starvation, and the loss of life in other districts also was appalling. As the efficiency of our administration increased and communications improved,* the efforts of Government have been directed with constantly growing success to reduce to a minimum the sufferings of the people during famine by providing employment for the able-bodied and gratuitous relief for the infirm and afflicted. And, as already stated, canals have been constructed in some parts to meet deficiencies in the rainfall by providing an artificial supply of water. The only districts which are considered free from all danger of famine are the 24-Parganas, Darjeeling and the districts of Eastern Bengal. In all other parts of the Province large areas are more or less liable to this calamity, but the extent to which relief is needed varies considerably, according to the character and density of the population and the nature of the neighbouring country. The classes that suffer most are the landless labourers and the petty artizans; the actual cultivators have usually a reserve of grain sufficient to save them from starvation.

DIVISION.	Total area.	Area liable to famine.	Population of area liable to famine.	Maximum number of persons likely to require relief in severe famine.
Burdwan	13,949	7,449	3,130,634	164,597
Presidency	12,966	4,355	2,565,241	131,437
Rajshahi	17,356	9,853	5,143,794	287,097
Patna	25,686	16,795	11,091,272	710,382
Bhagalpur	20,511	13,629	5,702,598	339,680
Orissa	9,841	4,342	1,396,149	248,985
Chota Nagpur	26,963	26,963	4,900,429	287,476
Total ...	124,372	83,386	33,930,117	2,169,564

37. During the decade preceding the census of 1901, relief operations on a considerable scale were necessary in 1891-92, in 1896-97 and again in 1899-1900. In the year 1891 the early close of the monsoon and the absence of the cold weather rains caused much damage to the winter rice and *rabi* crops, and relief operations were necessary in parts of Muzaffarpur, Darbhanga, Monghyr, Bhagalpur, Purnea and Dinajpur. The largest number on relief works on any one day was 83,016, and on gratuitous relief 4,699; the total cost of the relief operations was rather less than five lakhs of rupees. In 1899 the monsoon

* The importance of good communications as a palliative of famine is best illustrated by the experience of the Orissa famine of 1866, when the south-west monsoon prevented the supply of food to the starving people until hundreds of thousands had died.

was very capricious in parts of Chota Nagpur and Orissa. There was excessive rain in July but exceptionally little in August and September. The crops were very short throughout the area affected, but actual famine supervened only in about half Ranchi and a small part of the Palamau district. The famine of 1896-97 was far more serious. The causes of the crop failure were a very unfavourable distribution of the rainfall early in the season and the entire absence of rain after the early part of September 1896. There had been a very poor crop of winter rice in 1895, and in 1896 it was again this crop that suffered most. The brunt of the famine fell upon the districts of Champaran, Muzaffarpur, Darbhanga and Saran, and especially upon the tracts near the Nepal frontier, where the proportion of rice cultivation is greatest. The population is here dense, the tenantry are depressed, and the proportion of landless labourers is very high. West of Darbhanga the distress was less marked; it was severe only in parts of the Supaul and Madhipura subdivisions of Darbhanga and Purnea escaped altogether. In South Bihar the stress of famine was confined to parts of the Bhabua and Sasaram subdivisions of Shahabad. Elsewhere the protection afforded by the Sone Canal system, and the *ahars* or reservoirs constructed by the cultivators themselves, helped to avert a serious crop failure. In the Chota Nagpur plateau, Palamau, Hazaribagh, Manbhum, and two tracts in the Sonthal Parganas were seriously affected. In West Bengal, Bankura alone suffered severely, while in Central Bengal the crop failure was most marked in a lowlying tract known as the Kalantar, situated partly in Murshidabad and partly in Nadia, and in the Satkhira subdivision of Khulna. The famine in Orissa was confined to some parts of Puri, chiefly in the neighbourhood of the Chilka Lake.

38. Relief works were opened in November 1896, and by the close of the year 44,901 persons were employed on them. In March 1897 the distress deepened rapidly and the numbers on relief rose steadily until the monsoon had fairly set in, when they again quickly diminished, and during September and October all famine operations were brought to a close. The net result of relief operations reduced to terms of one day shows that over 61 million persons attended the works, and over 71 millions received gratuitous relief. The total expenditure was nearly 110 lakhs of rupees, in addition to advances to cultivators aggregating nearly 3 lakhs, donations of nearly 20 lakhs from the charitable relief fund, the outcome of voluntary subscriptions in this country and England, and private relief by zamindars and others. Foremost amongst the latter was the late Mahárája Báhádur of Darbhanga, who spent more than 3½ lakhs on relief operations and distributed nearly 6 lakhs in advances. The effect of the famine on the population will be discussed when dealing with the variations that have occurred during the decade, but it may be stated here that everything tends to show that although much suffering was unavoidable, the number of actual deaths from starvation must have been very small indeed, and that, having regard to the wide-spread nature of the calamity, a far greater measure of success was attained than on any previous occasion.

Month.	Numbers on—	
	Relief works.	Gratuitous relief.
March	302,789	225,748
April	365,512	312,955
May	402,346	426,369
June	384,252	457,782
July	166,429	295,505
August	81,997	253,789

Political Divisions of Bengal.

39. For administrative purposes Bengal is divided into nine large tracts officially called Divisions, each of which is superintended by a Commissioner. Of these, five are within the limits of Bengal Proper, and two are in Bihar, while Orissa and Chota Nagpur each forms a separate Commissionership. The average area of a Commissioner's Division is rather less than 17,000 square miles, and the average population a little more than 8 millions. These Divisions are again subdivided into districts, each under a Magistrate and Collector or Deputy Commissioner. Including Angul and the Chittagong Hill Tracts the total number of districts is 47. The largest districts are Hazaribagh and Ranchi each of which is about half as large again as Wales. The greatest number of people is found in Mymensingh whose population of nearly 4 millions does not fall far short of that of the whole of Upper Burma. The average area of a district exceeds 3,200 square miles, and the average population is more than 1½ millions.

The number of districts in a Commissioner's Division varies from four in the case of Dacca, Chittagong and Orissa, to seven in the Rajshahi and Patna Divisions. The latter with its population of over 15 millions or about that of the Bombay Presidency excluding Sindh, is overgrown, and its partition is a reform that cannot be much longer delayed. The Feudatory States attached to Bengal comprise Kuch Bihar, Hill Tippera, Sikkim and two groups of Tributary States known respectively as the Tributary States of Orissa and Chota Nagpur. The former group includes seventeen, and the latter nine, States.

UNITS ADOPTED FOR CENSUS STATISTICS.

40. The political divisions of the present day are not always conterminous with those indicated by the physical features of the country or the ethnic distribution of the people. This is especially the case in respect of Commissioners' divisions. There is no natural dividing line between the Patna and Bhagalpur Commissionerships, while the latter includes Malda, the greater part of which is in all respects similar to the adjacent districts of the Rajshahi Division to which it formerly belonged, and the Sonthal Parganas, which appertains both physically and ethnically to Chota Nagpur. It is, again, impossible to distinguish between Noakhali in the Chittagong Division and the adjoining district of Backergunge in the Dacca Commissionership.

It often happens also that the districts of our administration contain divergent features. The Mahánandá is the true ethnic and linguistic boundary between Bengal and Bihar, but nearly a quarter of the Purnea district lies to the east of that river. The Himalayan district of Darjeeling includes a considerable tract in the plains, and the southern boundary of Gaya and Shahabad is well within the limits of the Chota Nagpur Plateau. In discussing the results of the census, it is for several reasons inadvisable to break up districts. The district is the unit of our administration, and it is desirable that the results of the census should be viewed for each such unit as a whole. Moreover, in the census returns of birthplace, only the district of birth is given, and it is thus impossible to distribute the population according to any smaller area. In the case of Commissioners' Divisions, however, the same objections do not apply. In the Imperial Tables the districts have been grouped according to Commissionerships, but in the subsidiary tables in the body of this report they will usually be arranged by the following natural Divisions, viz.—

(1) *West Bengal*—The Burdwan Division.

(2) *Central Bengal*—The Presidency Division, excluding Khulna.

(3) *North Bengal*—The Rajshahi Division, Malda, Kuch Bihar, and Sikkim.

(4) *East Bengal*—The Dacca and Chittagong Divisions, Khulna and Hill Tippera.

(5) *North Bihar*—Muzaffarpur, Darbhanga, Champaran, Saran, Bhagalpur, Purnea.

(6) *South Bihar*—Patna, Gaya, Shahabad, Monghyr.

(7) *Orissa*—The Orissa Division, excluding Angul.

(8) *Chota Nagpur Plateau*—The Chota Nagpur Division, the Sonthal Parganas, Angul, and the Tributary States of Orissa and Chota Nagpur.

WEST BENGAL.

41. West Bengal thus corresponds to the ancient Rárh; it includes all the districts of Bengal Proper west of the Bhágirathi, the earliest known channel of the Ganges, and is peopled mainly by castes closely allied to the tribes of Chota Nagpur, such as the Bágdi, Báuri, Korá, Mál, Kaibartta, and Santál. It is also the home of several distinctive castes with claims to a higher rank in the hierarchy of Brahmanism, such as the Aguri, Sukli, Sadgop, Kástha, and Raju. The ethnic differences due to the old river barrier of the Bhágirathi are still recognised by Hindus, and to this day Bráhmans, Baidyas and Káyasths, and various lower castes, such as Nápits, Sutradhars, and Kámárs, who trace their origin to Rárh, will not intermarry with persons of the same castes who hail from Banga and Bárendra.

Though outside the Ganges delta, the eastern portion of this tract is low and of alluvial formation. Further west, laterite begins to predominate, and the surface rises and becomes more and more undulating and rocky until at last it merges in the uplands of Chota Nagpur.

CENTRAL BENGAL.

42. Central Bengal, which is bounded on the west by the Bhágirathi, on the north by the Padma, and on the east by the Madhumati, was formerly the Ganges delta, but it has gradually been raised above flood level, and the great rivers which formerly flowed through it, depositing their fertilising silt, yielding an ample supply of wholesome drinking-water and draining it, have shrunk to insignificance. Their mouths have been silted up and their beds are often higher than the surrounding country, which they are no longer able to drain. The country has thus become less healthy, and is far less fertile than it was formerly. The district of Khulna, which lies within the boundaries of Central Bengal, as defined above, is an exception to these conditions and still forms part of the true delta. It has therefore been excluded and treated as a district of East Bengal. Ethnically Central Bengal possesses comparatively few distinctive features save that its southern portion is the main habitat of the Pods who are closely allied to the Chandáls, and who, with them, are probably the descendants of the first of the Mongolian invaders from the north-east. The Kaibarttas and Bágdis have overflowed from West Bengal and the Chandáls from the east.

NORTH BENGAL.

43. North Bengal, or Bengal north of the Padma, corresponds very closely to the ancient Bárendra. The greater part is an alluvial formation, but it contains in its centre the Bárind, an elevated tract of *quasi*-laterite belonging to the same formation as the Madhupur jungle and the western part of Burdwan. The alluvial portion suffers, as does Central Bengal, from obstructed drainage due to the silting up of its rivers and the gradual raising of their beds above the general level of the country. In the north are the Himalayan State of Sikkim and the territory acquired from Sikkim, which now forms the greater part of the district of Darjeeling. These two tracts should, strictly speaking, be placed in a division of their own, but their population is so small (barely a quarter of a million) that it is not worth while to treat them separately. Malda now forms part of the Bhagalpur Division, but, as explained above, the greater part of it is in all respects an integral part of North Bengal. North Bengal is the home of the Mech and Koch tribes, the last of the invaders from the north-east, whose nominal strength is still nearly a million and-a-half, in spite of the fact that large numbers have embraced Muhammadanism.

EAST BENGAL.

44. East Bengal, or the districts of the delta and the country east of the Jamuna, with its numerous rivers, is the most fertile, healthy, and progressive portion of the Province. It is mainly alluvial, but in the south-east the hill range that divides Assam from Burma projects into it, and Hill Tippera, the Chittagong Hill Tracts, and parts of the districts of Tippera and Chittagong rest upon this formation. On the confines of Dacca and Mymensingh, the Madhupur jungle, a tract with a stiff clay soil resembling that of the Bárind, rises above the alluvium. The most marked characteristic of East Bengal is its extraordinarily large number of Muhammadans, nearly two-thirds of its population being followers of the Prophet. Its principal caste is the Namasudra or Chandál, whose home is in the swamps of the delta, and whose numbers, in spite of wholesale conversions to Muhammadanism, still exceed a million. To the north are found various Koch and Gáro tribes and to the south-east Tiparas, Kukis, and Maghs.

NORTH AND SOUTH BIHAR.

45. The division of Bihar into north and south with the Ganges as the boundary is based mainly on physical considerations.* The country north of that river is a flat alluvial formation, rising very gradually towards the foot of the Himalayas, and it enjoys in ordinary years a comparatively copious rainfall,

* Bhagalpur lies on both banks of the Ganges, but the inconvenience of breaking up districts has led to the inclusion of the whole area in North Bihar. For the same reason the part of Monghyr north of the Ganges has been treated as part of South Bihar.

increasing towards the north. South Bihar contains a strip of alluvium on the bank of the Ganges, but further south the soil changes and the surface becomes more undulating, and gradually rises until the Chota Nagpur plateau is reached. North Bihar is the great centre of indigo cultivation, and possesses many tracts of great natural fertility. On the other hand, it is peculiarly liable to failure of crops in seasons of deficient rainfall. In South Bihar a large area is protected by the Sone Canal system, and elsewhere the undulating surface enables the raiyats to construct small reservoirs on their own account from which to water their fields. The country is thus comparatively secure from famine. Ethnically there is no great difference between North and South Bihar, so far as the tracts near the Ganges are concerned, but further away, the people of South Bihar show unmistakable signs of their connection with the Dravidian tribes of Chota Nagpur, while in the north of North Bihar there has evidently been an intermingling with the Himalayan tribes, and in the extreme east there is a strong admixture of the Koch element.

ORISSA.

46. Orissa lies between West Bengal and Madras, the Chota Nagpur Plateau and the Bay of Bengal. The whole area is alluvial, and it possesses a language of its own and a system of castes differing alike from those of Bengal and of Madras. To the north and south there is no well-defined boundary separating Orissa from Bengal on the one side and from Madras on the other, and formerly the Province included part of the present district of Midnapore in Bengal and Ganjam in Madras. Even now Oriyá is spoken over a considerable area in the northern districts of Madras, and has largely modified the Bengali of South Midnapore, while the distinctive Oriyá castes are well represented for a considerable distance beyond the present political boundaries. To the west the change from alluvium to gneiss marks the boundary between Orissa and the Chota Nagpur Plateau, and the inhabitants of the latter tract, though doubtless very closely allied by origin, are distinguished from the inhabitants of the low country by their comparative want of civilisation and freedom from the trammels of the Bráhmanical hierarchy.

CHOTA NAGPUR PLATEAU.

47. The Chota Nagpur Plateau comprises the districts of the Chota Nagpur Division, the Sonthal Parganas, Angul, and the Tributary States of Orissa and Chota Nagpur. The whole area belongs to the same geological formation, and consists of a rugged tract of hill and jungle and there are frequent changes in elevation, and deep river valleys often separate the hills. Speaking generally the level rises towards the north and west, but some of the highest peaks are in the south.* It is the home of numerous non-Aryan tribes who were never properly subjugated either by the early Aryan invaders or by the Pathán and Moghal Emperors, or indeed by any outside power until the advent of the British. They have thus preserved in their mountain fastnesses an individuality in respect of tribal organisation, religion and language, which their congeners in the plains have long since lost.† They are gradually abandoning their tribal dialects in favour of the nearest Aryan form of speech—Hindi to the north and west, Oriyá to the south, and Bengali to the east, but a large number still speak their own languages. These are divided by philologists into two great families—the Munda and the Dravidian, but the distinction is merely an indication of some earlier political condition similar to that which is now leading to the adoption of Aryan languages, and does not represent any corresponding divergence in physical type, which has been proved by Mr. Risley's measurements to be fairly uniform throughout.

* The word plateau is used for want of a better designation for this tract of elevated country, but it is not intended to imply that the area referred to forms an open tableland like that to the north of Cape Colony. There are three plateaux in stricter acceptation of the term, one in Ranchi and two in Hazaribagh. Elsewhere the country is often very broken and there are numerous ranges or groups of steep hills intersected by deep ravines and occasionally by open valleys.

† Some of the present inhabitants of the plateau have traditions of former rule in the plains. The Oráons, for instance, say they were driven from the south of Shahabad by the Muhammadans—some to the north-west of the Chota Nagpur plateau and some to the Rájmahál Hills, where they became known as Malé or Sauria. The Cheros too claim to have ruled in Shahabad. Among the castes of the plains of pure aboriginal descent may be mentioned the Musahars, who are identified by Mr. Risley with the Bhuiyás. Instead of seeking their independence in the hills, they remained and submitted to foreign rule, and were rewarded with the opprobrious epithet of Musahar, or rat-eater.

AREA, POPULATION AND DENSITY.

GENERAL REMARKS.

48. The general statistics of the area and population of each district will be found in Imperial Table I. Table I in the Provincial volume contains similar information for thanas, and at the end of this chapter there are three subsidiary tables, showing (I) the density of the population, (II) its distribution between towns and villages, and (III) the average number of persons per house. The discussion in this chapter will be confined to a consideration of the population as it stood on the 1st March 1901, the day when the census was taken. The variations that are disclosed by a comparison with the results of previous enumerations will be considered in the next chapter, and as this aspect of the question is the one which possesses the greatest practical importance such descriptive matter as may be necessary to elucidate the statistics will be held over for incorporation in that chapter.

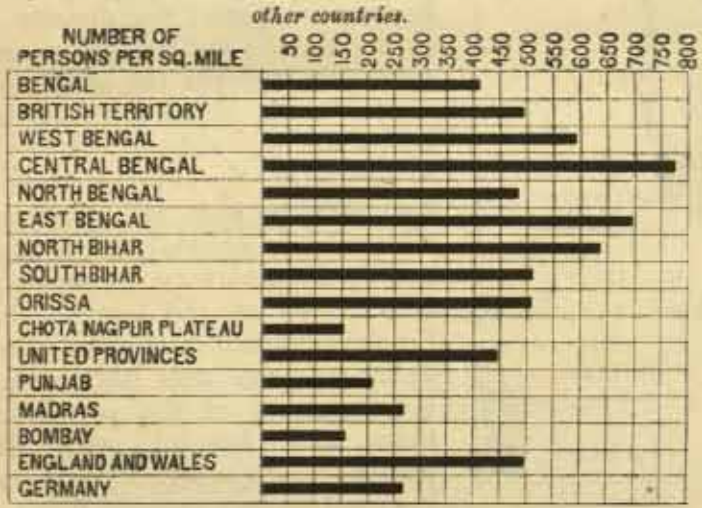
The area and population of each natural division as described in paragraph 40 above are noted in the margin. The Chota Nagpur Plateau has by far the largest area but it is very sparsely inhabited and its population is exceeded not only by that of East Bengal, which stands first in this respect, but also by that of North Bihar and North Bengal.

LOCALITY.	Area.	Population.
PROVINCE ...	*189,537*	*78,495,410*
West Bengal ...	13,940	8,240,076
Central ,,* ...	9,989	7,739,985
North ,, ...	23,389	10,005,177
East ,, ...	32,976	16,068,087
South Bihar ...	15,082	7,716,418
North ,, ...	21,746	13,831,120
Orissa ...	8,160	4,151,259
Chota Nagpur Plateau.	64,555	9,851,308

* Exclusive of Sundarbans.

The greatest density of population, viz., 775 persons to the square mile, is found in Central Bengal. Then follows North Bihar with 636, and then West Bengal with 591 and East Bengal with 514. The low position of the last mentioned tract is owing to the inclusion of the Chittagong Hill Tracts and Hill Tippera, which differ altogether from the rest of East Bengal and should, strictly speaking, form a separate division of their own. If these be excluded, the density of population in East Bengal rises to 701 persons per square mile, which exceeds that of any division except Central Bengal. Then follow in close proximity South Bihar with 511 and Orissa with 508 and then North Bengal with 428 persons per square mile. North Bengal has suffered from the inclusion of Sikkim, Darjeeling, and the newly acquired part of Jalpaiguri. If these areas be left out of account its density rises to 551 per square mile. The Chota Nagpur Plateau with less than a third of the average density of North Bengal, stands at the bottom of the list. The relative density of various parts of Bengal and of certain other provinces and countries is shown in the marginal diagram.

Diagram showing the density of Population in Bengal and certain other countries.



NOTE.—For the purpose of this diagram, Hill Tippera and Chittagong Hill Tracts have been excluded from the calculation of density in East Bengal and Sikkim from that of North Bengal.

49. The pressure of the population on the soil is far from uniform and a reference to the map* of Bengal at the commencement of this chapter will show that there are great variations even between districts of the same natural division.

* According to the map, the 24-Parganas with 986 persons to the square mile, is more thickly populated than any district in Bengal, except Howrah, but this is not really the case. The area on which the calculation was made is that shown in the records of the Jurisdiction Department of the Secretariat, and is exclusive of an extensive tract in the Sunderbans. The population of this tract, however, is included in the figures for the 24-Parganas, and if its area be similarly included, the density per square mile falls to 415.

The density of the population in Rangpur exceeds by more than 50 per cent. that of its neighbour Dinajpur and the disproportion in the case of Darbhanga and Bhagalpur is even greater. Nor are the conditions of each district uniform throughout, and the circumstances affecting density are frequently quite independent of district boundaries. It will, therefore, be desirable to consider the figures for each natural division in some detail, but before doing so we may glance briefly at the general results for districts taken as a whole.

The greatest density of population is found in Howrah, where there are 1,668 persons per square mile.* Even if Howrah City and Bally be excluded the figure still stands at 1,351. Its nearest rival is Dacca with 952 persons to the square mile, and then follow Muzaffarpur and Saran with 917 and 907 respectively, Hooghly with 881, Darbhanga with 873 and Tippera with 848 and Faridpur with 849. The scantiest population is found on the outskirts of the Province in Changbhakar and Korea to the west, Sikkim to the north, and the Chittagong Hill Tracts to the east; in none of these does the density exceed 24 persons to the square mile.

West Bengal.

50. In West Bengal the density is greatest in the metropolitan district of Howrah; then follows Hooghly, and then Burdwan, Midnapore, Birbhum, and Bankura. Howrah is a small district and the conditions throughout are very uniform. Dumjor thana, which adjoins Howrah town, is the most thickly inhabited, but no police circle in the district has less than 1,219 persons to the square mile. The Hooghly district is alluvial and the soil is fertile. But it is also indebted to some extent to the neighbourhood of the metropolis. Excluding the Hooghly and Serampore thanas the density is greatest in the south and decreases gradually towards the north and west, where alone does it fall below the high ratio of 800 to the square mile. Speaking generally, the eastern part of Burdwan, which has an alluvial soil, is more thickly peopled than the western, which rests on the laterite, but the development of the coal mines in the extreme west has led to a rapid growth of the population there, and Asansol, with 839 persons to the square mile, is now the most densely peopled thana in the district. The fewest inhabitants are to be found in Kaksa and Ausgram, which have a laterite soil but no coal-mining industry.



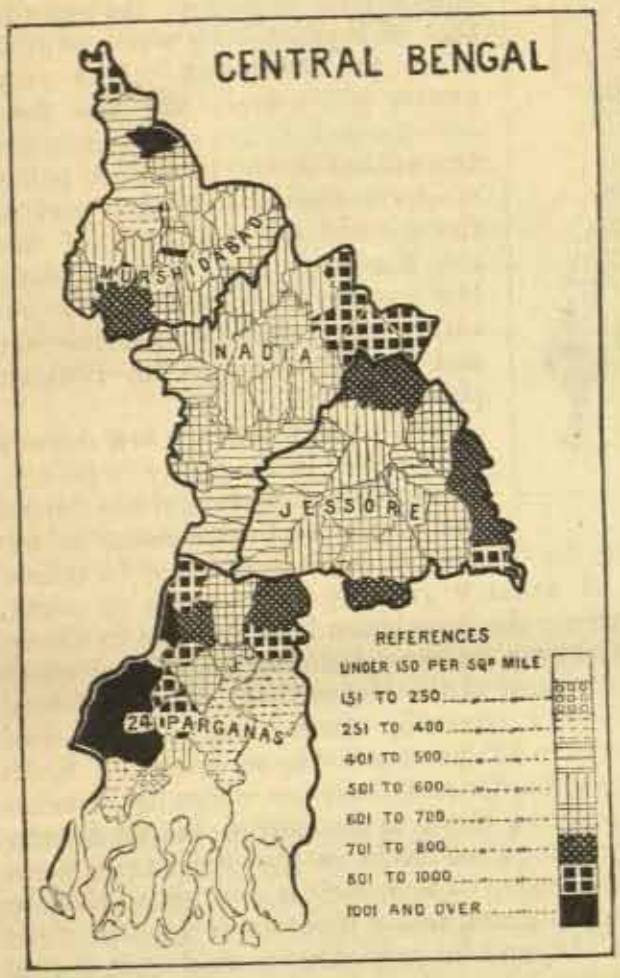
In Midnapore the pressure of the population is greatest along the bank of the Rupnarayan and the estuary of the Hooghly, the maximum density being found in Tamluk, where there are, 1,156 persons to the square mile. The town after which this thana is named was once a famous seaport; the sea has long since left it, but it is still a place of considerable importance as the centre of the boat traffic on the Rupnarayan. Further inland the soil is still fertile, but the climate is bad, and the population gradually decreases. The eastern half of the district has a laterite soil; the cultivable area is small, and the population steadily diminishes until, in the extreme west, on the confines of Singhbhum and Mayurbhanj, it is less than a quarter as dense as it is in Tamluk.

* In subsidiary Table I, at the end of this chapter, the population of cities has been excluded, but in mentioning in the text the density of population in a district, the whole area, urban as well as rural, is taken. In the case of thanas, however, the figures refer only to the rural area, and towns lying within the boundaries of the thana are excluded.

In Birbhum the local variations are slight, but the north of the district, where the soil is alluvial, and the proportion of cultivable land is highest, is more closely inhabited than the rest. Bankura has a very sparse population in the south and west, where the land is undulating, rocky and barren; towards the east, in the Vishnupur Subdivision, where the soil is alluvial and the conditions resemble those of South Burdwan, its population is more than twice as dense as it is in the south of the district.

CENTRAL BENGAL.

51. In the 24-Parganas, the pressure of the population is greatest in the riparian thanas, in most of which the density exceeds 1,000 to the square mile. In the other non-Sundarban thanas it ranges from 602 in Bhangar to 1,105 in Magra Hât. In the Sundarbans, owing to the large tracts of waste to the south the number of persons to the square mile is very small.* The Kushtia Subdivision is by far the most populous portion of Nadia, and in a great part of it, there are more than 900 persons to the square mile. The Ranaghat Subdivision and a great part of the Sadar, on the other hand, barely support half this population. The low density is here due, partly to the silting up of the rivers which has deprived the country of the silt it used to receive and of a means for carrying off the surplus water, partly to the bad system of land tenure, under which the cultivators are mere tenants-at-will, and partly to long-continued unhealthiness.



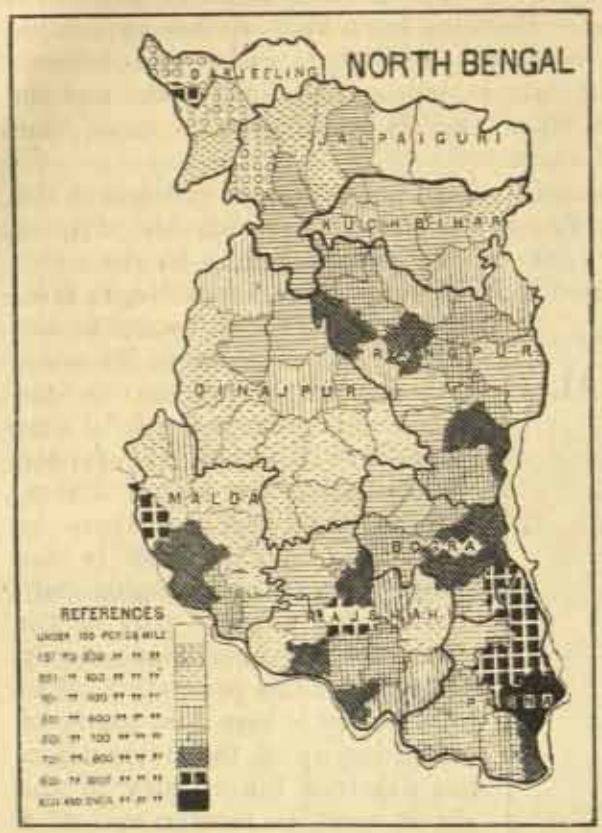
The population of Jessore is greatest in the east, where the soil is most fertile and still receives occasional deposits of silt, and least in the Bongaon Subdivision in the west, where the conditions are very similar to those in the south of Nadia to which it formerly belonged. The part of Murshidabad lying to the east of the Bhágirathi is wholly alluvial and its density considerably exceeds that of the western part of the district. In the latter tract the average density is reduced by the figures for the central thanas, which though now growing rapidly, are still very sparsely inhabited.

NORTH BENGAL.

52. Northern Bengal is an extensive area and its general conditions vary considerably. The density is greatest in the east and decreases towards the west and north. The centre of this tract contains the sparsely populated Bárind. Pabna which stands first in respect of density of population owes its position mainly to the rich jute fields of the Sirajganj Subdivision. It is also probable that the true area is greater than would appear from the figures based on the survey, and that in some parts the district has gained by alluvion. This would seem to be the case in the Shahzadpur thana where the nominal area gives a density of 1,235 to the square mile. Bogra is most thickly peopled in the east, between the Karátoyá and the Jamuna, where jute is the main staple. Towards the north-west the alluvium gives way to the *quasi*-laterite of the

* The high density shown against the Sundarban thanas in the Provincial tables for 1891 is due to the same reason as that already given for the density attributed to the district as a whole in the map at the beginning of this chapter. The area on which it was calculated did not include extensive tracts in the Sundarbans which appertain to these thanas and are responsible for a large proportion of the population shown against them.

Bárind and the population becomes more sparse. The unhealthy and jungly thana of Sherpur in the south-west has also a relatively small number of inhabitants. Thanks to its very fertile soil, Rangpur, in spite of long-continued unhealthiness, has a far greater population than Dinajpur to the west, Kuch Bihar and Jalpaiguri to the north or Goalpara in Assam to the east. Its density is very little less than that of Bogra and is almost the same as that of Mymensingh which adjoins it on the south-east. The only places where there are less than 500 persons to the square mile are two unhealthy and ill-drained police circles in the south-central part of the district on the bank of the old Karátoyá and a tract along the eastern boundary which includes in its nominal area the bed and sandy *chars* of the Brahmaputra.

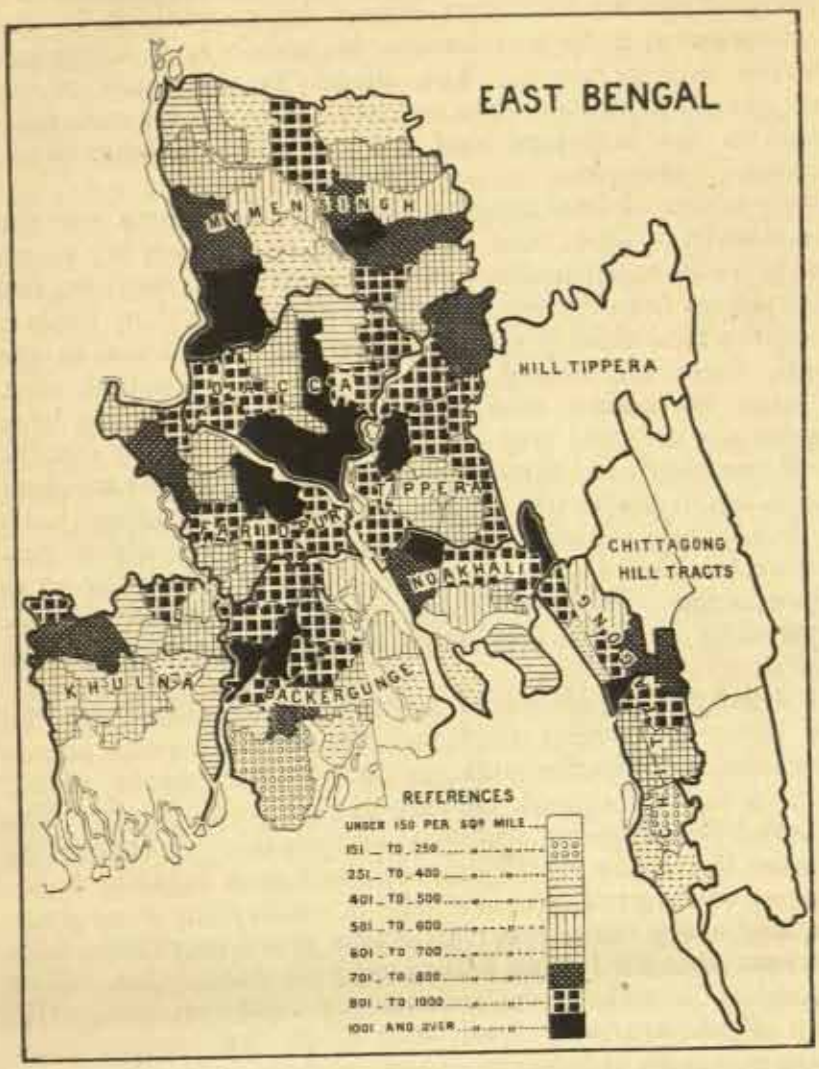


53. The relatively low density in Rajshahi is due partly to its containing a large portion of the Bárind and partly to the presence of numerous marshes and lakes, including the Chalan *bil*, the largest sheet of inland water in Bengal. There is a belt of country running from north to south, through the centre of the district, where the population is as great as in almost any part of North Bengal. Malda also owes its low position to the Bárind which extends over a great part of the district east of the Mahánandá. In the alluvial tract west of that river, the population is generally dense, and in one police circle, Manikchak, there are no fewer than 935 persons to the square mile. Kuch Bihar is most populous on the Rangpur border and least so where it marches with the Western Duars. Dinajpur owns a share of the Bárind, but its density is low throughout, and the most populous police circle in the district can boast of only 531 persons to the square mile. Jalpaiguri is even more sparsely populated. In only one police circle (Patgram, where there are a number of old families) does the population exceed 500, and in only three more does it exceed 400 to the square mile. The fewest inhabitants are in the Duars. Towards the west this tract has filled up rapidly owing to the extension of tea cultivation, but in the east, the population is still very sparse, and in Alipur it averages only 89 persons to the square mile. Darjeeling is throughout very thinly peopled. In the Terai the density is about the same as in Jalpaiguri, but it gradually diminishes towards the north, and in the head-quarters subdivision there are only 184 persons to the square mile. Sikkim, which comes last, owes its very low density to the mountainous region in the north which is practically uninhabited. The southern part of the State is less elevated and contains more cultivable land. The population, though still sparse, is here far greater than in the north, especially in the tract west of the Tista, where there are numerous settlers from Nepal.

54. The greatest density in East Bengal is found in Dacca, which lies in the angle formed by the confluence of the Padma and Megna, and next to Dacca, in Faridpur on the bank of the Padma and in Tippera on that of the Megna. The districts in question are sufficiently near the mouth of these great rivers to benefit to the full by their silt-laden floods, while they are far enough from the sea to escape damage by cyclones and storm waves, and they are not weighted, as are the districts on the coast, by the inclusion in their area of land not yet sufficiently elevated to be fit for permanent occupation. In Mymensingh to the north and Khulna,

East Bengal.

Backergunge and Noakhali to the south, the density is very uniform. It rapidly diminishes in the south-east where the alluvial plain is replaced by the rocky and jungle clad spurs of the hill range that separates Assam from Burma.



Great as is the density of the population of Dacca it would be still greater were it not for the inclusion within its borders of the southern part of the Madhupur jungle, where the population, though rapidly growing, is still comparatively sparse. The pressure is greatest in the Munshiganj subdivision which contains the famous Bikrampur pargana, the great home of the educated classes of East Bengal, whose sons are found all over Bengal and Assam, and even further afield, practising as pleaders, or holding posts in Government service, or in private employ as clerks. The proportion of the population dependent on means of livelihood other than agriculture is thus relatively high,* but even so the density of the population in some thanas is astonishing. In Srinagar there are 1,787, and in Munshiganj 1,526 persons to the square mile.

55. The population of Tippera is greatest in the fertile tract along the bank of the Megna (except in the extreme north, where there are numerous unreclaimed marshes) and in the old settled tract to the east, north and south of the head-quarters station. It is least in the centre of the southern part of the district, but as will be seen in the next chapter this tract is now rapidly being developed. Faridpur has about the same apparent density as Tippera, but it is probable that its area has grown since the time of the survey owing to the vagaries of the Padma, and that some of the thanas, such as Sibchar and Bhanga which, according to the survey figures, have respectively 1,406 and 1,223 inhabitants to the square mile, are in reality less crowded than these figures would show. In the south-west of the district the cultivable area is smaller than elsewhere and in the north-west the lower density is due to long continued unhealthiness. Bhushana which now has only 708 persons to the square mile had 843 at the time of the census of 1872.

56. Fluvial action has affected the survey areas in Noakhali even more than in Faridpur, and much stress cannot be laid on variations in the apparent

* In Srinagar 53 per cent. of the population are dependent on non-agricultural means of subsistence and in Munshiganj 30 per cent. In Kapasia the corresponding figure is only 13.

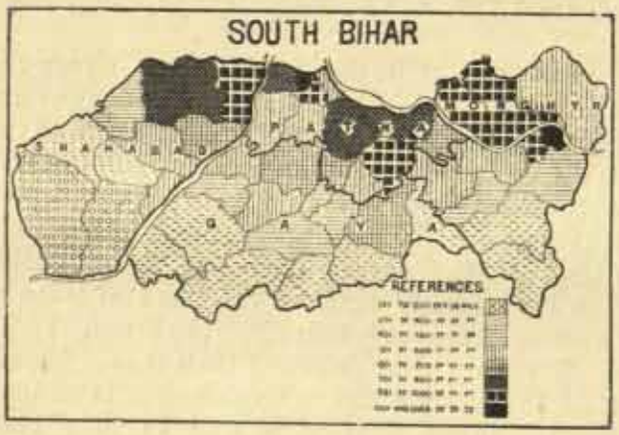
density of different parts of the district. Companyganj thana, for example, is shown as having a lower density than in 1891, but its loss of population is due to diluvion, the effect of which on the area cannot be ascertained in the absence of a new measurement and its true density is probably, if anything, greater than it was at the earlier census. The islands to the south of the district have a relatively small population because many of the cultivators have their permanent homes on the mainland and only go to the islands when agricultural operations are in progress.

The relatively low position of Backergunge is due to the figures for the south and west of the district. This tract is exposed to damage by storm waves; a large part of it is still uncultivable, and it was laid waste less than two centuries ago by Magh pirates from Arakan. To the west and north of Backergunge town are some of the most thickly-populated tracts in Bengal and in one police circle, Jhalakati, there are nearly 1,200 persons to the square mile. Khulna also has a large Sundarban area, but here a great part has been excluded from the generally accepted area of the district on which the density has been calculated. Mymensingh has three thanas with a population exceeding 1,000 to the square mile, but in the north, under the Garo Hills, many parts are very sparsely inhabited, and so also is the Madhupur jungle which projects through the centre of the district from the Dacca boundary almost as far as the town of Mymensingh. In the west the arable area is reduced by numerous *háors* or depressions in the surface which are filled with water for the greater part of the year.

57. The average density of population in Chittagong is reduced by the figures for the Cox's Bazar subdivision which includes the storm-swept islands of Maheshkhal and Banskhali and, in the south, an inhospitable region of hill and jungle with only a few infrequent patches of cultivable land. The Chittagong Hill Tracts and Hill Tippera are even less capable of supporting a large population than the south of Chittagong, and their inhabitants are mostly migratory savages, whose primitive methods of cultivation necessitate the clearing of new land every other year; after two years' crops have been obtained, the fields become unfit for further tillage until the weeds that follow in the wake of cultivation have been exterminated by reafforestation. Hill Tippera includes a strip of comparatively level land along its western boundary, and here cultivation is more permanent and the population is less sparse.

South Bihar.

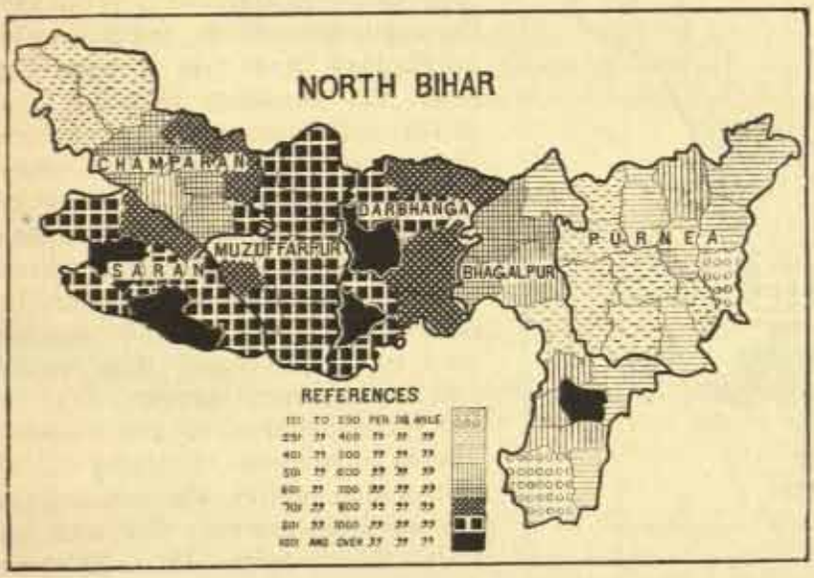
58. South Bihar is fairly populous along the bank of the Ganges, but further away, the surface rises and there is less cultivation, and the population becomes more and more scanty. Patna has a larger cultivable area than the other South Bihar districts, and its density is greater, but if its two large towns be excluded it has less than 700 persons to the square mile, which would not be considered a very high figure in East Bengal or North Bihar. Monghyr which comes next, owes its position to the tract north of the Ganges where there are 693 persons to the square mile compared with only 412 in the South Gangetic part of the district. In the south, on the border of Hazaribagh, there are barely 250 persons to the square mile. In Shahabad the population is greatest in the north and east, on the banks of the Ganges and the Sone, where much of the land is irrigated, and decreases rapidly towards the south and south-east, where the Kaimur Hills afford but small space for cultivation. The Bhabua thana, with 181 persons to the square mile, has the scantiest population of any tract in South Bihar. In



Gaya also the population is very sparse along the southern boundary of the district where, as in the south of Shahabad, a considerable area ought really to be included in the Chota Nagpur Plateau. It is greatest in the north, but even here the absolute density is not excessive. Compared with the productiveness of the soil, however, it appears to be very considerable, and there are numerous emigrants from this district who seek a livelihood in the more favoured tracts of Bengal Proper.

59. Although exceeded by the figures for a few individual districts elsewhere the portion of North Bihar which comprises the districts of Saran, Muzaffarpur and Darbhanga has a more teeming population than any other tract of equal size in the province. Towards the east and north-west the population rapidly declines. In the most populous district, Muzaffarpur, the inhabitants are very evenly distributed; in only a small tract to the west does the average number per square mile fall below 900, while in no part of the district does it exceed 1,000. Saran contains three police circles where this high ratio is exceeded, but it also contains five where it falls below 900. The population of this district is greatest in the south and centre and least in the north and north-east. Darbhanga, which lies between Muzaffarpur and Bhagalpur, partakes in the west, of the character of the former, and in the east, of that of the latter district. In only one of the western thanas does the density of population fall below 900 per square mile, while in none of the eastern thanas does it reach 800. So far as can be judged from the results of the recent census these inequalities in the distribution of the people are gradually being obliterated.

North Bihar.



The most populous part of Champaran is east of the Buri Gandak, where the soil is the stiff clay of the older alluvium. Then comes the tract between the two Gandaks, and lastly the north-western corner where the surface rises towards the Somesvar hills on the Nepal frontier, and a great part of the country is still covered with forest. This tract is very unhealthy and the aboriginal Thárus alone seem able to inhabit it with impunity. The district generally is less healthy than those previously described; it is also less fertile and has more recently been reclaimed from forest.* It needs more irrigation, and when the canals recently sanctioned have been completed, it seems probable that the less settled parts of the district will be rapidly filled up.

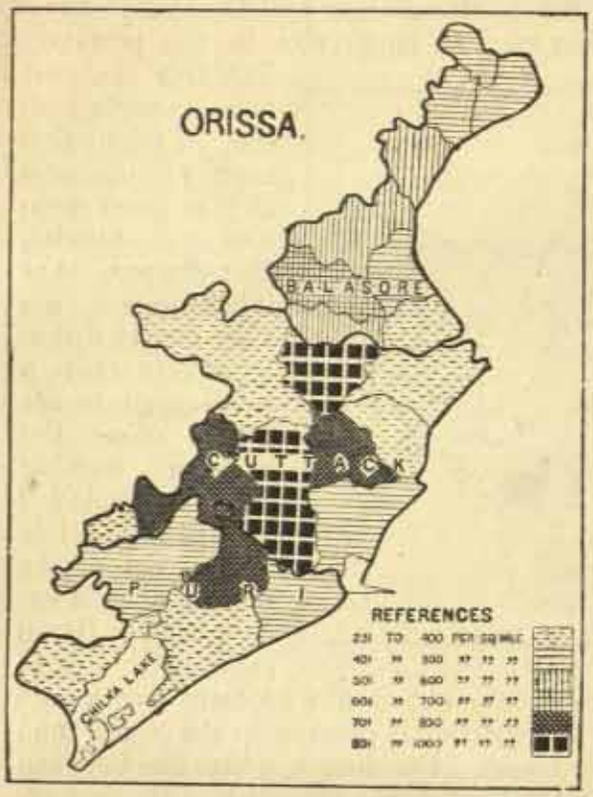
60. About half of Bhagalpur lies south of the Ganges, but, except for one thana in the extreme south-west, there is not much difference between the density of the population on the north bank and that on the south. The diminished density compared with the districts to the west of it may be ascribed to a less fertile soil and less healthy climate, combined in parts with occasional floods from the Kosi which leave behind them a barren sediment of sand. The low density in Purnea is due mainly to the unhealthiness of the climate and to the infertility of the western half of the district, where the soil consists mainly of sand deposited by the Kosi which, moreover, frequently causes

* Champaran is derived from *Champá áranya*, the forest of Champa trees.

disastrous floods. In the south-east there are numerous marshes which are not fit for cultivation. Further north, on both sides of the Mahánandá there is a somewhat greater population, but in only one thana, Bahadurganj, does it exceed 500 to the square mile.

Orissa.

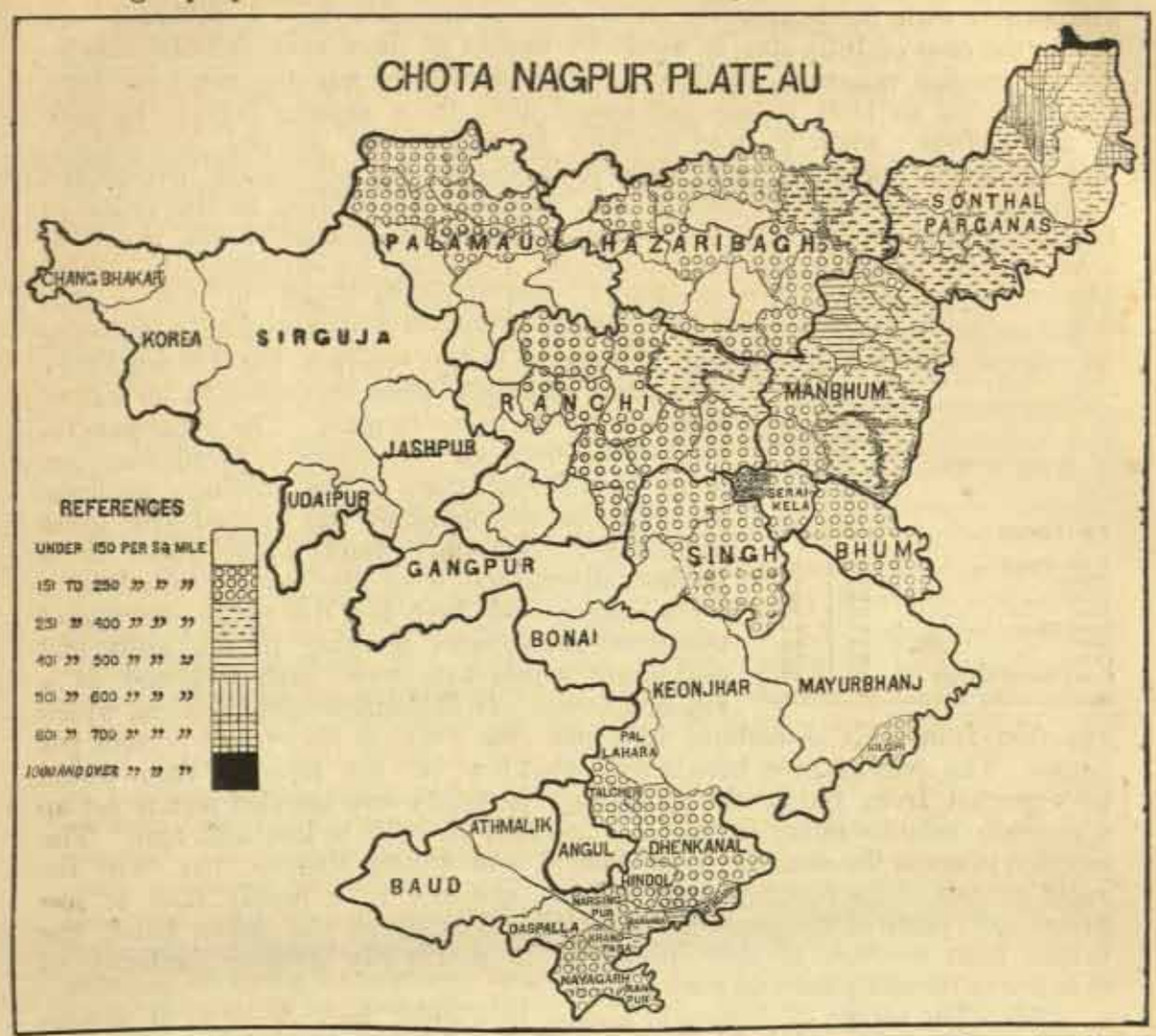
61. Orissa lies between the hills of the Tributary States and the sea. Its population is sparse on the lower slopes of the hills and also on the sea-coast where there is a strip of uncultivable land impregnated with salt; for some distance beyond it there are numerous marshes, and cyclones occasionally inflict great damage. Between these two extremes the area under cultivation is great and the population dense, but it is only in Cuttack that this is apparent from the statistics showing the density of thanas; the other two districts are so narrow that every thana contains a tract of coast or hill or both. Cuttack enjoys an extensive system of irrigation which protects the crops from failure in seasons of drought, and enables land to be cultivated that would otherwise remain barren. One of the central thanas of this district has a population of nearly 1,000 to the square mile, whereas Aul on the sea-coast has only 287, and the hilly Banki only 319. Balasore and Puri, having a smaller space intervening between the hills and the coast, and being for the most part dependent solely on the rainfall—the south of Balasore enjoys a certain amount of irrigation—the population is less dense. In Puri, moreover, the extensive Chilka lake occupies a large area and keeps down the average density. Its most closely-inhabited thana is Pipli with 767 persons to the square mile. In Balasore the greatest density, 674 persons to the square mile, is found in Bhadrak.



Chota Nagpur Plateau.

62. The Chota Nagpur plateau has an average population of 152 persons to the square mile. Speaking generally, the population tends to decrease from the east, where the general elevation is low and the transition from plains to hills is gradual, to the west, where the country is everywhere more hilly and broken. The most densely inhabited district is the Sonthal Parganas which projects beyond the rest of the formation and is surrounded on three sides by the plains of Bengal Proper and Bihar. On the north-east and north-west the administrative boundary of the district includes a considerable area of low and level country which carries a fairly dense population. The *Damin-i-koh*, or reserve for aboriginal tribes in the centre of the district, is a typical part of Chota Nagpur, and is very sparsely inhabited. The adjoining district, Manbhum, ranks next in point of density, chiefly on account of the alluvial tract along the course of the Damodar. In the north-west and south the country becomes more broken and the inhabitants are fewer, except in the neighbourhood of the great Jheria coal-field, where the mines attract large numbers of coolies. Hazaribagh borders on Gaya, but it derives no benefit from its proximity to the plains, as the natural dividing line lies far to the north of the district boundary and the lower slopes of the hills lie within the district of Gaya. The population is greatest in the west in the valley of the Barakar river, where there is a fair extent of level country, and the coal mines support a considerable number of

labourers. The country west and south-west of the central plateau contains the larger proportion of hill and ravine and has very few inhabitants.



63. Ranchi, which has almost the same general density as Hazaribagh, exhibits a steady decrease of population from the north-east to the west and south-west. In Singhbhum the density is fairly uniform except in the south-west corner bordering on Gangpur which consists of a mass of mountains rising to the height of about 3,500 feet. The Palamau district has a fair population in the alluvial valleys along the course of the North Koel river and on the right bank of the Sone, but away from these rivers, and especially in the south and west, the country is wild and inhospitable and its inhabitants are few in number.

The Orissa States have a slightly greater density than Palamau, but this is due to the figures for the eastern States which march with the Orissa plains. Some of the latter include a considerable area of level country and support a considerable population, amounting in the case of the little State of Tigaria to 492 to the square mile. Further west the density rapidly falls and Pal Lahara and Athmallik have respectively only 49 and 56 inhabitants to the square mile. Angul is surrounded by these States, and its circumstances are the same. At the bottom of the list are the Tributary States of Chota Nagpur with an average of only 62 inhabitants per square mile. The figure would be even lower but for the inclusion of the political States of Kharsawan and Seraikela which lie within the confines of Singhbhum. If these be excluded, the greatest density is found in Gangpur, and the least in Korea and Changbhakar.

HOUSES AND HOUSE-ROOM.

DEFINITION OF HOUSE AND VALUE OF THE RETURN.

64. In 1872 and 1881, a house was defined as the dwelling place of one or more families having a separate independent entrance from the common way. In Bengal, this definition was open to several objections, from the point of view of the actual enumeration, and it was therefore abandoned in

1891, when a house was defined as the residence of a commensal family.* The same definition was repeated at the present Census. There can be no comparison with the figures for 1881, when the unit was a different one and in the case of 1891 also it would be unsafe to draw very definite conclusions from the variations disclosed. The definition was the same on both occasions, but in 1891 it was not issued until three months before the date of the Census, when it was too late to ensure its general application, and in several of the Bihar districts it was to a great extent overlooked in favour of the older definition. It was also not applied in the cities of Calcutta and Howrah. On the present occasion the definition was issued at a very early stage in the operations; the necessity for enforcing its careful observance was constantly inculcated, and there is reason to believe that it has on the whole been properly applied. There was occasionally a tendency to conceal separate messes with a view to escape taxation, but it is not likely that such attempts were successful to an extent sufficient to vitiate the returns. The great general uniformity between the average population per house in different parts of the province, confirms the view that the definition was applied with great uniformity. The only tract where there is a marked divergence from the general standard is West Bengal, and here it is due to the presence of numerous immigrants working in the mills and coal mines whose huts were each reckoned as a separate house. In Midnapore and Bankura, which are free from this disturbing influence, the average rises to 4·8 persons per house. The average even here is somewhat low, but not more so than would be expected from the tendency in these parts of every married man to set up a separate establishment as soon as he takes his wife to live with him. The practice is much the same in East Bengal and Chota Nagpur, but here the rapid growth of the population gives more children to a family than in less progressive parts of the province. In North Bengal, on the other hand, the rather high average is due probably to a slightly greater tendency of families to remain joint and commensal.

Natural Division.	Number of persons per house.	
	1891.	1901.
PROVINCE	*5·4*	*5·2*
West Bengal	4·8	4·4
Central ,, ...	5·3	5·1
North ,, ...	5·4	5·3
East ,, ...	5·3	5·4
South Bihar	5·7	5·2
North ,, ...	5·8	5·2
Orissa	5·2	5·1
Chota Nagpur Plateau ...	5·5	5·3

65. The return of houses in Europe is of great importance as it throws light on the question of overcrowding. But in Bengal, except in a few large cities, the statistics have no bearing on this subject. The unit is social, not structural, and while in many cases a census house may comprise several distinct buildings, in others one building may be divided into several 'houses.' At the same time the figures have a certain value of their own.

SIGNIFICANCE OF FIGURES SHOWING NUMBER OF PERSONS PER HOUSE.

66. If it were not for the joint family system, *i.e.*, if every man living with his wife had a house of his own, the varying size of the house, which would then represent the family in the ordinary acceptation of the term, would afford a good index to the progressiveness or decadence of the population. As matters stand, however, these differences may equally well be due to the varying extent to which, in different districts or at different times, married sons remain in, or leave, the parental home. Some idea of the practice prevailing in different districts may be gathered by comparing the number of houses in a district with the number of married females over 15 years of age, but here too the comparison is obscured by the fact that in some parts the proportion of married females only slightly over 15 years of age, is higher than in others. This is especially the case in parts of Bihar, but even if this be allowed for, it is clear that in this part of the province, sons do not so readily leave their parents' house for a new one of their own, as they do elsewhere. The setting up of a separate house is a less simple matter in the crowded village sites of Bihar than it is in Bengal Proper; the people, moreover, are poorer, and many of the men who go to Bengal for work leave their wives in the charge of relatives who will look after them.

Natural Division.	Number of houses per 100 married females aged 15 and over.
PROVINCE	*96*
West Bengal	119
Central ,, ...	109
North ,, ...	101
East ,, ...	97
South Bihar	84
North ,, ...	85
Orissa	91
Chota Nagpur Plateau ...	96

* This subject has been more fully dealt with in the Administrative Report on the Census Operations.

TOWNS AND VILLAGES.

DEFINITION AND NUMBER OF TOWNS.

67. For the purpose of the census the word town was held to include all municipalities and cantonments and such other continuous collections of houses as it might be decided to treat as towns. The total number of towns, as thus defined in Bengal, is 199, of which 190 are in British Territory and 9 in the Native States. Of the former number, 157 are municipalities and 9 are cantonments, and 24 are other places treated as town on account of their urban character. The statistics regarding these towns are contained in Imperial Tables IV and V. The combined number of towns and villages and their distribution according to population will be found in Imperial Table III. The corresponding proportional figures are given in Subsidiary Table No. II at the end of this chapter.

DISTRIBUTION OF POPULATION BETWEEN TOWNS AND VILLAGES.

68. In the province as a whole, out of every 100 persons, 95 live in villages and only 5 in towns. Bengal is a distinctly agricultural country, and many even of the so-called towns are merely overgrown villages. The urban population is considerable only in Central Bengal where the inclusion of Calcutta and its environs brings the proportion of urban population up to 19 per cent. If they be excluded, it is only a little more than 7 per cent. The second place is shared by West Bengal, with its flourishing industrial centres at Howrah, Bally, Serampore and Raniganj, and South Bihar, with its ancient towns of Patna, Gaya and Bihar; in both these tracts 7 per cent. of the inhabitants live in urban areas. Orissa follows with an urban population of four per cent., then North Bihar and North Bengal with three per cent. and lastly East Bengal and the Chota Nagpur Plateau with only two per cent. The order in which the different tracts stand is sufficient to show the want of any connection between the prosperity of the people and the growth of towns. The general standard of comfort is highest in Eastern Bengal, although it has the smallest proportion of persons living in towns. South Bihar ranks comparatively high in this respect and yet it includes the poorest part of the province. The older towns, which usually owed their origin to the presence of a Native Court and its entourage, have few industries, and such as they possess are for the most part decadent, while in the newer towns the industries are carried on by foreign capital, and even the employés come from other parts of the country. The mills of Howrah and the coal mines of Asansol are alike worked with British capital by coolies from Bihar and the United Provinces, and the shop-keepers who are enriched by the trade they bring, are also for the most part foreigners. The district-born, as a class, have so far benefited but little by the growth of new industries. In the general account of towns which follows it will be convenient to deal not only with their present condition but also with the progress they have made in the past, instead of leaving the latter subject for discussion in a separate chapter as has been done in the case of variations in the population generally.

TOWNS IN WEST BENGAL.

69. Burdwan has six towns, all municipalities. The largest is Burdwan, itself, with 35,022 inhabitants. It has no special industries and owes its position to the fact that it is the head-quarters of the district and of the Burdwan Raj. It is growing in importance as a railway centre and appears to be fairly prosperous, but in spite of this the population has grown but little of late years. It has been deserted since 1891 by the Commissioner of the Division, who has removed with his establishments to Chinsura. Kalna and Katwa are on the Bhágirathi and enjoyed a considerable trade before the advent of the railway, but their prosperity has long been waning. The only progressive towns are Raniganj and Asansol in the west of the District which owe much of their development to the neighbourhood of the coal mines. Raniganj has, besides, large pottery works and is one of the busiest places in Bengal, while Asansol is an important railway junction; both are growing rapidly. The only town in Birbhum is Suri, the district head-quarters. It is small and has no special industries. Bankura has three towns with an aggregate population of more than 53,000; they are of small commercial importance and are unprogressive, but Vishnupur, the ancient capital, is still noted for its embroidered silk scarves and shawls.

Midnapore contains seven towns, none of which are of much moment. The largest is the head-quarters station with 33,140 persons, but it has no great industry or trade and shows no tendency to grow. Tamluk, the head quarters of the boat traffic on the Rupnarayan, has added 22 per cent. to its population, but it still has barely 8,000 inhabitants. There are eight towns in Hooghly of which by far the most important is Serampore on the bank of the Hooghly, which contains many important mills and is growing very rapidly, having added 40 per cent. to its population between 1881 and 1891 and another 24 per cent. during the decade preceding the present census; its population is now 44,451. Bhadreswar, also on the river, though barely a third the size of Serampore, has of late grown more rapidly, and now contains 57 per cent. more inhabitants than in 1891. This is due to the growth of a local jute mill and to the fact that the operatives in some of the mills on the opposite side of the river reside in this town. Hooghly itself (including Chinsura) is decadent. Bally in the Howrah district is prosperous, but has not grown much during the last decade. Howrah city shows a remarkable development, and is now, next to Calcutta, the largest in the Province. In view of its great and growing importance, it seems desirable to consider its history and statistics in some detail, and this I am enabled to do, thanks to a very full and interesting report from Mr. Duke, the Magistrate, from which the greater part of the following three paragraphs has been extracted almost *verbatim*.

HOWRAH.

70. Howrah, which stretches for seven miles along the Hooghly and comprises an area of about $8\frac{1}{2}$ square miles, is a town of entirely modern growth. In 1785 it was a small village held by one Mr. Lovett who found it so unprofitable that he petitioned to be allowed to relinquish it. The artillery part of the Nawab Nazim was at one stage located there, and the artillery practising ground is shown in old maps, north of the railway where the most densely populated part of the town now is. The Government salt golas were located within the limits of the town, and it gradually grew in importance until, in 1841, it became the seat of a separate Magistracy. In 1853 a further impetus was given to its development by the establishment there of the head-quarters of the East Indian Railway, the first part of which was opened to traffic in 1856. In 1874 the pontoon bridge was opened, and the improved communication with Calcutta thus afforded enabled many of the people employed in the metropolis to reside on the right bank of the Hooghly. Meantime there had been a continuous development of various large industries conducted according to European methods, of which the rope works at Ghuseri and Shalimar, founded a century ago, are probably the oldest. Then followed iron foundries and engineering works, and then the rise of the jute and cotton spinning industries, until at the present time there are no less than 40 registered factories working within the municipal limits. Excluding 13,815 persons in Bally, which then belonged to Howrah but has since been made a separate municipality, the population in 1872 was 83,969. In 1881 it was 90,391 and in 1891, 116,606; at the present census it is 157,594, or 35 per cent. more than it was ten years ago, and nearly 88 per cent. more than in 1872.

71. This rapid expansion is due almost entirely to the great industrial development that has taken place. The growing demand for labour has been met by the immigration of labourers from outside the district and about two-thirds of the total number of inhabitants are immigrants, chiefly from up-country. More than 33,000 come from the United Provinces and about 25,000 from Bihar. Amongst these foreigners there is an enormous excess of males who outnumber the females in the ratio of 2 to 1. As already stated they are for the most part operatives in the mills, who look forward to returning to their homes as soon as they have accumulated sufficient funds. In the meantime they live huddled together in crowded lodging-houses as close as possible to the mills and factories where they work. This over-crowding is not a necessary condition in Howrah, as there is ample room for building at no

DISTRICT.	Number of persons.
UNITED PROVINCES ...	*33,142*
Patna	5,905
Gaya	2,824
Shahabad	6,389
Saran	6,772
Muzaffarpur	1,822
Monghyr	2,293
Cuttack	3,166
Hooghly	16,939
Midnapore	6,276
Calcutta	4,459
24-Parganas	3,566
Burdwan	1,953

great distance from the centres of industry. It proceeds partly from the desire of the operatives to live as near as possible to their work, partly from their poverty which leaves them little to spare for rent, and partly from the pressure of municipal taxation which falls heaviest on huts and discourages the construction of new ones, unless there is a certainty of their being kept full of lodgers. The *basti* clauses of the Municipal Act enable roads to be opened out and drainage effected; but there is at present no law under which it is possible to prevent over-crowding, which sometimes attains truly astonishing proportions. The density of the population in each ward is noted in the margin; but as many of the wards contain large areas of uninhabited land, the figures afford no index to the density of the population in the neighbourhood of the mills where over-crowding chiefly occurs. Fortunately the lodging-houses are of very flimsy materials, and there is much natural ventilation, so that the effects are probably less harmful than they would be in the case of masonry buildings.

Ward number.	Number of persons per acre.	Ward number.	Number of persons per acre.
1	36	6	16
2	14	7	62
3	58	8	33
4	60	9	9
5	83	10	8

72. The rapid growth of the population has led to a great increase in the value of house property and land, and the municipal income in 1901 exceeded five lakhs of rupees, compared with less than 2¾ lakhs ten years previously. The greatest recent improvement was the introduction, in 1896, of a filtered-water supply. The town had previously suffered very severely from water famines and cholera, but since the opening of the water-works the former have of course ceased, and the latter has not appeared in epidemic form. On the other hand the general death rate has risen, and it is sometimes asserted that this is due to the water-works, as the drains are now seldom dry, and breeding grounds for mosquitoes have consequently been multiplied. The town, moreover, is low and naturally exposed to water-logging. What it now needs, more than any thing, is an efficient system of drainage.

CENTRAL BENGAL TOWNS.

73. Excluding Calcutta there are twenty five municipalities in the 24-Parganas. Of these the three suburban municipalities, Cossipore-Chitpur, Manicktala and Garden Reach were enumerated as part of Calcutta, and have been dealt with in the Calcutta Census Report. Most of the towns in this district lie on the bank of the Hooghly, and owe their prosperity to the jute mills and other industries which are carried on mainly by European capital. The most progressive are Naihati and Bhatpara, which together have grown by more than 50 per cent. during the decade. Budge-Budge is also a growing town, but as it was only created a municipality in 1900 the figures for previous enumerations are not available. The same remarks apply to Titaghar and Panihati, which were formed into municipalities in 1895 and 1900 respectively. North and South Dum-Dum are slightly decadent and so are several others, including the head-quarters of the Baraset subdivision. Of the inland towns Basirhat alone shows a considerable gain of population. Nadia has nine towns, but only one, Ranaghat, can boast of an increase. The apparent loss of population in Kushtia and Kumarkhali is due to the exclusion of a considerable area from the municipal limits; Kushtia, at least, on its present area, has probably gained rather than lost ground. Santipur was once the centre of a flourishing weaving industry and its muslins had an European reputation, but the modern machine-made article has driven them out of the market; the weavers are no longer prosperous, and in many cases they have been driven to supplement the earnings from their looms by agricultural pursuits. There was also at one time a considerable trade in date-sugar, but this too is becoming less profitable. The earthquake of 1897 destroyed many of the largest buildings, and these the impoverished owners have been unable to replace. The result of these adverse conditions is a decline of 11½ per cent. Krishnagar, the district head-quarters, is also decadent, chiefly on account of malarial fever which is very prevalent in the older part of the town. The decline in Nabadvip is to a great extent fictitious. It is celebrated amongst Baishnabs as the birth-place of Chaitanya, and is a favourite place of pilgrimage. There happened to be very few

pilgrims on the date of the present Census, whereas in 1891 and 1881 a great number were present, especially in the former year, when the Dhulut ceremony was in progress at the time when the Census was taken.

74. The towns of the Murshidabad district are either stationary or decadent. Berhampur, the district head-quarters, and Jangipur have gained, and Kandi has lost slightly, while Murshidabad and Azimganj, taken together (they were not separately enumerated in 1891) show a heavy falling off. Trade has declined, and of late years the ivory work and bell-metal manufactures, for which Murshidabad is famous, have become less profitable. The earthquake of 1897 destroyed a great number of houses which have not been rebuilt, and these two municipalities now wear a gloomy aspect of neglect and decay. The Jessore district has an urban population of only 1 per cent. which is collected in three small towns, all of which have lost ground since 1891. Kotechandpur has a considerable trade, chiefly in date-sugar: but this seems to be becoming less profitable.

North Bengal Towns.

75. North Bengal has no important towns. The head-quarters of the Dinajpur district and Rampur Boalia and Nator in Rajshahi have a slightly smaller population now than they had in 1872. Rampur Boalia enjoyed a considerable trade when the Padma flowed beside it, but its prosperity has departed now that the river has receded to a distance of several miles. English Bazar and old Malda in the Malda district have lost ground owing to the diversion of trade to other routes. Nawabganj on the Máhanandá, near its junction with the Ganges, has the largest population of any town in this district; but it is not a municipality, and comparative figures for earlier enumerations are not available. Jalpaiguri is progressive but still has less than 10,000 inhabitants. Rangpur has grown steadily since 1881, but its population is still only slightly greater than in 1872. The other places treated as towns in this district are subdivisional head-quarters or railway centres, but they are not municipalities and their population in 1891 is not known. Bogra is growing slowly, but Sherpur in the same district is stationary. In Pabna the head-quarters station has grown by nearly 12 per cent., while the flourishing jute mart at Serajganj which, with 23,114 inhabitants, is the largest town in North Bengal, has lost ground, though to a very slight extent.

76. The town of Darjeeling is peculiarly circumstanced, as it owes its existence not to local requirements but to the fact that it has been selected as the summer head-quarters of the Bengal Government, and is also largely resorted to during the hot season by Europeans, whose permanent residence is in the plains. Its inhabitants in 1872, before the days of the railway, numbered only 3,157. During the next nine years the population increased by more than 100 per cent., and it again doubled itself between the years 1881 and 1891. Since 1891 the growth has been less rapid, and at the present Census an increase of rather less than 20 per cent. has been recorded. Being a hill station its population is at a minimum during the cold weather months, when the general Census was taken. In order to ascertain its population during the season when it is chiefly resorted to, a special enumeration was effected on the 21st September 1900. This disclosed a population of 23,852, or nearly 50 per cent. more than that at the time of the regular Census taken about 5 months later. Kurseong is another hill station, also in the Darjeeling district, but its elevation is considerably lower, and it does not enjoy the same reputation as a sanitarium. Its population is small, and it shows no signs of any great development.

East Bengal Towns.

77. Although the urban population of East Bengal is relatively very small, this tract contains one of the largest towns in the Province. Dacca was a flourishing city long before the days of British rule, and for many years it was the capital of the Nawabs. It was subsequently the emporium of the well-known muslins which were greatly in demand in Europe, and especially in France. In 1801 it was estimated to have a population of 200,000. Its prosperity was seriously affected by the French wars, and in 1814 the police tax was levied only on 21,361 houses, which would indicate a population of less than 110,000. The place continued to decline, and an enumeration effected in 1830 disclosed a population of only 66,989 persons. In 1872 when the growth of the jute

trade had begun to cause a revival, the number of inhabitants was 69,212. The town has grown steadily since that date and now has a population of 90,542, or more than 30 per cent. greater than it was 29 years ago. The progress of Narayanganj which is quite a modern town and owes its progress entirely to modern trade and industrial developments, has been much more rapid, and its present population of 24,472 is about double that recorded in 1881.

78. Seven of the eight municipalities in Mymensingh have improved their position since 1891, the increase being most marked in Nasirabad, Sherpur, and Kishorganj, and in Jamalpur, the terminus of the railway and an important station for the river steamers. These towns are all small, and in none of them does the population reach 18,000. The Faridpur district contains Madaripur whose present population (17,463) shows an increase compared with 1891, of about 22 per cent.; it is favourably situated at the junction of the Arielkhán and Kumár rivers, and is the centre of a flourishing jute trade. Faridpur town has grown slightly, but it is still small and has no manufactures or general trade. In Backergunge all the towns are progressive and Barisal with nearly 19,000 inhabitants has grown by 22 per cent. in the course of the last decade. Its most flourishing town, however, is Jhalakati which is one of the largest marts in East Bengal. It is still small but shows a very rapid rate of expansion, and its population has considerably more than doubled itself during the last ten years.

79. In Tippera the population is growing rapidly and the towns are keeping pace with the general progress of the district. The capital, Comilla, has been tapped by the railway since 1891 and the population, which now stands at 19,169, has grown by more than 30 per cent. Chandpur is a branch terminus of the railway, a port for river steamers and the local head-quarters of the jute trade. It is a very rising town, but as it does not find a place in the returns for previous enumerations, it is not possible to give an exact measure of its rate of expansion. Akhaura is coming into importance as the railway station for the Brahmanbaria subdivision, and Laksam as an important junction, but neither of these places has as yet attained the rank of towns. Noakhali has practically no urban population, and its one small town is Sudharam, the head-quarters station. In spite of the advent of the railway and the development of its port, the town of Chittagong shows an apparent loss of population compared with 1891. This is due partly to the fact that the railway quarter, with a population of 1,289 persons, lies beyond the municipal limits and partly to the census having been taken during some holidays when large numbers of the usual residents were absent at their permanent homes a few miles away. The Magistrate estimates that the number of these temporary absentees must have reached nearly 3,000. Including them and the railway people, the population would be about 26,500, instead of 22,140 at which it stands in the census tables.

80. South Bihar contains the oldest towns in the Province and Patna, Gaya, Bihar, and Monghyr have a very ancient history. Buchanan Hamilton estimated the population of Patna at 312,000, but his calculation referred to an area of 20 square miles, whereas the city, as now defined, extends over only 9 square miles. The population returned in 1872 was 158,900, but the accuracy of the enumeration was doubted, and it was thought that the real number of inhabitants was considerably greater. It is thus probable that the growth indicated by the census of 1881, which showed a population of 170,654, was fictitious. There was a falling off of 5,462 persons between 1881 and 1891, while the present census gives a population of only 134, 785, which represents a further decrease of more than 18 per cent. This is due mainly to the plague which was raging in the city at the time of the census, and not only killed a great number but drove many more away; no one who could go elsewhere remained in the stricken city, and many of the private houses and even shops were left empty or in charge only of a care-taker. The census thus entirely failed to afford a true indication of the normal population of the city at the present time. To ascertain this a fresh count was effected at the end of July, *i.e.*, about five months after the general census, when the plague scare had passed away and the people had once more settled down.

South Bihar Towns—Patna.

This second enumeration disclosed a population of 153,739 or about 19,000, more than on the 1st March, but still nearly 7 per cent. below the corresponding figure for 1891. The relative deficiency of females on this occasion seems to indicate that some of those who had left during the plague scare had still not returned, but on the other hand there would in any case be more males in July than in March; fewer would then be absent in Bengal, and the schools, which were closed in March but open in July, would also swell their number. On the whole, then, except for the loss due to actual deaths from plague, which according to the returns amounted to 4,688, the population may be assumed to have regained its normal strength at the time of the second enumeration. The decrease due to causes other than deaths from plague which, allowing for an incomplete return of plague deaths is probably about 3 per cent., may be ascribed to the continued operation of the causes that led to a loss of population between 1881 and 1891, the chief of which was declining prosperity due to the gradual decay of the river-borne trade.

Census of	Population.		
	Total.	Males.	Females.
March 1901	134,785	67,038	67,747
July "	153,739	79,369	74,370

81. Of the other towns in the Patna District, Barh is stationary and Bihar has lost slightly, while in Dinapur the population is less by nearly 32 per cent. than it was in 1891. These variations are probably due entirely to the plague epidemic, and the heavy fall in Gaya and Tikari is attributable to the same cause. Plague appeared in Arrah shortly before the census and it is probably on this account that its population is slightly less than that returned ten years previously but, with the solitary exception of Sasaram all the towns in Shahabad seem to be decadent. Bhabua shows a heavy decrease during the last decade but it still has about as many inhabitants as in 1881. In Monghyr, as in Patna and Gaya, plague was raging when the census was taken, and the population returned in Monghyr town (35,880) was 37 per cent. less than it had been 10 years earlier. A second census, taken at the end of July when the plague had disappeared, gave a population of 50,133 or only 6,944 less than in 1891. Of this decrease 2,885 represents the number of reported deaths from plague. The railway town of Jamalpur in the same district suffered from plague, and some of the railway offices have been removed to Calcutta; its population has thus declined during the decade, but it still contains more inhabitants than it did in 1881.

Other Towns in South Bihar.

82. All the municipal towns of Saran have lost ground. Chapra, which showed a steady growth between 1872 and 1891, has now suffered a decrease of about 20 per cent. Revelganj has lost even more heavily. There was no second census in these towns, but there can be no doubt that plague is the principal, if not the only, cause of the falling off. At the same time it should be noted that Siwan, where there had been no plague, also shows a decline. Motihari and Bettiah, in Champaran, have both gained slightly, but in Muzaffarpur, Sitamarhi alone is progressive. Muzaffarpur town is less populous by 9 per cent. than it was ten years ago but it has still 3,000 more inhabitants than in 1881. Its decrease is to a great extent only apparent and, but for the exclusion of one of the old wards from the municipal limits and the temporary absence of a large number of people in connection with marriage ceremonies, it would probably have returned at least as many inhabitants as in 1891. In Darbhanga town also the fact that the date was an auspicious one for weddings led to the population being unusually small on the day of the census. Moreover, plague had just appeared and some of the residents whose permanent homes were elsewhere, had already begun to move away. The population of this town in 1891 was abnormally large on account of the presence of some 5,000 Brahmans who had come to partake of a feast given by the Maharaja. The decrease, which amounts to nearly 10 per cent., is attributable partly to these causes and partly to unhealthiness in recent years. In 1900 there were 924 deaths from cholera alone. Rosera, on the bank of the Gandak, is slightly decadent, as the railway has diverted some of the traffic that was formerly borne by the river. Samastipur and Madhubani, on the other hand, have benefited by the railway, especially the former. Comparative figures are not available, but it is known to be growing rapidly. Of the larger Bihar

North Bihar Towns.

towns Bhagalpur alone is progressive. It has grown steadily since 1872, but the increase of 9·6 during the last decade is greater than that brought out by the two previous enumerations put together. This is due mainly to the great development in the export trade in agricultural produce which has led to the opening of a second railway station and to a great extension of the accommodation provided for goods. On the south, moreover, the municipal limits have been enlarged.* The town is a very healthy one with a good natural system of drainage and a filtered water-supply which has been greatly increased and extended during the decade.

Purnea and Kishanganj both show a falling off in sympathy with the general decline of population in the Purnea district.

Orissa Towns.

83. There are only five towns in Orissa of which Cuttack, with 51,364 inhabitants, is the largest. It has grown by 9 per cent. since 1891 while the district taken as a whole has added only 6·4 per cent. to its population. The town is noted for its filagree work, but the advent of the railway has doubtless been the main factor in its somewhat rapid increase. At first sight the growth of Puri from 28,794 to 49,334 seems phenomenal, but it is due mainly to the presence of an unusually large crowd of pilgrims on account of the Gobind Duadasi festival. When preparing the preliminary totals, the Magistrate, Mr. Luson, caused these pilgrims to be separately counted, and they were thus found to number 17,085. If these be deducted, the population of the town falls to 32,259 or only 12 per cent. more than in 1891. There were doubtless some pilgrims present in the town at the last census also; it is impossible now to ascertain their number, but as there was then no great festival in progress, it was probably not very large.

The other Orissa towns call for no special remarks. Balasore and Jajpur are almost stationary while Kendrapara has lost population. Comparative figures for Bhadrak are not available.

Towns of the Chota Nagpur Plateau.

84. The towns of the Chota Nagpur Plateau are for the most part small and unimportant. Sahibganj, in the Sonthal Parganas, shows a serious decrease for which the plague scare is responsible; its normal population is probably quite as large as in 1891. Hazaribagh is slightly decadent, but on the other hand large increases have been registered in Ranchi and Purulia, amounting respectively to 28 and 42 per cent. In the case of Purulia the opening of the railway and the growth of the coolie recruiting business have added greatly to the population, but the reasons for the great expansion in Ranchi are less apparent. The other towns scarcely need separate mention. Chaibassa has grown and Raghunathpur and Lohardaga show a decrease. The population of the towns of the Tributary States was not shown on previous occasions, but they have no commercial importance, and it is not likely that there has been much change in recent years.

Summary.

85. If places now treated as towns for the first time be left out of account the urban population at the present census shows an increase of 5·4 per cent. over that recorded in 1891. The apparent rate of progress is greatest in Orissa where, however, it is to a great extent fictitious, being due mainly to the crowd of pilgrims collected at Puri in connection with a religious festival. In East Bengal which comes next, most of the towns are growing. The country is prosperous and trade is increasing, and the most progressive towns are those connected with the export trade in jute. The high rate of increase in Central Bengal is due to the expansion of Calcutta and the modern industrial towns on the banks of the Hooghly. In the other parts of this tract the old native industries have ceased to be profitable and the urban population is declining. West Bengal also owes its high position to the rapid expansion of new industries fostered by European

Natural Division.	Percentage of variation.
West Bengal	+12·5
Central	+12·6
North Bengal	+ 4·4
East Bengal	+15·7
South Bihar	−16·0
North Bihar	− 4·9
Orissa	+17·9
Chota Nagpur	+ 5·1
TOTAL ...	+5·4

* The true boundary had been lost sight of and a tract lying beyond it was treated as part of the Municipality, but the addition has not yet been legalised.

capital in Howrah and its environs, and at Raniganj and Asansol. The older towns show no tendency to grow. The progress of the urban population in North Bengal and Chota Nagpur is normal, and presents no points of interest, though it may be noted that, here too, the greatest advance has taken place in towns that have sprung up under British rule, such as Darjeeling, Purulia, and Ranchi. Throughout Bihar the urban population is stationary or decadent, save only in Bhagalpur and one or two smaller towns that owe their prosperity to the construction of new lines of railway. Few of the other towns contain any of the elements that make for progress, and many of them have suffered by the diversion of the traffic from the rivers to the railways. At the same time their position is not by any means so bad as the figures for the present census would indicate. The decrease in Muzaffarpur and Darbhanga is mainly fortuitous; and that in Chapra, Revelganj and many of the towns of South Bihar is chiefly on account of the plague epidemic. The second count taken in Patna and Monghyr, a few months after the regular census, showed that even then a great part of the loss had been made good, and unless plague or some other calamity intervenes it is probable that the next enumeration will show a general recovery throughout the plague area.

In India generally the urban population is small but it is particularly so in Bengal. In Madras, the Punjab and the United Provinces, the people who live in towns are more than twice, and in Bombay (excluding Sind) they are are nearly four times, as numerous, in proportion to the total population, as they are in this province. The province is also poor in respect of large cities. Calcutta, of course, is the largest city in India, but Howrah stands only fourteenth on the list; its population is exceeded by that of five cities in the United Provinces of Agra and Oudh, three in the Punjab, two in Bombay, one in Madras and one in Burma.

VILLAGES.

86. A village may be either a collection of houses bearing a separate name, or else the mauza, *i.e.*, the area treated as a village at the survey. The latter has the advantage of being a definite unit regarding which there can be no manner of doubt so long as the boundaries laid down at the survey are known, but it does not at the present time necessarily correspond to the residential village. One of the latter may spread over two mauzas or survey villages, or two may lie in the same mauza or the mauza may be altogether uninhabited. On the other hand, if the survey unit be neglected, it is very difficult to say with any precision what constitutes a village. There are many collections of houses which would by one person be called separate villages, while another would treat them as hamlets of other villages in the neighbourhood. In Bengal the records of the revenue survey have not been kept up to date and in most districts the survey mauza is no longer clearly traceable. Generally, therefore, a village was taken for census purposes to mean a residential village, *i.e.*, as a collection of houses bearing a separate name with its dependent hamlets. Where, however, there has been a recent cadastral survey, as in parts of Bihar and Orissa, the survey mauza was taken as the census unit.

Having regard to the indeterminate character of the villages, it is unnecessary to spend much time in discussing the statistics relating to them. The total number of villages is 222,656* and the average number of inhabitants is 355. Eleven per cent. of the rural population live in villages with a population exceeding 2,000; 42 per cent. in villages with a population of from 500 to 2,000 and 47 per cent. in villages with a population of less than 500. Assuming each village to be a point, the average distance between each would be rather less than a mile. The corresponding distance between each house would be about 212 yards.

* The figure 222,855 in Table III includes Towns.

SUBSIDIARY TABLE I.—DENSITY OF THE POPULATION.

NATURAL DIVISIONS, DISTRICTS, AND CITIES.	Mean density per square mile.				Variation Increase (+) or Decrease (−).			Net Variation, 1872-1901 (+) or (−).
	1901.	1891.	1881.	1872.	1891 to 1901.	1881 to 1891.	1872 to 1881.	
1	2	3	4	5	6	7	8	9
BENGAL	***413***	***393***	***366***	***328***	***+ 20***	***+ 27***	***+ 38***	***+ 85***
West Bengal	***391***	***351***	***350***	***345***	***+ 40***	***+ 21***	***− 15***	***+ 46***
RURAL	***375***	***329***	***333***	***469***	***+ 36***	***+ 16***	***+ 52***	***+ 110***
Howrah	1,351	1,263	1,142	1,078	+ 88	+ 121	+ 64	+ 273
Hooghly	846	841	800	923	+ 5	+ 41	− 123	− 76
Burdwan	570	519	518	553	+ 51	.....	− 35	+ 17
Midnapore	538	507	485	490	+ 31	+ 22	− 5	+ 48
Birbhum	515	456	452	486	+ 59	+ 4	− 34	+ 29
Bankura	426	408	397	369	+ 18	+ 11	+ 28	+ 57
CITIES	***14,714***	***11,284***	***8,746***	***8,148***	***+ 3,430***	***+ 2,538***	***+ 598***	***+ 6,566***
Howrah (Howrah and Bally)	16,023	12,119	9,603	8,889	+ 3,904	+ 2,516	+ 714	+ 7,134
Hooghly (Serampore)	11,112	8,988	6,390	6,110	+ 2,124	+ 2,598	+ 280	+ 5,002
Central Bengal	***608***	***578***	***560***	***501***	***+ 30***	***+ 18***	***+ 59***	***+ 107***
RURAL	***536***	***519***	***506***	***445***	***+ 16***	***+ 13***	***+ 61***	***+ 90***
24-Parganas	408	372	330	307	+ 36	+ 42	+ 23	+ 101
Murshidabad	622	584	572	566	+ 38	+ 12	+ 6	+ 56
Jessore	620	648	663	496	− 28	− 17	+ 167	+ 124
Nadia	597	589	586	537	+ 8	− 6	+ 58	+ 60
CITIES	***22,661***	***24,057***	***22,055***	***22,763***	***+ 5,604***	***+ 2,002***	***− 708***	***+ 6,898***
Calcutta	42,390	34,115	30,615	31,650	+ 8,275	+ 3,500	− 1,035	+ 10,740
24-Parganas (Cosipur-Chitpur, Garden Reach, and Manicktala.)	8,445	7,262	7,787	7,961	+ 1,183	− 525	− 174	+ 484
North Bengal	***428***	***458***	***459***	***417***	***+ 25***	***+ 19***	***+ 22***	***+ 66***
RURAL	***483***	***458***	***459***	***417***	***+ 25***	***+ 19***	***+ 22***	***+ 66***
Pabna	772	740	712	658	+ 32	+ 28	+ 54	+ 114
Bogra	629	563	505	472	+ 66	+ 58	+ 33	+ 157
Rangpur	617	591	600	616	+ 26	− 9	− 16	+ 1
Rajshahi	564	555	559	549	+ 9	− 4	+ 10	+ 15
Malda	466	429	374	356	+ 37	+ 55	+ 18	+ 110
Kuch Bihar	434	443	461	407	− 9	+ 18	+ 54	+ 27
Dinajpur	397	376	365	362	+ 21	+ 11	+ 3	+ 35
Jalpaiguri	206	230	196	141	+ 36	+ 34	+ 55	+ 125
Darjeeling	114	192	134	81	+ 22	+ 58	+ 53	+ 133
Sikkim	21	11	.....	.....	+ 10	.....	.....	.....
CITIES	.....	.....	.....	.....	.....	.....	.....	.....
East Bengal	***514***	***465***	***408***	***367***	***+ 49***	***+ 57***	***+ 41***	***+ 147***
RURAL	***511***	***463***	***405***	***365***	***+ 48***	***+ 58***	***+ 40***	***+ 146***
Dacca	923	834	725	634	+ 89	+ 109	+ 91	+ 289
Faridpur	849	800	727	671	+ 49	+ 73	+ 56	+ 178
Tippera	848	713	606	562	+ 135	+ 107	+ 44	+ 286
Noakhali	694	614	499	511	+ 80	+ 115	− 12	+ 183
Backergunge	628	591	521	518	+ 37	+ 70	+ 3	+ 110
Mymensingh	618	548	482	371	+ 70	+ 66	+ 111	+ 247
Khulna	603	567	520	504	+ 36	+ 47	+ 16	+ 99
Chittagong	543	518	454	452	+ 25	+ 64	+ 2	+ 91
Hill Tippera	42	34	23	8	+ 8	+ 11	+ 15	+ 34
Chittagong Hill Tracts	24	21	20	13	+ 3	+ 1	+ 7	+ 11
CITIES	***10,060***	***9,147***	***8,786***	***7,690***	***+ 913***	***+ 361***	***+ 1,096***	***+ 2,370***
Dacca (Dacca)	10,060	9,147	8,786	7,690	+ 913	+ 361	+ 1,096	+ 2,370
North Bihar	***636***	***635***	***600***	***526***	***+ 1***	***+ 35***	***+ 74***	***+ 110***
RURAL	***636***	***634***	***600***	***517***	***+ 2***	***+ 34***	***+ 73***	***+ 109***
Muzaffarpur	903	888	847	736	+ 15	+ 41	+ 111	+ 167
Saran	892	909	847	766	− 17	+ 62	+ 81	+ 126
Darbhanga	855	820	770	626	+ 35	+ 50	+ 144	+ 229
Champaran	507	527	488	408	− 20	+ 39	+ 80	+ 99
Bhagalpur	477	465	450	417	+ 12	+ 15	+ 33	+ 60
Purnea	375	389	370	343	− 14	+ 19	+ 27	+ 32
CITIES	***8,340***	***8,900***	***8,154***	***7,273***	***− 560***	***+ 746***	***+ 881***	***+ 1,067***
Bhagalpur	8,470	8,638	8,529	8,172	+ 832	+ 109	+ 357	+ 1,298
Darbhanga	9,463	10,508	9,422	7,677	− 1,045	+ 1,086	+ 1,745	+ 1,786
Muzaffarpur	7,682	8,198	7,076	6,373	− 926	+ 1,122	+ 703	+ 1,299
Chapra	6,557	8,193	7,381	6,618	− 1,636	+ 812	+ 769	− 55

SUBSIDIARY TABLE I.—DENSITY OF THE POPULATION—*concluded.*

NATURAL DIVISION, DISTRICTS, AND CITIES.	Mean density per square mile.				Variation increase (+) or Decrease (−).			Net Variation, 1872-1901, (+) or (−).
	1901.	1891.	1881.	1872.	1891 to 1901.	1881 to 1891.	1872 to 1881.	
1	2	3	4	5	6	7	8	9
South Bihar	***511***	***531***	***517***	***466***	− ***20***	+ ***14***	+ ***51***	+ ***45***
RURAL	*492*	*509*	*495*	*445*	− *17*	+ *14*	+ *50*	+ *47*
Patna	694	758	747	659	− 64	+ 11	+ 88	+ 35
Monghyr	519	505	484	448	+ 14	+ 21	+ 41	+ 71
Shahabad	449	471	443	391	− 22	+ 28	+ 54	+ 58
Gaya	423	437	402	401	− 15	+ 2	+ 35	+ 22
CITIES	*9,349*	*10,617*	*10,649*	*9,999*	− *1,277*	− *32*	+ *687*	− *762*
Patna (Patna and Bihar) ...	10,981	11,829	12,201	11,289	− 843	− 372	+ 912	− 308
Gaya (Gaya)	10,184	11,485	10,916	9,549	− 1,299	+ 567	+ 1,367	+ 635
Monghyr (Monghyr) ...	4,486	7,134	6,921	7,462	− 2,649	+ 213	− 541	− 2,977
Orissa	***508***	***475***	***445***	***378***	+ ***33***	+ ***30***	+ ***67***	+ ***130***
RURAL	*508*	*475*	*445*	*378*	+ *33*	+ *30*	+ *67*	+ *130*
Cuttack	568	534	494	425	+ 34	+ 40	+ 69	+ 143
Balasore	550	463	450	374	+ 87	+ 24	+ 85	+ 146
Puri	411	392	359	311	+ 19	+ 33	+ 48	+ 100
CITIES	.....	.....	.....	.....	.....	.....	.....	.....
Chota Nagpur Plateau	***152***	***141***	***124***	***94***	+ ***11***	+ ***17***	+ ***30***	+ ***58***
RURAL	*152*	*141*	*124*	*94*	+ *11*	+ *17*	+ *30*	+ *58*
Santhal Parganas	331	321	296	230	+ 10	+ 25	+ 66	+ 101
Manbhum	314	288	253	198	+ 26	+ 35	+ 57	+ 116
Hazaribagh	168	166	157	110	+ 2	+ 9	+ 47	+ 58
Ranchi	167	158	148	114	+ 9	+ 10	+ 34	+ 53
Singhbhum	163	145	121	85	+ 18	+ 24	+ 36	+ 78
Orissa Tributary States ...	135	118	98	77	+ 17	+ 20	+ 21	+ 58
Palamau	126	121	112	86	+ 5	+ 9	+ 26	+ 40
Angul	114	101	90	46	+ 13	+ 6	+ 49	+ 68
Chota Nagpur Tributary States.	62	55	42	31	+ 7	+ 13	+ 11	+ 31
CITIES	.....	.....	.....	.....	.....	.....	.....	.....

SUBSIDIARY TABLE No. II.—Distribution of Population between Towns and Villages.

Natural Divisions and Districts.	Average population.		Percentage of population living in—		Percentage of Urban population in Towns of—				Percentage of Rural population in Villages of—			
	Per town.	Per village.	Towns.	Villages.	20,000 and over.	10,000 to 20,000.	5,000 to 10,000.	Under 5,000	5,000 and over.	2,000 to 5,000.	500 to 2,000.	Under 500.
1	2	3	4	5	6	7	8	9	10	11	12	13
BENGAL	**20,504**	**335**	**5**	**95**	**62**	**24**	**12**	**2**	**1**	**16**	**42**	**47**
WEST BENGAL	**20,323**	**309**	**7**	**93**	**59**	**25**	**16**	.....	.....	**6**	**42**	**52**
Burdwan	11,454	395	5	95	40	36	24	.....	.....	7	52	41
Birbhum	8,602	239	1	99	.....	.....	100	.....	1	5	37	57
Bankura	17,758	120	5	95	39	61	.....	.....	.....	4	28	68
Midnapore	12,859	219	3	97	37	28	35	.....	.....	3	40	57
Hooghly	16,736	384	13	87	55	34	11	.....	.....	11	48	41
Howrah	88,129	465	20	80	90	10	.....	.....	.....	13	51	34
CENTRAL BENGAL	**33,455**	**369**	**19**	**81**	**78**	**15**	**6**	**1**	.....	**5**	**49**	**46**
24-Parganas	15,995	351	19	81	50	38	11	1	.....	3	43	54
Calcutta	847,796	.....	100	.....	100	.....	.....	.....	.....	.....	.....	.....
Nadia	10,595	461	5	95	64	11	16	9	.....	8	57	35
Murshidabad	13,181	345	5	95	32	68	.....	.....	.....	5	47	48
Jessore	7,999	366	1	99	.....	.....	80	20	.....	4	48	48
NORTH BENGAL	**8,586**	**295**	**3**	**97**	**22**	**51**	**15**	**12**	**2**	**9**	**36**	**53**
Rajshahi	15,121	226	2	98	71	.....	29	.....	.....	1	29	70
Dinajpur	13,430	198	1	99	.....	100	.....	.....	.....	3	20	77
Jalpaiguri	5,144	1,914	1	99	.....	.....	94	6	17	37	33	13
Darjeeling	10,636	400	8	92	.....	79	.....	21	2	13	45	40
Rangpur	4,914	408	1	99	.....	54	20	26	.....	16	43	41
Bogra	5,579	218	1	99	.....	.....	63	37	.....	3	29	68
Pabna	20,769	371	3	97	56	44	.....	.....	1	6	47	46
Malda	11,475	239	4	96	.....	89	.....	11	1	6	33	60
Kuch Bihar	3,315	467	2	98	.....	74	.....	26	6	9	54	31
Sikkim	.....	472	.....	100	.....	.....	.....	.....	0	32	34	25
EAST BENGAL	**5,177**	**398**	**2**	**98**	**53**	**32**	**14**	**1**	**2**	**12**	**46**	**40**
Khulna	8,078	357	2	98	.....	45	37	.....	1	6	47	46
Dacca	57,507	340	4	96	100	.....	.....	.....	1	9	48	42
Mymensingh	13,174	390	2	98	.....	94	6	.....	.....	7	44	49
Faridpur	14,556	361	1	99	.....	100	.....	.....	1	9	46	44
Backergunge	9,115	457	2	98	.....	72	22	6	1	16	54	29
Tippera	16,150	386	2	98	.....	81	19	.....	1	6	45	48
Noakhali	16,530	431	1	99	.....	.....	100	.....	7	21	42	30
Chittagong	12,992	915	2	98	86	.....	.....	14	9	37	42	12
Chittagong Hill Tracts ...	.....	421	.....	100	.....	.....	.....	.....	.....	2	34	64
Hill Tippera	9,513	112	5	95	.....	.....	100	.....	.....	.....	13	87
NORTH BIHAR	**22,312**	**602**	**3**	**97**	**66**	**20**	**14**	.....	**4**	**20**	**51**	**25**
Saran	29,280	397	3	97	57	19	24	.....	.....	9	46	45
Champaran	19,213	668	2	98	64	36	.....	.....	6	22	52	20
Muzaffarpur	22,014	647	3	97	76	13	11	.....	6	19	54	21
Darbhanga	25,848	669	3	97	64	27	9	.....	7	30	50	13
Bhagalpur	40,749	655	4	96	93	.....	7	.....	5	23	52	20
Purnea	10,480	240	1	99	.....	44	56	.....	1	15	53	31
SOUTH BIHAR	**22,203**	**543**	**7**	**93**	**70**	**19**	**9**	**2**	**1**	**12**	**42**	**45**
Patna	35,873	327	15	85	85	10	3	1	.....	3	36	61
Gaya	14,303	247	5	95	62	.....	32	6	.....	2	34	64
Shahabad	19,684	334	6	94	59	36	5	.....	.....	11	44	45
Monghyr	17,859	794	3	97	50	50	.....	.....	9	25	51	15
ORISSA	**27,908**	**332**	**4**	**96**	**73**	**27**	.....	.....	**1**	**6**	**46**	**47**
Cuttack	26,240	360	4	96	65	35	.....	.....	1	6	49	44
Balasore	19,699	307	3	97	53	47	.....	.....	1	4	45	50
Puri	42,334	312	5	95	100	.....	.....	.....	1	5	42	52
CHOTA NAGPUR PLATEAU ...	**8,343**	**190**	**2**	**98**	**16**	**26**	**46**	**12**	.....	**3**	**26**	**71**
Hazaribagh	11,943	129	3	97	.....	74	26	.....	.....	1	17	82
Ranchi	10,202	261	2	97	64	.....	28	8	.....	2	40	58
Palamau	4,723	192	2	98	.....	.....	61	39	1	5	23	71
Manbhum	8,779	231	2	98	.....	66	.....	34	.....	3	24	73
Singhbhum	8,653	192	1	99	.....	.....	100	.....	.....	1	21	78
Sonthal Parganas	7,745	195	1	99	.....	.....	100	.....	.....	6	27	67
Angul	.....	132	.....	100	.....	.....	.....	.....	.....	2	37	61
Chota Nagpur Tributary States	.....	214	.....	100	.....	.....	.....	.....	.....	1	28	71
Orissa Tributary States ...	5,635	163	1	99	.....	.....	80	20	.....	3	21	76

SUBSIDIARY TABLE III.—HOUSES AND HOUSE-ROOM.

Natural Divisions, Districts or Cities.	Average number of persons per House.			Average number of Houses per square mile.		
	1901.	1891.	1881.	1901.	1891.	1881.
1	2	3	4	5	6	7
BENGAL	**5·2**	**5·4**	**6·3**	**79**	**73**	**59**
WEST BENGAL	**4·4**	**4·5**	**5·2**	**131**	**121**	**102**
Burdwan	4·2	4·2	4·8	135	121	107
Birbhum	4·3	4·3	4·3	119	107	109
Bankura	4·5	4·9	6·1	88	82	64
Midnapore	4·5	4·9	6·0	111	104	82
Hooghly	3·9	3·6	4·08	223	233	196
Howrah	4·5	5·2	6·9	372	307	230
CENTRAL BENGAL	**5·1**	**5·3**	**6·0**	**151**	**136**	**119**
24-Parganas	5·4	5·7	5·4	182	157	145
Calcutta	6·7	10·1	17·7	6,274	3,376	4,317
Nadia	4·7	4·9	4·6	135	119	106
Murshidabad	4·7	4·4	4·7	152	131	120
Jessore	4·9	5·3	5·5	126	120	99
NORTH BENGAL	**5·3**	**5·4**	**6·0**	**90**	**83**	**72**
Rajshahi	5·1	5·7	6·4	111	106	93
Dinajpur	5·4	5·3	5·8	73	67	65
Jalpaiguri	5·0	5·4	6·1	53	43	33
Darjeeling	4·0	4·9	5·8	52	38	23
Rangpur	5·6	5·6	6·2	111	105	96
Bogra	5·2	5·3	6·0	105	98	66
Pabna	5·5	5·5	6·5	140	133	108
Malda	5·4	5·4	5·6	86	78	66
Kuch Bihar	5·1	4·9	5·2	85	90	88
Sikkim	5·3	.....	.....	.....	.....	.....
EAST BENGAL	**5·4**	**5·3**	**7·3**	**106**	**96**	**62**
Khulna	5·4	5·6	6·9	110	99	73
Dacca	5·6	5·4	6·7	169	157	110
Mymensingh	5·9	5·6	7·5	101	97	64
Faridpur	5·2	5·3	7·4	163	150	99
Backergunge	4·8	4·9	8·5	129	119	61
Tippera	5·8	5·7	8·4	144	125	72
Noakhali	5·5	5·3	9·4	127	114	53
Chittagong	4·9	4·9	5·3	110	101	82
Chittagong Hill Tracts	5·8	5·1	6·7	4	4	3
Hill Tippera	5·6	.....	.....	7	.....	.....
NORTH BIHAR	**5·2**	**5·8**	**6·6**	**120**	**109**	**90**
Saran	5·2	5·4	7·0	175	168	134
Champaran	5·7	6·2	6·08	89	84	80
Muzaffarpur	5·3	5·8	7·0	174	153	123
Darbhanga	5·0	6·1	7·2	173	137	108
Bhagalpur	5·5	5·7	6·1	99	83	75
Purnea	5·4	5·3	6·06	69	73	61
SOUTH BIHAR	**5·2**	**5·7**	**6·6**	**97**	**92**	**78**
Patna	5·2	6·1	6·2	150	140	134
Gaya	5·1	5·5	6·1	86	81	73
Shahabad	5·3	5·7	7·09	84	82	63
Monghyr	5·4	5·7	7·0	97	91	1
ORISSA	**5·1**	**5·2**	**5·9**	**99**	**90**	**75**
Cuttack	5·1	5·1	5·5	111	104	89
Balasore	5·9	5·4	5·8	89	89	78
Puri	4·6	5·2	6·2	99	72	51
CHOTA NAGPUR PLATEAU	**5·3**	**5·5**	**5·4**	**28**	**26**	**21**
Hazaribagh	5·3	5·9	5·9	33	37	26
Ranchi	5·3	5·0	5·6	31	31	24
Palamau	5·4	5·8	5·8	23	21	20
Manbhum	5·3	5·4	5·9	50	53	43
Singhbhum	5·3	5·3	5·2	31	27	23
Sonthal Parganas	5·8	6·1	6·3	57	51	45
Angul	4·9	9·0	9·07	23	11	29
Chota Nagpur Tributary States	6·1	5·3	6·0	16	10	7
Orissa do. do.	5·0	5·0	5·4	27	23	17

CITIES, 1901.

Cities.	Average number of persons per House.	Average number of Houses per square mile.	Cities.	Average number of persons per House.	Average number of Houses per square mile.
1	2	3	1	2	3
CALCUTTA	**6·7**	**6,274**			
Cossipur-Chitpur	6·9	1,948	Patna	4·1	3,573
Manicktollah	5·2	1,033	Bihar	4·9	1,019
Garden Reach	4·3	2,207	Gaya	5·0	1,999
Serampore	5·2	3,470	Darbhanga	4·7	1,987
Howrah	3·4	5,086	Muzaffarpur	4·7	1,000
Bally	3·9	2,377	Chapra	4·4	1,481
Dacca	5·2	1,907	Bhagalpur	5·2	1,713
			Monghyr	4·1	1,074

Figures for occupied houses for the cities for 1891 and 1881 being not available their average Nos. have not been given. The figures for Calcutta have been shown under Central Bengal. The number of houses has been taken from the first total reported by the District Officers.

Chapter II.

VARIATIONS IN THE POPULATION.

Preliminary Discussion.

87. In the previous chapter the statistics of the population as it stood on the 1st March 1901 have been considered. The present chapter deals with the variations that have taken place since 1872. The information regarding the early population of Bengal is very scanty and unreliable. In 1787 Sir William Jones thought that it amounted to 24 millions including part of the United Provinces then attached to Bengal. Five years later Mr. Colebrook placed it at 30 millions. In 1835 Mr. Adams assumed it to be 35 millions, but this estimate was thought too high and was reduced to 31 millions in 1844. In 1870 the population was held to be about 42 millions or less by more than a third than the figure disclosed by the first regular census of the province which was taken in 1872. The changes between the latter year and 1881 and between 1881 and 1891 have already been dealt with in the reports on those censuses; they will, therefore, be treated of very briefly, and the discussion in this chapter will refer chiefly to the variations which have occurred during the last decade. The figures showing the variations in the population of districts are contained in Imperial Table II. Similar information for thanas is given in Table I of the Volume of Provincial tables. Proportional figures illustrating some of the more important features of the statistics will be found in Subsidiary Table I at the end of this chapter. The map reproduced in the margin shows the variations since 1891 in the population of districts taken as a whole. Similar maps in which the thana is the unit will be found in paragraphs 222 to 231 below.

Introductory Remarks.



88. As already stated the population of the whole of Bengal now amounts to 78,493,410, *viz.*, 74,744,866 in British Territory and 3,748,544 in the Feudatory States. The increase since 1891 is 3,819,612 or 5·1 per cent., *viz.*, 3,397,905 or 4·7 per cent. in British Territory and 421,707 or 12·6 per cent. in the Native States It will be convenient to begin by examining the variations in each individual district and state, and then to proceed to a consideration of the changes in each natural division and in Bengal as a whole. Apart from the gradually increasing accuracy of each succeeding census, which has now for the first time ceased to be a factor of any importance in these Provinces, the growth of the population depends on (1) the excess of births over deaths and (2) migration. Migration will be dealt with in the next chapter, and it will be referred to here only so far as is necessary in order to gauge its effect on the variations in the population. Returns showing the number of births and deaths in each district are

Year of Census.	Population.	Percentage of variation since previous Census.
1872 ...	66,535,217	.....
1881 ...	69,533,940	+ 11·5
1891 ...	74,673,799	+ 7·3
1901 ...	78,493,410	+ 5·1

published in the annual reports of the Sanitary Commissioner for Bengal, but before any use is made of them, it is desirable to examine the figures in some detail in order to arrive at a conclusion as to their absolute and relative value.

THE VITAL STATISTICS RETURNS.

89. Births and deaths are recorded throughout the Province, except in Angul, the Chittagong Hill Tracts and the Feudatory States. The present system of mortuary registration was introduced in 1869. The duty of reporting deaths was imposed on the chaukidars, or village watchmen, and not on the relations of the deceased. In 1876 the system was extended to births but the returns received were so incomplete that they were soon discontinued and, except in towns for which special legislation was undertaken in 1873, deaths alone were registered until 1892. In that year the collection of statistics of births as well as of deaths was ordered, and the system now in vogue was introduced. In the Chaukidari Amendment Act of 1892 the reporting of vital occurrences was made one of the legal duties of the chaukidars. The births and deaths occurring in each chaukidar's beat are entered on leaflets by the chaukidar or, if he be illiterate, by the panchâyat, and taken by the former to the Police Station when he attends his weekly muster. A consolidated monthly return is compiled at the Police Station and submitted to the Civil Surgeon who prepares a similar return for the whole district. The accuracy of the reporting is checked by the police and other local officers, but the most valuable testing agency is that of the vaccination establishments, who are required to enquire regarding vital occurrences when on their rounds to test the vaccination operations. Errors and omissions thus brought to light are communicated to the District Magistrate and the chaukidars at fault are punished. In 1900, 276,077 births and 211,618 deaths were enquired into by inspectors and sub-inspectors of vaccination and the omission from the returns of 4,345 births and 2,910 deaths was thus detected. These omissions represent respectively 1·57 and 1·37 of the total number of births and deaths reported. The corresponding proportions in the previous year were 1·13 and 1·14. Under the special Act for towns above referred to, the reporting of births and deaths by the nearest male relative was made compulsory. The information was collected for some time by the municipal authorities, but the results were not satisfactory and the duty was subsequently transferred to the police.

COMPARISON WITH ESTIMATED BIRTH AND DEATH-RATES.

90. The above measures have led to a great improvement in the accuracy of the vital statistics. Prior to 1892, the returns of deaths, which alone were collected, were held to be highly unreliable, and it was thought that about half the total number were unreported. The latest authoritative pronouncement on the probable birth and death rates in Bengal is that of Mr. Hardy, F.I.A., F.S.S., the Actuary who was retained by the Census Commissioners of 1881 and 1891 to deal with the age returns prepared in connection with those enumerations. After an elaborate examination of the age statistics for both censuses he estimated that the birth and death rates in Bengal in 1891 were respectively 51·8 and 44·8 per 1,000. The rates calculated on the number of births and deaths actually reported in each year are noted in the margin. The figures are far below those of Mr. Hardy's estimate and, so far as this test goes, it is clear that they cannot yet pretend to a very close approach to accuracy. The returns for the latter half of the decade are, however, better than those for the earlier half. Moreover, it is not yet certain that the estimates based on the age statistics are altogether reliable. The age return is admittedly most inaccurate, and although errors due to the tendency of the people to mention one or other of certain favourite numbers as their age can be eliminated by means of elaborate methods of adjustment, it is not so easy to correct any general tendency to over-state or under-state the real age. Some check is afforded by the actual vairation in the population from one census to another, but this has hitherto

YEAR.	RATE PER 1,000 OF POPULATION OF 1891 OF REPORTED		
	Births.	Deaths.	Excess of births over deaths.
1892 ...	28·1	31·9	— 3·8
1893 ...	35·7	28·2	+ 7·5
1894 ...	32·8	34·9	— 2·1
1895 ...	34·6	31·4	+ 3·2
1896 ...	38·0	34·2	+ 3·8
1897 ...	37·0	32·9	+ 4·1
1898 ...	35·8	26·6	+ 9·2
1899 ...	43·0	31·2	+ 11·8
1900 ...	38·7	36·6	+ 2·1
Average of nine years	35·8	31·8	+ 4·0

NOTE.—Still-births are not included in the above figures. In 1899 when the largest number of still-births was recorded, the rate was about 1·2 per 1,000 of the population of 1891; 57 per cent. of the total number were males.

been obscured by the greater accuracy of each succeeding enumeration which has caused the growth of the population to seem greater than it really is. According to Mr. Hardy's estimated birth and death rates, the population should have grown at the rate of 7 per 1,000 per annum, *i.e.*, the 71,069,617 persons in the areas for which vital statistics have been recorded should have risen in the course of ten years to 76,204,194, but the census shows that the population of these areas in 1901 was only 74,428,193. The actual increase is thus 3,358,576, or at the rate of 4·6 per 1,000 per annum.* The net addition to this population indicated by a comparison of the returns for births and deaths during the decade is 3,159,200. For the purpose of this comparison the actual number of deaths reported in each year has been taken, but on the side of births the number returned in the years 1892 to 1900 inclusive has been increased by one-ninth to allow for the year 1891 when births were not registered. It will thus appear that there is a very close approximation between the actual growth of the total population and that indicated by the vital statistics of the Province.

EFFECT OF MIGRATION ON POPULATION AND VITAL STATISTICS.

91. In Subsidiary Table II the variations in the population of each district are compared with the excess of births over deaths, but before considering these figures it is necessary to point out that the census variations and the vital statistics are both complicated by the movements of the people from one part of the country to another. A district may have gained or lost largely owing to migration and the actual variation in such a case will not of course correspond with that indicated by the excess of births over deaths. It is sometimes assumed that the effect of migration can be discounted and the true growth ascertained comparing not the actual, but the district-born or natural population, which is arrived at by deducting from the actual population at the time of each census the immigrants from other districts and adding the number of persons born in the district who were enumerated beyond its borders. It is also assumed that the excess of births over deaths in any given area affords a direct measure of the natural growth of the population of that area. Neither of these assumptions is correct, and it is, therefore, necessary to examine briefly the real effect of the movements of the people on the population and the vital statistics.

92. For the purpose of this enquiry permanent and non-permanent migrants must be considered separately; the permanent migrants again must be subdivided into two groups, those who had changed their abode before the commencement of the decade under consideration, and those who did so during the said period. The former, having already severed all connection with their district of origin and having been counted at the earlier of the two enumerations in the district where they have made their home, can have no further effect on the population or on the birth and death returns of their old district. When they die, the death is registered in the district of adoption and the births of their children are also registered there. Neither class of events in any way affects the vital statistics of their district of origin. These persons, therefore, should be considered an integral part of their district of adoption and their number should not be deducted from the figures showing the actual population of that district nor added to those of the district of origin.

The circumstances of the second group are identical so far as their history after entering the new district is concerned. They should, like the first group, be treated as part of its population, and not as part of that of their old district. But the movement having taken place after the first and before the second enumeration, allowance must be made for it, when estimating the natural growth with reference to the census figures, by deducting from the actual population of the new district, and adding to that of the old, the number of permanent migrants who have come to the one from the other in the course of the decade. In this connection it must be remembered that the volume of the migration is not to be gauged by the mere difference between the figures for two successive enumerations; but allowance must also be made for deaths. Thus if 1,000 permanent emigrants born in district A were found in district B in 1891, and the same number were again found there in 1901, this would indicate, not that there had been no further movement, but that, on

* The growth of the population is, of course, progressive and if "r" be the annual rate of growth, the sum will be:—Population of 1901 = Population of 1891 × $(1 + r)^{10}$.

the average, just enough persons had gone from the one district to the other to fill death vacancies. If the death-rate be taken at 40 per 1,000, this would involve a migration of 40 persons a year or 400 persons in the decade. If the number of emigrants had risen from 1,000 to 1,500 persons then, in addition to this migration of 400 persons required to keep up the original number, there must have been a further settlement of about 62 persons a year to produce the increase of 500,* and the total number of persons going from district A to district B during the ten years will be 1,020.

The case of non-permanent migrants is quite different. These persons are usually males. They leave their wives behind them and return home themselves at frequent intervals. It is not probable that their absence has any appreciable effect on the size of their families. Their children are born, and the births of their children are registered, in their native district. On the other hand, the deaths occurring amongst them while absent from home are included in the mortuary returns of the district where they may happen to be at the time. Consequently, if the birth and death rates be calculated on the total population, the districts which attract a large number of temporary settlers will have a much smaller birth rate and a much higher death rate than those from which they come. In order to ascertain the true progress of a district from its vital statistics, a proportionate deduction must be made from the mortality returns on account of deaths amongst temporary settlers and the same number must be added to the figures of the district whence these people have come.† When making a similar calculation on the basis of the census figures, these temporary migrants should be replaced in their district of birth.

93. The census makes no distinction between permanent and temporary migration, and although a careful consideration of the birth place, of the occupations followed by the foreign-born in the district of enumeration, and of the proportion of the sexes might enable an approximate estimate to be arrived at of the number of migrants of each class, both at the present census and in 1891, the enquiry would take more time than can be devoted to it in the present report. It must suffice to have pointed out reasons why the results of the census necessarily seem to differ from those of the returns of births and deaths and to note in general terms whether the actual differences are capable of explanation on these grounds or not. Almost all the districts of Bengal Proper contain a large floating population from Bihar and the United Provinces who have no effect on the number of births, while the deaths that occur amongst them go to swell the mortality returns. As a rule, therefore, the increase in the actual population is considerably higher than would appear from a comparison of the number of births and deaths shown in the vital statistics returns. There will necessarily be some discrepancy, even where the number of these temporary settlers is about the same as in 1891, but it is, of course, proportionally greater where it has grown during the decade. This has been the case in Burdwan, Hooghly, the 24-Parganas, Calcutta, and other districts, and the recorded excess of births over deaths is, therefore, far less than the difference between the population as now ascertained and that of 1891. Permanent migrants affect the birth and death rates equally, and in their case there will only be a discrepancy when there has been fresh migration since 1891, as in the case of Dinajpur, Darjeeling, Sikkim and Jalpaiguri. In districts where there are comparatively few temporary settlers and whose population has not been greatly affected by migration since 1891, such as Midnapore and Backergunge, the difference between the census results and the vital statistics is less marked. It may, therefore, be concluded that in Bengal Proper the discrepancies are in most cases not greater than the movements of the people would account for. In Orissa and Chota Nagpur also the discrepancies admit of a similar explanation. Many of the inhabitants of these divisions are constantly to be found in Bengal Proper whence they return to their homes at more or less frequent intervals. In Bihar, however, and especially in the Patna Division, the difference is so marked that

EXAMINATION OF DISCREPANCIES BETWEEN VITAL STATISTICS AND CENSUS.

* For the method of arriving at this figure see footnote on page 102.

† Thus Puri with its vast number of pilgrims and a very small volume of permanent migration shows an actual increase between 1891 and 1901 of 72,286, the vital statistics returns show an excess of only 48,749 births over deaths. The difference is due doubtless to the mortality amongst pilgrims who have nothing to do with the district.

further explanation is necessary. The number of its inhabitants enumerated in other parts of the Province, taking the mean of the two last censuses, is 563,076. Assuming this to be the average number absent at one time, the number of deaths amongst them at the rate of 40 per 1,000 would be 21,523 yearly or 215,230 in the decade. The division has also sent out 15,373 emigrants to the colonies, who have necessarily been omitted from the census returns, and also a considerable number to Upper Burma. But even so, a large divergence remains. The causes will be discussed when dealing with the figures for each district, but it may be stated generally that a great part of it is due to the plague. The returns of plague deaths were very incomplete and the 20,075 deaths that were reported during the first two months of 1901 have not been included in the figures in the Subsidiary Table. Moreover, apart from the mortality, reported and otherwise, the plague frightened away most of the shop-keepers and others whose permanent homes were elsewhere and it also disorganised the arrangements for taking the census and interfered to some extent with the accuracy of the enumeration. The true population of the plague districts was doubtless somewhat greater on the 1st March 1901 than the returns would indicate. It is probable that, but for these disturbing causes, the difference between the recorded births and deaths in Bihar would have afforded as near an approximation to the census results as it has been shown to do in other parts of the Province.

CONCLUSIONS AS TO VALUE OF VITAL STATISTICS.

94. Generally, therefore, it appears that the difference between the total number of births and deaths shown in the vital statistics returns affords a very close approximation to the actual growth of the population. Why this should be so when the number of each class of occurrences reported is considerably below the estimated birth and death rates, is a question that will be again referred to in the chapter on Age. It is usually supposed that deaths are more fully reported than births, but assuming that the estimated rates are not excessive, the result indicated above can only be due to a slightly greater inaccuracy on the part of the death returns. The understatement of deaths in the case of plague is well-known, but this is due to special reasons. It may be that when any epidemic disease is specially prevalent, many deaths occur which are not reported, owing to the general demoralisation and alarm and, in some cases, to the death or flight of the village official who is responsible for preparing the return. The results of testing by vaccination officers do not disclose any special tendency to leave deaths unreported, but it is not likely that their testing is as thorough in villages where epidemic disease is prevalent as it is elsewhere. The natural tendency would be to avoid such villages. Apart from epidemics it has been suggested that there is a tendency to ignore the deaths of infants under 1 year of age, who are buried without any special ceremony and whose decease is thus not likely to attract much notice. This also may possibly help to account for the result.

REPORTED CAUSES OF DEATH.

95. This discussion has already reached a much greater length than I had intended, but before leaving the subject a brief notice of the reported causes of death may not be uninteresting. The first thing that strikes one is the high proportion of deaths ascribed to fever. More than 70 per cent. of the total mortality is returned under this head. This is due mainly to the difficulty of diagnosing all but a few well-defined diseases. Cholera, dysentery and small-pox are known, but most other complaints are classed indiscriminately as fever. It is impossible to say what proportion of the total is attributable to malarial affections, but it may safely be assumed that wherever the mortality entered under this head is unusullay high, the greater part of the excess over the normal is due to their prevalence. On an average about one-twelfth of the total mortality is due to cholera, but the prevalence of this disease varies greatly from year to year and from district to district. In 1898 it was responsible for less than 1 death per 1,000 of the population of the province, but in 1900 the mortality from it rose to nearly 5 per 1,000. In the latter year it killed off nearly 24 persons in every 1,000 in Purnea, while in Bankura only 1 person in 4,000 died from the disease. Dysentery and diarrhœa account for barely a quarter as many deaths as cholera, while small-pox claims only 1 victim in every 5,000 persons yearly.

WEST BENGAL.

BURDWAN.

96. The greater portion of Burdwan has a rich alluvial soil, but towards the west the surface becomes more undulating and the alluvium gives place to laterite. At the time of the Census of 1872 the fever epidemic, to which the district has given its name, was at its height, and for many years the resulting mortality was enormous. It is unnecessary to describe in detail the origin and course of this fatal epidemic, as a full narrative up to 1881 was given in the Census Report of that year and its subsequent history was described in the Report on the Census of 1891. It will suffice to say that the disease is supposed to have originated in the Jessore district whence it spread gradually westwards, and appeared for the first time in Burdwan in 1862. Between 1872 and 1881 the population declined by more than 6 per cent. During the next decade, except in the Raniganj subdivision, there was a further decrease due to the same cause, but it was noted in the last Census Report that though the disease was still virulent in parts of the district, it seemed at last to be dying out. This expectation has been fulfilled. The district is still subject to fevers, especially to those of a remittent type, but the real "Burdwan fever" which often proved fatal within one or two days is now very rarely met with. Cholera is seldom absent from the district, and is markedly endemic in Kalna, but there have been no outbreaks of special virulence during the last decade. Crops have been good as a rule, and were short only in 1895 and 1896, and the cultivators have benefited by the rise in prices that has taken place. There are very few landless labourers except such as obtain remunerative employment in the coal mines. The consequence is that the district has made a rapid recovery, and has added ten per cent. to its population, which now exceeds by more than 30,000 the number recorded in 1872. This satisfactory result is due partly to an increase in the number of immigrants, partly to a reduced flow of emigration and partly to the excess of births over deaths. The increase is greatest in the Raniganj subdivision, which rests on the laterite, and is not only the healthiest part of the district, but has also gained most by immigration. More than half the total number of immigrants in the district were enumerated in this subdivision, whither they have been attracted by the growing demand for labour in the coal mines and other industries that are rapidly being developed.

Thana.	Population. 1901.	Percentage of variation. 1891–1901.	1881–1891.
DISTRICT TOTAL ...	***1,532,475***	***+10·1***	***−0·16***
Sadar Subdivision ...	***679,412***	***+ 9·6***	***− 3·1***
Burdwan	69,816	+ 8·4	+ 14·3
Sahibganj	80,641	+ 7·7	− 1·3
Khandaghosh	61,747	+ 10·4	+ 1·2
Raina	105,363	+ 10·5	− 1·8
Satgachia	108,699	+ 9·8	− 3·1
Jamalpur	72,947	+ 1·6	+ 0·1
Galsi	93,239	+ 14·8	− 15·2
Ausgram	87,560	+ 12·4	− 11·6
Raniganj Subdivision ...	***370,988***	***+19·6***	***+ 8·9***
Raniganj	165,911	+ 12·1	+ 8·9
Asansol	167,895	+ 31·8	+ 28·9
Kakса	37,182	+ 6·3	− 9·8
Katwa Subdivision ...	***248,806***	***+ 8·1***	***− 1·0***
Kougnon	85,287	+ 10·8	+ 1·0
Katwa	85,607	+ 4·8	− 1·5
Mangalkot	77,912	+ 8·8	− 2·4
Kalna Subdivision ...	***233,969***	***+ 0·8***	***− 2·5***
Kalna	90,813	+ 0·8	− 2·4
Purbasthali	74,794	− 3·0	− 6·0
Manteshwar	67,962	+ 5·2	+ 1·9

Population.	1901. Male.	1901. Female.	1891. Male.	1891. Female.
Actual population ...	764,742	767,733	692,872	709,008
Immigrants	84,208	74,139	51,050	52,342
Emigrants	43,537	50,877	66,758	59,709
Natural population ...	724,071	744,471	689,580	716,435

97. The growth of the population in the Asansol thana is phenomenal; it has increased by more than 30 per cent. since 1891 and by 130 per cent. since 1872. Nearly one-third of the present inhabitants were born in other districts, and of the district-born many have doubtless migrated from other thanas. Raniganj shows an increase of 12 per cent. since 1891, which is somewhat surprising in view of the fact that its coal mines are nearly worked out and that the people find more remunerative employment in the new mines further west. The town of Raniganj, however, is a great trading centre, and is probably one of the busiest places in Bengal. The increase is also considerable in the Burdwan and Katwa subdivisions, where there has been comparatively little

immigration. It is most noticeable in the Ausgram and Galsi thanas which were mentioned in the last Census Report as then forming with Kaksa the focus of the fever. Apart from the fever these thanas are naturally healthy, and the disappearance of the epidemic has been followed, as is usual in such circumstances, by a rapid recovery in the population. That the improvement is not equally marked in Kaksa is due to the poverty of the soil in that thana and to its proximity to the coal mines where good wages are obtainable. The only part of the district that has failed to share in the general revival are two of the three thanas of the Kalna subdivision, Kalna and Purbasthali, where the Burdwan fever first appeared forty years ago. These thanas lie along the bank of the Bhágirathi; the soil is water-logged, and they are full of jhils and jungle; they are thus more unhealthy than any other part of the district.

BIRBHUM.

98. Birbhum like Burdwan suffered most from the great fever epidemic between 1872 and 1881, during which period its population declined to the extent of nearly 60,000 persons. The epidemic continued its ravages during the earlier years of the next decade, especially in the south of the district, and although there was an improvement during the following years the Census of 1891 showed a further decrease of nearly 4 per cent. in the population of the head-quarters subdivision. This result was due to the high mortality in the Bolpur and Sakulipur thanas which adjoin the Ausgram thana of Burdwan that has already been mentioned as one of the tracts where the fever was still prevalent in 1891. The loss in the south of the district was counterbalanced by an increase of 10 per cent. in the Rampur Hát subdivision, and the net result for the whole district was a small increase, amounting to barely one-tenth of the loss registered ten years earlier. Since 1891 there has been a great improvement in the health of the people. The fever epidemic has disappeared, and although cholera has often broken out, especially in the south-eastern thanas, there have been no serious epidemics. The district is wholly agricultural, and owing to its undulating surface, the crops can never fail altogether. There were short crops in 1891, 1895 and 1896, but there was no serious distress; in other years the outturn was good, and the cultivators have benefited by the rise in prices. Their material condition has thus improved considerably. The only classes that have suffered are the respectable poor with fixed incomes, and the landless labourers; the latter, however, can always obtain remunerative employment in the coal mines round Asansol.

Thana.	Population.	Percentage of variation.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL ...	***902,280***	***+13·0***	***+0·78***
Sadar Subdivision ...	***535,928***	***+14·0***	***- 3·8***
Suri	140,033	+ 10·9	- 0·4
Dubrajpur	138,628	+ 15·5	- 5·6
Bolpur	118,849	+ 17·3	- 8·5
Sakulipur	77,749	+ 14·1	- 8·9
Labpur	64,281	+ 11·5	+ 9·8
Rampur Hat Subdivision	***366,352***	***+11·7***	***+ 8·0***
Rampur Hat	102,810	+ 10·03	+ 6·6
Mayureshwar	93,839	+ 8·6	+ 6·0
Nalhati	83,831	+ 11·9	+ 10·3
Muraroi	86,182	+ 17·2	...

99. The result of the prosperous condition of the district and of its comparative freedom from disease is an increase of 104,240 persons or 13 per cent. The population now exceeds by about 6 per cent. that recorded at the first census, 29 years previously. There has been some immigration of Santáls for cultivation and of up country men in connection with the railway, but the total number of foreign settlers is only 14,000 greater than it was in 1891. This, moreover, is to a large extent counterbalanced by a greater amount of emigration, and it is thus clear that the increase is due mainly to the natural growth of the population. Excluding Muraroi in the extreme north of the district, where the immigration of Santáls has been greatest, the improvement is most marked in the south of the district where

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	444,689	457,591	380,842	408,412
Immigrants	27,194	33,412	20,744	25,804
Emigrants	20,876	26,485	17,128	22,449
Natural population ...	438,371	450,664	386,226	404,967

it represents a recovery from the losses recorded at the last census. The continued advance of the Rampur Hát Subdivision is attributable partly to its fertile soil and partly to the fact that it is tapped by the railway. The least progressive thana is Muraroi, where the soil is comparatively infertile and there is a large proportion of unculturable waste.

BANKURA.

100. The greater part of the Bankura district is high and undulating. The soil is poor, but the country is well drained, and the people suffer very little from malarial affections. Towards the east, in the Vishnupur subdivision, the land is low and alluvial; the soil is very fertile, but the climate is unhealthy and malarious. The "Burdwan" fever was introduced from the adjoining thanas of Galsi and Khandaghosh in Burdwan, and caused a very heavy mortality, but its westward course was checked on reaching the high ground in the west of the Vishnupur subdivision. The head-quarters subdivision never suffered from the disease, and its population increased by 21 per cent. between 1872 and 1891, while that of the Vishnupur subdivision, in spite of the superior fertility of the soil, declined by more than 8 per cent. As in Burdwan and Birbhum the great fever epidemic has now died out, and the past decade has been a fairly healthy one. Between 1894 and 1897 the mortality was comparatively high, and cholera was unusually prevalent, but in spite of this, the recorded birth-rate has throughout exceeded the death-rate. The people suffered considerably from scarcity during 1896 and 1897, but on the whole the crops have been good, and the material condition of the people has improved. The Magistrate, Mr. De, writes on this subject as follows:—

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,116,411***	***+4·37***	***+2·7***
Sadar Subdivision ...	***712,055***	***+ 2·8***	***+ 6·8***
Bankura	148,870	+ 5·9	+19·1
Onda	122,917	+ 0·4	+ 2·4
Gangajalghati with I. O. Burjora.	185,411	+ 4·9	+ 3·2
Raipur	101,435	− 0·8	+ 6·5
Simlapal Independent outpost.	38,109	+ 3·0	+13·0
Khatra	115,313	+ 3·0	+12·8
Vishnupur Subdivision	***404,356***	***+ 7·2***	***− 4·4***
Vishnupur	142,873	+ 5·1	− 3·4
Sonamukhi	82,488	+ 9·3	− 5·8
Kotalpur	96,652	+ 5·9	− 3·9
Indas	82,343	+ 10·4	− 5·3

"There has been, on the whole, an increase in the prosperity of the people. They evince a growing desire to provide themselves with better food, better clothing, and better appliances generally. Gold and silver ornaments are more common than they were ten years ago, brass utensils have usurped the place of earthen pots, and shoes, umbrellas and better articles of dress are more extensively used. New brick-built houses are springing up everywhere, and articles of food which were formerly luxuries are now in common use."

101. In view of these favourable conditions it is somewhat surprising to find that the population is only 4 per cent. greater than it was in 1891, and that the increase is far less than the vital statistics would indicate. The Vishnupur subdivision has increased by 7 per cent., so that it has now nearly recovered the combined losses of the two previous decades, but the head-quarters subdivision has added less than three per cent. to its population. This is due to the movements of the people. The immigrants from outside the district are fewer by about 13,000 than they were ten years ago, while the emigrants have increased by more than 38,000. The emigrants are for the most part hardy aborigines from the south and east of the district who find the high pay obtainable on the coal fields of Asansol or the distant tea gardens of Assam a more attractive prospect than a penurious livelihood laboriously extracted from the unwilling soil of their native uplands. There is also a considerable amount of temporary migration on the part of the semi-Hinduized tribes in the south and west of the district, who supplement their scanty harvests by working as labourers in the metropolitan districts when they have no crops to look after. They leave home in December, after the winter-rice has been reaped, and do not return till the monsoon breaks. This temporary emigration was greatly stimulated in the

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	542,484	568,927	525,941	543,727
Immigrants	11,740	17,794	17,194	25,459
Emigrants	75,289	71,249	57,665	54,373
Natural population ...	613,013	620,382	566,412	572,641

cold weather, when the Census was taken, by the short harvest of that year, and this accounts to a great extent for the falling off in the population of Raipur and the very small increase in the other thanas in the south of the district. But for these movements of the people the growth of the population would have been two or two and-a-half times as greater as that recorded at the Census.

MIDNAPORE.

102. Midnapore is an extensive district and comprises tracts of very different characteristics. The subdivisions of Contai and Tamluk are on the sea-coast and the estuary of the Hooghly and contain the mouths of the Rasalpur and Haldi rivers. They are comparatively free from malaria and produce very rich crops of rice. The Ghatal subdivision further north, slopes back from the bank of the Rup Narayan; the soil is a rich alluvium, but much of its area is liable to floods, and though excellent crops are obtained, the inhabitants suffer greatly from malarial affections. The head-quarters subdivision consists in the north and west of thinly wooded and rocky uplands. The climate is good, but the laterite soil is dry and infertile. Towards the east and south the level dips, and a swampy hollow is formed between the elevated country to the west and the comparatively high ground along the coast; the conditions in this tract are very similar to those in the Ghatal subdivision which it adjoins.

Thana.	Population.	Percentage of variation.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL.	***2,789,114***	***+ 5·99***	***+ 4·6***
Sadar Subdivision ...	***1,277,749***	***+ 4·5***	***+ 1·3***
Kharakpur	111,016	+ 10·5	+ ·2
Midnapore	78,321	+ 9·1	
Jhargaon	76,463	+ 7·1	+ 17·4
Binpur	104,982	+ 2·4	+ 13·9
Salbani	60,850	+ 9·7	+ 2·3
Debra	67,872	— 1·3	— 4·0
Sabang	163,805	+ ·6	— 1·5
Narayangarh	118,441	+ ·8	— 3·0
Garhbeta	133,122	+ 5·3	+ 5·4
Keshpur	86,580	+ 3·3	— 1·3
Dantan	123,541	+ 2·3	— 5·8
Goipballabpur	163,156	+ 7·4	+ 9·1
Ghatal Subdivision ...	***324,991***	***— 0·9***	***+ 4·6***
Ghatal	92,082	— 0·5	+ 10·1
Daspur	130,664	— 2·4	+ 7·0
Chandrakona	102,245	+ 0·7	— 2·3
Tamluk Subdivision ...	***583,238***	***+ 9·0***	***+ 11·6***
Tamluk	147,919	+ 10·5	+ 8·6
Maslandapur	96,995	+ 8·3	+ 10·1
Sutahata	70,644	+ 11·0	+ 13·8
Panskura	143,356	+ 4·1	+ 1·7
Nandigram	124,344	+ 12·6	+ 27·5
Contai Subdivision ...	***605,136***	***+ 10·6***	***+ 6·1***
Khajri	87,867	+ 14·7	+ 10·0
Contai	172,643	+ 17·1	+ 12·8
Ramnagar	75,020	+ 8·3	+ 11·5
Bhagwanpur	130,728	+ 8·6	— 5·3
Egra	77,884	+ 5·5	+ 13·1
Pataspur	99,294	+ 6·2	+ 1·0

Thirty years ago the whole of the north-eastern portion of Midnapore suffered severely from "Burdwan fever," and in 1881 the district showed a loss of 1·07 per cent. of its population as compared with that found in 1872.

Since 1881 the health of the district has been fair and the population on a whole has made satisfactory progress. Although much ordinary fever exists in the badly drained and flooded tracts, in other respects the health of the people shows a marked improvement, and during the last decade the district has been peculiarly free from cholera and small-pox epidemics. This is due in recent years to the opening of the railway through the district which carries the crowds of pilgrims to Jagganath, who previously plodded wearily on foot and spread disease in all directions along their line of march. The railway has benefited the district in many other respects. By facilitating the disposal of the produce, prices have risen, and the cultivators, who enjoy fixity of tenure, are very well off. It has opened up several of the jungle thanas and stimulated trade. The decade has been a prosperous one, and in 1897, when the pinch of famine was keenly felt elsewhere, the birth-rate was unusually high—a circumstance attributed by the Magistrate to the prosperity of the people, who disposed of their hoards of rice at famine prices.

103. In the district as a whole, there has been an advance of about 6 per cent. in the population since 1891, as compared with a gain of 4·6 per cent. in the previous decade, and a decrease of 1 per cent. in 1872—1881. The Contai subdivision leads the way with an increase of 11 per cent. All the thanas in this subdivision have gained considerably, but especially Contai itself, which has added nearly a sixth to its population of 1891, and the other three thanas on the coast, which contain the great temporarily-settled estate of Majnamutha. The Ghatal subdivision has lost nearly 1 per cent. of the population recorded at

the last census. This decrease, as well as one of 1·3 per cent. in thana Debra and insignificant increases of ·6 and ·8 per cent. respectively in thanas Sabang and Narayangarh, all in the Sadar subdivision, is largely due to the movement of a portion of the population from the densely populated and lowlying tracts in the north-east and centre of the district to the reclaimed *jalpai* lands along the coast and tidal rivers in the Contai and Tamluk subdivisions. From the times of the Muhammadans these lands had been reserved by Government for the accumulation of salt and for the supply of fuel to boil the brine. The manufacture of salt by Government was stopped about forty years ago, and the lands, which are very extensive, were settled with various persons. After some time they began to be cleared and to be surrounded with embankments to keep out the salt-water. Thus protected, they yield abundant crops and are still an attraction to cultivators from distant parts of the district. Unfortunately the embanking of these lands is said to have caused deterioration in the beds of various tidal rivers and *khals*, and so to have rendered more frequent the flooding of the lowlying tracts inland which have been previously referred to. Considering its size the volume of emigration and immigration from and to the district is small. The immigrants aggregate less than 50,000, and the emigrants about 134,000. The number of females on both sides of the account is very nearly the same as it was ten years ago, but the number of males who have come to the district is greater by about 7,000, and that of those who have left it by about 10,000 than it was in 1891. On the one hand the railway has attracted a number of coolies and railway employés, and on the other it has induced a greater exodus than before to Calcutta and Hooghly for temporary employmet during the early months of the year. The net result of these movements is a small loss amounting probably to something less than 1 per cent.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	1,390,233	1,398,881	1,398,674	1,393,392
Immigrants	27,754	22,107	20,405	21,956
Emigrants	72,129	62,116	62,374	65,541
Natural population ...	1,434,608	1,438,890	1,440,643	1,436,977

104. The Hooghly district suffered almost more than Burdwan itself from the great fever epidemic, and in 1881 the Census Superintendent estimated the total loss of population due to this cause at 650,000. Between 1872 and 1881 there was a decrease in the population amounting to 12½ per cent. During the next decade, there was a recovery of 6 per cent., chiefly owing to the rapid expansion of the Serampore subdivision. During the last decade there have been no specially violent outbreaks of cholera or small-pox. Cholera was bad in several years, but the mortality so caused was but a small fraction of that due to fever. The country is flooded yearly by the spill of the Damudar. Its surface is but little above sea level, and the drainage is bad and is yearly getting worse, as the silting-up of the old streams and water-courses continues. The soil is thus water-logged to an exceptional extent. The peculiarly malignant "Burdwan" fever has disappeared, but even now the fevers of the district are of an unusually virulent kind. The death-rate is consequently high, and in only two years of the decade have the recorded births been more numerous than the deaths.

HOOGHLY.

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,049,282***	***+ 1·4***	***+ 6·0***
Sadar Subdivision	***308,715***	***- 0·3***	***+ 0·4***
Balagarh	46,229	- 1·1	- 3·2
Pandua	68,656	+ 4·1	- 0·3
Dhaniakhali	93,594	- 0·1	+ 2·3
Polba	43,462	- 3·0	+ 2·0
Hooghly	54,774	- 3·0	+ ·05
Serampore Subdivision ...	***413,178***	***+ 3·3***	***+ 13·6***
Haripal	96,910	+ 1·9	+ 11·1
Kristanagar	57,694	- 3·7	+ 8·6
Serampore	93,611	+ 12·3	+ 16·7
Singur	65,517	- 3·1	+ 14·1
Chanditala	99,446	+ 1·9	+ 15·6
Arambagh Subdivision ...	***327,389***	***+ 0·8***	***+ 3·4***
Arambagh	121,347	+ 2·3	+ 6·7
Goghat	106,788	- 1·3	- 1·5
Khanakul	99,254	+ 1·9	+ 4·8

But if the health of the people has been bad, they have very little to complain of in other respects. The crops have been good in most seasons, and prices have ruled high. The cultivators are thus so prosperous that there was no serious distress even in 1897 when the rice crop was little more than a third of a normal one. There is a large and growing demand for labour in the mills and brick-fields, but the natives of the district are so well off that not only do they not, as a class, take employment in the mills, which obtain their labour force chiefly from Bihar and Orissa and from Bankura, but they themselves employ imported labour to do the work which the same class in other parts of India are accustomed to do unaided. Writing on this subject the Magistrate (Mr. Inglis) says:—

"There is no doubt that the uneducated classes are remarkably well off. The cultivators find ready markets for their jute, potatoes and other crops, and make large profits. The ordinary ryot or cultivator can afford to hire other men to do most of his field work. This is evidenced by the large number of foreign labourers who have settled in the district or who visit it at the harvest season. Wages have risen greatly, both for skilled and unskilled labour, and in the towns all classes are well off, specially the coolies, who will not work regularly, and the shop-keepers. The mills in the Serampore subdivision and those near the head-quarters station (across the river) pay a large amount in wages, and most of this is spent locally."

105. The result of the recent Census is a small increase of $1\frac{1}{2}$ per cent., but this is due entirely to a rise in the number of immigrants, and a diminished loss from emigration. The number of persons from elsewhere who were enumerated in the district was greater by nearly 40,000 than in 1891, while the number of emigrants was less by more than 33,000. The number of the latter has been diminishing steadily since 1881. So much of the migration to and from this district is of a temporary nature that it is impossible to gauge its effect on the population with any approach to accuracy. If the whole of the movements of the people were permanent and distributed evenly over the decade, the census figures would indicate an annual exodus of some 2,358 persons coupled with a settlement of 8,935 new comers,* or a net gain by migration of about 65,000 persons since 1891. The actual gain is probably considerably below this figure, but even so, it is evident that it far more than accounts for the small addition to the population which has been recorded. The same conclusion is obvious from the fact that the only part of the district which has shown any marked advance is the great industrial centre, Serampore, where it amounts to more than 17 per cent., the actual addition to its population being about the same as that for the district as a whole. Two other thanas in this subdivision, Haripal and Chanditala, show a nominal increase, due in the latter to the importation of coolies to work in the brickfields. The head-quarters subdivision shows a general decline except in Pandua, where there is an advance of 4 per cent. Two of the three thanas of Arambagh show an increase, slight but none the less welcome, because quite unexpected. This subdivision is generally looked on as the unhealthiest part of the district, and its crops are often damaged by floods from the Damodar, whose waters are now allowed to spill over its right bank in order to obviate the risk of the embankment on the left side being breached. In spite of the immigration that has taken place, the district is still less populous by 70,000 than it was in 1872 and even then the district had suffered terribly from the fever epidemic for nearly a decade. It seems very doubtful whether it will ever fully recover its losses until the drainage problem is solved.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	528,279	521,003	509,185	525,111
Immigrants	81,823	87,891	49,662	50,332
Emigrants	70,290	53,551	84,581	71,660
Natural population ...	516,746	515,663	544,104	546,439

106. The district of Howrah occupies a peculiar position, owing to its proximity to Calcutta, and to the presence within its limits of a great industrial town. Its inhabitants

HOWRAH.

* This is on the basis of a death-rate of 40 per 1,000, the slight difference between this and the normal estimated death-rate being due to the consideration that there are very few children amongst migrants. The way in which this and similar calculations have been made is explained in the footnote on page 102.

even in the rural areas are dependent on agriculture to a smaller extent than those of any other district in Bengal, and many of them go daily to their work in the metropolis, returning home in the evening. The communications between the rural areas and the town have been greatly improved during the decade by the opening of the Bengal-Nagpur Railway through Bagnan and the construction of the light railways from Howrah to Amta and to Sheakhali in the Hooghly District. Howrah suffered to a far less extent than its neighbours, Hooghly and Burdwan, in the fever epidemic, and even in 1881 the only thanas that showed a decrease were Jagatballabhpur and Amta in the north-west of the district. With the exception of Bagnan, which is bounded on the west by the Rup Narayan, and Jagatballabhpur, the other thanas are on the bank of the Hooghly, and though not very high, are better drained than the country behind them. The last ten years have not been healthy, and, except in 1897 and 1898, fever has been very prevalent. Prior to 1896 cholera broke out constantly, but the construction of the water-works in that year, in addition to improving the general health of Howrah city has resulted in a far smaller mortality from epidemic disease. The crops were bad in three years owing to deficient rainfall, and in two years they suffered from floods, but the people are so well-off that relief works are never necessary. The mills offer ample employment to those who need it, and the only class that has not improved its position is the poorer middle class, who find the cost of food constantly increasing, while their incomes remain the same. The district now contains 86,889 persons more than it did in 1891, but nearly half of this is due to the phenomenal expansion of Howrah city which has already been referred to. Excluding Howrah city, the growth has been 6½ per cent. Along the river the increase has been about 9 per cent.; in Bagnan 5, and in Amta 2½ per cent. The relatively small increase in Amta is surprising, in view of its greater accessibility since the construction of the Howrah-Amta light railway.

Thana.	Population.	Percentage of variation.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL ...	***850,514***	***+ 11·4***	***+ 15·06***
Sadar Subdivision ...	***431,257***	***+ 17·7***	***+ 17·5***
Bally	18,662	+ 11·8	+ 12·7
Howrah	157,594	+ 35·15	+ 28·4
Golabari			
Sibpur			
Dumjor	102,431	+ 9·9	+ 14·9
Jagatballabhpur	92,570	+ 8·6	+ 9·9
Ulubaria Subdivision ...	***419,257***	***+ 5·5***	***+ 9·6***
Amta	106,939	+ 2·4	+ 9·6
Bagnan	73,439	+ 5·2	+ 8·2
Ulubaria	95,396	+ 8·8	+ 7·5
Syampur	84,487	+ 8·7	+ 13·0

107. The number of immigrants has increased by rather more than 34,000 according to the returns, but it is probable that the true increase is really greater. Howrah, though a separate district for ordinary administrative purposes, is in revenue matters a subdivision of Hooghly, and the villagers often describe themselves as born in Hooghly. The inspection of the schedules was more thorough on this occasion than in 1891, and mistakes of this nature were corrected to a much greater extent. At the present Census only 26,120 persons were returned as born in Hooghly compared with 34,459 in 1891. It is probable that the whole of the excess in 1891 was due to this misdescription of birth-place, and that the immigrants, as compared with 1891, were fewer by 8,000 than the returns would indicate. If so, the gain by immigration at the present Census probably accounts for about 7 per cent. of the total increase. Two-thirds of the total number of persons born out side the district were found in Howrah city, whither they had been attracted for work in the numerous mills and other industrial undertakings. Of the remainder, excluding accidental migration across the border line of adjacent districts, the great majority were doubtless attracted by the prospect of employment in the brick-fields, or on railway works. The terminus of the Bengal-Nagpur railway at Shalimar and extensive sidings were being constructed at the time of the Census, and afforded employment to thousands of coolies from Midnapore, Bankura and Ranchi.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	439,525	410,989	384,844	378,781
Immigrants	98,641	45,979	68,071	42,131
Emigrants	14,884	8,512	12,430	7,094
Natural population ...	355,768	373,822	329,203	343,744

CENTRAL BENGAL.

108. The 24-Parganas includes several tracts of entirely different characteristics.

They are divided by the Magistrate, into the following five groups:—

(1) The thanas on the bank of the Hughli in which there are mills. These are Naihati, Nawabganj, Barrackpore, Khardaha, Barnagore, Budge-Budge. The suburbs also fall within this category, but they have been treated for census purposes as part of Calcutta. These thanas on the whole are more healthy than those further inland, being higher and enjoying a better supply of drinking water, which, in some cases, is filtered by the management of the mills.

24-Parganas.

(2) The northern and central thanas, Habra, Diganga, Barasat, Dum-Dum, and Tollygunge. The drinking water is here very bad, being derived mainly from tanks polluted by surface drainage; the drainage channels are blocked and there are numerous swamps, and the homesteads are surrounded by dense jungle. Malaria is very prevalent.

(3) The eastern thanas, Baduria and Basirhat. The inhabitants are for the most part sturdy Muhammadans; the country is now healthy and the main crop is jute which yields a handsome profit to the cultivators.

(4) The southern thanas, Haroa, Bhangar, Sonarpur, Baruipur, Vishnupur, Fultah, Diamond Harbour and Magrahat. These thanas are salubrious owing to better drainage, the comparative absence of noxious undergrowth and the sea-breeze that blows almost continuously during the south-west monsoon.

(5) The Sundarban thanas, Hasanabad, Canning or Matla, Jaynagar, Mathurapur and Kalpi. Cultivation is here spreading rapidly, and reclamation is extending southwards.

The census of 1881 showed a net increase of nearly 6 per cent. but there was a decline in the north and east of the district due to the prevalence of malaria. The so-called "Burdwan fever" spread thither from Nadia about 1861, but is said to have died out in 1864. However that may be, this tract continued to be very unhealthy, and there was a loss of population amounting to 9 per cent. in the Barrackpore subdivision and to 10½ per cent. in the Naihati thana. In 1891, excluding the suburban municipalities, the district showed a further increase of 11·8 per cent., but several of the northern and central thanas were either stationary or decadent, the worst being Habra, where there was a decline of 5·4 per cent.

Thana.	Population.	Percentage of variation.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL ...	***2,078,359***	***+ 9·9***	***+ 11·8***
Sadar Subdivision ...	***709,917***	***+ 11·9***	***+ 15·4***
Tollygunge	105,827	+ 1·4	+ 29·1
Sonarpur	43,646	+ 3·3	— ·4
Baruipur	90,786	+ 4·6	+ 14·3
Jaynagar	107,969	+ 18·6	+ 19·3
Matla (Canning) ...	80,210	+ 41·9	+ 18·9
Vishnupur	83,994	+ 6·2	+ 10·2
Bhangar	76,483	+ 12·6	+ 4·4
Budge-Budge	82,352	+ 13·9	+ 15·5
Barnagore	38,648	+ 12·7	+ 14·3
Barasat Subdivision ...	***451,963***	***+ 4·6***	***+ 6·8***
Dum-Dum	54,666	+ 1·4	+ 18·8
Barrackpore	35,630	+ 28·1	— 9·2
Nawabganj	16,984	— 7·9	+ 62·2
Khardaha	26,019	+ 7·5	+ 15·9
Barasat	76,596	+ 1·3	+ 3·4
Diganga	55,268	— 1·0	+ 4·2
Habra	70,938	— 0·9	— 5·4
Naihati	96,232	+ 11·8	+ 11·8
Basirhat Subdivision ...	***372,187***	***+ 7·2***	***+ 9·6***
Basirhat	86,925	+ 10·7	+ 2·8
Baduria	132,236	+ 4·6	+ 2·1
Haroa	61,554	— 2·0	+ 27·4
Hasanabad	91,472	+ 15·3	+ 18·0
Diamond Harbour Subdivision	***460,748***	***+ 14·4***	***+ 17·0***
Diamond Harbour ...	70,585	+ 11·4	+ 20·0
Fultah	52,403	+ 5·2	+ 14·6
Magrahat	130,424	+ 4·8	+ 16·0
Kalpi ...	133,099	+ 26·8	+ 18·8
Mathurapur	72,157	+ 23·9	+ 16·1

Since 1891 the condition of the people has, on the whole, been prosperous, but some distress was caused by the floods of September 1900 which destroyed the crops in a great part of the Sundarban area. There has been a considerable industrial development and the demand for labour in the mills lining the Hughli has been growing steadily. The public health is said to have been bad

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throughout the decade, save only in 1893, 1897 and 1898. The most unhealthy year was 1896, when the recorded deaths from cholera numbered 11,043 and from fever 37,919. The reported deaths exceeded the births in this year and also in 1895.

109. Excluding the suburbs of Calcutta, the census of 1901 shows a net increase of 9·9 per cent., or, if the suburbs be included, of slightly more than 10 per cent. The total number of immigrants has increased by nearly 64,000, and as most of these new arrivals are males, it is clear that a very large proportion of them are temporary settlers attracted by the high wages obtainable in the mills. But immigration only accounts for about a third of the total gain, and there is a loss of more than 20,000 persons by emigration (chiefly to Calcutta) which must be set off against it. The greater part of the increase is, therefore, due to natural growth.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	1,092,916	985,443	980,278	902,755
Immigrants	154,953	70,414	106,381	55,290
Emigrants	63,017	55,043	51,061	47,127
Natural population ...	1,000,980	970,072	933,958	894,592

The variations in each of the five groups of thanas into which the district is divided are noted in the margin. Speaking generally it may be said that the growth of the riparian thanas is due to immigration to the mills and that of the Sundarban thanas to new reclamations by imported cultivators. Elsewhere the changes are due mainly to natural conditions. The only thana in group (1) that has lost ground is Nawabganj. The rural area is here very unhealthy, and a large cotton mill has been partially closed. Group (2) shows a stationary population. The number of persons enumerated in Calcutta, but born in the 24-Parganas, has grown by 24,000 during the decade, and it is mainly from this group of thanas that they have gone, attracted partly by the pleasures of town life and partly by the superior healthiness of Calcutta compared with their own fever-stricken homes. In group (4) a decrease is returned from Haroa where several large estates have been abandoned owing to the breach of the embankments constructed to keep out the salt water. The greatest expansion in the Sundarban area is in Canning, where large reclamations have been effected by the Land Improvement Company, but all the Sundarban thanas show a remarkable development.

Group.	Percentage of variation.
THANAS.	
(1) Riparian	+ 12·4
(2) North & Central	+ ·5
(3) Eastern ...	+ 6·9
(4) Southern ...	+ 5·8
(5) Sundarban ...	+ 24·2

110. The census of Calcutta and the three Suburban Municipalities, Cossipore-Chitpur, Manicktala, and Garden Reach, was taken under the supervision of the Deputy Chairman of the Calcutta Corporation, who has discussed the results in a separate report. It will suffice to note here that the total population of the metropolis, including its suburbs, is about 950,000, and that if Howrah, which is really as much a part of Calcutta as Southwark is of London, be included, the population is very nearly 1,107,000, which is greater than that of any European city except London, Paris and Berlin, and of any city in America, except New York, Chicago and Philadelphia. Excluding China, the population of whose cities is uncertain, the only city in Asia with more inhabitants than Calcutta is Tokio. According to this census returns the expansion of the city of Calcutta during the last decade exceeds 24 per cent., but this is due no doubt in part to a more accurate enumeration, facilitated by an excellent set of maps which were prepared at my suggestion. No fewer than two-thirds of the inhabitants of Calcutta, are immigrants, and of these barely one quarter are females. Amongst the persons born in Calcutta the sexes are more evenly represented and 46 per cent. of the total are females. The increase in the number of immigrants is about 83,000 and in that of the Calcutta-born about 72,000. In the Suburbs the proportion of immigrants is even greater than in Calcutta itself.

CALCUTTA AND SUBURBS.

	1901.	1891.	1881.	1872.
Calcutta	847,796	682,305	612,307	633,009
Cossipore-Chitpur ...	40,750	31,423	26,204	25,369
Manicktala ...	32,387	28,161	48,125	53,992
Garden Reach ...	28,211	Not available.		
Total ...	*949,144*	*741,889*	*686,726*	*712,379*

NADIA.

111. The Nadia district is a part of the old delta, but its rivers have gradually dried up and it no longer receives the annual deposits of silt which formerly renewed its fertility. In the greater part of the district the soil is sandy, and will not retain the water necessary for the growth of winter rice. Its main crops, therefore, are early rice, pulses and oilseeds. Late rice is generally grown only in the Kalantar, a lowlying tract of black clay soil stretching from the adjoining part of Murshidabad through the Kaliganj and Tehatta thanas, and in the country north and east of it in the thanas of Meherpur, Damurhuda, Gangni, Nowpara and Kushtia. Formerly indigo was extensively cultivated, and the ruins of the old factories are to be seen all over the district. But the indigo planters have well nigh disappeared, and although in their time complaints of their highhanded behaviour were frequent, there is little doubt but that the condition of the raiyats is on the whole less favourable than it was in their time. They are for the most part tenants-at-will, a circumstance which the planters made use of to procure the cultivation of indigo, while their new landlords have taken advantage of it to force up rents, and their landlords' underlings to levy blackmail on their own account.

The district was once famous as a health resort, and it is said that Warren Hastings had a country house at Krishnagar. But it has long since lost this reputation. Before attacking Burdwan, the fever, subsequently called after that district, had devastated Nadia and Jessore, between 1857 and 1864, at which time it was known as "Nadia fever." A fresh outbreak occurred in 1880 and continued for five years, causing a terrible mortality, especially in the southern half of the district. A Commission appointed in 1881 concluded that the epidemic had its origin, partly in the insanitary conditions which prevail in native villages, but chiefly in the silting up of the rivers which had become "chains of stagnant pools and hot-beds of pestilence in the dry season". There were two destructive floods, in 1885 and 1890, and it is thus small wonder that at the Census of 1891 a decrease of rather more than 1 per cent., should have been recorded.

112. The decade which has just passed has witnessed no such widespread calamities as that which preceded it, but the conditions have not been favourable to the growth of the population. Fever has been very prevalent in the south of the district, especially in Krishnagar town and in the old jungle-smothered villages of the Ranaghat subdivision. In the extreme north-east of the district also, obstructed drainage has favoured the spread of a virulent form of malarial fever which has caused a very heavy mortality.* Cholera also was very prevalent, especially in 1891, 1892 and 1896. The only two healthy years of the decade were 1897 and 1898. The seasons were on the whole unfavourable to the crops, especially those of 1895 and 1896, in which years the early rice crop was little more than a half and a third, respectively, of the normal outturn. The winter rice suffered even more, yielding less than half of an average crop in 1895 and barely a seventh in 1896. Distress was severe throughout the district and deepened into famine in the tracts where late rice is the staple crop. The

Thana.	Population.	Percentage of variation.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL ...	***1,667,491***	***+ 1·4***	***- 1·1***
Sadar Subdivision	***361,333***	***+ 3·5***	***— 6·9***
Krishnagar	105,887	— 5·7	— 6·1
Chapra	65,477	+ 3·1	...
Hanskhali	42,777	+ 14·5	— 17·6
Kaliganj	52,792	+ 12·7	— 15·4
Nakasipara	86,719	+ 0·6	— 5·6
Kissengunge	37,681	+ 15·4	— 0·1
Ranaghat Subdivision	***217,077***	***— 5·6***	***— 4·6***
Ranaghat	76,144	— 1·3	— 8·4
Santipur	49,559	— 8·2	+ 0·1
Chakdaha	91,374	— 7·48	— 4·0
Kushtia Subdivision ...	***486,368***	***+ 0·7***	***+ 6·4***
Kushtia	113,499	+ 7·2	+ 10·7
Nowpara	143,934	+ 3·2	+ 2·5
Daulatpur	84,653	— 0·1	+ 4·2
Kumarkhali	144,882	— 4·7	+ 8·4
Meherpur Subdivision	***348,124***	***+ 5·4***	***— 0·5***
Meherpur	73,865	+ 6·3	— 3·0
Kurimpur	100,944	— 5·5	+ 2·2
Gangni	79,467	+ 8·5	— 2·3
Tehatta	93,848	+ 7·7	— 2·2
Chuadanga Subdivision	***254,589***	***+ 5·7***	***— 5·5***
Chuadanga	57,151	+ 1·6	— 8·1
Damurhuda	69,503	+ 9·6	+ 0·5
Alamdanga	91,231	+ 1·7	— 6·2
Jibannagar	36,704	+ 1·5	— 9·9

* The mortality was so high in some of the villages in Kumarkhali in 1897 when I was Magistrate of the district, that I suspected the presence of plague. A special inspection was made by the Civil Surgeon who reported that the deaths were due to a bad outbreak of malarial fever.

relief afforded by Government was eminently successful in preventing loss of life, and the deaths reported in the years 1896, 1897 and 1898 were less numerous by nearly 50 per cent. than those of the preceding triennium. The statistics were tested, but the results showed that the reporting was quite as accurate as usual, and the fact that during the same period the reported births exceeded by 7 per cent. those of the previous three years, points to the same conclusion.

113. The net result of the present census is an increase of 23,383 persons or 1·4 per cent. The Ranaghat subdivision in the south of the district shows a loss of 5·6 per cent., or 1 per cent. more than that recorded ten years ago. The Krishnagar thana, in the head-quarters subdivision which adjoins this tract, has also lost ground, though not quite to the same extent as in the previous decade. The whole of this tract is, as has been already noted, malarious and unhealthy, and its continuous decline must be ascribed mainly to this cause. The decadent condition of the weaving industry of Santipur may also have contributed to the falling off in that direction. Calcutta and the mills in its neighbourhood attract immigrants from this neighbourhood, but the total number of Nadia-born settlers in Calcutta and the 24-Parganas was less in 1901 than it had been ten years previously. The district has lost by migration during the decade to the extent probably of about 1 per cent.

Population.	1901.		1891.	
	Males.	Females.	Males.	Females.
Actual population ...	827,509	829,982	802,147	841,061
Immigrants	30,226	28,784	35,408	37,447
Emigrants	71,160	53,577	77,696	57,223
Natural population ...	868,443	863,775	844,345	861,769

The two eastern thanas of the head-quarters subdivision, Hanskhali and Kissengunge, show the greatest increase of any in the district. These thanas lie in the hollow across which the floods of the Bhágirathi sweep whenever the great Lalitakuri embankment in the Murshidabad district gives way, and they suffered severely in the floods of 1885 and 1890. The population of Kissengunge was stationary in 1891, while that of Hanskhali showed a great decrease. In the two thanas together, the present figures represent a very slight improvement on the population recorded twenty years ago. The only other part of the district that shows a satisfactory rate of progress is the line of thanas stretching through the centre of the district from Kaliganj to Kushtia, which corresponds very closely to the area in which famine relief operations were found necessary. This area, taken as a whole, is probably at the present time the healthiest part of the district. The decrease in Kumarkhali, in the north-east, is due to malaria which, as will be seen further on, has caused an even greater loss of population in the adjoining parts of Jessore and Faridpur. The falling off in Karimpur to the north-west is less easy to explain.

MURSHIDABAD.

114. The district of Murshidabad is bisected from north to south by the Bhágirathi, the ancient course of the Ganges, and the character of the country on the two banks is very different. East of this river the soil is lowlying and alluvial, and forms a part of the old delta. It is fertile, but is liable to be flooded by the spill of the Bhágirathi and other rivers, to prevent which numerous embankments have at various times been erected.* The climate is damp and malaria is prevalent. On the western side, the land is high and undulating; the soil is a hard clay on which winter rice alone grows well, and the climate is comparatively dry. The Bhágirathi is more than a mere physical boundary. It was the ancient dividing line between Rárh to the west and Bárendra to the east. West of the river Hindus predominate, while to the east Muhammadans are more numerous. The population is comparatively dense on the eastern side of the river, but is sparse in the central portion of the Rárh country. Shortly before 1881 the Burdwan fever spread to the district, and devastated not only the lowlying waterlogged eastern tract, but also the elevated country to the west. For some years the death-rate was exceptionally high, and the growth of the population was checked, the variation between 1872 and 1891,

* The most important is the line of embankments along the left bank of the Bhágirathi. The propriety of maintaining all these embankments has often been called in question. The land that would otherwise be flooded is thereby deprived of its supply of fertilizing silt, and the river, being confined to its bed, deposits its silt there, and thus gradually raises itself above the level of the surrounding country.

being barely 2 per cent. Since 1891 there has been a great improvement and the death-rate has fallen considerably, especially in the Rárh country, which is reputed to be the healthiest part of the district. The reported birth-rate has been higher than in most other parts of Bengal and in 1899, it exceeded 48 per 1,000, a figure which shows that the vital statistics of the district have attained a fair standard of accuracy. According to these returns the excess of births over deaths during the eight years 1892—1900 exceeds 98,000. The gradual decay of the silk and indigo industries has affected the prosperity of the people, and in 1897 the district was visited by famine. The distressed area comprised the tract east of the Bhágirathi and a small strip in the Kandi subdivision, but relief operations were not found necessary on so large a scale as elsewhere. The scarcity had no apparent effect on the death-rate which was considerably lower in 1897 than in any other year of the decade, except 1898, while the average birth-rate of 1897 and 1898 was higher than that of any other consecutive two years. Except for the famine year, the crops have been generally good, and the condition of the cultivators is satisfactory.

115. The present census shows a net increase of 6·6 per cent. but there are considerable local variations. Two tracts have sustained a loss of population, viz., Suti thana in the Jangipur subdivision, and the country bordering on the left bank of the Bhágirathi from Azimganj to Berhampore, including Manullabazar, Shahanagar, Sujaganj and Daulatbazar. Suti is surrounded by progressive thanas, and the falling-off is due probably to temporary causes connected with the movements of the Ganges. A good deal of land has been washed away and thrown up on the opposite bank of the river in the Malda district, and this has led to a considerable amount of emigration. The other tract has been decadent ever since 1872. It is to a great extent urban, and its prosperity has been steadily declining. Trade has been diverted to other channels, and the ivory work and bell metal manufactures for which it is noted have become less profitable. The Lalbagh subdivision was closed for some years, and this also must have had some effect in reducing the population. The prosperity of this locality, moreover, is closely connected with the fortunes of the Nawáb whose circumstances have not been improving. The earthquake of 1897 overthrew a great number of buildings which the people have not been able to replace, and the untouched ruins give to the visitor an uncomfortable impression of neglect, decay and poverty. Throughout the rest of Eastern Murshidabad there has been a fair increase, which is greatest in Raghunathganj in the north, and in Burwa in the extreme south. In this part of the district as a whole, the rate of growth is 3·1 per cent. In Western Murshidabad, on the other hand, it amounts to 12·9 per cent. In no thana in the latter tract is it less than 9, while in Sagardighi and Kalianganj it amounts to 26 per cent. These thanas, which are still very sparsely populated, attract a large proportion of the immigrants from Birbhum and the Santhal Parganas. Sagardighi showed a fair increase at the two previous enumerations, but Kalianganj was almost stationary in 1891, while in 1881 it showed a decrease of 11·7 per cent. The thana that has

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,333,184***	***+6·6***	***+ 1·9***
Sadar Subdivision ...	***471,962***	***+3·7***	***+ 2·5***
Sujaganj	9,377	− 0·3	+ 12·7
Gorabazar	12,921	+ 0·02	− 4·8
Berhampore	24,397	+ 3·7	− 0·3
Burwa	74,418	+ 8·3	+ 1·1
Daulatbazar	37,542	− 4·2	− 1·4
Hariharpara	57,181	+ 2·9	− 3·6
Nowda	51,817	+ 6·3	+ 2·5
Gowas	88,044	+ 5·9	+ 9·0
Jalangi	116,465	+ 3·2	+ 4·0
Lalbagh Subdivision ...	***192,978***	***+6·2***	***− 1·2***
Shahanagar	25,380	−19·9	+ 17·4
Manullahazar	12,927	−18·6	+110·4
Asanpur	14,723	+ ·6	− 9·3
Bhagwangola	56,093	+ 4·7	− 6·3
Sagardighi	37,078	+26·0	+ 5·3
Kalianganj	47,858	+26·3	+ 1·3
Jangipur Subdivision ...	***334,191***	***+5·4***	***+ 4·1***
Raghunathganj	73,509	+10·7	− 9·8
Shamshirganj	88,840	+11·9	+ 14·0
Suti	64,432	− 6·4	+ 4·5
Diwan serai	61,640	+ 1·5	+ 10·3
Mirzapur	45,761	+ 9·3	+ 2·8
Kandi Subdivision ...	***334,053***	***+12·4***	***+0·6***
Kandi	31,334	+12·5	− 1·7
Barwan	69,806	+ 9·8	+ 3·5
Khargaon	63,772	+13·9	+ 3·6
Bharatpur	121,947	+11·6	+ 1·9
Gokarna	46,894	+19·7	− 5·8

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	653,346	679,838	605,695	645,281
Immigrants	35,548	35,248	44,843	35,429
Emigrants	33,469	40,227	39,981	41,911
Natural population ...	655,967	684,817	600,803	651,763

progressed most rapidly since 1872 is Shamshirganj, which is nearly 50 per cent. more populous than it then was. During the last decade the district has lost by migration more than it has gained but the changes have not been very great, and have probably not affected the rate of growth by more than 1 per cent.

JESSORE.

116. The conditions adverse to health that exist in the south of Nadia prevail over almost the whole of Jessore. The banks of the rivers are higher than the country behind them, and depressions are thus formed between the main water courses. The drainage of these depressions was always difficult, and it has now become almost impossible owing to the silting-up of the mouths of the rivers and drainage channels. Stagnant swamps are thus formed, while good drinking water is scarce and the homesteads are enveloped in dense jungle. The district has long been notoriously unhealthy, and it was here that cholera first appeared in a violently epidemic form in 1817.* Here, too, twenty years later, originated that terribly fatal kind of fever, subsequently known as "Nadia," and then as "Burdwan," fever, which decimated the population of the country from Jessore westwards as far as the Vishnupur subdivision of Bankura. The first known outbreak occurred near Muhammadpur amongst a body of some six hundred prisoners working on the road from Jessore to Dacca. In 1843 the epidemic seemed to disappear, but it again broke out in 1846 and spread as has already been described. At the present time this malignant type of fever is not noticeable, but a milder form is very prevalent, and although the mortality is not sufficiently striking to attract attention, it is silently and relentlessly at work, destroying many and sapping the vitality of the survivors and reducing their fecundity. The soil of Jessore is fertile, but the crops during the decade have not been very good. The condition of the cultivators appears, however, to be satisfactory, and they have benefited by the rise of prices. There was some distress in 1897, and a few thousand rupees were expended in helping the very poor, but regular relief operations were not needed. The decade has been consistently unhealthy and the returns of vital statistics for the nine years ending with 1900 show an excess of deaths over births aggregating 70,934.

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,813,155***	***—4·0***	***—2·6***
Sadar Subdivision ...	***561,242***	***—5·6***	***—5·3***
Jessore	146,089	— 7·5	— 5·6
Kaliganj	81,385	— 2·1	— 6·9
Bagharpara	54,256	— 5·9	— 9·8
Gadkhali	60,751	— 4·6	— 5·5
Manirampur	142,278	— 4·9	— 3·8
Keshabpur	76,583	— 7·5	— 3·0
Jhenida Subdivision ...	***304,899***	***—2·3***	***—4·5***
Jhenida	84,680	— 5·8	—12·3
Kotchandpur	40,287	— 2·5	— 6·1
Salkopa	179,932	— 0·4	+ 0·4
Magura Subdivision ...	***277,381***	***—8·5***	***+3·4***
Magura	149,889	— 9·7	+ 4·3
Muhammadpur	83,031	— 8·5	+ 8·3
Salikha	44,461	— 4·3	— 8·0
Narail Subdivision ...	***352,281***	***+1·1***	***+6·2***
Narail	154,171	+ 1·7	+ 0·1
Lohagara	112,611	— 2·5	+11·2
Bar Kalia	85,499	+ 4·9	+11·7
Bangaon Subdivision ...	***317,352***	***—3·9***	***—8·7***
Bangaon	106,619	— 4·6	— 7·4
Maheshpur	99,282	— 3·6	— 8·2
Sarsha	68,199	— 2·9	—11·1
Goighata	43,252	— 4·4	— 9·7

117. The census of 1881 brought out an apparent increase in the population, but this is attributed by the Magistrate (Mr. Hallifax) to the inaccuracy of the census of 1872. In 1891 there was a decrease of 2·6 per cent., and this has now been followed by a further decline of 4 per cent. In the head-quarters subdivision the loss of population is about the same as in 1891, and it is considerably less in Jhenida and Bangaon; it is greatest in Magura (8·5 per cent.) where there was an improvement of 3·4 per cent. in 1891. The only subdivision which shows a slight advance is Narail in the south-east. The great falling-off in Magura is due to the prevalence of malarial fever, which has caused such a marked diminution in the population of the adjoining thanas of Kumarkhali in Nadia, and Pangsa, Balia-kandi and Bhushana in Faridpur. The subdivision has the advantage of better drinking water (from the Gorai and Nabaganga) than any other part of the district except Narail, but malaria is not dependent

* Westland's Report on Jessore, pages 179—183. The disease had previously been known, but only in a milder endemic form.

on the drinking water; and Muhamadpur, where the great fever epidemic first broke out, lies within the limits of this subdivision.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	914,085	899,130	941,334	947,493
Immigrants	24,485	22,551	24,083	27,817
Emigrants	45,775	27,359	37,814*	36,490*
Natural population ...	935,315	903,938	975,065	956,146

*Corrected figures.

The above changes in the population are practically independent of migration. The total amount of inter-district movement is small and immigrants and emigrants have alike decreased in number since 1891.

NORTH BENGAL.

118. The district of Rajshahi is composed of three entirely distinct tracts. The first, or Bárind, is elevated and undulating. The soil is a stiff red clay or quasi-laterite, and, where not cultivated, is covered with brushwood interspersed with large trees, the remains of extensive forests. The only crop that can here be grown is winter rice. The population, though once plentiful, is now sparse. The climate is healthy. This tract comprises the whole of Godagari, the greater part of Tanor, Manda and Mahadebpur, and the north of Singra. The Gangetic thanas, Rampur Boalia, Charghat and Lalpur, form another well-defined area. It has a grey sandy soil and a variety of crops are grown. The level is relatively high, and the climate is moderately healthy. The population is fairly dense and includes a considerable number of landless labourers who were originally attracted by the silk industry. The remaining thanas, Naugaon, Bagmara, Puthia, Panchupur, Nator, Singra and Burigaon, constitute the third area, a swampy depression, waterlogged and abounding in *jhils*. The rivers that once drained it have been cut in halves by the Padma, and their mouths have been silted up. The soil is a black loam, and is most fertile. But malaria is very prevalent, especially during the winter months, and the death-rate is high. The population, however, is dense except in Singra, which divides with thana Raiganj in the Pabna District the distinction of containing the Chalan bil, the largest sheet of inland water in Bengal. This tract may be again subdivided into areas growing and not growing ganja. The former category includes the Naugaon and Panchupur thanas which supply the whole of Bengal with ganja. The crop is a most profitable one and is a great attraction. The land is here somewhat higher and the drainage less obstructed than elsewhere in this group of thanas.

RAJSHAHI.

119. During the nineteen years between 1872 and 1891, the population of the district as a whole was almost stationary. The thanas to the north and west increased rapidly, while those in the centre and south declined. The latter part of the district suffered so terribly from fever that in 1883 and 1884 Rajshahi held the first place in the list of fever-stricken districts. The unhealthiness still continues, and the annual reports of the Civil Surgeon are most depressing. In seven out of the last ten years it has held a place amongst the six most feverish districts in Bengal, and in 1892 and 1893 it stood at the top of the list. In only two years of the decade, 1897 and 1898, did the reported births exceed the deaths. These years were comparatively healthy owing to the low rainfall which caused many of the shallower bils to dry up. The crops have been good since 1897, but prior to that year they had been short for several years in succession. The famine did not touch this district, though the people suffered from the high prices due to scarcity elsewhere. The condition of the cultivating classes is said to be satisfactory, especially in the Naugaon Subdivision. The decade, however, has seen the practical extinction of indigo cultivation in the district, and the decline of the silk industry has impoverished silk-worm rearers and silk-workers and the cultivators of the mulberry.

120. The net result of the last ten years in the district as a whole is a slight increase of 1·6 per cent. As on previous occasions, there has been an increase in the Bárind and in the ganja-growing thanas and a decrease in the centre and south of the district. The most progressive thanas are Naugaon and Mahadebpur, both of which have been growing rapidly during the whole of the last thirty years. Manda, in the Bárind, and Panchupur, in the Naugaon Subdivision, also show a considerable increase. The

stationary condition of Tanor and Godagari is surprising, as the immigration of Santáls and other tribes continues. Bagmara thana shows an increase for the first time since 1872, and Puthia and Lalpur, though they have again declined, have done so at a far less rapid rate than on previous occasions. In Nator, Burigaon, Charghat and Boalia, on the other hand, the decrease is greater now than it was in 1891. There seems to be little ground for expecting any early improvement in the condition of the central and southern part of the district. Since 1872 the population has decreased by 12·8 per cent., while that of the thanas in the Bárind and in the ganja-growing tracts has grown by 25·6 and 59·3 per cent., respectively. These two tracts owe a great deal of their development to immigration—from other districts in the case of the Bárind, and from other parts of Rajshahi itself, chiefly from Puthia, Baghmara and Nator, in the thanas where ganja is grown. The total number of immigrants from other districts has increased by more than 9,000 during the decade, but it is very uncertain how many have come to stay. A great many are palki-bearers, earth-workers and field labourers who visit the district during the cold weather and leave again before the rains set in. Probably not more than a third are permanent settlers. Of these, the great majority are Santáls, Mundas and Oráons who have made clearances in the jungles of the Bárind. The zamindars allow newly-cleared land to be held rent-free for the first three or four years, and this exactly suits the taste of these unsophisticated aborigines, who do not mind the physical labour involved in breaking down the jungle, but have a very great aversion to the payment of rent. They remain until rent is demanded and then move on, leaving the land they have brought under cultivation to be occupied by the less hardy and less industrious Hindu cultivators who would shrink from undertaking on their own account the irksome task of reclamation.

Thana.	Population. 1901.	Percentage of variation. 1891–1901.	1881–1891.
District total ...	***1,462,407***	***+ 1·6***	***– 0·8***
Sadar Subdivision ...	***363,936***	***– 1·3***	***– 3·1***
Boalia	95,065	– 3·9	– 2·6
Tanor	88,487	+ 1·8	+ 1·6
Godagari	47,371	+ 1·3	+ 6·7
Puthia	92,330	– 3·6	+ 21·0
Charghat	116,999	– 5·8	+ 6·1
Bagmara	123,704	+ 3·9	– 2·1
Naogaon Subdivision ...	***476,072***	***+12·1***	***+10·8***
Naogaon	170,540	+ 14·2	+ 13·1
Manda	126,155	+ 10·8	+ 10·1
Panchopur	95,774	+ 8·7	+ 11·7
Mahadebpur	83,598	+ 14·1	+ 8·9
Nator Subdivision ...	***422,399***	***– 4·8***	***– 6·2***
Nator	127,753	– 10·1	– 7·5
Singra	119,418	+ 1·8	+ 4·1
Burigaon	108,684	– 6·8	– 4·1
Lalpur	66,544	– 1·3	– 20·7

Population.	1901. Male.	1901. Female.	1891. Male.	1891. Female.
Actual population ...	741,690	720,717	719,289	720,345
Immigrants ...	46,815	32,364	43,710	28,054
Emigrants	11,941	11,671	19,026	19,257
Natural population ...	706,816	709,024	695,605	711,568

Dinajpur.

121. The whole of Dinajpur is alluvial, with the exception of four or five thanas in the south of the district which lie in the Bárind. The surface is here elevated, undulating and well drained, and even in the alluvial parts of the district the proximity to the foot of the hills and the consequent greater speed of the rivers has given the country a more rapid slope than is the case further south, and it is thus far better drained. In spite of this the district was for many years exceedingly unhealthy, and in 1878 a Committee was appointed to enquire into the causes. The only remedial measure adopted was a scheme for draining the neighbourhood of the chief town. The census of 1881 showed a gain of barely 1 per cent. which was more than accounted for by the greater accuracy of the enumeration. The district continued to be unhealthy for some years longer, but it then took a turn for the better, and in 1891 there was an increase of rather less than 3 per cent., of which, however, a considerable part was due to immigration. Since 1891 the health of the district has continued to show some further improvement, but it is even now far from satisfactory, and malarial fevers are still very prevalent. In every year of the decade, Dinajpur has been one of the six districts with the highest recorded mortality from fever, but this may be due in part to the greater accuracy of the returns, as its reported birth rate is exceeded in only two districts in the Province. A comparison of the births and deaths reported during the nine years 1892-1900 shows a net excess of 21,080 births. The crops have been good on the whole. There were

partial failures in 1891 and 1897, and the scarcity in the latter year was aggravated by the high prices which prevailed throughout India. The good harvests of subsequent seasons have restored the prosperity of the cultivators, but they are wanting in industry. The Magistrate says:—

"They are an idle lot. In spite of the fertility of the soil, the low rates of rent, and the opening up of the country by railways, their material condition is still much the same as it was many years ago. The reclaimers of jungle and waste land, a large portion of the reapers during harvest time, the coolies, the domestic servants, the skilled labourers, such as carpenters, etc., the wholesale traders, all, or nearly all, come from other districts. The standard of comfort of the ordinary cultivator is low. He is content with a house which can but indifferently protect him and his family from stress of weather, and he does not possess any warm clothing for the winter. The standard of comfort has not shown any tendency to rise during the last ten years. The people are improvident and spend a large amount of money every year in buying useless articles of rural luxury at the fairs held all over the district during the dry months."

122. During the last decade the population has grown by 84,510 or 5·7 per cent., but this is to a great extent due to immigration from other districts. The increase in the immigrant population is nearly 20,971, but this, of course, does not represent the real number of new arrivals. There were already 112,000 immigrants in 1891, and at the assumed annual death rate of 40 per 1,000, about 45,000 new settlers must have come to the district during the decade to keep up this number. A further immigration of 2,486 persons yearly, or of about 25,000 during the decade, would be necessary to produce the excess of 20,000 immigrants recorded at the present census. There is some uncertainty as to the actual number of emigrants in 1891* but in any case the loss by fresh departures during the decade cannot well have exceeded 6,000. These figures indicate a net gain owing to migration of about 64,000 persons, which leaves less than 21,000, or say 1½ per cent., as the gain due to the excess of births over deaths. This result agrees very closely with that indicated by the returns of births and deaths. I have discussed this matter at some length, not because it is claimed that a very close approximation to the true growth of the population has been arrived at, but because the volume of immigration is here exceptionally large and the mere percentage of increase in the population actually enumerated in the district thus differs to an unusual extent from that indicating the true growth. Moreover, where a district has so long been decadent, it is specially desirable to know whether the conditions adverse to health still continue or not.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	823,972	743,108	774,380	708,190
Immigrants	80,780	51,214	66,032	46,024
Emigrants	8,420	8,723	16,946	16,735
Natural population ...	751,603	700,667	725,314	678,901

123. An examination of the figures for individual thanas shows, as was to be expected, that the increase is most rapid in the south of the district, especially in Patnitala, Patiram, Chintaman and Nawabganj. These thanas lie in the Bárind. Magnificent tanks and scattered bricks afford ample evidence of a considerable population at some former period but at the present time the country is very sparsely inhabited, and it is for the most part covered with scrub jungle. About fifty years ago it occurred to the manager of a Government estate that the waste land might be reclaimed if Santáls were imported and settled there. The experiment was made and

Thana.	Population.	Percentage of variation.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL ...	***1,567,080***	***+5·7***	***+2·8***
Sadar Subdivision ...	***1,023,994***	***+7·7***	***+3·6***
Dinajpur	209,248	+ 0·3	− 1·8
Kaliganj	105,336	+ 4·7	+ 6·9
Raiganj	85,872	+ 1·6	− 9·8
Bunshihari	86,457	+ 3·8	+ 1·5
Patnitala	93,900	+19·0	+14·9
Parsa	54,887	+ 8·4	+ 4·1
Patiram	93,243	+23·9	+ 6·6
Gangarampur ...	85,513	+ 4·9	+ 7·9
Chintaman	61,027	+12·7	+ 4·3
Parbatipur	74,978	+ 8·4	+ 3·7
Nawabganj	73,473	+13·7	+ 7·7
Thakurgaon Subdivision	***543,086***	***+2·2***	***+0·4***
Thakurgaon	227,423	+ 1·3	+ 1·8
Ranisankail	73,712	− ·4	− 1·3
Pirganj	98,211	+ 3·7	+ 1·0
Birganj	143,740	+ 4·2	− 1·3

* In 1891 more than 10,000 persons censused in the Sonthal Parganas were returned as born in Dinajpur. There would seem to have been some mistake about this. There is no known trend of emigration, from Dinajpur to the Sonthal Parganas, and a movement on such a scale as this could not have escaped observation. At the present census only 67 persons in the Sonthal Parganas returned Dinajpur as their birthplace. The Deputy Commissioner of the Sonthal Parganas is of opinion that the birthplace table of 1891 for that district is quite unreliable.

proved such a success that the influx has continued ever since. The total number of Santáls in the district now amounts to 74,101. Their lead has been followed by a few Mundas and Oráons from Ranchi. The zamindars welcome the advent of these hardy pioneers, but they do not usually take any active steps to import them, as they are of a roving disposition and readily move on to fresh clearances on other estates. Outside the Bárind the Santáls seem averse to settling and the increase in the population is consequently smaller. The only thana showing a decrease is Ranisankail. The opening of the Bihar section of the Eastern Bengal Railway does not, so far, appear to have had much effect in developing the tract through which it runs, and both Raiganj and Dinajpur show a less rapid rate of increase than the thanas on either side of them.

JALPAIGURI.

124. Jalpaiguri comprises two distinct tracts, viz., (1) the regulation portion which was formerly a subdivision of Rangpur. This tract includes thanas Jalpaiguri, Rajganj, Titalya, Boda and Pathgram, all of which lie west of the Tista, except the last which is an enclave of the Kuch Bihar State; (2) the Western Duars, taken from Bhotan after the war of 1864-65, comprising the thanas of Damdim, Mainaguri, Dhupguri, Falakata and Alipore, all east of the Tista. The regulation tract has been long settled, and except in the north, it has a fairly dense population. The non-regulation thanas, on the other hand, were very sparsely populated when first acquired. The former tract is decadent, while the latter is very progressive. Its two western thanas, Damdim and Mainaguri, contain much land very suitable for the cultivation of tea. The first garden was opened in 1874, and others followed so rapidly that in 1881 there were 55 tea estates with 6,230 acres under tea. In 1891 there were 79 gardens with 35,683 acres of tea, and in 1901, 103 gardens with 76,158 acres. Apart from the tea gardens, the settlement of land for ordinary cultivation is progressing rapidly; the rates of rent are very low, and cultivators are attracted not only from the thanas west of the Tista, but also from Rangpur and the Kuch Bihar State.

The census of 1872 was not very accurate, even in the regulation part of the district, and for the Western Duars an estimate by the Settlement Officer was accepted in lieu of a detailed enumeration. There were also various changes of jurisdiction between 1872 and 1881, and the effect of these on the population was not ascertained. It is thus impossible to give an accurate idea of the variations that took place between 1872 and 1881. So far as the figures go, the population of the district as a whole rose by nearly 40 per cent., and that of the Damdim and Mainaguri thanas was more than nine times as great in 1881 as it was in 1872. The Deputy Commissioner (Mr. Forrest), however, calculates that the population of the district in 1872 must have been at least 70,000 greater than that returned at the census in which case the growth of the district between that year and 1881 would be only about 16 per cent.

125. During the next ten years the registered increase was 17·2 per cent. The regulation portion of the district showed a loss of 3·5 per cent., which was shared by all thanas except Jalpaiguri, where an advance of 10 per cent. was recorded. The Western Duars continued to show a phenomenal rate of development, especially Damdim, Dhupgari and Falakata. Since 1891 the area under tea has more than doubled. The industry has now fallen upon hard times and its further development has been checked, but the effect of this will not be apparent until 1911. The settlement of lands in the Duars for ordinary cultivation has continued to progress. The crops have been good and the growing demand for labour has been met, as in previous years, by extensive importation from other parts of the Province. Amongst the natives of the district the landless labourer does not exist. There have been considerable extensions of the Bengal Duars and Kuch Bihar State Railways, and a workshop has been opened in connection with the former, employing nearly 1,000 workmen. There have been no specially serious outbreaks of epidemic disease, but fever is always prevalent, and in eight out of the ten years the district has figured amongst the six districts with the highest recorded mortality from fever in the Province. The births reported by the police have exceeded the deaths only in four years, and on the average the deaths have outnumbered the births by nearly 3 per 1,000 per annum. Even if we allow for a slightly more defective registration of deaths than of births the natural population must still be slightly decadent. The registered mortality is greatest in the Duars

where the reported deaths give an average of 39·9 per 1,000 compared with only 33·1 in the regulation tracts. The birth-rate, on the other hand, is here 34·3 per mille, as against 33·3 in the Duars.

Thana.	Population.	Percentage of variation.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL ...	***787,380***	***+ 15·6***	***+ 17·2***
Sadar Subdivision ...	***668,027***	***+ 9·8***	***+ 11·8***
Jalpaiguri	82,324	— ·7	+ 10·0
Rajganj	51,976	— 3·9	— 6·0
Titalya	20,542	— 1·8	
Boda	165,007	— 0·8	— 6·8
Pathgram	56,925	— 5·3	— 7·1
Maynaguri	119,426	+ 20·5	+ 27·9
Dhupgari	64,260	+ 57·1	+ 77·7
Damdim	107,567	+ 28·3	+ 157·4
Alipore Subdivision ...	***119,353***	***+ 64·7***	***+ 29·2***
Alipore	69,746	+ 70·0	+ 29·2
Falakata	49,607	+ 57·9	+ 77·7

126. The census of 1901 shows that the population has grown by 15·6 per cent. The rate of progress has fallen slightly but the actual addition to the population is greater than it has ever been before.* The regulation tract is still decadent, while the Western Duars have again made a great advance. The most progressive thana on this occasion is Alipore, followed by Falakata and Dhupgari. Damdim, which showed the greatest development in 1891, has increased far less rapidly during the last decade, owing probably to the fact that there was less scope for further expansion. Most of the land suitable for tea had been taken up prior to the last census. Throughout the district the variations in population are due almost entirely to migration, which accounts alike for the falling-off west of the Tista and for the rapid growth of the population east of that river. The western thanas are the more healthy, and they, if any, would show an increase if migration were left out of account. These, however, are the thanas where alone a loss of population has taken place, and it is clearly due to the movement of the population within the district from west to east which has been going on since 1881, and probably from a much earlier date, and will doubtless continue so long as the supply of culturable waste in the Western Duars remains unexhausted, unless in the interval the zamindars reduce their rates to the level of those payable on the Government estates in the Duars. There is also a slight balance against these thanas in the migration between them and Siliguri in the Darjeeling district.

127. In the Duars, on the other hand, the natural population has probably declined, and the whole of the increase is due to the growth of the immigrant population. There are now 188,223 immigrants in the district, compared with 143,922 in 1891, and most of them were enumerated in the Duars. The proportion borne by the foreign-born to the total population in each thana in the Duars is noted in the margin. It is highest in Falakata, where more than half the inhabitants were born outside the district. About half of the immigrants are tea-garden coolies from Chota Nagpur and the Sonthal Parganas. Many of these stay permanently. The number of settlers from Kuch Bihar and Rangpur is less than it was in 1891; many of the older settlers have died, and the new-comers have not been sufficient to replace them. Assuming the death-rate amongst the immigrant population to be 40 per 1,000 per annum, an annual supply of nearly 6,000 new-comers would be needed to keep up their original number, and of nearly 5,000 to raise it to the present figure. The countervailing loss owing to departures from the district has been very small, and the net gain on account of migration must have considerably exceeded 100,000. The district-born population has increased from 552,856 to 616,583, but this is due not so much to the excess of births over deaths as to the fact that the ranks of the district-born are swelled by the children of immigrants who have made their homes in the district. There were 60,613 immigrant females in the district in 1891, and if the births amongst

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	422,877	364,503	364,319	316,417
Immigrants	104,914	83,309	87,914	60,613
Emigrants	7,555	9,871	9,631	11,016
Natural population ...	325,518	291,065	286,036	266,820

Thana.	Percentage of immigrants.
Maynaguri ...	28
Damdim ...	48
Dhupgari ...	38
Alipore... ...	40
Falakata ...	55

* I assume that the Deputy Commissioner's estimate of the 1872 population is correct. If the census figures be taken, the increase between 1872 and 1881 was greater than that between 1891 and 1901.

them were at the rate of 80 per 1,000 yearly, this alone would account for a great part of the increase that has taken place.

DARJEELING.

128. Darjeeling consists of three distinct tracts, viz., (1) the thanas of Darjeeling, Jorbangala and Kurseong, *i.e.*, the whole of the hills except the Kalimpong outpost; most of this was ceded by the Raja of Sikkim in 1835; (2) the Kalimpong outpost taken from Bhutan in 1865; and (3) the Terai, *i.e.*, the Siliguri thana, annexed from Sikkim in 1850.

At the time of its cession, the first mentioned tract was almost wholly under forest, and it is said that there were not twenty resident families in the whole area.* Its rapid growth is due partly to the establishment of the hill stations of Darjeeling and Kurseong and partly to the development of the tea industry. The first garden was opened in 1856, and proved such a success that within thirty years the greater part of the cultivable area was under tea. Excluding the municipality and cantonment, more than two-thirds of the population is resident on the tea gardens. The Kalimpong outpost contains only one or two gardens, and the greater part of the area is reserved for native cultivation. Five-sixths of the inhabitants are settled on the Government *khás mahál*s. The aborigines of Kalimpong are Lepchas and Limbus, but many Nepalis are now settling there. The Terai contains a number of tea plantations along the foot of the hills, but there are also extensive areas under ordinary cultivation, and the tea-garden population is barely one-fifth of the total.

The hills, including Kalimpong, are very healthy, and the death-rate is comparatively low. The Terai, on the other hand, is notoriously malarious, and the mortality is always very heavy. In the district as a whole there was an increase of 63 per cent. between 1872 and 1881 and of 43 per cent. in the next decade, but the rate of expansion was far more rapid in the hills than in the plains. In the former the growth in these nineteen years was 222 per cent., while in the Terai it was only 52 per cent. The census of 1872 was admittedly inaccurate, and in 1881 also there were probably a good many omissions, especially of women, owing to the alarmist rumours that were spread amongst the ignorant hill men. The census of 1891, on the other hand, is believed to have been very accurate, and some part of the difference between that and the previous enumerations must be set down to the greater accuracy of the work, but even if liberal allowances are made for the errors on previous occasions, it is still clear that the growth of the population during these nineteen years was phenomenal.

129. During the last ten years the health of the hilly part of the district has been normal. In Siliguri, however, the mortality, always high, has been exceptionally heavy. The recorded deaths for the decade as a whole give an average death-rate of 59·8 per 1,000 per annum, and in 1900 it exceeded 71 per 1,000. The average birth-rate, on the other hand, was only 19·4 per annum. The ordinary crops have been good, and in Kalimpong the number of settlers has been rapidly increasing. There is a great demand for labour of all kinds and wages are far higher than in any other part of Bengal. As observed by the Deputy Commissioner, there is no district in the Province where the people are so well off. The tea industry, however, on which the growth of the district mainly depends, has been passing through a serious crisis. Prices have fallen greatly during the last four or five years, and many gardens are no longer able to work at a profit. Some few have already been closed and others have reduced their labour force, so that the increase due to extension of cultivation during the earlier years of the decade has been to a great extent discounted by subsequent reductions of establishment.

Thana.	Population 1901.	Percentage of variation.	
		1891-1901.	1881-1891.
DISTRICT TOTAL ...	***249,117***	***+11·5***	***+43·4***
Sadar Subdivision ...	***133,386***	***+26·2***	***+62·6***
Darjeeling	69,302	+ 13·8	+ 62·6
Kalimpong	41,511	+ 55·9	
Jorbangala	22,513	+ 24·5	
Kurseong Subdivision ...	***115,731***	***— 1·6***	***+29·8***
Kurseong	45,265	+ 1·4	+ 58·5
Siliguri	70,466	— 3·3	+ 17·9

The population of the district as a whole has grown by 11·5 per cent. during the decade. The increase is greatest in Kalimpong, where the waste

* This however seems doubtful. In the course of enquiries made regarding the alleged gradual disappearance of the Lepchas it was stated to me that many of the hill-slopes now occupied by tea-gardens were formerly the sites of Lepcha villages.

land is rapidly being brought under cultivation by new settlers, chiefly from Nepal. In Darjeeling and Jorbangala there has been a fair growth, and the tea gardens in this tract have added 5,000, or more than 12 per cent., to their population since 1891. In the Kurseong subdivision the tea-garden population has been slightly reduced, but the loss has been more than counterbalanced by the growth of the town of Kurseong and by new settlers for ordinary cultivation. Of the decrease of 2,464 persons in the Siliguri thana, about two-thirds is accounted for by the fact that in 1891 a large number of temporary immigrants were employed there in the construction of a road. The tea gardens have added 11 per cent. to their population, but the settlers in the *khás mahâls* have decreased by nearly 5 per cent. Having regard to the heavy death-rate, the loss of population in this thana must have been far greater than it is but for immigration from outside. If we accept the recorded death-rate as correct and raise the birth-rate from 19·4 as reported, to 25 per 1,000, the annual decrease, but for immigration, would, at the rate of 35 per 1,000, exceed 2,500. There must thus apparently have been an immigration to the thana of more than 25,000 persons. This supposition is corroborated by the returns of birthplace which show that no less than 36,907, or 52 per cent. of the inhabitants of the thana were born elsewhere.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	133,005	116,112	125,646	100,208
Immigrants	71,289	53,102	78,978	58,079
Emigrants	3,128	2,568	3,298	1,526
Natural population ...	64,844	65,576	47,966	43,715

The total volume of immigration is less than it was in 1891, but the foreign-born still number nearly half the total population of the district. A very large proportion of them are permanent settlers on the tea gardens and *khás mahâls*. The earlier immigrants are gradually dying out and their place is being taken by their children born in Darjeeling; the reduction in the foreign-born population, therefore, merely means that the flow of fresh immigrants is growing less and not that it has ceased altogether. If there had been no new settlers since 1891 the number then enumerated would have fallen to about 91,000, or 32,000 less than the actual strength of the immigrant population. These figures indicate an average annual arrival of some 4,000 new-comers. The corresponding loss by emigration is inappreciable.

RANGPUR.

130. The soil of Rangpur is fertile and the population is very dense compared with other districts in North Bengal. A wave of unhealthiness swept over the district some time before 1872. Between that year and 1891 malarial fever was very prevalent and the district lost more than 4 per cent. of its population. The thanas that suffered most before 1881 were Rangpur, Mitapokhur and Pirganj in the centre of the district. In 1891 these three thanas taken together recovered some of their losses, but there was a general decadence in all the thanas to the east and north of them and in Badarganj which adjoins them on the west. The death-rate from fever was very high throughout the decade, and there were frequent outbreaks of cholera which was imported by coolies passing through the district *en route* for Assam. The prevalence of malaria was attributed at the last census to the obstruction of the old drainage channels south and west of the Tista. The greatest loss of population occurred in the thanas on the north-east of that river, but there can be no doubt that the drainage is obstructed throughout the district.

Since 1891 there has been a great improvement in the health of the people, and the births reported by the police since birth registration was introduced in 1892 have exceeded the deaths by more than 29,000. The crops on the whole have been good, and even in 1897 the distress was comparatively slight. No relief works were found necessary, and the amount distributed in charitable doles was very small. Railway communications have been greatly improved. The Gaibanda subdivision has been rendered more accessible by the line running from Santahar to Fulchari on the Brahmaputra; the Kuch Bihar Railway has been completed on the north; the railway to the Jalpaiguri Duars running from Lalmonir Hát through the north of the district has been opened, and the Tista river has been bridged. These operations have not only opened out the country, but they brought many labourers into the district, some of whom were

still there at the time of the census. Speaking of the prosperity of the people, the Magistrate (Mr. P. C. Mittra) says:—

"The material condition of the people has greatly improved during the last decade. They generally reap good harvests, and now derive larger profits than formerly by selling jute and tobacco, owing to the facilities for export afforded by the new lines of railway. Agriculturists may now be seen using umbrellas and other imported things which were formerly considered luxuries. They are now better fed, better clothed and better educated. It is true that they are sometimes in difficulties on account of failure of crops owing to deficient or excessive rainfall, but they very soon recover, as the land is very fertile and rents are lower than in Central Bengal."

The earthquake of 1897 did a great deal of damage to roads and buildings, but it is not certain that its effects were altogether harmful. It caused marked changes in the level of the country and it is possible that it has improved the drainage. The Magistrate reports that since the earthquake malaria has abated, and his opinion derives some support from the vital statistics of the district according to which there were 190,883 deaths during the three years 1898 to 1900 compared with 223,034 during the previous triennium.

131. The census shows a net increase of rather more than 4 per cent., and the population is now almost exactly what it was thirty years ago when the first census was taken. This result is due in a great measure to immigration, and if this were left out of account the increment would not much exceed 1 per cent. This, though small, may perhaps be taken as showing that the general opinion regarding the improvement in the health of the district is well founded, in which case a marked increase may be looked for in 1911. The greatest expansion has occurred in the Gaibanda subdivision where it is due partly to the advent of the railway, and partly to its comparative salubrity and to the extension of jute cultivation, which has attracted settlers from the unhealthy thanas to the north-west, and also from Pabna and Mymensingh. Nilphamari shows a fair rate of growth, due in part to the railway settlement at Saidpur and the flourishing trade centre at Darwani, and in part to the healthiness of the locality and the extension of sugarcane and tobacco cultivation. The only thanas which show a marked decadence are Mitapokhur which lost heavily between 1872 and 1881, Mahiganj, where the decrease in 1891 was greater than in any other part of the district, and Kurigram, where it seems to be due chiefly to diluvion and to outbreaks of cholera which is said to be imported by coolies passing through the subdivision on their way to Assam.

POPULATION.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	1,125,109	1,029,072	1,061,812	1,003,632
Immigrants	80,429	28,996	43,070	14,188
Emigrants	26,692	27,470	38,679	36,085
Natural population ...	1,071,354	1,027,381	1,057,421	1,025,549

THANA.	POPULATION.	PERCENTAGE OF VARIATION.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL ...	***2,154,181***	***+ 4·3***	***— 1·5***
Sadar Subdivision ...	***658,291***	***+ 1·8***	***— 2·9***
Rangpur	142,033	+ 7·5	+ 6·1
Mahiganj	106,382	— 3·4	— 14·9
Kaliganj	151,911	+ 1·7	— 4·1
Badarganj	94,704	+ 8·0	— 1·1
Mitapokhur	93,648	— 5·7	+ 0·2
Pirganj	69,913	+ 2·8	— 1·6
Nilphamari Subdivision	***461,314***	***+ 5·0***	***+ 0·4***
Nilphamari	160,563	+ 9·6	+ 2·2
Dimla	133,289	— 2·5	— 1·5
Jaldhaka	167,462	+ 1·8	+ 0·6
Kurigaon Subdivision	***514,392***	***+ 1·3***	***— 5·0***
Kurigaon	56,069	— 4·8	...
Sarabari	125,678	+ 4·3	— 6·2
Nageshwari	142,010	+ ·7	— 8·3
Ulipur	190,605	+ 1·8	— 3·4
Gaibanda Subdivision ...	***520,184***	***+ 19·2***	***+ 2·8***
Gaibanda	200,779	+ 14·7	— 2·8
Gobindganj	234,183	+ 16·0	+ 6·6
Sundarganj	85,222	— 1·6	— 5·3

BOGRA.

132. Bogra is a small district, but it is one of the most prosperous in Bengal. The soil is fertile and the great majority of the inhabitants are sturdy, industrious and prolific Muhammadans. It is bisected from north to south by the Karátoyá. East of that river the land is alluvial and lowlying and is subject to annual inundations. On the west the surface is somewhat higher and towards the north-west the soil is a hard clay and is covered, where not reclaimed, by a thick undergrowth. This is in fact a part of the Bárind or old tertiary formation already described in the accounts of Rajshahi and Dinajpur. The district appears to be

more healthy than either Rajshahi to the west or Rangpur to the north. The drainage is somewhat better and malaria is less prevalent. It is only along the bank of the moribund Karátoyá, and especially in the towns of Bogra and Sherpur, that serious unhealthiness exists. Between 1872 and 1881 the population grew by 6·5 per cent., and this was followed during the next decade by a further increase of 11·2 per cent. The great general prosperity of the district is shown by the fact that on both occasions every thana in the district shared in the general improvement.

"Since 1891," says the Magistrate, Mr. Sen, "there has been no disaster of any kind, either in the shape of famine or pestilence, to impede progress. There was some scarcity in 1897, but there was no loss of life and local help sufficed to allay it. The inhabitants are well off, and most cultivators hire foreign labour at the time of reaping and sowing. Wages are high and the local people will rarely condescend to work as coolies." The opening of the railway from Santahar to the Brahmaputra towards the close of the decade promises to still further increase the prosperity of the district, and a great amount of produce is already being exported by means of it. The earthquake of 1897 overthrew most of the brick buildings in the towns of Bogra and Sherpur, including the Government offices, and struck a serious blow at the prosperity of Sherpur town which was already decadent; but otherwise no serious harm was done. The vital statistics show a net excess of about 30,000 births since 1892, and the Bogra and Sherpur towns and Sherpur thana are the only places where the registered deaths have outnumbered the births.

133. The census of 1901 shows a further increase of 11·8 in the population, and again every thana in the district contributes to the general result. The growth is greatest in the Panchabibi thana in the Bárind, which adjoins the highly progressive thanas of Patiram and Patnitala in Dinajpur and its development is due to the same cause, viz., the clearance of the jungle by immigrant Santáls and Mundas, who remain only so long as the landlord demands no rent, and then move on elsewhere with their portable huts, leaving the land already reclaimed to be occupied by the settled cultivators of the district. The thana is sparsely populated compared with the rest of the district, and there is still ample room for further expansion. The next most progressive thana, Dhunote, presents a complete contrast to Panchabibi, as it is already the most densely populated part of the district. The soil is a grey sandy loam of great natural fertility and it receives annual deposits of silt from the Jamuna. The principal crop is the highly profitable jute which finds a ready market at Serajganj, on which the numerous water channels which intersect it converge. It lies along the bank of the Jamuna and the climate is good. Very similar conditions prevail in Sibganj and Shariakandi, but here jute is less extensively grown. The smallest increase is in the Bogra and Sherpur thanas. These are less healthy than the rest; Sherpur in particular is jungly and the silting up of the Karátoyá has led to obstructions in the drainage. As already observed, the vital statistics of this thana show an excess of deaths over births. Bogra thana is already somewhat densely populated, but in Sherpur the population is sparse.

Than.	Population 1901.	Percentage of variation.	
		1891-1901.	1881-1891.
DISTRICT TOTAL	*854,533*	*+ 11·8*	*+ 11·2*
Bogra ...	258,104	+ 6·2	+ 9·3
Shariakandi ...	134,586	+ 16·2	+ 11·6
Sibganj ...	76,522	+ 13·4	+ 13·9
Panchabibi ...	89,026	+ 22·1	+ 9·1
Khetlal ...	81,477	+ 11·2	+ 16·1
Adamdighi ...	110,967	+ 9·5	+ 15·2
Sherpur ...	54,032	+ 7·1	+ 7·5
Dhunote ...	79,820	+ 19·0	.....

134. The variations above noticed are due in the main to natural causes. There is a considerable immigration from Pabna and Rajshahi on the south and south-west, and some emigration to Rangpur and Dinajpur on the north and north-west. There are about 5,000 immigrants from the Sonthal Parganas and Chota Nagpur and more than twice as many from Bihar and the United Provinces. The former

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	437,349	417,184	391,632	372,829
Immigrants	22,358	15,529	26,683	17,931
Emigrants	8,137	7,619	6,301	5,908
Natural population ...	423,128	409,264	361,250	360,816

are mostly settlers in the Bárind, while the latter are nearly all temporary visitors who carry palkis, dig earth and serve as field labourers during the dry months. Since 1891 the number of immigrants has fallen off, especially those from Rajshahi, Chota Nagpur and the Sonthal Parganas, while the number of emigrants is greater. It is thus probable that although the immigrants exceed the emigrants very nearly in the ratio of 5 to 2, the balance of migration during the decade under review has not been very much in favour of this district, and the greater part of the improvement that has taken place may be attributed to natural causes.

PABNA.

135. The head-quarters subdivision, which forms the south-western half of Pabna, like the adjoining portion of Rajshahi, is an area of silted up river-beds, obstructed drainage and marshy swamps. It is fertile, but abounds in old villages surrounded by jungle, and malaria is very rife. From 1880 to 1884 it shared in the excessive mortality from fever that distinguished the Rajshahi district during these years. The Serajganj subdivision, on the other hand, resembles the neighbouring half of Bogra. It is lowlying, but except in Raiganj the drainage is not impeded by the high banks of dead rivers. It thus receives the benefit of an annual deposit of silt from the Jamuna, but at the same time, when the floods subside, the water readily flows off and does not stagnate as it does further east. The climate is consequently far more healthy than that of the south-western half of the district. Jute is the main crop and the people are very prosperous. Apart from the climate and the fertility of the soil, the variations in the population depend largely on fluvial action. The district lies in the angle formed by the confluence of the Padma and the Brahmaputra, or Jamuna as it is here called. Both these rivers, but especially the latter, are constantly oscillating, cutting away land in one direction and adding it in another. For a series of years one bank is gradually cut away and the other added to, and then for another series of years the opposite process goes on.

The census of 1881 showed an increase in all parts of the district, especially in the Dulai thana, where it was probably due to alluvion. Then came the fever epidemic in the south-west of the district and during the early part of the next decade there was a very heavy mortality. The Padma cut away its northern bank and spread over the country, leaving in many parts a deposit of sand that effectually checked cultivation. At the same time the Jamuna washed away a considerable area in the Mathura thana. The result was that at the end of the decade the population of the head-quarters subdivision was less by 1·1 per cent. than it was at the beginning. In Serajganj none of these adverse conditions prevailed and the population grew by 8·3 per cent.

136. During the decade that has just passed the circumstances have been very similar to those of the previous ten years. The south-western part of the district has been uniformly unhealthy and the vital statistics show an excess of deaths over births in every thana. In Serajganj, on the contrary, the public health has been good and in every thana, except Raiganj, where the figures approach equality, the births greatly exceed the deaths. In the head-quarters the process of erosion is still going on. The Padma has caused some further diluvion in Dulai and a considerable portion of Mathura has been cut away by the Jamuna and thrown up on its opposite bank in the Dacca and Mymensingh districts. Owing to the difficulty of defining a village with precision, it is impossible to lay very much stress on variations in the number of inhabited villages, but it is none the less significant that in this thana the number has fallen since 1891 from 257 to 216. Further north the river has been making amends for these ravages, and in the Shahzadpur and Serajganj thanas extensive *chars* have been formed which are now sufficiently high to admit of cultivation and permanent habitation. The crops have been good in most years of the decade, especially in Serajganj. In 1897 there was a partial crop failure, but the only tract at all seriously affected was a small strip of about 70 square miles between Sara and Pabna with a population of 35,000. Test works were opened, but the demand for employment was very trifling and no regular relief works were found necessary.

137. The outcome of the above conditions is a net gain of 4·3 in the population of the district as a whole, the resultant of a further fall of 2·1 per cent. in the head-quarters, and a further increase of 9·4 in the Serajganj subdivision. In the former tract all the thanas show a decrease except Pabna, which includes Sara. The growing importance of this place as the terminus of the northern section of the Eastern Bengal State Railway has here led to a rapid growth of the population. The decadence elsewhere has already been sufficiently accounted for. In Serajganj the greatest development has taken place in Raiganj where the so-called Bunas are at work clearing jungle. The immigration of Santáls and other aboriginal tribes has not greatly increased since last census, and as the number of births only slightly exceeds the number of deaths, it is probable that there has been a movement to this thana of ordinary cultivators from other parts of the district to take up land cleared by these pioneers, but vacated by them as soon as rent was demanded. Some natives of Saran and Champaran have also settled down in this thana as cultivators. The number of immigrants and emigrants alike has decreased since 1891, and it is not probable that the movements that have taken place during the decade have had much effect on the population. The losses on the one side would seem to have been nearly balanced by the gains on the other.

Thana.	Population.	Percentage of variation.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL ...	*1,430,461*	*+ 4·3*	*+ 3·9*
Sadar Subdivision ...	*586,749*	*— 2·1*	*— 1·1*
Dulai	170,882	— 7·2	+ 0·9
Pabna	197,973	+ 6·3	— 1·8
Mathura	88,244	— 6·3	— 6·6
Chatmohar	129,350	— 3·8	+ 0·6
Serajganj Subdivision ...	*833,712*	*+ 9·4*	*+ 8·3*
Serajganj	264,180	+ 10·9	+ 8·6
Shahzadpur	261,896	+ 8·3	+ 9·2
Raiganj	110,368	+ 11·6	+ 7·1
Ulapara	197,268	+ 7·2	+ 7·1

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	709,396	711,065	677,252	683,971
Immigrants	34,158	14,882	37,710	15,790
Emigrants	30,917	24,388	46,216	27,977
Natural population ...	716,155	720,571	685,758	696,158

138. The Mahánandá river flows through Malda from north to south and divides it into two nearly equal parts which present very different characteristics. West of the river the soil is alluvial and comparatively low and a great deal of it has been subject to fluvial action in very recent times. The Ganges once washed the walls of Gaur, but it now flows sixteen miles further east. Except between Gaur and the Máhanandá where there are extensive undrained swamps, the land on this side of the Máhanandá is very fertile and admirably adapted for the cultivation of rice, mulberry and the celebrated Malda mangoes. The eastern half of the district lies in the Bárind, and has a high undulating surface and a stiff clay soil. It once bore a dense population, but is now very thinly inhabited and is covered with thorny tree-jungle locally known as *kátál*. It is well suited to the growth of winter rice and it is now being rapidly opened out in the manner already described in the case of Dinajpur, Rajshahi and Bogra. Towards the south, in Nawabganj, the land becomes alluvial and here, as in the other portions of the district bordering on the Ganges, the cultivable area varies according to the trend of the river.

Malda.

Between 1872 and 1881 there was an increase of 5·0 in the district population. Nawabganj, Kharba and Ratua showed a rapid expansion, but the southern half of the tract west of the Mahánandá was found to contain fewer inhabitants than it had done at the earlier of the two enumerations. This was attributed to the prevalence of malarial fevers during the latter half of the decade. During the next ten years the district prospered greatly. The opening out of the Bárind thanas by Santáls which had barely commenced in 1876,* made great strides, especially in Gajol and Old Malda. There was also a great growth of population along the south of the district which was attributed to the advent of Musalman cultivators from Murshidabad, on the other side of the Ganges. Since 1891 the general health of the

* In the Statistic Account of Malda, which was issued in this year, it is stated that "some efforts are being made to reclaim the borders of the jungle. But little progress, however, has been made, for the ravages of wild beasts of all kinds cause great devastation, and the population is both scanty and unprogressive."

district has not been satisfactory. Cholera was rife in several years, and in 1900 there was a specially bad outbreak in English Bazar. Malarial fever generally breaks out on the cessation of the rains, and in six years out of the last ten the district has been amongst the six with the greatest reported mortality from fever; in 1899 it headed the list with a recorded fever mortality of 44·7 per 1,000. In spite of this, the vital statistics for the last nine years show an excess of births over deaths of nearly 20,000. The crops on the whole have been good and there has been no famine. The cultivation of jute is extending, and the agricultural community is well off.

Thanas.	Population.	Percentage of variation.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL ...	*884,030*	*+ 8·5*	*+14·5*
Gumashtapur	60,316	+ 6·6	+ 7·0
English Bazar	90,449	— 6·2	+13·5
Nawabganj	78,393	— 1·6	+12·3
Kaliachak	148,201	+14·3	+12·9
Sibganj	127,906	+11·2	+13·5
Kharba	70,810	+10·4	+ 8·5
Ratua	68,743	+ 7·8	+33·0
Tulsihatta	48,134	+ ·3	(a)
Old Malda	72,348	+22·9	+15·2
Gajol	79,776	+25·9	+17·5
Manikchak	43,955	— ·3	(b)

(a) Included in Kharba.
(b) Included in Ratua.

139. The increase in the population during the decade has been 8·5 per cent. The thanas that have grown most are Old Malda and Gajol in the Bárind where Santáls are still settling in large numbers. In 1891 there were 19,457 immigrants from the Sonthal Parganas, and there are now no less than 42,575. The number of Santáls in the district is 52,126, compared with 20,989 ten years ago. There were then only 7,369 female immigrants and there are now 21,180. In spite of the great development that has taken place during the last twenty years, these two thanas are still the most sparsely inhabited in the district, and unless something unforeseen happens, they will probably continue to grow rapidly for many years to come. The next most progressive thanas are Kaliachak and Sibganj in the south-west of the district where new *chars* have attracted a number of Muhammadan cultivators from English Bazar and Nawabganj and from Murshidabad on the other side of the river. English Bazar shows a considerable decrease which is attributed partly to declining trade, partly to unhealthiness, including several outbreaks of cholera, and partly to migration to the adjoining thanas. The slight falling off in Manikchak and Nawabganj is due probably to diluvion. Since 1891 the immigrants have increased from 86,124 to 97,887, while the number of emigrants has fallen from 32,699 to 26,764. If allowance be made for deaths amongst the migrants enumerated in 1891 the district would appear to owe considerably more than half of its total increase of population to immigration, and the amount to natural growth is probably not quite 4 per cent.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	437,839	446,391	399,917	415,002
Immigrants	55,070	42,817	49,793	36,331
Emigrants	13,220	13,544	16,545	15,854
Natural population ...	395,789	417,118	366,969	394,525

140. The Kuch Bihar State is a lowlying waterlogged plain, the whole of which has at one time or another been subject to fluvial action. Major Colvin, the Superintendent of the State, says:—

Kuch Bihar.

"It is intersected by several large rivers, but except in the cold weather when they are at their lowest, it is impossible to get any fall from the surrounding country to the rivers. Even in the cold weather the fall to the bed level of the rivers is very small, and any attempt to cut drainage channels to the rivers would lead in the rains to an inundation rather than to the drainage of the area they might be constructed to serve. The State generally is in fact hopelessly waterlogged. In the rains it is not uncommon to see the wells overflowing. ....There can be no doubt as to the general unhealthiness of the State."

The census of 1881 disclosed a gain of 13·1 per cent., but much of this seems to have been owing to improved methods of enumeration. The census of 1872 was taken by the Settlement Department and was spread over more than three months. Ten years later there was a loss of 3·9 per cent., due mainly to the unhealthiness of the climate, and, to a smaller extent, to emigration. The only thana that showed an increase was Fulbari in the north-east of the State. There was a severe epidemic of cholera in 1891, and fever has always been

present. The condition of the people in other respects is satisfactory. They get three crops a year, and if one falls short, they have the others to fall back upon. There was scarcity in 1892 and 1897, but it did not amount to famine, and in other years the outturn has usually been good. The opening of the Kuch Bihar State Railway and the Bengal-Duars Railway has done much to develop the resources of the country and has greatly facilitated the disposal of produce of all kinds.

141. If only the climate were more salubrious, a rapid expansion might be expected, but, as matters stand, the unhealthiness of the climate has more than counterbalanced the productiveness of the soil, and the recent census shows a further decline of 2 per cent. The only thana that can boast of an increase is Haldibari. This is the principal centre of the jute trade and is yearly growing in importance. It is on the Eastern Bengal State Railway, and it enjoys with Mekliganj, the reputation of being the healthiest portion of the State. The falling-off is greatest in the head-quarters thana, where it is due not only to unhealthiness, but also to migration to Fulbari; the figures for the last census moreover included several large gangs of labourers employed on public works. There has been a decrease both of immigrants and of emigrants since 1891, especially of the latter, and the net result seems to be a slight gain of nearly 15,000 persons. These variations which have occurred are therefore due mainly to natural causes.

Thanas.	Population 1901.	Percentage of variation.	
		1891-1901.	1881-1891.
STATE TOTAL	*566,974*	*−2·05*	*−3·9*
Kotwali	121,609	− 4·4	− 4·0
Dinhata	143,990	− 2·0	− 5·5
Mekliganj	50,203	− 1·0	− 2·2
Haldibari	57,641	+ 9·9	− 1·7
Mathabhanga	146,183	− 3·5	− 6·7
Fulbari	73,438	− 1·5	+ 3·7

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	301,382	265,592	302,457	276,411
Immigrants	27,884	16,859	27,990	19,219
Emigrants	16,761	15,783	24,962	21,199
Natural population ...	290,259	264,415	290,429	278,391

142. Sikkim has been described as the catchment area of the head-waters of the Tista. It lies between the Singilela and the Chola Ranges, and is bounded on the west by Nepal, on the north and north-east by Tibet, on the south-east by Bhotan, and on the south by the district of Darjeeling. By the treaty of Titalya in 1817, the independence of Sikkim, which the Gurkhas had begun to menace was guaranteed, and the settlement of Nepalese in Sikkim was thus prevented from that date until 1889. In the latter year the aggression of the Tibetans led to a war which was succeeded by the more active intervention of the British Government. A Political Officer was appointed; communications were greatly improved by the construction of roads and bridges, and the settlement of Nepalese was permitted in certain parts of the State. These measures were followed by a rapid development of the country. Settlers from Nepal flocked in and the land revenue, which was Rs. 29,294 in 1891, rose to Rs. 61,879 ten years later. The census of 1891 indicated a total population of 30,458, and this has now grown to 59,014, an increase of 93·7 per cent. The enumeration of 1891, following as it did so soon after the introduction of British methods of administration, was admittedly incomplete, and some of the increase must be ascribed to the greater accuracy of the present census. The two main indigenous castes, Lepchas and Bhotias, now number 7,982 and 8,184 (including 7,253 Sikkim-Bhotias), as compared with 5,762 and 4,894, respectively, in 1891. The climate is good; there have been no serious epidemics; the people have been prosperous, and they are naturally very prolific, the crowds of children being a very striking feature of every Sikkim hamlet. But even so, it is impossible that these tribes should have developed to the extent the above figures would indicate in the short space of ten years. The bulk of the total increase, however, is in the immigrant population. Of the total inhabitants, no fewer than 22,720 or 38·5 per cent. were born in Nepal. There was no return of birthplace in this State in 1891,

Sikkim.

Caste or tribe.	Total number in—	
	1901.	1891.
Khambu ...	9,668	5,985
Limbu* ...	5,336	3,356
Murmi ...	5,267	2,887
Gurung ...	4,503	2,931
Khas ...	3,260	829
Kami ...	2,838	1,670
Mangar ...	2,445	901
Newar ...	1,618	729
Damai ...	838	287

* The Limbus are natives of Sikkim as well as of Nepal, but they are shown in this category as their increase is probably due mainly to immigration from the other side of the boundary.

but there can be no doubt that the majority of these immigrants must have come in since that year. The caste return at that census was very rough, but a comparison of the figures for some of the main Nepalese castes shows how greatly they have increased in number. There is still a great quantity of waste land fit for cultivation, and it is probable that for many years to come the population will continue to grow at a very rapid rate.

EAST BENGAL.

143. The Dacca district is one of the most favourably situated in Bengal. Lying between the Jamuna and the Megna, and, watered by numerous other streams, it is blest with a pure and abundant supply of water. The greater part of the district lies low and is flooded every year; but after the rains the flood water is drained off, and leaves the land enriched with a thick deposit of silt, while in the absence of large stagnant marshes it is very free from malaria. Jute is very extensively grown and yields a handsome profit to the cultivators. North of Dacca, towards the Mymensingh border, the character of the country changes. The surface rises and becomes undulating, and a stiff clay takes the place of the alluvium. This part of the district is still somewhat sparsely populated, but it is now rapidly being opened out by various Mongoloid tribes: the main crop here is the winter rice and, unlike other parts of the district, its success is dependent solely on the rainfall. More than three-fifths of the people are Musalmans, who are hardier and more prolific than their Hindu neighbours. A large proportion of the population derives a livelihood from fishing in the great rivers which adjoin the district, and boatmen from Dacca are to be found on every waterway in the Province. The Bikrampur pargana in Munshiganj, where Ballála Sena once held his Court, is the great home of many respectable and well-to-do families, and its sons are to be found all over Bengal and Assam, holding appointments under Government or in private service, or practising as pleaders in the courts. The jute industry attracts numerous natives of Bihar and the United Provinces, and many of the domestic servants, street coolies, palki-bearers, &c., also come from up-country.

DACCA.

Thana.	Population.	Percentage of variation.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL	***2,649,529***	***+10·6***	***+14·5***
Sadar Subdivision	***881,517***	***+11·4***	***+13·8***
Kotwali	93,682	+ 12·0	+ 4·8
Keraniganj	206,591	+ 10·7	+ 13·7
Kapasia	174,435	+ 22·3	+ 18·9
Nawabganj	170,855	+ 9·1	+ 9·7
Sabha	235,954	+ 13·7	+ 18·3
Narayanganj Subdivision	***660,712***	***+15·0***	***+22·0***
Narayanganj	157,993	+ 14·3	+ 33·2
Rajpura	276,827	+ 15·7	+ 18·0
Rupganj	225,892	+ 14·6	+ 20·2
Munshiganj Subdivision ...	***638,351***	***+ 9·9***	***+16·7***
Munshiganj	300,506	+ 10·2	+ 20·2
Srinagar	337,789	+ 9·5	+ 13·7
Manikganj Subdivision ...	***468,949***	***+ 4·5***	***+ 4·6***
Manikganj	207,772	+ 3·9	+ 8·4
Saaloo Aircha	159,929	+ 7·5	+ 3·4
Harirampur	101,250	+ 1·1	...

144. In these circumstances a rapid growth of the population is but natural. In 1881, and again in 1891, an increase of more than 14 per cent. was recorded, and this has now been followed by a further gain of 10·6 per cent. The slight diminution in the rate of progress is not to be attributed to any falling-off in the prosperity of the people or in the healthiness of the district. The crops and the public health have alike been good during the decade; even in 1897 there was no crop failure, and the only sufferers were people with small fixed incomes who felt the pinch of the high prices caused by the famine in other parts of the Province. The riparian tracts are subject to changes by the action of the rivers, especially those on the Padma, in the manner described in paragraph 135. There is thus from time to time a considerable movement of the riparian population from one side of the river, where land has been diluviated, to the other, where new *chars* have been thrown up. Since 1891 the Padma has cut away a considerable area along the western boundary of the district, especially in Nawabganj and Harirampur, and has thus checked their growth. Moreover, the population in some parts is already so dense that further progress at the rapid rate which prevailed prior to 1891 is impossible. Munshiganj thana,

which showed an advance of 20·2 per cent. in 1891, has now grown by only 10·2 per cent., but even this rate of expansion is extraordinary, having regard to the fact that the thana has a density of 1,526 persons to the square mile. Narayanganj, with 1,362 persons to the square mile, has increased 14·3 by per cent., compared with 33·2 per cent. in 1891. The circumstances of Raipura, Rupganj, and Srinagar are very similar. The rate of increase is considerably less than in 1891, but even now it is very high, having regard to the great density of the population, which, in Srinagar, reaches the extraordinary average of 1,787 per square mile. In the part of the Manikganj subdivision, south and west of the Dhalesvari the quality of the land is said to be deteriorating. The district has also lost by migration. Immigrants now number only 85,399, compared with 101,841 ten years ago, while the number of emigrants has risen from 108,300 to 128,381. This would indicate a net loss of nearly 45,000 if the whole of the migration were of a permanent nature, but the great excess of male emigrants seems to indicate that the majority of them are only temporarily absent.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	1,313,417	1,337,105	1,187,739	1,207,691
Immigrants	56,767	28,532	68,903	32,938
Emigrants	94,842	33,646	80,506	27,956
Natural population	1,350,492	1,342,219	1,199,342	1,202,709

MYMENSINGH.

145. Until the beginning of the last century the main channel of the Brahmaputra flowed through the middle of the Mymensingh district, and although it now passes along the western boundary and the old Brahmaputra has shrunk to a mere fraction of its former volume, there is still a marked difference between the country on either bank. The people to the east of it resemble those of Sylhet in their dialect, social customs, and observances, while those to the west are like the inhabitants of Pabna, Faridpur and Dacca.* To the east the country is intersected by marshes or *hâors* where large herds of buffaloes are grazed in the cold weather. In the rains the whole country is submerged, except the crowded village sites which are artificially raised above the ordinary flood level. The general elevation of the country west of the old Brahmaputra is higher, and it contains a great part of the formation known as the Madhupur jungle, which stretches northwards from the boundary of the Dacca district almost as far as the town of Mymensingh. The climate is generally salubrious, and the Durgapur thana at the foot of the Garo Hills alone has a bad reputation for its unhealthiness. The soil is rich and large areas are under jute, the most profitable of all the main crops. The great majority of the inhabitants are prolific Muhammadans. The census of 1881 showed an increase in the population of 29·9 per cent.; but a great deal of this must have been due to the inaccuracy of the enumeration of 1872. The next decade resulted in a further gain of 13·4 per cent. Since 1891 there has been no crop failure. In 1896 the rice crop was short, but this was compensated for by a good yield of jute. The public health has been fairly good. Cholera was more prevalent than in the previous decade, and there was a comparatively high death-rate in 1895, 1899 and 1900, but even in these

Thana.	Population.	Percentage of variation.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL ...	***3,915,068***	***+12·8***	***+13·6***
Sadar Subdivision	***977,476***	***+14·6***	***+14·6***
Nasirabad	204,753	+ 14·4	+ 10·4
Phulbaria	110,547	+ 19·0	+ 26·0
Ghafargaon	162,454	+ 15·5	+ 19·1
Nandail	115,773	+ 9·2	+ 8·6
Iswarganj	160,560	+ 12·1	
Phulpur	163,389	+ 17·7	+ 22·1
Netrakona Subdivision ...	***574,771***	***+ 7·1***	***+11·3***
Netrakona	271,037	+ 8·6	+ 20·2
Kendua	189,421	+ 10·6	+ 9·2
Durgapur	114,313	— 1·2	— 0·6
Jamalpur Subdivision ...	***673,398***	***+16·1***	***+16·2***
Jamalpur	282,477	+ 15·9	+ 16·2
Nalitabari	99,352	+ 16·0	+ 24·5
Dewanganj	145,067	+ 11·9	+ 28·0
Sherpur	146,502	+ 21·2	+ 1·6
Tangail Subdivision ...	***970,939***	***+12·9***	***+14·2***
Tangail	467,739	+ 10·6	+ 18·2
Kalihati	230,807	+ 10·4	+ 28·3
Gopalpur	271,702	+ 19·4	+ 4·4
Kishorganj Subdivision ...	***719,184***	***+11·8***	***+11·1***
Kishorganj	297,378	+ 10·1	+ 12·3
Katiadi	154,387	+ 11·0	+ 13·5
Bajitpur	267,419	+ 14·2	+ 9·4

* Under the Muhammadans the tract west of the old Brahmaputra belonged to Sirkar Bazuha, which included also portions of Rajshahi, Bogra and Pabna, while that to the east of this river was included in Sirkar Silhat.

years the births outnumbered the deaths. The average of the returns for the decade shows a yearly excess of births over deaths amounting to 10 per 1,000.

146. The result of these favourable conditions is a further addition to the population, exceeding that of the preceding decade, though it represents a slightly smaller percentage of increase. The only tract which has not shared in the general advance is the swampy terai in Durgapur, the unhealthiness of which has already been alluded to. This thana was slightly decadent in 1891, and the small increase of 1881 appears to have been due to the inaccuracy of the previous census. The adjoining thanas in Sylhet are equally unhealthy. Further west the country at the foot of the Garo Hills seems less injurious to health, though it is more suited to the residence of people of Gáro origin than to that of the ordinary Bengali. The thanas in this tract all show a great development, especially Sherpur, which has a larger proportional increase than any other thana in the district. Then comes Gopalpur on the Padma, where the growth may be due partly to the formation of new *chars*, and then the sparsely populated thanas in the Madhupur jungle, Ghafargaon and Phulbaria, where the Koch Mande and other aboriginal tribes are pursuing their congenial task of converting the forest into cultivated fields.

There has been an increase of 13,020 immigrants, and of 5,785 emigrants since 1891. The former include 36,891 persons from the United Provinces, and 18,604 from the Patna Division. A large proportion of these are merely temporary visitors, employed in earth-work, palki-bearing, domestic service, and the like. They form their matrimonial connections in their own country, and have no permanent effect upon the population of the district. The net gain by migration since 1891 has been less than 17,000 and forms an inappreciable proportion of the total variation in the population.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	2,814,803	1,900,363	1,788,616	1,683,370
Immigrants	82,760	32,650	78,783	23,207
Emigrants	45,971	34,594	43,577	31,203
Natural population ...	1,978,016	1,002,007	1,753,410	1,691,566

FARIDPUR.

147. The whole of Faridpur is an alluvial formation. To the north and east the land is comparatively high, but in the south the whole country is one vast marsh intersected by strips of high land along the banks of the numerous rivers that once flowed through the tract. The inhabitants are mostly Chandáls (Namasudras), who build their houses on the river banks or on mounds from twelve to twenty feet in height laboriously thrown up during the dry months when the water disappears. The main crops are jute and a long-stemmed variety of winter rice which grows as the flood increases and will thrive in any depth of water provided that it does not rise too suddenly. Subject to this condition the higher the water rises the better; the silt deposit is proportionately increased, the sewage and other impurities which collect in the cold weather are more completely removed, and there is less difficulty in obtaining potable water during the winter months. These marshes are slowly but steadily being silted up, and a constantly growing area is being reclaimed for cultivation. The Padma forms the northern and north-eastern boundary of the district and here changes are constantly taking place. Sometimes large areas are diluviated, while at other times extensive accretions are formed. In recent years accretions have been the order of the day, and several thanas, notably Sibchar and Bhanga, are believed to possess an area considerably in excess of that with which they are credited in the records of the Survey Department. The western boundary is formed by the Gorai and its continuation, the Madhumati, which affords an outlet for much of the water brought down by the Padma during the early part of the monsoon, when the comparatively high level of the Brahmaputra prevents an exit by the ordinary channel. The population has been growing steadily since the time of the first census. "The material condition of the people" says the Magistrate, Mr. K. C. De, "is very good. Wealth is very equally distributed, and there are no very rich or very poor people. The majority live by cultivation or by fishing. The land is very fertile and yields rich harvests with very little toil. The rivers and other inland waters are richly stocked with fish and it would be a very abnormal state of things that could cause a famine."

The health of the district does not appear to be so satisfactory as its material condition. Malarial fever is prevalent especially in the north-western thanas. The years 1892, 1899 and 1900 were very unhealthy, particularly 1900, when special steps were taken to afford medical aid to the sufferers. In the district as a whole the vital statistics indicate an excess of 61,144 births over deaths, but in the Goalundo subdivision and the Bhushana thana the deaths outnumbered the births by 32,709.

148. The census of 1901 shows a net increase of 6·2 per cent., but this is the outcome of very different figures for the various thanas. The north-western part of the district, including the whole of the Goalundo subdivision and Bhushana thana in the head-quarters subdivision, which marches with the decadent thanas of Kumarkhali in Nadia and Magura and Muhammadpur in Jessore, has lost heavily, and Awanpur which adjoins this area is practically stationary. The decrease in the Bhushana thana has been continuous since 1872. Pangsa showed an increase in 1881, but in 1891 it lost more than it had gained in the previous decade. In Goalundo thana the decline is to some extent apparent only, the population having been swollen at the time of the previous census by a great number of pilgrims from the districts further east who had visited Calcutta for the Ardhodaya Yoga and happened to be at Goalundo on their way home on the date when the census was taken. There is another decadent tract in the Palang thana to the west of the Madaripur subdivision which has also suffered much from malarial fever. The decrease would here have been greater but for the additions to its area owing to alluvion. The other thanas in Madaripur all show a great increase due, in the case of Sibchar, to new settlements on alluvial accretions, and in the rest of the subdivision, to the natural growth of the population. The climate is here more healthy, the inhabitants are prolific, the soil is fertile and the gradual reclamation of the swamps still affords ample room for expansion. Trade also is developing, and the boat traffic with Calcutta now passes this way, and not, as formerly, by channels further north. The increase in the Bhanga and Faridpur thanas, like that in Sibchar, is probably due in part to immigration to new *chars*.

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,937,646***	***+ 6·2***	***+ 9·9***
Sadar Subdivision ...	***712,226***	***+ 6·8***	***+ 6·6***
Faridpur	106,452	+ 9·6	+ 5·8
Bhanga	214,064	+ 13·9	+ 11·4
Maksudpur	191,801	+ 8·8	+ 12·4
Awanpur	104,271	+ 1·3	+ 2·6
Bhushana	95,548	— 6·5	— 4·0
Goalundo Subdivision	***319,285***	***— 9·2***	***+ 9·0***
Goalundo	120,520	— 5·3	+ 18·3
Baliakandi	88,532	— 9·8	+ 50·2
Pangsa	110,233	— 12·9	— 15·8
Madaripur Subdivision	***906,135***	***+ 12·5***	***+ 13·0***
Madaripur	218,345	+ 12·8	+ 16·6
Palang	295,819	— 1·2	+ 8·2
Sibchar	180,846	+ 38·9	+ 12·9
Gopalganj	110,897	+ 14·5	+ 13·4
Kotalipara	91,228	+ 15·3	+ 19·5

149. The total number of immigrants is about the same as in 1891, but the number of females is nearly 5,000 less, which seems to indicate a considerable decrease in the number of permanent settlers from other districts. It must however be remembered that many of the foreign-born enumerated in the district in 1891 were pilgrims on their way home from the Ardhodaya Yoga, amongst whom the females far outnumbered the males. The number of emigrants, both male and female, is far smaller than it was ten years ago. The district may have gained slightly by the movements of the people during the decade, but not sufficiently to materially affect the variation in the population which has taken place. By far the greater part of increase must be ascribed to natural causes.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	970,164	967,482	966,594	917,131
Immigrants ...	46,673	26,810	41,637	31,695
Emigrants ...	54,190	21,020	63,807	23,712
Natural population ...	977,681	963,292	928,764	914,138

150. Khulna was formed into a district in 1882. Prior to that date the head-quarters and Bagerhat subdivisions belonged to Jessore, and Satkhira to the 24-Parganas. The northern part of the Satkhira subdivision is a densely populated tract, resembling in its general physical characteristics the adjoining thanas of Jessore; the drainage is bad and there are numerous swamps, and malaria is always present. The other northern thanas are also lowlying, and *bils* are large and numerous,

Khulna.

but the country is more open and there is less jungle, while the stagnant pools and tanks, which are so common in North Satkhira, are rarely to be seen. In these thanas there is still room for expansion and much of the *bil* land is capable of reclamation. The southern thanas include large areas in the Sundarban forests where there is an immense quantity of fertile land awaiting the axe and the plough. The jungle is steadily being pushed back, and every year more land is being brought under cultivation. A great deal of the work of reclamation is carried out by persons whose permanent homes are elsewhere, but the number of regular settlers is gradually growing. The only thanas which showed a decrease between 1872 and 1881 were the Sundarban thanas, Asasuni and Morellganj, for which the sale of the Morell Estate, which led to the abandonment of many of the clearances, was held responsible. During the next ten years both thanas recovered their losses and there was a general advance throughout the district, save only in Kalaroa and Baitaghata, where there was a slight falling-off, which was attributed by the Magistrate to the fact that the population had reached the limit which the soil was capable of supporting.

The health of the people during the decade ending in March 1901 has not been specially good, but no serious epidemics have been reported. The crops were short in 1896 and 1897 owing to drought, and in 1900 a great part of the winter rice was destroyed by the disastrous downpour in September of that year. The cultivators, however, seem on the whole to have been fairly prosperous.

Thana.	Population.	Percentage of variation.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL ...	***1,253,043***	***+ 6·4***	***+ 9·0***
Sadar Subdivision ...	***401,785***	***+17·7***	***+10·0***
Khulna	153,999	+ 8·4	+ 11·5
Dumuria	82,262	+ 4·1	+ 5·3
Baitaghata	40,615	+ 0·8	− 1·2
Paikgachha	124,869	+ 49·7	+ 18·2
Bagerhat Subdivision ...	***363,041***	***+ 6·6***	***+13·3***
Bagerhat	155,806	+ 2·6	+ 19·1
Mollahat	62,460	+ 5·6	+ 17·1
Rampal	61,637	+ 10·7	+ 11·8
Morelganj	83,138	+ 12·4	+ 18·1
Satkhira Subdivision ...	***488,217***	***− 1·5***	***+ 5·7***
Satkhira	101,819	− 1·7	+ 7·2
Kalaroa	71,743	— 10·1	— 2·0
Magura	91,326	+ ·3	+ 10·7
Asasuni	55,227	— 11·9	+ 14·6
Kaliganj	168,102	+ 6·1	+ 2·3

151. The census of 1901 shows that the population has grown by 6·4 per cent., compared with 9 per cent. in the previous decade, and 3·1 per cent. in the nine years prior to 1881. This is the net result of an increase of 17·7 per cent. in the head-quarters subdivision, and of 6·6 in Bagerhat, coupled with a falling-off of 1·5 per cent. in Satkhira. In the latter subdivision again the decrease is practically confined to two thanas, Kalaroa and Asasuni. In Kalaroa it is due to the prevalence of malaria and to repeated attacks of fever. In this thana the vital statistics show a considerable excess of deaths over births, while in the district at large the number of births reported exceeds that of the deaths. There has also been some emigration to the clearances in the Sundarbans where there is a great demand for labour, and wages are two or three times as high as in Kalaroa. Asasuni, which also shows a large decrease, has a very fluctuating population. The cause of the falling-off has not been clearly ascertained. For some reason the extension of cultivation in the Sundarbans is proceeding far more slowly in this subdivision and in Bagerhat than in the adjoining thanas of the 24-Parganas, and of the Khulna head-quarters subdivision. Satkhira itself, which shows a slight decline, has suffered from the diversion of the boat traffic between Calcutta and East Bengal to channels further south; the health of the people has also been unsatisfactory. The growth of the Bagerhat subdivision is normal and calls for no special comment. Cultivation is being steadily extended into the shallow *bils* which form so marked a feature of this part of Bengal. In the district head-quarters the most noticeable item in the statistics is the increase in the population of Paikgachha thana which has grown by nearly 50 per cent. during the decade. This is owing to the

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	653,470	599,573	617,261	559,671
Immigrants	45,097	22,020	51,872	28,007
Emigrants	14,068	11,815	13,785	13,580
Natural population ...	622,441	589,368	579,194	545,244

progress made in pushing back the jungles of the Sundarbans, and to the settlement of cultivators on the new clearances, which attract cultivators not only from other parts of the district, but also from Nadia, Jessore, Faridpur and other districts. The total number of immigrants has fallen off considerably since 1891 but they are more numerous by nearly 27,000 than they would have been, had no fresh settlers come in to the district. Taking the ebb and flow together, the district seems to have gained by the movements of the population during the decade to the extent of from 20,000 to 25,000 persons. It should be noted, however, that many of the Sundarbans cultivators are not permanent settlers, but continue to reside in their old homes and only visit the Sundarbans when ploughing or harvesting operations are in progress.

BACKERGUNGE.

152. Backergunge lying between the Haringhátá and the Megná is an unmistakeable part of the true delta, and its surface nowhere rises much above high water-mark. The whole district is intersected by innumerable rivers and water-channels, whose banks, being comparatively high, are lined with village sites. To the north the country is full of low swamps like those of South Faridpur, where the Namasudras live an almost amphibious life; here much land is still too low to be cultivable, but the process of reclamation is steadily going on as fresh deposits of silt gradually replace water by mud. To the south there are extensive areas of waste land covered with forest, part of which has never yet been brought under the plough, while part was formerly cultivated, but was devastated by the Maghs during the early part of the 18th century. At the present time cultivation in this direction is being rapidly extended, chiefly by Muhammadans, who are constantly throwing out new colonies further and further into the jungle.

Between these two tracts, the marshy swamps on the north and the Sundarban forests in the south, a strip of comparatively high land intervenes which has been longer settled. The population has here attained a density that does not permit of further rapid growth, save only in the east, in the Bhola and Barmuddin thanas, where the effects of Magh incursions have not yet altogether passed away, and there are still considerable stretches of cultivable waste land. The main crop is the long-stemmed winter rice. The land is everywhere most fertile and owing to the extensive areas of waste and the independent character of the cultivators, especially of those who are Muhammadans, they have succeeded in making very favourable terms with their landlords. They are thus exceptionally prosperous even for this part of Bengal. Thanks to the strong winds of the south-west monsoon, fever is far less prevalent than further inland. The water-supply, however, is bad during the winter months and epidemics of cholera are not uncommon.

153. The district is most fortunately situated in every way but one; it is peculiarly exposed to the devastating effects of cyclones, and from time to time, when an exceptionally strong south wind happens to coincide with the bore or tidal wave which sweeps up the Megna at the time of the full and the new moon, storm-waves break over the country, drowning men and cattle and destroying crops and houses. Nor is the harm done by these storm-waves confined to the immediate loss which they cause. The inrush of salt-water renders the water of the rivers undrinkable and the people have resort to stagnant tanks and *bils*. This and exposure inevitably lead to disastrous epidemics of cholera and other diseases. The worst of the recent calamities of this nature was in 1876, when a tidal wave submerged a great part of the district to a depth of from 10 to 45 feet. Nearly 74,000 persons in the Backergunge district alone were drowned, and the cholera epidemic which followed carried off nearly 50,000 more. The census of 1881 showed a nominal increase, but it seems probable that this was due to the inaccuracy of the previous census, and that there had in reality been a considerable loss of population. During the next decade the district made a rapid recovery and added 13·3 per cent. to its population. The increase was enormous in Galachipa, Barmuddin and other thanas that had suffered most seriously from the storm-wave. By 1891, therefore, it may be said that the district had fully recovered from the disaster of 1876.

During the last decade the crops have been uniformly good, but there have been three cyclones, in October 1893, October 1895, and June 1896,

which though not in any way comparable to the terrible disaster of 1876, did a considerable amount of damage, especially in the Bhola and Barmuddin thanas, where they were accompanied by floods of salt-water. These cyclones were followed as usual by epidemic diseases, but in spite of this the average reported birth-rate was nearly 43 per 1,000 compared with a death-rate of rather less than 37. The decade, in the district as a whole, may therefore be held to have been fairly healthy, but some parts suffered from fever, especially the Bauphal and Matbari Police Circles; in the former the reported deaths exceeded, and in the latter they very nearly equalled, the births.

154. The census of 1901 shows a net gain of 6·4 per cent., or almost exactly the same as that indicated by the vital statistics of the district. There has been a decline in the Matbaria thana which has suffered from fever, but the other two Sundarban thanas—Amtali and Galachipa—have grown rapidly, as also have the *bil* thanas, Gaurnadi and Sarupkati. The remaining thanas all show a fair rate of increase, with the exception of Nalchiti, Backergunge, Patuakhali, and Bauphal in the centre of the district, the population of which is almost stationary. This appears to be due, to a great extent to migration to the waste lands in the north and south of the district, and, in Bauphal to a high rate of mortality. The eastern thanas in the Dakhin Shahbazpur subdivision would doubtless have shown a much greater increase but for the damage done by cyclones.

Thana.	Population.	Percentage of variation.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL ...	***2,291,752***	***+6·4***	***+15·3***
Sadar Subdivision. ...	***945,567***	***+7·5***	***+7·9***
Barisal	143,573	+ 7·7	+ 5·1
Gaurnadi	238,034	+14·3	+ 12·9
Mehdiganj	165,675	+ 6·2	+ 7·0
Jhalakati	174,184	+ 6·5	+ 14·0
Nalchiti	80,406	+ 2·5	− 0·7
Backergunge	143,695	+ 2·1	+ 3·1
Pirojpur Subdivision	***553,494***	***+6·5***	***+16·2***
Pirojpur	136,835	+ 5·3	+ 5·8
Sarupkati	206,113	+13·7	+ 12·9
Matbari	114,105	− 4·3	+ 24·3
Bhandaria	102,551	+ 8·0	+ 29·2
Patuakhali Subdivision.	***522,658***	***+5·2***	***+16·4***
Patuakhali	127,283	+ 1·4	+ 11·9
Bauphal	110,583	− 0·4	+ 16·6
Amtali	124,690	+11·3	+ 13·4
Galachipa	90,102	+13·8	+ 33·8
Dakhin Shahbazpur Subdivision.	***270,933***	***+4·6***	***+21·7***
Bhola	150,485	+ 4·2	+ 17·2
Barmuddin	119,748	+ 5·0	+ 28·1

There has been a considerable falling-off in the number of female immigrants, while amongst emigrants the decline is chiefly in the number of males who have left the district. The immigrants greatly outnumber the emigrants, but this is due to a large excess of males who, when not accompanied by females, are usually only temporary settlers. So far as permanent migration is concerned, the balance seems to be against the district, and the natural increase of the population is probably somewhat greater than the variation in the actual population would indicate.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	1,175,903	1,115,849	1,104,443	1,043,522
Immigrants	51,085	5,000	51,725	12,588
Emigrants	24,535	14,477	22,173	15,068
Natural population ...	1,149,353	1,124,426	1,081,901	1,051,992

NOAKHALI.

155. The Noakhali district comprises a tract of mainland, the whole of which, with the exception of a narrow strip in the extreme east, where it borders on Hill Tippera, is low-lying alluvium, and several islands in the mouth of the Megná, the largest of which are Sandip and Hatiya. The houses are built, as in many other parts of East Bengal, on mounds of earth. Each house stands in the middle of its occupant's fruit trees and cultivation, and there are few of the crowded village sites so common further west. The district is still being added to by alluvion, and though parts are occasionally washed away, its area is steadily growing. The Megná, which once flowed past the town of Noakhali has now receded to a distance of more than eight miles. The country is intersected by numerous water channels. The soil is an alluvial clay and is extraordinarily fertile; as long ago as 1625 the island of Sandip was described by Sir Thomas Herbert as "one of the fairest and most fruitful spots in all India." It is estimated that in normal years the district produces twice the quantity of rice required for local consumption. Large quantities of betel-nuts and cocoanuts are also grown for export. The health of the people is generally fair, but the water-supply is very inferior; it is usually derived from the shallow tanks or

ditches from which earth has been taken to raise the land on which the houses are constructed. The district is liable, like Backergunge, to inundation from the bores or tidal waves which sweep up the Megná at the full and new moon, especially at the time of the equinox, and when assisted by a strong south wind occasionally submerge the islands and considerable areas on the main land. Noakhali suffered even more than Backergunge from the bore of 1876; its recorded decrease of 2·3 per cent. between 1872 and 1881 probably represents only a small portion of the actual loss of population, but even according to the census figures the islands of Hatiya lost a quarter and Sandip a sixth of its inhabitants. During the next ten years, everything was favourable to a rapid increase of the population, and the census of 1891 showed an increase of 23·0 per cent., which was wholly due to natural growth, a remarkable instance of the way in which a community can recover from the effects of a catastrophe of the kind without any assistance from outside, and a proof that in some way or other, positive or preventive checks on the growth of population in India, operate to a far greater extent than is usually imagined. It is possible that to some extent the deaths from drowning and disease occurred mainly amongst the old and feeble, *i.e.*, the part of the population whose share in the reproductive process was already at an end, and whose loss would be noticeable only for the few years which would otherwise have elapsed before their decease. But other causes also must have operated. It is often the case that a severe epidemic is followed by a cycle of healthy years when the death-rate is much below normal. Moreover the great reduction in the population must have relaxed the pressure on the land, thereby enabling young men to set up separate establishments at an earlier date and encouraging the Muhammadans, who comprise three quarters of the population, to accept in their harems the widows of their deceased neighbours, who in leaner years would have remained unappropriated and infertile.

156. During the last ten years the crops have been uniformly good and communications have been greatly improved. As stated by the Magistrate, Mr. Cargill:—

> The district has rapidly advanced in the scale of civilisation during the decade Telegraphic communication to Noakhali has been introduced. Roads have been improved Steamer communication has been opened up between various parts of the district and Backergunge. The Assam-Bengal Railway, which runs though the east of the district, has been constructed. These improved communications have facilitated exports and the standard of living has thus been raised. An ordinary raiyat wearing a shirt and a pair of shoes with an umbrella over his head, is a common sight.

The district was visited in 1893 by a cyclone which destroyed a fifth of the rice crop and up-rooted nearly half the betelnut trees. The loss caused by the destruction of these valuable trees has not yet been made good, but in other respects the people soon recovered from their losses.

The health of the district is said to be deteriorating, but this conclusion seems to be based on the gradually growing number of deaths reported by the police, which in 1900 reached a ratio of 42·1 per 1,000 calculated on the population of 1891. The gradual increase, however, seems attributable rather to improved reporting and, if the ratio be calculated as it should be, on the population of 1901, it falls to 37·2 per 1,000, which, if correct, is considerably below the estimated actual death-rate in the province as a whole. That the reporting is now very accurate seems to be proved by the extraordinary number of births shown in the return for 1900 which represent a ratio of 59·3 per 1,000 on the population of 1891 or 52·3 on that of the present census.

157. The growth of the population during the last decade has been 13·1 per cent., and the whole of this has been due to natural development. The volume of migration has not varied to any great extent, and so far as it goes, the balance is against the district. The amount of permanent movement to and from the district is about equal, but amongst temporary migrants the number of persons born in Noakhali who were absent from the district was far greater than that of

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	563,777	572,351	508,727	500,966
Immigrants	10,775	8,568	10,976	7,741
Emigrants...	26,139	7,444	21,469	7,626
Natural population ...	584,141	571,227	519,220	500,843

persons born elsewhere who were found there at the time of the census; it was also considerably in excess of the corresponding number in 1891. The fact already alluded to, viz., that three quarters of the inhabitants are Muhammadans is no doubt largely accountable for the continued rapid growth of the population. An examination of the details reveals a decrease only in one thana, Companyganj, where the Megná, or Bamni as it is here called, has cut away a considerable area of land, and has caused many people to move to other parts of the district. The increase is moderate in Sudharam where there has probably been some diluvion, and in the eastern thana, Chhagalnaia; it is somewhat greater in Fenny and in the islands of Sandip and Hatiya, and is greatest of all in the north-west of the district, in the Begamganj, Ramganj and Lakhmipur thanas. These thanas escaped the cyclone of 1893; their climate is particularly good, and they adjoin the thanas in the Tippera district where the greatest development in that rapidly growing district has taken place. They have also benefited by the great expansion in jute cultivation which has taken place since 1891.

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,141,728***	***+15·1***	***+25·0***
Sadar Subdivision ...	***822,891***	***+14·4***	***+24·2***
Sudharam	139,942	+ 8·8	+ 18·7
Companiganj (old thana Bamni).	45,168	— 8·1	+ 17·7
Lakhmipur	166,526	+ 18·9	+ 23·3
Begamganj	190,065	+ 18·8	+ 24·0
Ramganj	119,673	+ 20·4	+ 23·0
Sandip	115,127	+ 14·0	+ 39·5
Hatiya	55,390	+ 12·2	+ 22·4
Fenny Subdivision ...	***318,837***	***+ 9·7***	***+20·0***
Chhagalnaia	135,316	+ 8·9	+ 13·7
Fenny	183,521	+ 10·4	+ 25·4

TIPPERA.

158. Tippera is bounded on the west by the Megná, and on the east by the native State of Hill Tippera. The greater part of the district is a low alluvial flat, with a light and somewhat sandy, but very fertile soil. Towards the eastern boundary the surface becomes undulating and gradually rises; the soil is here a deep alluvium, alternating with bands of clay and sand. The great majority of the inhabitants are Muhammadans, who are known to be more prolific than their Hindu neighbours. The district was still very sparsely inhabited at the time of the first census in 1872, but since that date it has enjoyed uninterrupted prosperity and its population has grown rapidly. Between 1872 and 1881, the increase was only 7·8 per cent., but during the next decade it rose to 17·8. The growth was greatest in the Chandpur subdivision, which added nearly 31 per cent. to its population in the course of the ten years, a result which was at the time thought impossible, in the absence of any marked immigration from outside, except on the hypothesis that the count of 1881 were defective.

Since 1891 the harvests have been good in seven years, poor in two, 1895 and 1896, and bad in one, 1893, when heavy floods destroyed a great part of the crop and drowned many cattle, and relief works on a small scale were found necessary. The peasantry soon recovered from this temporary calamity, and they are now, says the Magistrate, as prosperous as any in India. The construction of the Assam-Bengal Railway, which traverses the district from north to south and has a branch running westwards through the southern part of the district to Chandpur on the Megná, has greatly improved the communications, and has assisted in the rapid development of the jute trade, which is the other main feature of the decade. The Civil Surgeon reports that the health of the people has not been satisfactory, but here, as elsewhere, the gradual rise in the reported death-rate appears to be due to the greater accuracy of the vital statistics rather than to a growing unhealthiness. And even now the reported death-rate is extraordinarily low, that in 1900 being at the rate of only 28 per 1,000 of the population according to the present census. The recorded birth-rate in the same year was 39·4 per 1,000. According to the returns, the net excess of births over deaths during the nine years preceding the year 901 was 215,933.

159. The actual increase as shown by the census is 335,056, or 18·8 per cent. This great development has taken place entirely through the procreative capacity of the people living in the district without any aid from migration. The number of immigrants slightly exceeds that of emigrants,

but whereas the former stand at almost the same figure as in 1891, the latter have increased by more than 10,000. The number of women, and therefore of permanent migrants, is also considerably greater amongst those who have left the district than amongst the new arrivals. As on previous occasions the growth is greatest in the south of the district and the Chandpur subdivision now possesses considerably more than twice as many inhabitants as it had in the year 1872. The rapid spread of jute cultivation, the formation of new accretions along the bank of the Megná, and the development of trade in Chandpur town account in part for the exceptionally rapid rate of progress in the Chandpur thana. Matlab Bazar has also benefited by alluvion, but in Hajiganj no such explanation is possible, and the reasons for its rapid growth must be sought in the extension of jute cultivation and the improvement of communications by the construction of the Assam-Bengal Railway and the enlargement by natural processes of the channel cut some years ago to connect the Dakatia river with the Megná. Here and in the Laksam and Chandina thanas which adjoin it, the population is still far less dense than in any other part of the district, and there is even now ample room for further expansion. In view of the fact that the present rate of progress has been continuous since 1872, too much stress cannot be laid on the improvement of communications during the decade, and the main reason for the increase which has taken place is doubtless the fact that the tract is very fertile and is capable of supporting a much larger population than it possessed in 1872, and that it has enjoyed great prosperity unbroken by any serious crop failure or wave of unhealthiness. To the east and north of this area of maximum development the rate of growth gradually diminishes. It is least in Comilla and the thanas to the north of it which adjoin the Hill Tippera boundary. This is the oldest part of the district; there is less room for further expansion, the soil is not so well adapted to the cultivation of jute, and there is a tendency to migrate to Hill Tippera where a considerable amount of cultivation is already carried on by persons resident in the adjacent parts of the Tippera district. It seems probable also that there has been some migration from these thanas to those in the south of the district.

Population.	1901.		1891.	
	Males.	Females.	Males.	Females.
Actual population	1,085,989	1,032,002	911,799	871,136
Immigrants	37,095	19,637	37,965	18,697
Emigrants	30,367	25,182	26,199	18,892
Natural population	1,079,261	1,037,507	900,033	871,331

Thana.	Population.	Percentage of variations.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***2,117,991***	***+18·8***	***+17·8***
Sadar Subdivision ...	***957,699***	***+16·6***	***+16·7***
Comilla	177,458	+ 12·4	+ 13·3
Daudkandi	206,414	+ 19·3	+ 20·7
Moradnagar	204,715	+ 15·7	+ 16·1
Chandina	110,707	+ 18·6	+ 18·5
Chauddagram	133,558	+ 14·5	+ 13·1
Laksam	124,847	+ 20·7	+ 19·0
Brahmanbaria Subdivision	***677,084***	***+14·7***	***+11·9***
Brahmanbaria	300,934	+ 14·4	+ 11·2
Kasba	143,638	+ 12·8	+ 11·5
Nabinagar	232,512	+ 16·3	+ 13·6
Chandpur Subdivision ...	***483,208***	***+30·0***	***+30·8***
Chandpur	208,944	+ 36·8	+ 23·6
Matlab Bazar	135,654	+ 23·9	+ 34·3
Hajiganj	138,610	+ 26·8	+ 22·5

CHITTAGONG.

160. Chittagong differs from the portions of Eastern Bengal hitherto dealt with in that it is not, in the main, alluvial. It consists of a long narrow strip of coast, valleys and low ranges of hills, lying between the Bay of Bengal and the Hill Tracts. Its average breadth is about 15 miles and its length about 165 miles. The general trend of the hills is north and south. The islands of Maskhal and Kutubdia, a small tract opposite to them in the centre of the district and a very narrow line along the coast are the only parts of the district where the soil is alluvial. The people generally are prosperous. They get two crops of rice yearly and building materials are cheap and easily procurable. The poorer classes obtain remunerative employment in Arakan during the paddy harvest when they earn a rupee a day. The south of the district is occupied largely by Maghs who took refuge there in the latter years of the 18th century after the conquest of Arakan by the Burmese. In Chittagong town the climate is feverish, but in most parts of the district it is fairly healthy. As in Noakhali and Backergunge so also in the alluvial part of Chittagong, the great scourge of the country are the inundations of sea water which occasionally take place. The tidal wave of 1876 afflicted Chittagong almost as much as Backergunge and the

census of 1881 would have disclosed a decrease of population but for the omissions that took place in 1872. The next ten years were healthy and prosperous and a considerable increase of population was recorded in 1891.

In the earlier years of the last decade the crops and the public health were good, but in 1896 and 1897 the outturn of rice was considerably below the average. The opening of the Assam-Bengal Railway not only vastly improved the local communications but also gave a great stimulus to the development of the Port of Chittagong, which has now become the natural harbour for South Assam and a great part of Eastern Bengal.

161. But on the 24th October 1897 the district was devastated by a cyclone worse than any that it had suffered from since 1876. The hurricane reached its maximum intensity in the small hours of the night when a series of storm waves swept over the islands of Kutubdia and Maskhal and the villages on the mainland near the coast, drowning many thousands of men and cattle, sweeping away homesteads and destroying the standing crops. The loss of life by drowning alone was estimated at 14,000 souls. But this was by no means all. The district had suffered from two bad seasons and the total loss of the rice crop in the tracts chiefly affected caused intense distress. The houses in which the people lived had been blown down and where the storm waves had passed, the very materials had disappeared. In his report on the disaster, the Magistrate says that in some places, only the stumps of broken fruit trees and, here and there, the remnants of the posts of houses remained to show that a village had ever existed. Active relief operations were taken in hand immediately and it is believed that there was no direct loss of life from starvation. But want and exposure must have lowered the general health and rendered the people liable to the attacks of disease. The country was covered with corpses of men and animals and the water-supply was polluted. Cholera broke out with appalling intensity, and in Kutubdia alone it was estimated that 1,300 persons, or more than 11 per cent. of the population, died during the epidemic. The total number of deaths from cholera in the district in 1897 and 1898 aggregated 21,001 against 4,234 in the two preceding, and 1,340 in the two following, years. The excess over the average of these two periods or about 18,000 deaths may not unreasonably be ascribed to the effects of the cyclone.

To fully appreciate the effect of this catastrophe on the population of the affected tracts it is necessary not only to compare the figures for the recent census with those for 1891 but also to note the progress made during the previous decade. These tracts form the most fertile part of the district and in 1891 the Maskhal thana, which includes the island of that name and Kutubdia, showed an increase of 29·5 per cent. Banskhali increased by 25·4 and Chakaria by 20·3 per cent. It may be assumed that, but for the cyclone, an equally rapid rate of progress would have occurred between 1891 and 1901. Maskhal however shows a decrease of 7·3 per cent. and the other two thanas are practically stationary. The difference between their present population and that which they would have attained if they had continued to grow at the same pace as in the previous decade is about 54,000. The whole of this difference may be laid to the account of the cyclone. There has been some loss in other thanas also, due partly to damages caused by the cyclone and partly to subsequent emigration to Kutubdia and Maskhal which has concealed to some extent the true extent of the loss sustained by the latter tracts. It is probably to these causes that the decline in the population of the Satkania thana must be ascribed.

Thana.	Population.	Percentage of variation.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL ...	***1,353,250***	***+4·9***	***+13·9***
Sadar Subdivision ...	***1,153,081***	***+4·6***	***+13·5***
Chittagong	99,343	+10·8	+ 10·3
Mirsarai	102,259	+ 9·3	+ 21·8
Sitakund	74,048	+ 0·9	+ 15·0
Hathhazari	94,571	+ 5·8	+ 0·1
Phatikchari	112,308	+ 2·4	+ 4·7
Raojan	164,501	+ 2·9	+ 10·6
Patiya	221,275	+ 7·1	+ 14·5
Satkania	156,613	— 1·1	+ 7·3
Banskhali	128,163	+ 0·3	+ 25·4
Cox's Bazar Subdivision ...	***200,169***	***+6·5***	***+16·8***
Cox's Bazar	75,909	+ 9·9	+ 10·2
Maskhal	34,991	— 7·3	+ 29·5
Chakiria	47,288	+ 1·3	+ 20·3
Teknaf	42,000	+21·3	+ 13·3

162. The net increase during the decade in the district as a whole is 63,083 or 4·9 per cent. It seems probable that this is only about half of that which would have taken place but for this disastrous cyclone. The greatest growth has occurred in the thanas along the coast which

escaped the brunt of the cyclone, *viz.*, Teknaf and Cox's Bazar in the south, and Chittagong, Sitakund and Mirsarai in the north. The two former are still very sparsely inhabited. In the three latter, the soil is more fertile and there is a much smaller proportion of uncultivated waste than in the inland thanas, and they are now traversed by the railway.

There is very little immigration to Chittagong and the emigration which takes place is to a great extent of a temporary character. Such emigrants are mostly men who leave their wives in Chittagong and go for a few months to Arakan to cut paddy or serve as laskars on steamers. Owing to the poor rice crop in 1900-1901, the number of these temporary emigrants was greater than usual from Satkania thana which furnishes the greater number of these harvest labourers. The number of females amongst the Chittagonians enumerated elsewhere, however, shows that there is also a good deal of permanent emigration, especially to Arakan, where nearly 14,000 of the total number were found. Most of these are Maghs, whose ancestors sought refuge in the district a century ago and who are now gradually finding their way back to their old home. The Magistrate reports that a number of families left the district to settle permanently in Arakan after the cyclone of 1897. The net loss by the movements of the people during the decade would be about 37,000 if all were permanent migrants, but as so many of them are only temporary, the real loss is much less, possibly not more than from 20,000 to 25,000.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	641,392	711,858	615,868	674,290
Immigrants	7,029	4,310	7,824	3,879
Emigrants	85,804	20,233	78,633	24,087
Natural Population	720,167	727,781	686,077	694,514

163. The Chittagong Hill Tracts lie to the east and north-east of the regulation district of Chittagong of which they formed part until 1860. The greater part of the country consists of hills and ravines covered with dense tree-jungle. The main rivers are the Karnaphuli, Sangu, and Mamori. The inhabitants are Maghs, Chákmás, Tiparas, and Kukis. They build their houses on bamboo platforms raised ten feet from the ground, and cultivate on the *jhum* system; that is to say, they make clearances in the jungle, and when the trees and undergrowth they have cut down become sufficiently dry, they burn them; then after the ground has been softened by rain, they dibble in seeds of rice, cotton, maize, melons, and yams, all mixed together. The ashes of the trees furnish a rich manure, and the cultivator thus obtains a bountiful return from the various seeds sown. After one or two years, cultivation becomes impossible on account of the choking weeds that spring up, and the cultivator moves on to a fresh clearance. The district is divided into three circles, each under its own chief or rájá,—the Bohmong in the south, the Chákmá in the centre and north, and the Mong in the north-west. The Chákmá Circle is inhabited mainly by Chákmás, the Bohmong by Maghs, and the Mong by Tiparas. The settlement of the Maghs dates from the Burmese occupation of Arakan, rather more than a century ago. During the decade preceding the census of 1891 the district showed a net gain of 5·2 per cent. There was a great decrease in the Mong Circle, and a still greater increase in the Bohmong, whither many families had migrated from the Mong Circle, but the differences are so great that it seems doubtful if the areas of the two circles were the same on both occasions. The Chákmá Circle showed an increase of 6·6 per cent.

CHITTAGONG HILL TRACTS.

164. Since 1891 the history of the district has been quiet and uneventful. The crops have been good, there have been no widespread epidemics, and the population has grown rapidly, especially that of the Bohmong Circle in the south, where the increase exceeds 40 per cent. This is due probably to migration from other circles or to differences in the dividing line as drawn at the two enumerations. The district lost about 1,500 persons by the adjustment of the Lushai boundary in 1898. But for this the net increase would have been 17·7 per cent. This is due entirely to natural

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	*194,762*	*+16·3*	*+ 5·5*
Chakma	48,780	+ 7·1	+ 6·6
Mong	31,996	+ 40·4	− 40·2
Bohmong	44,075	+ 12·9	+ 86·8

development, coupled perhaps with a more accurate enumeration. The number of immigrants is slightly less than in 1891. On both occasions males greatly preponderated over females; many of the former were temporary visitors who had come to cut timber in the forests or to serve in the local police force. The emigrants are slightly more numerous than they were ten years ago, but even now their number is very small. Two-thirds of the total number were found in Hill Tippera. Thirty years ago there was a considerable movement from that State to the Chittagong Hill Tracts, but since then the pendulum seems to have swung in the other direction, and the present tendency is to leave these hills and settle in Hill Tippera.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	68,238	56,324	59,566	47,720
Immigrants	6,537	527	7,384	985
Emigrants	1,189	703	516	297
Natural population ...	63,010	56,759	52,488	47,032

HILL TIPPERA.

165. The State of Hill Tippera comprises several ranges of low hills running from north to south and the narrow valleys between them. The country, covered for the most part with dense jungle, resembles very closely the Chittagong Hill Tracts. Towards the north the valleys become wider, and along the northern and western boundary there is a strip of comparatively low, level land very similar to the adjoining parts of Noakhali and plains of Tippera. In the hills the *jhum* system of cultivation prevails, but in the lowlands the plough is used and the land is held permanently. Here cultivation is rapidly extending, and the waste land is being brought under cultivation by immigrants from British territory, including, it is said, absconding coolies from the tea gardens of Sylhet.

Thanas.	Population 1901.	PERCENTAGE OF VARIATION.	
		1891-1901.	1881-1891.
DISTRICT TOTAL	*173,325*	*+26·1*	*+43·8*
Agartala	65,615	Not available.	
Belonia	27,343		
Sonamura	30,229		
Kailashor	20,673		
Khowai	18,296		
Dharmagram ...	10,179		

The census of 1872 was very inaccurate, and in 1881 also it is doubtful if a complete count was obtained. In 1891 the arrangements were less elaborate than in British territory, and less information was asked for but it is probable that so far as the bare numbers are concerned, the enumeration was very fairly accurate. On the present occasion the full schedule was used and the procedure differed but little from that laid down in the Census Code for Bengal. The increase of 43·8 per cent. recorded in 1891 was probably due, to a considerable extent, to the greater accuracy of the census in that year. The increase of 26·1 per cent. now recorded appears to be due mainly to the growing immigration from the neighbouring districts of Sylhet, Tippera, Chittagong, Noakhali and the Chittagong Hill Tracts. With a death-rate of 40 per 1,000 per annum, the total number of new settlers requisite to fill death vacancies in the foreign-born population of 1891 would be rather more than 13,000, and about the same number would be required to raise their strength to the number now returned. This leaves 10,000 or rather more than 7 per cent. to be accounted for by the natural growth of the population.

Population.	1901.		1891.	
	Males.	Females.	Males.	Females.
Actual population ...	92,495	80,830	71,598	65,846
Immigrants	24,828	19,066	19,262	14,066
Emigrants	88	64	904	811
Natural population ...	67,755	61,828	53,238	52,591

SOUTH BIHAR.

PATNA.

166. The Patna district stretches along the southern bank of the Ganges. Except in the extreme south the land is a dead level and the soil alluvial. It is watered by various rivers, and practically the whole area is under cultivation; the western part is irrigated by the Sone Canals. Before the era of railways the trade followed the course of the Ganges, and its banks are lined

with towns that once throve on the river traffic, but the trade has now been diverted and the towns are decadent. Twenty years ago the agricultural population had apparently already reached the limit which the land could support, and in 1891 the increase recorded was purely nominal. During the last decade the conditions were normal up to 1900. There was a heavy mortality in 1892, 1894 and 1896, but in other years the public health was fairly good, and the crops were on the whole satisfactory. There was a short outturn in several years, notably in 1891 and 1896, but even in the latter year, though the landless classes suffered from the prevalent high prices, the district escaped the stress of famine. The recorded births during the eight years 1892—1899 inclusive exceeded the deaths by 22,762, and but for the plague, which appeared in epidemic form in January 1900, there is no reason why the district should not have at least maintained its position.

But the plague wrought terrible havoc. It broke out in January 1900, and in the course of that month the number of reported deaths was 2,127. The recorded mortality rose to 4,461 in February and 8,486 in March. In April the epidemic began to subside, and only 3,229 deaths were reported. During the next few months the disease was comparatively quiescent, but on the advent of the cold weather it again began to assume serious dimensions. In November 814 deaths were recorded, and in December 2,886. The total reported mortality from plague during the year was 23,022. The deaths from all causes aggregated 86,996, and exceeded the births by 17,946. In January and February 1901 the epidemic continued to grow in virulence, and 11,510 deaths from plague were reported. The epidemic was at its height at the time when the census was taken. The inhabitants were in a wild state of alarm. Most of the people whose homes were in other districts had fled, and wherever the disease broke out, those who did not go away altogether removed *en masse* to temporary sheds in their fields. Even if the census staff had escaped the general panic and the ravages of the disease, the work of enumerating a population that was constantly on the move would have been a very difficult task. But they did not escape. Many of them were stricken, and many fled, often at the last moment when it was next to impossible to replace them and to prepare afresh the preliminary record, which as often as not had disappeared. Everything possible was done to overcome these difficulties, and in the places where the epidemic was worst, the census was taken in the day time instead of at night. At the same time it cannot be contended that the enumeration was as accurate as it would otherwise have been, and it is inevitable that some of the people who were absent from their own houses must have been left out of account.

167. The net result of the census was a decrease of 148,425, or 8·4 per cent. The decline was greatest amongst immigrants, who were fewer by 53,052, or 36 per cent., than at the previous census. It may be assumed that this result was entirely due to the plague scare which caused many of these temporary settlers to return to their homes. There does not appear to have been any great exodus of the district-born and the number of such persons who were enumerated in other districts was almost the same as in 1891. After allowing for the loss due to the absence of a great part of the foreign-born population there is still a net decrease of 95,373 to be accounted for. For this it would seem, the plague is mainly to blame. There is no doubt that the general tendency in Bihar is to a diminished rate of reproduction, and it will be shown in another chapter that the proportion of children is slowly, but steadily, falling. But the fact remains that, until the plague epidemic, the vital statistics showed a slight excess of births over deaths, and even if we allow for deaths in other districts amongst the large number of Patna people who seek work elsewhere during the dry season, it is not likely that the total number of deaths would greatly exceed that of births. To what extent this decrease of more than 95,000 is to be ascribed to actual mortality, and how far it was due

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	804,383	820,402	868,026	905,384
Immigrants	32,262	50,178	57,408	78,024
Emigrants	73,982	68,334	72,717	69,327
Natural population	46,303	838,558	883,275	896,687

to omissions from the Census Record, it is impossible to say. The total reported mortality from plague was less than 35,000, but a very great number of deaths must have occurred which were not included in the returns. The people feared the remedies that it was sought to apply almost as much as they did the plague itself, and it was often found that plague had been raging for weeks in villages where no deaths had been notified. The reporting agency moreover suffered as much as the general population and, apart from wilful suppression, the reporting must often have been very incomplete, owing to the general disorganisation from which the district administration suffered. It is probable that at least half the plague deaths escaped notice, but even so, it must apparently be admitted that some 25,000 persons were omitted from the census returns.

Thanas.	Population.	Percentage of variations.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL	***1,624,985***	***- 8·4***	***+ 0·9***
Sadar Subdivision	***504,666***	***-15·7***	***+ 0·7***
Firbahor	27,565	+ 7·1	+ 4·0
Sultanganj	1,746	- 90·3	+111·1
Alamganj	25,222	- 14·5	- 4·7
Khwaja Kalan	35,003	- 19·9	+ 2·7
Chauk Kalan	23,619	- 16·7	-12·0
Malsalami	27,742	- 20·3	- 3·6
Phulwari	90,839	- 9·9	+ 3·9
Bikram	165,612	- 9·4	...
Masaudhi Buzurg	109,228	- 11·4	+ 2·2
Dinapore Subdivision ...	***152,085***	***- 11·4***	***+6·6***
Dinapore	64,982	- 11·2	+ 6·6
Maner	87,103	- 11·6	- 3·4
Barh Subdivision	***365,327***	***-10·5***	***+8·5***
Barh	195,829	- 4·4	+ 5·5
Fatua	72,698	- 18·7	+ 1·4
Mokameh	96,800	- 15·1	+21·6
Bihar Subdivision	***602,907***	***- 0·9***	***-3·1***
Bihar	265,066	+ 0·8	- 6·7
Hilsa	99,529	+ 2·7	- 0·9
Chandi	70,760	- 8·8	
Islampur or Athasaria ...	79,482	- 2·4	+ 3·1
Silao	88,064	- 1·9	- 1·8

The thanas which show the greatest decadence are all, with two exceptions, on the bank of the Ganges or the Sone. The greatest falling-off is in Malsalami, Fatua and Mokameh. The only inland police circles where there has been a marked decline are Masaudhi Buzurg and Chandi. Excluding Chandi, the Bihar subdivision, as a whole, shows a slight increase. These variations follow very closely the course of the plague epidemic, and the greatest decline has occurred in the thanas where plague was most virulent.

GAYA.

168. Prior to 1865 the northern part of Gaya formed, with the south of Patna, the old district of Bihár, while the southern part constituted, with part of Hazaribagh, the *zila* Ramgarh. The characteristics of the two tracts are quite distinct. The northern part, which extends southwards to about 10 miles beyond Gaya town, is fairly level, but has a slope sufficient to enable the water to be collected in reservoirs, which the people construct themselves and from which they irrigate their fields. In the west a considerable area is irrigated from the Sone Canals. The soil is fertile and the population is fairly dense. The southern part which still bears the name of Ramgarh, is imperfectly irrigated, the surface is more elevated, and the soil is comparatively barren; the population is here sparse and a great part of the area is still covered with forest. The proportion of landless labourers in Gaya is somewhat high, and many eke out their local earnings by working as coolies in Calcutta and other parts of Bengal Proper during the dry months.

The population is not progressive. Between 1872 and 1881 there was an increase of 9·1 per cent., but much of this must has been due to better enumeration. The following decade showed an increase of only 0·6 per cent. This was explained partly by the unhealthiness that had prevailed and partly by the loss sustained by migration. Since 1891 the conditions have on the whole been unfavourable. During the earlier years fever was very prevalent, and in 1892 and 1894 the number of reported deaths exceeded that of the births. Later on there seems to have been less fever, but in 1897 a severe outbreak of cholera again caused the deaths to exceed the births. Taken as a whole, the births during the nine years ending with 1900, exceeded the reported deaths by about 60,000, but it must be borne in mind that a considerable proportion of the district population is absent for several months in

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	1,011,271	1,048,602	1,046,011	1,093,320
Immigrants ...	18,875	27,239	31,714	32,350
Emigrants ...	96,550	76,939	114,349	106,506
Natural population ...	1,088,996	1,098,302	1,127,646	1,167,306

the year, and if the deaths that occur amongst them while away from home, which are registered in other districts, be added to those recorded in Gaya, the total number would equal, if it did not surpass, the recorded excess of births over deaths. There was scarcity in 1891-92 owing to the failure of the winter-rice and spring crops, and again in 1896-97, when the short outturn was aggravated by the high prices of food grains consequent on famine elsewhere. In the Nawada Subdivision, the Sakri river overflowed its banks in 1896, carrying away several villages and covering the land with a deposit of sand. The condition of the poorer classes during the decade has, on the whole, been unsatisfactory.

169. At the same time, but for the appearance of plague in the latter part of 1900, there seems no reason why the population of the district should have decreased, as it has done, by 3·7 per cent. The disease broke out in Gaya town in October and spread rapidly during the next four months. By the day of the census the recorded deaths from plague aggregated 3,775, and it is probable that the actual mortality was far greater. The people of villages where plague appeared left their homes and took refuge in temporary sheds constructed sometimes near and sometimes far away from the old village sites. The difficulties which the prevalence of the epidemic threw in the way of the census operations in the Patna district have already been described. The conclusion arrived at, which is applicable also to Gaya, was that the census results were affected in three ways; firstly, by a mortality far in excess of that indicated in the death returns, secondly, by the departure to their own homes of temporary settlers from other districts, and, thirdly, by an incomplete census, due partly to deaths and desertions amongst the census staff and partly to the failure to enumerate the whole of the panic-stricken fugitives from villages where plague had broken out. The loss due to the flight of natives of other districts may be taken as the difference between the present immigrant population of the district and that recorded ten years ago, or about 8,000. The rest of the decrease or about 70,000 must apparently be attributed partly to plague mortality and partly to a defective census in the plague-stricken parts of the district. The Magistrate, Mr. Oldham, points out that the Nawada Subdivision which was remarkably free from plague up to the date of census, shows an increase in every thana. As regards the extent to which the two factors mentioned have contributed to the falling off in the district-born population it is difficult to form any definite opinion. Mr. Oldham has gone into the subject with some fulness, and the following remarks are extracted from his report:—

"As soon as the first of the charge totals came in, the decrease attracted my attention. The very marked decrease in the case of the Tikari thana charge led me to seriously doubt the correctness of the figures reported, although I knew that plague had been particularly bad in that jurisdiction.

"I immediately ordered that all supervisors and enumerators were to be brought into Tikari with all schedules, both draft and final, and kept present from the morning of the 6th, when I would go out and inspect the books and examine the staff. As soon as the provisional district totals were despatched, I proceeded to Tikari and made a systematic inspection and enquiry. I found in this way that *more than eleven thousand persons* had in this jurisdiction alone left their houses since the preliminary record. We then proceeded to cross-question the enumerators, especially those in whose books the largest number of red ink cuttings had been made, as to each individual case. I read out each name that had been scored through and questioned the enumerator as to where the person had gone to. I found the enumerators were invariably able to give a satisfactory answer. The persons had either run away or had died of plague. When the person was reported to have fled to another village in the same jurisdiction, I got out the schedule book of that village, and made sure that he had been entered in red ink in that village. In this way we checked a large proportion of the red ink cuttings, and I was agreeably surprised to find the general care and precision with which the work had been done and the general accuracy of the books."

170. It would thus appear that the decrease was due more to the deaths that occurred than to the disorganisation of the arrangements, but where such large numbers of persons were on the move it seems impossible that all could have been accounted for by the census staff. Even without the complication of plague, absentees are more readily noticed than new arrivals, and the final revision on the night of the actual census tends to reduce slightly the real population. This tendency must be greatly exaggerated when

the absentees and new arrivals are exceptionally numerous, and when the people are overcome with fear and the enumerators are performing a thankless task under exceptional difficulties, and when many of them are new men, hastily appointed at the last moment, to replace others who have died or disappeared. When we turn to the statistics for individual thanas, the responsibility of the plague for the loss of population that has occurred becomes very apparent. Up to the date of the census the epidemic had wrought most havoc in Tikari, and this thana has sustained a loss of 19·8 per cent.; then come Atari, Gaya Town and Gaya thana, with decreases of 14·9, 11·3 and 6·5 respectively. The three thanas in the Nawadah Subdivision which escaped the epidemic all show a slight advance. The decadence in Ramgarh, *i.e.* in the southern thanas, is due partly to long continued general unhealthiness and partly to emigration to the adjoining districts of Hazaribagh and Palamau. In Sherghati and Barachati the decrease has been continuous since 1881.

Thanas.	Popula-tion.	Percentage of variation.	
	1901.	1891–1901.	1881–1891.
DISTRICT TOTAL ...	***2,039,933***	***–3·7***	***+0·6***
Sadar Subdivision	***751,855***	***–9·7***	***–1·6***
Gaya	168,238	– 6·5	–10·9
Gaya Town	71,288	–11·3	+ 5·1
Tikari	148,131	–19·8	+ 4·3
Sherghati	132,023	– 1·4	– 1·3
Atari	105,269	–14·9	+27·9
Barachati	106,916	– 2·3	– 0·3
Nawadah Subdivision ...	***453,868***	***+3·2***	***–7·4***
Nawadah	296,039	+ 4·8	– 6·7
Rajauli	75,202	+ 0·5	–10·4
Pakribarwan	82,627	+ 1·0	– 7·9
Aurangabad Subdivision	***467,675***	***–1·0***	***+5·8***
Aurangabad	247,653	– 4·1	+ 5·1
Daudnagar	112,525	+ 6·1	+ 8·7
Nabinagar	107,497	– 0·5	+10·3
Jahanabad Subdivision ...	***366,535***	***–1·8***	***+2·2***
Jahanabad	247,879	– 3·9	– 1·6
Arwal	118,656	+ 3·2	+13·2

171. Shahabad, like Gaya, is divided into two distinct parts. The northern portion, comprising about three-fourths of the whole area, is a low-lying alluvial flat and is entirely under cultivation. It is extensively irrigated by canals and its crops are thus to a great extent protected from drought. The south of the district, comprising the greater part of the Bhabua and Sasaram thanas, is occupied by the Kaimur hills, an undulating plateau, unprotected by irrigation, and yielding poor and precarious crops. The population is here very sparse and much of the land is still covered with forest. The tenantry of Shahabad are sturdy and independent and enjoy a fixity of tenure which saves them from rack-renting. The proportion of landless labourers is small and the surplus inhabitants find ready employment in Bengal either in the police force or as peons and club-men under the zamindars; the district also supplies many recruits for the native army. The climate of the northern part of the district is said to be steadily deteriorating. There is no system of drainage and the surface is so flat and low that there is no outlet for the water which accumulates. The introduction of the canals is said to have raised the sub-soil water level and so encouraged malaria, but whether this be so or not, and the view is not supported by the results of recent research regarding the spread of malarial affections. The district has long been very unhealthy, and in 1891 a decrease was averted only by a large gain from migration. Fever began to make its ravages felt in 1879 and from that time the epidemic grew steadily worse until 1886, when the district was stigmatized as the worst in the whole province in respect of fever mortality.

Shahabad.

172. During the last decade there has been no marked change in the condition of the people. There was a partial crop failure in 1896 but it resulted in famine only in the southern part of the district where irrigation affords no protection against deficient rainfall. Wages have risen, and the prices of food grains are higher than they were ten years ago. The district has maintained its reputation for unhealthiness. The worst year was 1894 when the recorded death-rate exceeded 53 per thousand and the mortality from fevers was greater than in any other district in the Province. From 1892 to 1900 inclusive, the vital statistics show an excess of deaths over births amounting to more than 25,000. It would probably be double this amount if allowance were made on account of the deaths registered in other parts of Bengal amongst the

natives of this district who were temporarily away from it. The district escaped the ravages of plague until shortly before the census, when it broke out in the head-quarters station. The number of deaths reported was small, but the alarm which the epidemic created sufficed to drive to their homes most of the temporary settlers from other districts and the number of foreign-born males found in Shahabad at the time of the census was only 26,877 compared with 73,184 ten years previously. The movements of the population have resulted in a net loss to the district which is even greater than the figures would indicate, as there has been an emigration of 7,633 persons to the colonies, of whom the census necessarily took no count. Allowing for those who returned from the colonies, the net loss on this account may be placed at about 6,000.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	936,544	1,026,152	989,596	1,070,983
Immigrants	26,877	37,036	73,184	51,048
Emigrants	82,579	77,134	79,030	100,714
Natural population	992,246	1,046,250	995,442	1,120,649

173. The above adverse conditions have resulted in a decrease in the district population of 97,883 or 4·7 per cent.* The falling off is most marked in the Bhabhua subdivision where it is due to the unhealthiness of the climate and to the migration of the people, especially during the famine year, to more favoured parts of the district. Elsewhere the decrease is greatest in the Arrah thana where it may be wholly ascribed to the exodus on account of plague. The only thanas that have gained ground are three of the four thanas of the Sasaram subdivision. The construction of the railway accounts largely for this, coupled with the fact that the area irrigated by canals is greater here than in any other part of the district. Piaro thana in the head-quarters subdivision, which borders on this tract and has also a plentiful supply of canal water, is practically stationary.

Thana.	Popula-tion.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL	***1,962,696***	***– 4·7***	***+ 5·7***
Sadar Subdivision	***699,956***	***– 5·9***	***+ 8·5***
Arrah	314,168	– 9·4	+ 12·3
Shahpur (Belauti)	188,059	– 5·4	+ 1·5
Piaro	197,729	– ·1	+ 10·3
Buxar Subdivision	***416,704***	***– 5·0***	***+ 3·5***
Buxar	156,426	– 6·0	+ 2·0
Dumraon	260,278	– 4·4	+ 4·4
Sasaram Subdivision ...	***539,635***	***+ 1·2***	***+ 9·5***
Sasaram	166,359	+ 1·3	+ 5·3
Bikramganj	186,482	+ 2·3	+ 1·4
Dehri	90,167	+ 8·1	+ 5·7
Kharghar	96,736	– 6·6	— 1·0
Bhabhua Subdivision ...	***306,401***	***–11·2***	***+ 6·8***
Bhabhua	165,728	– 8·9	+ 5·8
Mohanea	140,673	– 13·7	+ 8·2

Monghyr.

174. Rather more than a third of the Monghyr district is on the north bank of the Ganges, and is a low but fertile alluvial plain which supports a dense population and differs but little from the adjoining portions of Darbhanga and Bhagalpur. The south of the district is also to a great extent alluvial, but the general level is higher and the surface is more undulating, and several ranges of hills, outliers of the Vindhyan series, enter the district from the south and gradually converge towards Monghyr town. The soil in the hilly tracts is comparatively barren and the inhabitants are few in number. Between 1872 and 1881 there was an increase of 8·5 per cent. in the population, and the next ten years showed a further advance of 3·3 per cent. The greatest gain was in the Begusarai subdivision. The north-eastern part of the Jamui subdivision, which had suffered a great deal from fever, lost heavily and the population of the subdivision as a whole was stationary.

During the last decade the public health has been fairly good, and in only two years, 1892 and 1894, did the deaths out-number the births. According to the returns the net excess of births during the nine years, 1892-1900, has been 126,872, but the Magistrate is of opinion that the reporting of the deaths of infants is very defective, and that the real difference between the birth and death rates is less than these figures would indicate. Plague broke out in January 1900, but subsided in May, only to reappear with renewed virulence in the ensuing cold weather. The total number of deaths from plague reported in 1900 was 2,052, and in the first two months of

* All the districts of the United Provinces which adjoin Shahabad, viz., Mirzapur, Benares, Ghazipur and Ballia, show a heavy decrease.

1901 it was 2,433. As in other districts, so also in Monghyr, the disease was far more prevalent than the reports would indicate, and much of the mortality was concealed. The parts of the district which suffered most were Monghyr town and Shaikhpura Thana. The material condition of the people seems to have improved somewhat. The crops were short in 1891, especially in the northern part of the district, and relief works were opened for several months. In 1896-97 the people suffered from high prices consequent on famine elsewhere, but the crops were fairly good. In other years the general result of agricultural operations has been satisfactory, and the wages of ordinary labourers have risen by about 33 per cent.

175. The census of 1901 shows an increase of 32,783, or 1·6 per cent. The improvement is considerably less than the vital statistics would lead one to anticipate. Directly and indirectly, plague probably accounts for a loss of from 20 to 25 thousand persons, and in addition to this there is a large adverse balance on account of migration. Assuming all movements to and from the district to be of a permanent nature, it may be noted that the immigrants now number 86,069, or about 19,000 more than would have been the case had there been no new comers since the previous census, but the emigrants exceed by about 58,000 the estimated number of survivors amongst the emigrants of 1891. These figures indicate a net annual loss of about 4,500 persons. The whole of the migration however is not permanent and the real loss on this account in the decade may be roughly estimated at from 30,000 to 40,000 persons.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	1,011,580	1,057,224	987,072	1,048,949
Immigrants	37,824	48,545	42,517	58,757
Emigrants	96,544	87,573	105,221	83,997
Natural population ...	1,070,300	1,096,252	1,049,776	1,074,189

The fertile Begusarai subdivision, on the north of the Ganges, again shows the greatest growth of population. The head-quarters subdivision is stationary, owing mainly to the out-break of plague and the consequent mortality and confusion, but the removal of certain offices of the East Indian Railway from Jamalpur to Calcutta has also affected the population some extent. The only thana in this subdivision which shows a marked increase is Kharagpur, which at the time of the census harboured a considerable number of plague refugees from Monghyr. In the Jamui subdivision also, the population is practically stationary; the sparsely inhabited and hilly thanas in the south-east, Jamui and Chakai, continue to show steady development, while Shaikhpura and Sikandra, in the north-east, have lost ground. The decrease is specially marked in Shaikhpura, which adjoins the Mokameh Thana in Patna and, like it, has suffered greatly from the ravages of plague. Both Shaikhpura and Sikandra were decadent in 1891 also, when the prevalence of fever was blamed for the result.

Thanas.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***2,068,804***	***+ 1·6***	***+ 3·3***
Sadar Subdivision ...	***874,611***	***+ ·4***	***+ 3·3***
Monghyr	129,064	− 9·3	+ 1·8
Gogri	308,913	+ 2·5	+ 4·3
Jamalpur	18,371	− 14·4	+ 9·4
Surajgarh	168,906	+ ·3	+ 1·3
Khargpur	159,137	+ 6·8	+ 4·5
Jamui Subdivision ...	***551,227***	***− 0·5***	***+ 6·5***
Shaikhpura	181,897	− 7·5	− 1·3
Jamui	133,979	+ 6·6	+ 4·4
Sikandra	112,622	− 1·6	− 6·4
Chakai	122,729	+ 4·8	+ 7·0
Begusarai Subdivision ...	***642,966***	***+ 5·2***	***+ 5·9***
Begusarai	408,110	+ 3·7	+ 5·3
Tegra	234,856	+ 7·8	+ 7·0

NORTH BIHAR.

SARAN.

176. Saran is a fertile tract of rich alluvial soil lying between the Gogra and Ganges on the south, and the Gandak on the north and east. It is very densely populated and highly cultivated. It grows an unusual variety of crops, including opium and indigo. There is a great deal of irrigation, partly from canals but mainly from wells constructed by the cultivators themselves. The increase of 10·5 per cent. between 1872 and 1881 was attributed mainly to better enumeration and the same explanation was given of part of the increase of 7·4 per cent. during

the next decade. The district already contains a larger population than it can support and it sends its surplus inhabitants into every part of Bengal in search of employment. Most of them go only for a time and return home after intervals of one or two years. The improvement in railway communications has greatly facilitated their movements, and while on the one hand more people leave the district, on the other, they return home at more frequent intervals than formerly.*

177. During the first half of the last decade the crops were fairly good, and the general condition of the people was satisfactory. The short rainfall of 1896 resulted in famine but, owing to the variety of crops grown, the district suffered less severely than its neighbours in North Bihar. The distress was greatest where the cultivation of rice predominated. The Gopalganj subdivision suffered most, but about two-thirds of the Siwan and half of the Sadar subdivision were also badly affected. Relief operations were commenced in November 1896 and continued for ten months, when the position was restored by a good *bhadoi* harvest (Indian corn and millets), and the prospect of a bumper crop of winter rice. The famine undoubtedly told severely on the people and there must necessarily have been some deaths amongst the aged and infirm owing to the want of proper nutrition, but it had no apparent effect on the reported death-rate for the district as a whole. The recorded mortality in 1897 was less by 11 per cent. than the average of the preceding three years. The birth-rate was also less, but only by 8 per cent. It may be argued that a diminution in both births and deaths points to defective reporting of vital occurrences in a season of special strain. But a reduction in the birth-rate, especially in the latter months of the year, would naturally result from the reduced vitality of the people. Moreover, defective reporting would reduce the returns for births and deaths in equal proportions so that even if it be assumed that the whole decrease in the birth-rate in 1897 was due to this cause, the fact remains that the deaths decreased in a greater proportion than the births, and that they were therefore really fewer in number than the average of the three previous years. Since the famine the harvests have again been good and by the time of the census the people had entirely recovered from its effects.

The decade was healthy up to 1899, when plague made its first appearance, and between 1892 and 1900 the reported births exceeded the deaths by 83,725.† The history of the plague is much the same as in Patna, but it was less widespread and the Gopalganj and Siwan subdivisions were very little affected. The disease assumed epidemic proportions during the cold weather of 1899-1900. It then gradually faded away, but appeared with renewed intensity in the succeeding cold weather. During the three months preceding the census 6,767 deaths from plague were recorded. The real number was doubtless far greater, and the epidemic had the usual effect of disorganising the arrangements for the census as already described in the case of Patna and Gaya.

178. There has been a loss of population during the decade of 57,968 persons or 2·2 per cent.‡ Owing to the unusual amount of emigration from this district it is very difficult to trace the causes of this most unsatisfactory result. At first sight it would seem that as immigrants have increased and emigrants have fallen off the district must have gained by the movements of the people, but this does not appear to be the case. There is of course no doubt that the district has gained by immigration. Just as prior to 1891 there had existed a

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	1,695,788	1,314,221	1,132,670	1,332,337
Immigrants	18,432	27,072	7,609	32,023
Emigrants	175,498	69,436	225,300	153,654
Natural Population ...	1,852,354	1,345,585	1,350,267	1,454,168

* It is interesting to notice that the remittances by money order to the Saran district aggregated nearly 35 lakhs of rupees in 1900 compared with less than 17 lakhs in 1892-93. This may be due partly to an increasing use of the post office for the purpose of remitting money, but it must also be attributed in part to the larger number of Saran people who go elsewhere to earn a livelihood.

† From this excess at least half should be deducted on account of deaths amongst temporary emigrants registered in other districts.

‡ The neighbouring districts of the United Provinces—Ballia, Azamgarh and Gorakhpur—also show a loss of population.

strong flow of population from Saran to its neighbours, so during the last decade the tendency seems to have been in the other direction, especially in the case of the adjoining districts of the United Provinces, which have given 32,064 persons to Saran, compared with only 14,992 in 1891. But on the other hand, in spite of the superficial conclusion indicated by the figures, it seems equally clear that there has been an even greater loss by emigration. The decrease in the census figures occurs in the districts adjoining Saran where the migration may be held in the main to be of a permanent character. Prior to 1891, in addition to accidental migration due to marriage, there had evidently been a strong overflow of the surplus population of Saran into Champaran, Gorakhpur and Shahabad, where there was a greater quantity of land available for cultivation.* Owing to bad crops and other causes the same inducement for migration no longer exists and the movement has died out. The decrease in the number of Saran people now found in the adjoining districts seems to be due, not so much to these settlers having returned to their old homes, as to the fact that many of them have died since 1891, and their place has been taken by their children born in the districts to which they migrated. If so, it will appear that even in the case of Bihar districts there has been some loss by migration†, and in Bengal Proper, the emigrants from Saran now number 114,193 compared with 85,536 ten years ago. Another reason for the decline is that there has been a reduction in the birth-rate. This subject belongs properly to another chapter, but it may be noted here that there has been a marked falling off in Saran in the proportion of married people and also in the number of children brought into the world.

District.	1891.		1901.	
	Male.	Female.	Male.	Female.
Shahabad	2,643	5,197	13,326	20,350
Muzaffarpur	7,991	4,646	7,969	6,009
Patna	4,131	4,741	7,684	16,894
Champaran	15,464	9,089	40,278	33,063
Gorakhpur	11,207	12,831	29,464	41,096
Ballia	1,622	9,587	1,574	13,061
Azimgarh	157	177	27	242
Total ...	46,925	59,218	99,566	130,621

179. The net result of the last census is, as already stated, a decrease of 2·2 per cent. For this the Sadar subdivision, which has lost 5·5 per cent., and Darauli thana in Siwan are responsible. This is the tract which was suffering from plague at the time of the census. The rest of the district has almost exactly the same population as it had ten years ago. The decrease is greatest where the plague was worst, *i.e.*, in Sonpur, Chapra and Parsa thanas. It may therefore be concluded that while the general want of progress is due to the adverse balance of migration and to the fact that the district is unable to support a much greater population than it already possessed in 1891, the plague is to blame for the greater part of the decrease that has occurred in the southern part of the district.

Thana.	Population.	Percentage of variation.	
	1901.	1891-1901.	1881-1891.
DISTRICT TOTAL ...	***2,409,509***	***-2·2***	***+ 7·4***
Sadar Subdivision ...	***979,718***	***-5·5***	***+ 2·7***
Chapra	353,078	- 7·3	+ 5·6
Manjhi	152,460	- 5·4	+ 4·3
Parsa	223,266	- 4·9	+ 2·0
Mashrak	143,503	- 1·6	- 3·7
Sonpur	100,411	-10·3	- 6·1
Gopalganj Subdivision	***635,047***	***+ ·1***	***+11·7***
Gopalganj	275,742	+ ·3	+ 5·6
Mirganj	359,305	- ·1	+ 16·9
Siwan Subdivision ...	***801,744***	***+ ·1***	***+10·2***
Siwan	337,926	+ 1·1	+ 11·1
Darauli	220,820	- 2·3	+ 5·8
Basantpur... ...	242,998	+ 1·1	+ 14·0

180. In the east and south, Champaran closely resembles the adjoining districts; the land is level, fertile and highly cultivated, and the population is dense. Towards the north-west the surface becomes more undulating and gradually rises as the Nepal frontier is approached. The population is here sparse and cultivation gives

CHAMPARAN.

* In the case of Champaran most of the immigrants took up land in the sparsely populated northern thanas, Bagaha, Shikarpur and Adapur. A similar movement within Saran itself was observed between 1881 and 1891 in the case of Mirganj, where the existence of more waste land there than in other parts of the district attracted settlers from other thanas.

† The decrease in the number of Saran-born persons enumerated in the Sonthal Parganas from 15,145 to 1,370 is probably fictitious. I have already had occasion to point out that the Sonthal Parganas return of birth-place in 1891 is untrustworthy. No less than 18,238 persons enumerated in the Sonthal-Parganas were shown as born in Gaya compared with 1,545 on the present occasion.

way to grassy plains and extensive tracts of forest, amongst which the scattered clearances of the aboriginal Thárus afford the only evidence of human occupation. The Gandak now forms the south-western boundary of the district, but it formerly flowed through the centre, along the course indicated on the maps as the Buri Gandak. The whole of the tract between the new and the Buri Gandak has been subject to fluvial action within comparatively recent times, and the soil is here light and sandy. On the other or north-eastern side of the Buri Gandak the prevailing soil is the *bhángar* or older alluvium, a stiff clay requiring irrigation, which is provided mainly from tanks and wells and by damming up the hill-streams. The rainfall in Champaran is heavier than in other Bihar districts, and the moisture of the atmosphere, the presence of numerous stagnant lakes formed in old river beds, and the proximity of the hills combine to give the district a far more unhealthy climate than any other part of Bihar. The census of 1881 showed an increase of population amounting to 19·5 per cent., due partly to improved enumeration and partly to immigration from the adjoining districts to the sparsely inhabited thanas in the north of the district. The next decade showed a further advance of 7·9 per cent., which was due mainly to a continuance of the stream of immigration; the total number of persons born elsewhere but censused in Champaran having reached the unprecedented number of 248,511.

181. Since 1891 the district has suffered from a succession of lean years. The harvest of winter-rice in 1891 and the spring crop of 1892 were poor and prices rose, causing the people to feel the pinch of scarcity. In 1892 the winter-rice crop was again below the average. The crops of 1893 and 1894 were fair, but they were followed by bad harvests in 1895, and the spring of 1896, and a general failure of the winter-rice crop in the autumn of that year. The general distress then deepened into famine. Practically the whole district was affected. The Dhaka thana suffered most of all, and next to that, the whole of the Bettiah subdivision, except a narrow strip on the banks of the Gandak. The relief operations were on a larger scale than in Saran, but their duration was about the same. The efforts of Government to save the people from starvation appear to have been successful, and the number of deaths reported was lower than the average of the three preceding years. The birth-rate fell in about the same ratio, but as already explained, a low birth-rate is a necessary sequel of a famine, and the number of births during the latter part at least of 1897 would naturally be below the normal. It follows that even if the reporting agency did their work less efficiently during the famine than at other times, there is no reason to suppose that there were more deaths in 1897 than in the average of the three preceding years. The death-rate which was low in 1897 was still lower in 1898 while, on the other hand, more births were reported in 1899 than in any other year since the registration of births was introduced. In 1898 and 1899 the people enjoyed better harvests, but it was not until the end of 1900 that their prosperity was generally restored.

It was reported in 1891 that the district was gradually losing its reputation for unhealthiness, but unfortunately the improvement has not been maintained. There have been frequent epidemics of cholera, and fever also has been very prevalent. In five of the nine years ending in 1900 the reported deaths have exceeded the births, and the net result is an excess of 1,059 deaths.*

182. The census of 1891 shows a decrease of 69,002 persons, or 3·7 per cent. The result is directly attributable to the short crops and unhealthiness from which the district has suffered. Not only has the arrival of new settlers ceased, but it seems probable that if the figures for 1891 are correct many of those who were then in the district must since have returned to their former homes. Even if a death-rate of 50 per 1,000 be allowed on account of the great unhealthiness of the decade, there would still be nearly 150,000 survivors from amongst the settlers enumerated in 1891 even if their ranks

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	885,607	904,856	936,135	923,330
Immigrants ...	55,047	51,734	140,992	107,519
Emigrants ...	16,442	19,633	80,896	19,288
Natural population ...	847,002	872,757	816,039	835,099

* The Magistrate thinks it probable that deaths due to epidemics of cholera are not fully reported.

had not been swelled by the arrival of any new comers. The actual number of immigrants, however, is now only 106,781.*

Part of the decrease may be due to the return home of some of the immigrants who were enumerated in Champaran in 1891, but the falling off is due mainly to the unhealthiness of the decade and the diminished fecundity of the people consequent on a series of bad years. The emigration to other districts of persons born in Champaran does not appear to have greatly affected the figures. The number of female emigrants remains practically unchanged, but that of males, most of whom are temporary absentees, has decreased. The Magistrate is of opinion that there has been some movement across the Nepal frontier, but of this no statistics are available. The only thana that has not contributed its quota to the general result is Adapur. This thana enjoys the advantage of irrigation and an exceptionally fertile soil, and in part of it a full rice crop was secured even in 1896, when there was a disastrous failure elsewhere. The thanas that have suffered most are Gobindganj and Motihari, but the reason for this is not apparent. They suffered from the famine far less than Dhaka, where the proportional decrease is only half as great as in the thanas to the north-west, where there has been only a very slight loss of population.

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891
DISTRICT TOTAL ...	***1,790,463***	***- 3·7***	***+ 8·0***
Sadar Subdivision ...	***1,040,599***	***- 5·4***	***+ 7·8***
Motihari	173,474	- 8·8	+ 6·4
Gobindganj	157,971	- 11·8	+ 8·4
Kesaria	177,917	- 3·1	+ 3·3
Madhuban	194,427	- 4·6	+ 4·1
Dhaka Ramchandra ...	258,898	- 4·0	+ 6·2
Adapur	167,882	+ ·2	+ 20·8
Bettiah Subdivision ...	***749,864***	***- 1·3***	***+ 8·0***
Bettiah	338,755	- 1·6	+ 3·3
Shikarpur	207,088	- 1·3	+ 15·1
Bagaha	214,021	- ·9	+ 9·3

183. Muzaffarpur is an alluvial plain, but the country north of the Bághmati is more marshy than that to the south of this river, and its climate is reputed to be less salubrious than the rest of the district, which is usually very healthy. The staple crop is rice. Between the Bághmati and the Buri Gandak the land though less marshy, is more liable to inundation. Further south again, in the Hajipur subdivision, the country is higher, but it is capable of irrigation from the numerous streams that intersect it. This part of the district is reputed to be the most fertile, and it has the further advantage of a variety of crops, so that it is less seriously affected by a failure of the monsoon rains. Prior to 1881 there was a considerable growth of population, which was attributed in part to the defects of the first census in 1872. There was a further increase of 5 per cent. in 1891; there was a falling off in Hajipur, but this was counterbalanced by a large increase in the northern subdivision of Sitamarhi. Since 1891, there have been crop failures in 1891-92, when relief operations on a small scale had to be undertaken, and again in 1896-97 when the whole district suffered from famine except the southern part of Hajipur. The Sitamarhi subdivision sustained the brunt of the distress. The relief operations were carefully planned and successfully carried out. The people, no doubt, suffered a good deal, but so far as can be gathered from the vital statistics returns, there was no loss of life directly attributable to want of food. The reported deaths fell short of the average of the three preceding years by about 25,000, while the deficiency of births was only 17,000. Thanks to a succession of good harvests, the effects of the famine were not long felt, and by the end of the decade the people had entirely recovered their normal condition. There were heavy floods in 1898, which caused considerable loss of cattle and damaged the standing rice crops, but no permanent injury resulted from them.

Muzaffarpur.

There were epidemics of cholera in 1892, 1894, 1896 and 1900, which carried off more than 76,000 persons, but in other respects the public health has been fairly good, and the returns for the nine years 1892—1900 show an excess of nearly 80,000 births over deaths.

* The great reduction in the number of immigrants attracted my attention as soon as the figures were reported, and their number was again counted direct from the schedules. This fresh count, which was carefully checked, brought out a slightly smaller number of immigrants than the original return. The figures for 1891 appear to be extraordinarily high. They include 83,241 immigrants from Saran, 56,076 from Muzaffarpur, 52,186 from the United Provinces and 34,626 from Nepal.

184. The recent census discloses a slight increase of 41,933, or barely half the amount that the vital statistics returns would indicate. The immigrant females are about as numerous as in 1891, and there is an increase of more than 4,000 male immigrants. The emigrants are less numerous than in 1891, the falling off being especially marked in the case of males. There is still, however, a heavy balance against the district and the figures do not include the persons who have migrated across the Nepal frontier.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	1,318,547	1,436,243	1,306,027	1,406,832
Immigrants	32,200	55,502	27,803	55,185
Emigrants	86,507	67,030	121,547	82,642
Natural population	1,372,854	1,447,771	1,399,771	1,434,287

It is a notable fact that every thana in the great rice-growing tract north of the Bághmati, which suffered most severely from the famine, has shown a marked increase, while every thana south of that river, except Hajipur in the extreme south, has lost population. The former tract has been growing steadily since the time of the first census in 1872, and it attracts settlers both from Nepal and from the southern part of the District. The progress is greatest in the Sitamarhi and Shuihar thanas which march with the Nepal frontier. The Hajipur subdivision, which is now stationary, sustained a slight loss of population in the decade preceding the census of 1891. The decline in the Muzaffarpur thana is said to be due to its having suffered most from cholera and to the fact that it is this tract which supplies the majority of the persons who emigrate to lower Bengal in search of work.

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***2,754,790***	***+1·5***	***+ 5·0***
Sadar Subdivision ...	***1,050,097***	***−2·3***	***+ 4·8***
Muzaffarpur	540,534	− 5·6	+ 5·3
Paru	301,717	− 2·8	+ 5·7
Katra	207,776	+ 8·7	+ 2·1
Sitamarhi Subdivision ...	***986,582***	***+6·7***	***+10·4***
Sitamarhi	346,790	+ 9·0	+ 8·1
Shiuhar	195,962	+ 7·2	+ 5·7
Pupri	244,554	+ 5·4	+ 12·4
Belsund	199,276	+ 4·1	+ 10·3
Hajipur Subdivision ...	***718,181***	***+0·6***	***− 1·0***
Hajipur	270,929	+ 2·6	− 2·3
Lalganj	153,162	− 1·0	− 1·9
Mahuwa	294,090	− 0·4	+ 0·6

185. Darbhanga greatly resembles Muzaffarpur, to which it appertained until 1875. The northern and central part of the district is devoted mainly to the cultivation of rice, and the bulk of the inhabitants are dependent on the *aghani* or winter rice crop. The south of the district possesses great natural fertility, and wheat, barley, oil-seeds, various kinds of pulses, indigo, and opium are grown. As in Muzaffarpur, the increase recorded in 1881 was attributed to a more accurate enumeration. In 1891 the district showed a further increase of 6·5 per cent.—a gain of 11·5 per cent., in the Madhubani subdivision in the north being to some extent counter-balanced by a loss of 2·3 in the southern or Samastipur subdivision. The decade peceding the present census has not been a prosperous one. In 1891, the crops failed over a considerable area in the sadar and Madhubani subdivisions, and relief operations were necessitated. In the next two years the crops were good and the people regained their prosperity, but in 1895 the harvest was again a short one, and this was followed by the great crop failure of 1896. This affected the whole district except two of the three thanas of the Samastipur subdivision; in the third thana, Warisnagar, the distress was less acute than in the rest of the district, and it was greatest in the western part of the sadar and Madhubani subdivisions. The number of persons requiring relief was greater in Darbhanga than in any other district, but the recovery after the famine was at least equally rapid. The Magistrate, Mr. Wheeler, says that "the end of 1897-98 saw the people almost restored to their normal condition." The prosperity of 1898-99 was marred by floods. There were floods in the south-west of the district in the following year also, but both then and in 1900, the crops were generally good.

DARBHANGA.

The decade has not been a very healthy one. The mortality was exceptionally heavy in 1892, 1894, 1896 and 1900; in the three former years the

deaths out-numbered the births, but taking the period as a whole (excluding 1891 when births were not registered) the reported births exceeded the deaths by 65,918. Plague broke out in the Dalsing Serai thana about the beginning of December 1900 and in February 1901 the number of reported deaths was 420, but the real number, says the Magistrate, was undoubtedly greater. The disease subsequently spread to other thanas, but not until after the census.

186. The census shows that since 1891 there has been an addition to the population of 110,656 persons or 3·9 per cent. The movements of the people do not appear to have materially affected the result. So far as permanent movements are concerned the balance of the account is in favour of the district. The number of emigrant females is nearly 4,000 less than that of those who have come in from other districts, and while the former are slightly less numerous than in 1891, the latter have grown by more than 9,000. There has been a loss on account of the migration of males, but this is mostly of a temporary character. There were 3,000 fewer male immigrants at the time of the present census and 3,000 more emigrants than there were ten years previously, and the excess of the latter over the former now amounts to about 25,000 or some 6,000 more than in 1891.

Population.	1901.		1891.	
	Males.	Females.	Males.	Females.
Actual population	1,416,474	1,496,137	1,370,985	1,430,970
Immigrants	53,673	57,197	56,381	47,799
Emigrants	58,443	53,069	55,228	53,962
Natural population	1,441,444	1,492,009	1,369,832	1,437,133

On examining the details of the variations for individual thanas we again fail to find any correlation between them and the famine. The increase is slight in the Samastipur subdivision which was least affected by famine, while it is greatest in Madhubani, where the distress was most severe both in 1892 and 1897. Nor do the vital statistics seem to account for the variations, and in the four unhealthy years, the reported death-rate was higher in Madhubani than any where else. The fact seems to be that while the other parts of the district have as dense a population as the land will support, there is still some room for expansion in Madhubani, where even now there are only 812 persons per square mile compared with 872 in the sadar, and 985 in the Samastipur subdivision. Madhubani, moreover is the great rice-growing subdivision, and although it is the rice crop which suffers most in seasons of deficient rainfall, it seems probable that it gives the greatest return to the husbandman when the seasons are normal. It would appear, therefore, that there has been some migration within the district from the more crowded thanas of the south to the rice-growing country in the north. The only thanas in the district that show a decline are Warisnagar and Darbhanga. In the former the death-rate was higher than the district average while the latter would have shown a slight increase but for the inclusion of the figures for Darbhanga town which, as explained elsewhere, has suffered a considerable loss of population. Dalsing Serai, where a decrease of 6·1 in 1891 was attributed to diluvion, would probably have shown a somewhat larger increase but for the advent of plague. The Ganges is still working its way further north, but not sufficiently so to appreciably affect the population.

Thana.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***2,912,611***	***+ 3·9***	***+ 6·5***
Sadar Subdivision	***1,065,595***	***+ 1·6***	***+ 8·0***
Darbhanga	435,453	− 1·4	+ 9·3
Bahera	355,024	+ 2·7	+ 11·5
Rosera	275,118	+ 5·1	+ 2·0
Samastipur Subdivision ...	***752,637***	***+ 1·9***	***− 2·3***
Samastipur	311,312	+ 2·0	− 0·3
Dalsingh Sarai	272,270	+ 3·8	− 6·1
Warisnagar	169,055	− 1·1	+ 6·6
Madhubani Subdivision ...	***1,094,379***	***+ 7·8***	***+ 12·0***
Madhubani	253,655	+ 3·9	+ 13·3
Khajauli	247,389	+ 8·0	+ 28·6
Baulpati	266,431	+ 8·8	− 1·0
Phulparas	326,904	+ 10·2	+ 10·9

187. Bhagalpur, like Monghyr, lies on both sides of the Ganges which divides it into nearly equal parts. The northern

Bhagalpur.

forms a continuation of the great alluvial plain of North Bihar, but it is reputed to be less fertile than the country further west. It is intersected with rivers and water channels which, except in seasons of very deficient rainfall, render irrigation easy, and rice is therefore the

principal crop. In the southern tract also, rice is the main staple. The riparian portion is fairly level but in some parts the surface is undulating, and here it is cleverly terraced and irrigated from reservoirs constructed on the higher slopes. In addition to rice, wheat, sugarcane, and other crops requiring a less ample and constant supply of water are extensively grown. Further from the river the surface has a greater elevation, and the soil becomes more shallow and rocky and comparatively barren. The inhabitants are fewer in number and their crops are scantier and more precarious. The growth of the population between 1872 and 1881 amounted to 7·7 per cent. In the next decade there was a further advance of 3·3 per cent., the out-come of a considerable increase in the western part of North Bhagalpur, and in Banka and Bhagalpur thanas in South Bhagalpur, combined with a decline in the Kishanganj and Bihpur thanas, due to the ravages of the Kosi, on the north bank, and in Sultanganj and Amarpur on the south.

Since 1891 the district has twice suffered from famine, in 1891-92 and again in 1896-97. The tract most affected was the western part of North Bhagalpur, in the Supaul and Bangaon thanas. South of the Ganges, the scarcity did not in either year amount to famine. Test works were opened early in 1897 but they failed to show any demand for labour. Since the last famine the crops, on the whole, have been good and the people have recovered from its effects. The vital statistics show an excess of births over deaths between the years 1892 and 1900, amounting to 76,604. The only year in which the deaths out-numbered the births was 1894. There was an epidemic of cholera which was especially bad in Amarpur, but the main cause of the high mortality was the excessive prevalence of fever which accounted for 42·23 deaths per 1,000 in Kishanganj and for 33·03 in the district as a whole. The healthiest years were the famine year, 1897, and the year immediately after it. In 1897, 13,000 fewer deaths were reported than the average of the three previous years, while the births were 7,000 above the same average.

188. The addition to the population brought out by the last census is 56,257 or 2·8 per cent. The general rate of increase is much the same as in the previous decade, but the figures for individual thanas vary greatly. Sultanganj, Bihpur, and Amarpur *cum* Katauria, which lost ground in 1891, now show an increase, while Banka which then was the most progressive thana in the district, is now slightly decadent. Madhipura has lost rather more than it gained in 1891. The two thanas of the Supaul subdivision and the Bangaon and Bhagalpur thanas show a continuous growth, and Kishanganj a continuous decline. Kishanganj suffers more from fever than any other part of the district, and it is liable to devastation from the sandladen floods of the Kosi. The same adverse conditions account also for the decline in Madhipura. The most progressive thana on this occasion is Pratapganj, which owes its access of population to the advent of the railway.

Thanas.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***2,088,953***	***+ 2·8***	***+ 3·3***
Sadar Subdivision ...	***583,944***	***+ 6·0***	***+ 1·1***
Bhagalpur	189,980	+ 8·5	+ 9·3
Colgong	204,088	+ 6·6	+ 0·6
Sultanganj	95,116	+ 5·1	− 6·3
Bihpur	96,060	+ 2·7	− 3·0
Banka Subdivision ...	***455,499***	***+ 2·4***	***+ 0·5***
Banka	161,602	− 1·1	+21·8
Amarpur	152,953	+ 5·6	− 9·3
Katauria	118,944	+ 3·6	
Madhipura Subdivision ...	***539,510***	***− 2·8***	***+ 2·1***
Madhipura	276,452	− 3·8	+ 2·7
Kishanganj	116,647	− 6·9	− 7·5
Bangaon	146,311	+ 2·0	+ 9·7
Supaul Subdivision ...	***510,000***	***+ 6·1***	***+ 9·9***
Supaul	359,535	+ 3·6	+12·5
Pratapganj	151,365	+12·4	+ 3·9

part of the district, and it is liable to devastation from the sandladen floods of the Kosi. The same adverse conditions account also for the decline in Madhipura. The most progressive thana on this occasion is Pratapganj, which owes its access of population to the advent of the railway.

The number of settlers from other districts, and chiefly from the adjoining portions of Darbhanga, Monghyr, and Purnea is far less than in 1891, but this is due probably to deaths amongst the original immigrants rather than to their subsequent return to their original homes. The gradual westward movement of the Kosi, however, may have driven some of the settlers from Purnea back

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	1,027,535	1,061,418	1,004,863	1,027,831
Immigrants	58,606	49,532	78,564	76,000
Emigrants	66,491	57,814	64,804	60,323
Natural population	1,036,020	1,069,700	991,105	1,012,154

to that district. The emigrants stand at almost the same figure as they did ten years previously but they include rather fewer females and rather more males. The district has thus sustained some loss by migration and it seems probable that, but for this, it would have shown an increase somewhat in excess of that recorded in 1891.

PURNEA.

189. The part of Purnea east of the Mahánandá is more nearly allied to Bengal than Bihar, and the bulk of the inhabitants are of Rájbansi Koch origin, though most of them are now converts to Muhammadanism. There are numerous swamps especially towards the south. The main crop is rice, but jute also is extensively grown. West of the Mahánandá the population is mainly Hindu, and the castes are the same as in the adjoining Bihar districts. A considerable portion of this latter tract, say from Purnea town westwards, has been greatly affected by the vagaries of the Kosi river, which is constantly changing its bed, and the soil, which elsewhere is a rich loam, is here very sandy, admirably adapted for pasture, but not so well suited for cultivation. The main crop is rice and though it seldom suffers from want of rain, there is constant danger of destruction by the floods from the Kosi.

Though their houses are poor and luxuries are rare, the inhabitants are not badly off. Rent is low and the people have as much food and clothing as they require. There are very few landless labourers, and unskilled labour has to be imported from outside. There was an increase of population between 1872 and 1881 amounting to 7·8 per cent., and a further increase of 5·1 per cent. between 1881 and 1891. The growth was most rapid in the great pastoral thanas, Dhamdaha and Matiari, which attracted numerous settlers from Bhagalpur, Monghyr and Darbhanga, and added a third and a seventh respectively to their population.

Since 1891 the crops have been fairly good. There was some scarcity in 1891-92 and again in 1896-97, but no famine. The communications have been improved by the construction of new railway lines. The district, however, is very unhealthy and the vital statistics show that the deaths outnumbered the births in every year except 1897, 1898 and 1899. The mortality was greatest in 1891 and 1900 when cholera was terribly prevalent; in the latter year, according to the returns, it accounted for no less than 46,240 deaths or 24 per 1,000, a rate of mortality which, with one solitary exception, was more than double that of any other district in any year of the decade. The total recorded death-rate in that year reached the appalling figure of 57 per 1,000. During the years 1892—1900 the reported deaths were more numerous than the births by 38,239. Statistics of births in 1891 are not available, but the year was a very unhealthy one, and it probably witnessed a considerable loss of population.

190. The result of this marked unhealthiness is reflected in the statistics of the present census which shows a decrease of 69,864 or 3·6 per cent. There is a decrease both in the number of persons coming from, and going to, other districts, but the inter-district movements appear to have been exceptionally great in the years preceding 1891; in 1881 the figures very closely resemble those of the recent census. At all three enumerations the number of persons who have come to the district has greatly outnumbered that of those who have left it, but a great many of the former are only temporary visitors who bring their cattle in the cold weather months to graze in the splendid pasture lands on the left bank of the Kosi. The fact that the fees for grazing are now more strictly realised coupled with the extension of cultivation may, perhaps, account for the smaller number of persons who now come with their cattle from other districts. Where so much of the migration is of a purely temporary character it is very difficult to frame an estimate of the extent to which a district has gained or lost by the movements of the people. The immigrants have decreased to a greater extent than the emigrants and if they were all only temporarily away from their native districts, the figures would indicate a small net loss. If on the other hand, they had all changed their residence

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	958,458	916,342	993,406	951,102
Immigrants	69,092	38,973	83,313	51,035
Emigrants	20,615	16,803	40,680	30,398
Natural population	910,035	894,172	950,872	929,025

permanently there must have been a considerable gain; the number of immigrants is greater by about 18,000 than it would be had there been no new arrivals while the shrinkage amongst emigrants is greater than deaths alone would account for, and it would seem as if some must have returned home since 1891.

The only thana in the whole district that shows an increase is Saifganj, which owes its development to the fact that it contains the important railway junction at Katihar.* Thanks to this thana the tract west of the Mahánandá has slightly gained in population since 1891, while that to the east of that river is responsible for the whole of the loss which has taken place. The decline is least marked in Purnea itself and in the thanas bordering on the Kosi, and if Saifganj be left out of account it becomes gradually greater towards the east. It is most considerable in Balarampur, which is the unhealthiest thana in the district. No reason beyond unhealthiness can be assigned for the changes which have taken place. The fatal epidemic of cholera in 1900 affected chiefly the western part of the district which, but for this, would doubtless have shown much better results.

Thanas.	Population, 1901.	Percentage of variation.	
		1891—1901.	1881—1891.
DISTRICT TOTAL	***1,874,794***	***−3·6***	***+ 5·1***
Sadar Subdivision	***838,333***	***−2·6***	***+ 6·0***
Purnea	155,894	− 0·6	+ 1·9
Kasba Amur	119,135	− 2·7	+ 0·6
Gondwara	109,268	− 5·5	− 6·4
Dhamdaha	144,691	− 0·1	+30·9
Kadwa	132,821	− 8·9	+ 3·8
Balarampur	78,174	−23·3	+ 7·0
Saifganj	98,280	+28·6	...
Kishanganj Subdivision	***619,476***	***− 4·8***	***+ 2·7***
Kishanganj	140,979	− 5·0	− 2·2
Bahadurganj	268,281	− 5·4	+ 3·6
Kaliaganj	266,216	− 4·3	+ 0·2
Araria Subdivision	***416,985***	***−3·6***	***+ 8·0***
Araria	201,028	− 4·6	+ 5·3
Matiari	128,906	− 2·4	+14·2
Ranigani	87,051	− 2·9	+ 5·0

ORISSA.

Cuttack.

191. Cuttack is divisible into three parts with very different characteristics. To the east, bordering on the Garjáts or Tributary States of Orissa, is a region of rocky hills and barren soil, supporting a scanty and semi-Hinduised population with a very low standard of comfort. To the west, on the seashore, is a low-lying tract of great natural fertility where protected from the action of the salt water. A great part, however, is unprotected and unfit for cultivation, and much of the rest is exposed to damage from storm-waves. The population is therefore sparse. Between these two extremes lies a fertile alluvial plain watered by three great rivers—the Mahánadi, Brahmani, and Baitarni—and protected from drought by an extensive system of canal irrigation. This tract is very highly cultivated and has in parts a density of population very little less than that of the most thickly inhabited parts of Eastern Bengal and Tirhut. It is this portion of the district which sends its surplus male population to Calcutta and other parts of Bengal in search of work as domestic servants, door-keepers, &c., and the local earnings from agriculture are supplemented by large remittances from relatives in service elsewhere. Thanks partly to these remittances the standard of comfort is a comparatively high one.

The censuses of 1881 and 1891 disclosed an increase of 16·3 and 7·8 per cent. respectively. The Magistrate thinks that the census of 1872 was incorrect and that the actual population exceeded the census figures by at least 100,000. This would reduce the rate of growth in the succeeding nine years to 13 per cent., which is about what might be expected during the period when the district was recovering from the terrible famine of 1866. The progress during the next decade would have been greater, but for the cyclone of September 1885 which destroyed 45 villages in Patamandi thana, most of whose inhabitants were either drowned or succumbed to fever and cholera which, as usual, followed in the wake of the flood.

* Saifganj has been formed into a thana since 1891 and there may possibly have been some mistake in working out the population from the tabulation registers of the last census. At the same time the great growth of the railway centre at Katihar, seems quite sufficient to account for the recorded increase in its population.

192. Since 1891 the district has on the whole enjoyed good health. Cholera has seldom been absent, but it assumed serious proportions only in 1892, when it accounted for 21,289 deaths and in 1900 when the mortality was at the rate of 8·5 per 1,000. Since 1896 the birth-rate has been uniformly high, and in 1899 the number of births reported represents a ratio of nearly 50 per 1,000 on the population of 1891. During the nine years preceding the present census the reported births exceeded the deaths by 108,540. The excess would probably have been greater had the figures been available for 1891, which was a particularly healthy year. There was a cyclone in 1891 which did considerable damage along the coast, and in this year and again in 1900 there were floods throughout the district. The famine of 1897 was felt in Cuttack but slightly, but the high prices affected the non-agricultural community. The advent of the railway has greatly improved the communications; it has not been open long enough to affect the area under cultivation, but it has greatly stimulated migration. Natives of the district employed in Bengal return home at much more frequent intervals than formerly, but on the other hand the number seeking employment elsewhere has increased. But the greatest immediate benefit from the railway is the comparative immunity which the district has secured from the dissemination of epidemic disease by the crowds of pilgrims travelling to and from the temple of Jagannath at Puri who now go by rail instead of on foot. The fact that the work on the railway embankments did not attract labour from within the district shows that the people are too well off to care about serving as coolies. The recent settlement operations, though possibly irksome at the time, have done much to teach the peasantry what their rights are and to protect them from harassment and illegal exaction.

193. The result of the recent census is an increase of 125,087, or 6·5 per cent. This is rather greater than the vital statistics would indicate, but the latter do not include the figures for 1891, which, as already stated, was a particularly healthy year. The number of immigrant females is about the same as it was ten years ago; male settlers are more numerous, chiefly on account of the railway. The number of females who have left the district is considerably less than in 1891, chiefly on account of the figures for the Orissa States, Singhbhum and the 24-Parganas, but that of males is nearly 20,000 greater. These males are mostly temporary absentees, but even so the district has suffered a net loss by the movements of the people since 1891 which is sufficient probably to account for the diminution in the rate of growth that has taken place as compared with that in the previous decade. The general increment is shared by all parts of the district and the rate of development is remarkably uniform throughout. The growth of population has been least in the already densely inhabited thanas, Cuttack, Salepur, Jajpur and Jagatsinghpur and greatest in the fertile but sparsely inhabited thanas, Patamandi and Aul, on the sea coast, where the construction of protective embankments and the offer of easy terms of settlement have led to considerable reclamations of land thrown out of cultivation by the salt water floods of 1885. Then follow the inland thanas which adjoin the Garját States, Dharmsala and Banki, which have also a very sparse population, and then Kendrapara, where the population, though more dense, is less so than in the other central thanas.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	995,409	1,027,349	940,557	997,114
Immigrants	11,606	21,338	8,817	21,117
Emigrants	81,283	35,476	61,463	51,163
Natural population	1,065,086	1,041,487	993,192	1,027,160

Thanas.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	*2,069,758*	*+6·5*	*+ 7·9*
Sadar Subdivision	*1,035,275*	*+5·4*	*+ 8·5*
Cuttack	213,238	+ 5·0	+ 8·6
Banki	74,077	+ 7·4	+ 0·8
Salepur	182,691	+ 4·5	+ 8·2
Tirtol	180,309	+ 6·8	− 8·1
Jagatsinghpur	284,860	+ 5·4	+ 25·5
Kendrapara Subdivision	*467,081*	*+8·7*	*+19·0*
Kendrapara	230,197	+ 7·1	+ 12·4
Patamandi	125,545	+10·7	+ 1·4
Aul	111,339	+ 9·8	+ 11·1
Jajpur Subdivision ...	*567,402*	*+6·6*	*+ 5·3*
Jajpur	269,419	+ 4·3	+ 3·0
Dharmsala	298,983	+ 8·8	+ 7·5

BALASORE.

194. Balasore consists of a long strip of land, mostly alluvial, between the Bay of Bengal and the Tributary States, very narrow in the centre but growing broader towards the north and south. Along the coast is a belt of land about three miles broad, which is impregnated with salt and unfit for cultivation. The western portion, which runs along the foot of the hills and borders on the Garjáts, is jungly and uncultivable. Between these two extremes lies the fertile, arable country which constitutes the greater part of the district. The climate is good, except in the north, whither malarial fevers have spread from the adjacent unhealthy thanas of Midnapore. Between 1872 and 1881 a great increase in the population was recorded, owing partly to improved enumeration and partly to a recovery from the losses caused by the famine of 1866. The next decade witnessed a greatly diminished rate of progress, and the general expansion was only 5·0 per cent. Jellasore, in the extreme north, had suffered from malaria and was stationary. Chandbali, in the south, with its scanty population and prosperous port, showed a rapid development. In the rest of the district the increase was very evenly distributed.

Since 1891 the north of the district has become less unhealthy. There have been frequent epidemics of cholera. The worst outbreak was in 1892 when this disease was responsible for a mortality of 15 per 1,000, a rate which was exceeded during the decade only in the terrible epidemic in Purnea in 1900. The reported deaths exceeded the births in this year and also in 1894 and 1896, when cholera was again prevalent, but in the nine years, 1892—1900, taken together, the births outnumbered the deaths by about 30,000. The people are fairly prosperous, and nearly every one has land of his own. The crops have been good, and the opening of the railway has greatly improved communications. The only disaster was a high flood in October 1900, which destroyed crops and cattle, though it caused very little loss of life. The area chiefly affected was in the Dhamnagar Thana.

195. The favourable conditions of the decade are reflected in the census figures, and in spite of some loss by migration the population of the district in 1901 is greater than that at the time of the previous census by 76,522, or 7·7 per cent. Amongst females emigrants and immigrants are both less numerous than in 1891, but the excess of the former over the latter remains about the same. The male immigrants are fewer and the emigrants more numerous than in 1891, and the latter now out-number the former by nearly 16,000. The loss owing to the movements of the people may thus be roughly estimated at from 1 to 2 per cent.

POPULATION.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	517,543	553,654	481,604	513,011
Immigrants	13,295	16,232	15,006	19,917
Emigrants	29,024	32,739	25,147	25,421
Natural population	533,332	560,161	491,905	518,515

The population of the Chandbali and Basudeb thanas shows the greatest development. They are on the sea shore; both contain much land fit for cultivation, and the absence of destructive cyclones has encouraged reclamation. Dhamnagar, in the south-east, shows a slight loss of population, which, however, is probably only temporary. The destruction of crops by the flood which occurred shortly before the census sent many of the males to other districts in search of work whereby to tide over their temporary difficulties. This explanation is confirmed by the figures for each sex; the falling-off is confined to males and the female population is greater than in 1891. Throughout the rest of the district the rate of increase is uniform, varying from 7·9 per cent. in Jellasore to 9 per cent. in Bhadrak.

THANAS.	POPULATION.	PERCENTAGE OF VARIATION.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,071,197***	***+7·7***	***+5·2***
Sadar Subdivision ...	***592,544***	***+8·3***	***+5·1***
Balasore	122,594	+ 8·0	+ 4·6
Soro	229,263	+ 8·1	+ 7·2
Jellasore	59,791	+ 7·9	+ 0·5
Baliagal	160,833	+ 8·7	+ 5·3
Basta	80,127	+ 8·0	+ 4·6
Bhadrak Subdivision ...	***478,653***	***+6·9***	***+5·1***
Bhadrak	193,410	+ 9·6	+ 4·3
Basudebpur	83,129	+ 12·0	+ 4·9
Dhamnagar	127,708	− ·4	+ 4·5
Chandbali	64,406	+ 11·5	+ 11·1

PURI.

196. The Khurda subdivision of Puri is a Government estate. The country is above flood level while the numerous rivers that intersect it afford ample means for irrigation. The rest of the district is a low alluvial plain, through which the torpid rivers find their uncertain way towards the sea; their final exit is often impeded by ridges of sand thrown up by the strong monsoon currents, and their banked-up waters often burst the embankments that have been erected to restrain them and cause disastrous floods. The district has grown steadily since 1872, and at the last census an increase of 6·4 per cent. was recorded, part of which was attributed to defects in the previous enumeration.

The public health has not been very good since 1891. Cholera is imported annually by pilgrims, and fever is prevalent during the cold weather. The reported deaths exceeded the births in 1892 and 1897, but in the nine years, 1892 to 1900, there was, according to the returns, a net excess of 32,153 births over deaths. In 1891 the crops suffered greatly from insufficient rain at the commencement of the monsoon, and the damage was aggravated by a cyclone in the following November, which brought on an exceptionally heavy flood. Most of the embankments were breached, the standing crops were destroyed, and cholera, dysentery and other diseases spread all over the district. The year 1897 was equally disastrous. The crops of 1896 were short everywhere and this resulted in a general scarcity, though relief operations were necessary only in the neighbourhood of the Chilka lake and in parts of the Khurda subdivision. The country round the Chilka was again subject to scarcity in 1900, when the rainfall was scanty and ill-distributed. So far, therefore, as agricultural operations are concerned the conditions have not been satisfactory. On the other hand the opening of the railway has benefited the people greatly. Its construction afforded profitable employment to the labouring classes, and now that the district has been brought into easy communication with Calcutta, emigration in search of employment has been greatly facilitated and wages have risen in consequence. Prices of produce also have risen owing to the new markets that have been brought within easy reach.

197. The actual increase brought out by the present census is 72,286, or 7·6 per cent., but part of this is due to the fact that a great religious festival was in progress at Puri at the time when the census was taken, and the immigrant population thus exceeded that of 1891 by more than 13,000. The number of persons born in the adjoining districts who were enumerated in Puri was less than in 1891, and the addition to the population due to pilgrims must therefore have been greater than the above figures would indicate.* The number of female emigrants has fallen off while that of males is about the same as it was ten years previously. But for these movements of the people the rate of growth would probably have been about the same as, or rather less than, that of the previous decade.

POPULATION.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	566,839	510,445	474,530	470,408
Immigrants	20,919	32,114	15,797	24,132
Emigrants	11,487	17,564	11,517	19,371
Natural population	497,407	495,835	470,250	465,707

The Sadar and Pipli thanas show the smallest rate of progress. This is due mainly to the fact that the embankment on the left bank of the Bhargavi river has been breached so often that it was decided in 1895 to leave it unrepaired, and the consequences, says the Magistrate, have been fatal to the prosperity of the country which this embankment had been designed to protect, and which was previously one of the richest and most populous parts of the district. The opening of the railway, moreover, has stimulated emigration in search of employment. Many of the

Thana.	POPULATION, 1901.	PERCENTAGE OF VARIATION.	
		1891—1901.	1881—1891.
DISTRICT TOTAL	***1,017,284***	***+7·6***	***+6·3***
Sadar Subdivision	***658,048***	***+7·2***	***+8·6***
Puri	217,125	+ 3·7	+ 6·6
Puri Town	49,334	+71·3	+12·1
Gop	142,459	+ 6·3	+15·3
Pipli	249,130	+ 5·2	+ 6·4
Khurda Subdivision ...	***359,236***	***+8·4***	***+2·3***
Khurda	254,295	+ 8·3	+ 6·4
Bhanpur	104,941	+ 8·5	+ 7·6

* The number of pilgrims in Puri town on the night of the census exceeded 17,000.

poorer Brahmans who live in these thanas, have taken to the profession of pilgrim-conductors and large numbers were absent at the time of the census. Gop thana shows a greater increase than the rest of the head-quarters subdivision, as it is less developed, and there is more room for expansion. The Khurda subdivision is more favourably situated than other parts of the district, and its somewhat more rapid development is therefore only natural.

CHOTA NAGPUR PLATEAU.

198. Hazaribagh, like the rest of Chota Nagpur, consists, to a great extent, of rock and ravine, and the cultivable area is comparatively small. On the north it rises somewhat abruptly from the Bihar plain, and a table-land is formed which stretches along the northern face of the district, having an average width of about 15 miles. South of this a further rise takes place and leads to a second plateau which occupies the west central part of the district. In the western part of the northern table-land the slope is towards the north and the rivers find an exit in the plains of South Bihar, but elsewhere there is a gentle declivity towards the south-east, and the drainage of the country converges on two main channels, the Barakar and the Damodar. The valleys of these rivers gradually become wider and in places highly cultivated and fairly extensive tracts of level alluvium take the place of the scattered and laboriously constructed terraces on the slopes of the hills which are to be seen elsewhere. The district is unprotected by irrigation. A large proportion of the inhabitants are Bhuiyás, Santáls and Kurmis, and other aborigines whose habits are migratory and who are in great demand as coolies on tea gardens. The first reliable census was in 1881, and between that year and 1891 there was an increase of 5·7 per cent. in the population. With the exception of the Hunterganj thana the whole of the north-west of the district lost ground, but this was more than compensated for by the increase elsewhere, especially in the Giridih subdivision. These variations were explained partly by the movement of the population within the district and partly by emigration to Assam and other parts of Bengal.

HAZARIBAGH.

199. Since 1891 the district has suffered three times from short crops, the result of deficient rainfall. In 1895 the harvests were deficient, and in the following year they failed almost completely. Owing to the partial failure in the previous year, the distress thus caused was general and widespread. The thanas most affected were Barhi, Koderma, Bagodar, Gumia, Ramgarh, Mandu and Hazaribagh. Relief works failed to attract labour owing, it is said, partly to the unwillingness of the wilder tribes to engage in unaccustomed forms of labour, and partly to a fear that the acceptance of famine rates of payment would tend to lower wages permanently, but a good deal of employment was afforded by the District Board, and gratuitous relief was given to beggars and destitute travellers. Two good years sufficed to restore prosperity, and when, in in the year 1899, there were again deficient harvests, the people were able, without serious distress, to tide over till better times. The decade was, on the whole, a fairly healthy one, and, according to the vital statistics returns, the births out-numbered the deaths by about 81,000. The unhealthy years were 1894 and 1897, especially the latter, when the deaths that were reported numbered 56,036, or 48 per 1,000. This was the famine year. The Sanitary Commissioner attributed the high death-rate to unhealthiness resulting from an excessive rainfall in 1896, but the Civil Surgeon of the district thinks that although fever and cholera were chiefly to blame, the high mortality was due partly to the indirect effects of the privations endured by the people. The tendency of the forest tribes who suffered most from the famine is to seek for sustenance in the jungles rather than go to the Government relief works. These jungle products when taken with other articles of food are not unwholesome, but as a sole dietary they are innutritious and indigestible. A long course of such food enfeebles the constitution and predisposes to disease. The birth-rate of 1897 and 1898 was very low, but in 1899 it suddenly rose to over 50 per 1,000. Plague appeared in several parts of the district in January 1901, but the total number of plague deaths reported prior to the census was only 309.

200. The recent enumeration shows a slight increase of 13,640, or 1·2 per cent., but the real growth of the population has been much greater than these figures would indicate. The balance of migration has gone heavily against the district. The immigrants are far fewer than in 1891, and the arrival of new settlers has, if anything, been more than discounted by the return to their old homes of persons from other districts who were enumerated in Hazaribagh in 1891. On the other hand, the number of persons who have left the district has greatly increased, and 150,356 persons born in Hazaribagh, were enumerated elsewhere compared with only 113,622 in 1891. If we take the death-rate at 40 per 1,000, and assume that an equal number of persons left the district in each year of the decade, this would seem to indicate an annual emigration of about 9,111 persons, or 91,110 during the decennium.* The real number is probably somewhat less than this, as the years when emigration was most active were in the latter part of the decade. But for this loss by migration the rate of increase would probably have been not less than 8 per cent.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	570,122	607,839	566,964	597,357
Immigrants	19,469	22,414	26,312	46,638
Emigrants	85,572	65,784	60,484	53,138
Natural population...	634,225	651,209	601,136	603,863

201. The local variations follow the same general direction as in 1891. The decadent thanas include the whole western half of the district except Hunterganj in the extreme north-west, and Mandu, Ramgarh, and Kasmar in the south. It is in this part of the district that recruiting operations for the tea districts are most active. The great increase in Kasmar is due in part to its proximity to the Jheria coalfield and to the construction of the Jheria-Katras railway. All the thanas in the Giridih subdivision show an improvement, but the growth is most marked in the case of Giridih itself where the coal mines of the East Indian Railway attract a steadily increasing number of labourers. In Ganwan and also in Kodarma, in the head-quarters subdivision, mica mining affords employment to many, and this has no doubt affected the population to some extent.

Thanas.	Population, 1901.	Percentage of variation.	
		1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,177,961***	***+ 1·2***	***+ 5·4***
Sadar Subdivision ...	***760,164***	***— ·3***	***+ 2·2***
Hazaribagh	92,923	— 8·3	— 1·3
Barhi	50,292	— 2·5	— ·6
Bagodar	64,490	— ·4	+ 8·9
Ramgar	80,781	+ 1·4	+ 9·5
Kodarma	79,413	+ 2·2	+ 5·3
Kasmar	57,741	+13·5	+ 7·4
Simaria	37,936	+ 3·7	— 1·0
Chorparan	60,858	— 3·3	— 10·6
Hunterganj	63,693	+ 8·9	+ 2·5
Gumian	29,813	— 2·7	+ 12·3
Mandu	29,908	+ 2·1	
Barkagaon	54,479	— 4·9	+ 2·6
Chatra	67,777	— 5·9	— 4·2
Giridih Subdivision ...	***417,797***	***+ 4·0***	***+ 12·0***
Giridih	116,903	+ 8·8	+ 25·1
Kharagdiha	87,595	+ ·6	+ 4·0
Dhanwar	71,252	+ 2·3	
Dumurhi	82,328	+ 2·5	+ 17·9
Ganwan	59,629	+ 4·1	+ 7·2

202. The greater part of Ranchi is an elevated undulating table land, where the terraces cut in the slopes of the depressions which lie between the ridges are almost the only places where permanent cultivation is feasible. Towards the west and south the surface becomes more broken; the hills are steeper and the valleys are replaced by ravines where no crops can be grown. The district is the great

RANCHI.

* The yearly number of deaths at 40 per 1,000 amongst the 113,622 persons enumerated elsewhere in 1891 would be 4,545, consequently 45,450 persons must have gone from the district during the decade to keep up the number of emigrants ascertained in 1891. The annual emigration necessary to produce the excess of 36,734 emigrants shown in the returns for 1901 is calculated thus:—

Let x=the number of emigrants in each year and ·040 (*i.e.* 40 per 1,000) the death-rate. Then the number surviving at the end of one year will be $x \times ·960$, at the end of two years $x \times ·960^2$, and so on. Hence the total number of new emigrants at the end of the decade will be—

$$x\ (·960^{10} + ·960^9 + ·960^8, \text{ etc. } + ·960) = 36,734.$$

$$\text{Hence } x = \frac{36,734}{8·044} = 4,566.$$

The approximate accuracy of this method is confirmed by a comparison of the results afforded by it with the statistics maintained under the Inland Emigration Act. According to these statistics 180,097 persons were imported to Assam as tea-garden coolies from Chota Nagpur Plateau during the decade. The number of persons enumerated in Assam but born in this tract was 209,068 in 1891 and 281,275 in 1901. At a death-rate of 40 per 1,000, the total number of emigrants needed to keep up the 1891 population would be 89,627, while to produce the excess of 72,207 shown in 1901, 89,765 new emigrants would be needed. The total of 173,392 falls very little short of the figure separately ascertained under the labour law.

recruiting ground for tea-garden coolies, and its aboriginal tribes, especially its Oráons and Mundas, are to be found on every tea-garden in Assam and the Western Duars. During the decade ending in 1891 the district added 6·7 per cent. to its population, in spite of a large and growing stream of emigration.

Since 1891 there have been four years of very deficient harvests, viz., 1891, 1895, 1896 and 1899. The crop failure was greatest in 1899, but the suffering was less than in 1896, partly because it came after two seasons of good crops, while the short harvest of 1896 followed immediately on that of 1895, partly because in the absence of general scarcity elsewhere prices never rose so high, and partly because the preparations for meeting the calamity were more complete. In 1897, as test works failed to attract labour, it was hoped that the people would be able to surmount their trouble without help from Government. The hope unfortunately was not realized, and great distress suddenly developed in the central part of the district, chiefly in the Toto, Sessai and Karra thanas. Relief operations were at once undertaken, but the acute stage of distress was of very short duration. Only three deaths from actual starvation were reported, but cholera broke out in epidemic form, and the mortality from bowel-complaints was also exceptionally high, owing, it is said, to the bad effects of eating *gondli* (*panicum miliare*) grain unmixed with rice on constitutions enfeebled by a low and insufficient diet of roots and fruits gathered in the forests. In 1899 the area that suffered most was much the same as in 1896. Relief works were opened in ample time, the attendance on them was far better than in the previous famine, and the distress that would otherwise have ensued was thus to a great extent averted.

203. The decade is reported to have been an unhealthy one, but if so, vital statistics returns must be exceptionally inaccurate. The returns show an average birth-rate of 38 and a death-rate of 29 per 1,000 and a net excess of births over deaths of 110,191. The mortality was highest in 1897, when it exceeded the average of the decade by 50 per cent. It is to be feared, as in Hazaribagh, that the famine was indirectly responsible for much of this mortality. In Bihar, as we have seen, the famine had no apparent effect on the mortality. Not only was the death-rate lower than usual during the famine year, but the census shows that the tracts which suffered most are precisely those where the greatest increase of population has taken place. The reason why the result should have been different in Chota Nagpur is explained as follows in the Government Resolution on the famine:—

"A considerable part of the food-supply consists of the edible forest products, which are resorted to largely everywhere as a supplement to the ordinary diet. The most important of these products is the flower of the *mahua* tree, but a large variety of jungle fruits and roots are eaten, both cooked and raw. This diet, though sufficient to sustain life, is deleterious if unaccompanied by rice or other grain, yet the aboriginal and semi-aboriginal tribes, who form a considerable portion of the population, are content to eke out a scanty subsistence upon it in their native jungles rather than do anything in the nature of a "task" upon relief works. Consequently, though the numbers on relief in the division were at no time large, there was more anxiety felt on its account than elsewhere. The inaccessible character of the country and the range of prices, which here reached a higher level than in any other portion of the Province, combined with the scattered character of the population and the restless and independent nature of the forest tribes, gave rise to great fears lest the food supplies should fail and the people die in the jungles rather than resort to relief works; and although only a few actual deaths from starvation are reported to have occurred, yet there is reason to apprehend that the sufferings of the poorer classes have been greater in this Division than elsewhere."

It should, however, be mentioned that in 1900, when famine conditions again prevailed, the reported death-rate was below the average for the decade.

204. The short harvests greatly stimulated emigration, and recruiting for the tea gardens of Assam and the Duars was very active.* The balance of

* The number of emigrants to Assam registered at Ranchi during the decade was 47,764 of whom nearly 20,000 were registered in the two years 1897-98 and 1899-1900. These figures do not include all emigrants to Assam but they may probably be taken to show the relative strength of the tendency to emigrate in these two years. The total number of Ranchi-born persons enumerated in Assam in 1901 was 91,794 compared with 70,049 in 1891 which would represent a total emigration of about 55,000 assuming an equal number of emigrants to have left yearly.

migration was heavily against the district in 1891 and it is still more so at the present census; emigrants now outnumber immigrants by 213,195 compared with 220,517 ten years ago. Assuming all these to be permanent migrants and assuming also that the volume of migration was the same in each year and that the annual death-rate was 40 per 1,000, these figures indicate a net loss to the district during the decade, of about 116,000 persons.* After allowing for temporary migrants and for the excessive emigration of 1897 and 1900 it would still seem that the true increase in the population must have been about 13 per cent., instead of 5·2 per cent. which is the actual variation in the number of persons enumerated in the district. It will be seen, when dealing with Palamau, that some of the emigrants from that district have probably been credited to Ranchi, but it is impossible to frame any estimate of their number.†

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	577,180	616,745	551,873	577,612
Immigrants	17,038	15,026	12,526	12,979
Emigrants	142,970	132,341	119,190	126,832
Natural population	703,060	728,060	656,537	690,865

NOTE.—Ranchi and Palamau formed one district until the 1st January 1892 and the statistics of migration collected in 1891 refer to the two districts combined. For the purpose of this and similar statements it has been assumed that the immigrants and emigrants were then distributed between them in the same proportions as at the present census.

205. The variations in individual thanas cannot always be clearly ascertained, owing to the numerous changes in jurisdiction that have taken place since 1891, and to the fact that the corrections made in the population according to the census of that year were in some cases rough estimates based on the assumption that the density of population was uniform throughout the thanas affected. This circumstance probably accounts for the great apparent increase in Chainpur, Kurdeg, Kolebira and Bano. The last two thanas with Bassia all formed one thana in 1891. In the three together the increase is 23·3 per cent. which is still sufficiently striking. There has been a good deal of migration within the district from the older thanas, where the landlords are very powerful, to the more jungly thanas along the south of the district, and this probably accounts for a good deal of the increase in Kurdeg, Kochdega, Kolebira, Bano, and Khunti, and for the slight decline that has taken place in Palkot ‡ The only thanas that show a more than nominal decrease are Toto, Sisai and Karra which suffered most from the famines of 1897 and 1900 and so furnished a specially large quota of emigrants to Assam as well as to the more favoured tracts in the south of the district.

Thanas.	Population 1901.	Percentage of variation.	
		1891—1901.	1881—1891.
DISTRICT TOTAL ...	*1,187,925*	*+5·2*	*+6·7*
Toto	45,553	— 4·1	+ 6·3
Khunti	113,660	+17·8	...
Mandar	75,307	+ 2·1	+10·6
Bishenpur	19,485	+ 1·2	Included in Chainpur.
Kurdeg	28,918	+22·6	Included in Kochdega.
Palkot	67,036	— ·2	+ 4·4
Ranchi	177,585	— ·01	+22·7
Tamar	111,747	+ 9·3	+ 9·2
Karra	90,346	— 6·5	—11·0
Silli	75,543	— ·9	+11·3
Kolebira	37,788	+55·1	+ 4·7
Basia	40,582	+ 2·6	
Bano	22,526	+30·7	
Kochdega	58,903	+13·7	+16·6
Lohardaga	100,049	+ 1·3	+ 1·0
Sisai	53,490	— 6·7	Included in Toto.
Chainpur	46,754	+29·9	+ 1·1

PALAMAU. 206. Palamau was separated from Lohardaga, now called Ranchi, and formed into a separate district in 1892. Unlike Ranchi and Hazaribagh, there is here but little table-land and the face of the country is wilder and more broken by rock and jungle. The only level areas of any extent are in the north of the district, on the right bank of the Sone, and in the valleys of the North Koel and Amanat rivers, where permanent rice cultivation is carried on by people of the same castes as are met with in the adjoining plains districts. Elsewhere the inhabitants are mainly forest tribes who eke out their precarious crops of oil-seeds, maize and cotton with the blossoms of the *mahuá* tree and other

* The method of arriving at this figure has been explained in the foot-note on page 102.

† There was also some confusion in 1891, due to the system of working out the results in the districts instead of in Central offices. The work was generally well done but in a few offices it was not properly supervised. In Hooghly for instance, the birthplace return shows no persons born in Hazaribagh, Ranchi or Singhbhum and 2735 in the Chota Nagpur States. This figure clearly refers to persons born in all parts of Chota Nagpur except Manbhum.

‡ The Deputy Commissioner thinks that in the jungle thanas to the south of the district and in Chainpur on the west, the census of 1891 was not altogether accurate.

products of the jungle. The district is not a healthy one. Malarial affections are prevalent, and also bowel-complaints, due to the unwholesome food on which the people live. The decade ending in 1891 disclosed an increase in the population of 8·3 per cent.

The district suffered like its neighbours from short crops in 1895, 1896 and 1899. The crop failure of 1896 resulted in severe distress, culminating in famine in a broad tract stretching across the centre of the district from east to west. In 1899 the Mahuadanr thana in the south-east of the district suffered most but, judging from the numbers on relief, the famine that ensued was less than a third as widespread as that of 1897. Both in 1897 and 1900 the recorded mortality was above the average, but while the death-rate in 1897 had been exceeded in 1892 and 1894, that of 1900 was by far the highest of the decade and reached the unprecedented rate of 48 per 1,000. Fever was very prevalent, and there was a severe epidemic of cholera. The reported birth-rate in 1899 had been exceptionally high (53 per 1,000), and a high rate of mortality is usual amongst infants. It is possible that the famine was, to some extent, responsible for the high mortality but probably not to any marked extent.

207. In the decade as a whole, the returns indicate an excess of births over deaths aggregating 21,099. This corresponds very closely with the result of the recent census, which shows an increase of 22,830 persons, or 3·8 per cent. The returns of migration are given in the margin. They disclose but little variation during the decade, and if they are correct, they show that the district gains rather more than it loses by the movements of the people. It is doubtful, however, how far the statistics of emigration are reliable. The district is a new one, and it is not impossible that its emigrants often described themselves as having been born in Lohardaga, and have thus been credited to Ranchi in our returns. It is certain that there was a good deal of emigration during the recent lean years, but this does not seem to be reflected in the census figures.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	306,203	313,397	294,320	302,450
Immigrants ...	19,809	18,269	19,351	16,147
Emigrants ...	14,689	17,521	12,251	16,702
Natural population ...	300,963	311,979	287,220	303,005

The increase of population is greatest in the northern part of the district, where there is the greatest proportion of land fit for permanent cultivation, and where the inhabitants belong to castes that do not emigrate to tea-gardens or readily leave their homes. The only thanas which show a decrease are Balumath and Latehar in the south-east, where the conditions are very similar to those in the decadent part of Hazaribagh that adjoins them. The Balumath thana is co-extensive with the estate Tosi which has suffered much from mismanage ment and has recently been brought under the provisions of the Chota Nagpur Encumbered Estates Act.

Thanas.	Population, 1901.	Percentage of variation.	
		1891—1901.	1881—91.
DISTRICT TOTAL ...	*619,600*	*+3·8*	*+8·3*
Daltonganj ...	105,479	+1·9	+33·5
Latehar ...	41,938	−1·9	− 3·5
Balumath ...	64,712	−3·9	+ 2·0
Ranka ...	36,161	+5·7	− 27·3
Mahuadanr ...	26,999	+3·2	+36·8
Chhatarpur ...	45,437	+5·8	+ 11·4
Husainabad ...	119,744	+7·0	+ 8·3
Garhwa ...	93,256	+7·1	+ 9·6
Patan ...	87,074	+6·1	+ 4·0

208. Manbhum occupies the declivity between the Chota Nagpur Plateau and Western Bengal. It is far more open than the districts hitherto dealt with; the area under cultivation is greater, and the inhabitants are more civilised. Towards the east there is but little to distinguish it from the adjacent parts of Burdwan and Bankura. In the uplands the soil is a ferruginous gravel which changes in the lower levels to a rich alluvium. The district is a healthy one and its population has grown rapidly since the first census, thirty years ago. The last decade has witnessed a marked improvement in the material condition of the people. The crops were bad in several years, as in the rest of Chota Nagpur, but the failure was not so great, and the people enjoyed advantages which, in the decade as a whole, more than compensated for their losses on account of bad harvests. The Bengal-Nagpur Railway traverses the headquarters subdivision from north-east to south-west, and the branch line to Katras has opened out

MANBHUM.

the Gobindpur subdivision, and caused a rapid development of the mining industry in the Jheria coalfield. The death-rate has been low and the birth-rate comparatively high. The return is no doubt still very defective, but a comparison of the births and deaths shows an excess of 107,744 births between 1892 and 1900.

209. According to the census there has been an increase of 108,036 persons or 9·1 per cent. during the decade. In spite of the attraction of the coal mines the district has lost by migration. The number of female immigrants is slightly greater than in 1891 and there are about 14,000 more male immigrants, chiefly from Hazaribagh, Bihar and the United Provinces, who have come to work in the mines. The emigrants, however, have grown in even greater proportions, and the net excess of emigrants over immigrants is now 73,853 compared with 71,812 in 1891.* It would thus appear that there has been a net loss of about 30,000 persons by migration, and if these be added, the natural increase in the district comes to between 11 and 12 per cent.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	653,336	648,028	593,199	600,129
Immigrants	37,301	24,728	23,720	25,362
Emigrants	64,915	71,057	59,354	59,530
Natural population	680,860	694,357	628,824	636,316

The figures for the different thanas show considerable variations. The development of the coal mines has caused a remarkable expansion of 75 per cent. in the population of Jheria, and of 30 per cent. in that of Topchanchi. Part of this is due to immigration from outside the district and part to the attraction of settlers from the surrounding thanas. The latter movement accounts for the slight decrease in Para and the stationary condition of Gaurandi, Jhalda, Chas and Raghunathpur. Away from the coal mines the growth of population is greatest in the Purulia and Manbazar thanas. The former contains the head-quarters of the district with its huge coolie recruiting business and an important railway station. The relatively high increase in Manbazar is due mainly to the fact that its inhabitants are for the most part Hindus, belonging to castes that do not emigrate to tea gardens.

Thanas.	Population.	Percentage of Variation.	
	1901.	1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,301,364***	***+ 9·1***	***+12·9***
Sadar Subdivision ...	***1,024,242***	***+ 5·4***	***+12·7***
Purulia	249,038	+ 11·8	+ 12·1
Barabhbhum	130,330	+ 4·9	+ 13·9
Manbazar	86,083	+ 10·4	+ 21·2
Chandil	98,710	+ 6·5	+ 14·0
Chas	101,266	+ ·8	+ 12·7
Raghunathpur	119,029	+ 2·6	+ 7·5
Gaurandi	52,741	+ ·2	+ 14·7
Para	42,554	− 2·1	+ 9·2
Baghmandi	40,582	+ 5·4	+ 15·2
Jhalda	103,009	+ ·7	+ 11·6
Gobindpur Subdivision...	***277,122***	***+25·1***	***+12·6***
Gobindpur	38,183	+ 4·4	+ 14·5
Tundi	30,776	+ 4·1	+ 6·0
Jharia	76,603	+ 75·1	− 5·4
Topchanchi	69,327	+ 30·2	+ 26·1
Nirsha	62,833	+ 7·1	+ 15·9

210. The general level of Singhbum is high, but the country is fairly open except towards the south-west, where the undulating uplands give way to a mountainous tract clothed in virgin forest. The population is very sparse, and there are large areas capable of reclamation, especially in Ghatsila and the Kolhan. The district has shown a continuous and rapid growth since 1872. The climate is healthy and the inhabitants are prolific, but at the same time it is probable that a great part of the apparent increase prior to 1891 was due to the imperfection of the earlier enumerations. Since that year the people have been fairly prosperous. The recent resettlement of the Kolhan has resulted in a revenue demand twice as great as that previously assessed, but the people have paid it with ease. There was scarcity in 1897 and in 1900, especially in the Kolhan, where the inhabitants are chiefly improvident Hos who spend all they get and never trouble to save. At the same time they are better able than their Hindu neighbours to endure the want of proper food and can subsist, without great suffering, on various kinds of roots, fruits and leaves gathered in the jungles. The opening of the railway through the district has done much to develop trade and wages, and the prices of grain have risen considerably.

SINGHBHUM.

* More than half the emigrants from Manbhum were enumerated in Assam.

211. According to the vital statistics of the district the births during 1892—1900 outnumbered the deaths by 56,604. The actual increase between 1891 and 1901, according to the census, is 68,091, or 12·5 per cent. The progress would have been much greater but for the adverse balance of migration. Immigrants are fewer by nearly 3,000 while emigrants exceed by about 20,000 the number recorded in 1891.*

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	302,425	311,154	271,417	274,071
Immigrants	18,556	18,044	19,535	19,774
Emigrants	31,293	32,627	21,824	21,707
Natural population	315,182	325,737	273,706	276,004

But for this loss by emigration the increase would probably have been about 18 per cent. The tracts that show the greatest development are Chakradharpur and Manoharpur through which the railway runs. Then comes Ghatsila which has also, but more recently, been tapped by the railway. The Kolhan which shows the least progress is away from the line of railway and has probably sent out more emigrants than any other part of the district.

Thanas.	Population, 1901.	Percentage of variation.	
		1891—1901.	1881—1891.
DISTRICT TOTAL ...	*613,579*	*+12·5*	*+20·2*
Chaibasa	250,404	+ 9·3	+ 18·1
Chakradharpur	102,568	+ 18·0	+ 13·7
Ghatsila	220,259	+ 13·3	+ 17·3
Manarpur	40,348	+ 15·7	+100·0

212. The Sonthal Parganas comprises (1) a narrow strip of flat alluvial country running along the loop line of railway; (2) a belt of hill and forest stretching from Sahibganj southwards beyond Naya Dumka, the greater part of which is included in the Damin-i-Koh, or reservation originally intended for the benefit of the Malé, but now also extensively occupied by Santáls, and (3) a rolling, open country stretching south and west from the Damin-i-Koh, resembling in its general features the neighbouring districts of Hazaribagh and Manbhum.

Sonthal Parganas.

The district is for the most part healthy, and the Santáls, who are the most numerous tribe, are remarkable for their fecundity; there is much waste land available for cultivation, and rents are light. The district showed a rapid growth of population in 1881 and again in 1891, but on both occasions improved enumeration accounted for a great part of the increase. The first reliable census was that of 1891. Since that year the district has enjoyed good health, except in parts of Rajmahal, where malaria is prevalent. The vital statistics returns are, however, too inaccurate to be worthy of examination. Two months before the census, plague broke out in Sahibganj. Two hundred deaths were reported, and there was a great efflux of population. The crops have been fair on the whole, but there was a serious failure in 1896, which resulted in famine in the Deoghur and part of the Jamtara subdivisions in the south-west of the district. In September 1899 a heavy storm burst over the northern slope of the Damin-i-Koh in the Godda subdivision. The precipitation of the rain was so great that the streams quickly overflowed their banks and flooded the valleys through which they flowed. Whole hamlets ware washed away, and over sixteen hundred human beings besides thousands of cattle and goats were drowned. The flood subsided as quickly as it came, and the crops escaped with comparatively slight damage.

213. The recent census shows a net increase of 55,962, or 3·2 per cent., a surprisingly small rate of development for a healthy district with a prolific population. The statistics of migration supply the necessary explanation. The number of persons born in the Sonthal Parganas but enumerated elsewhere was 117,142 in 1891, and their number has now risen to 226,008, an increase of 108,866. Most of these have left the district permanently for the Bárind and other parts of Bengal, or for Assam. If the death-rate amongst emigrants be assumed to be 40 per 1,000, an annual exodus of 4,685 persons would be needed to maintain the emigrants of 1891 at their

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	896,873	913,364	870,364	883,411
Immigrants	53,191	58,334	77,396	75,691
Emigrants	115,894	110,114	61,678	55,464
Natural population	959,076	965,144	854,646	863,184

* Most of the emigration is to the Orissa States, Assam and Jalpaiguri.

original strength, and a further exodus of 13,534 would be required to raise it to the figure recorded at the recent census.* In other words, at least 182,190 persons must have left the district during the decade. The immigrants are about half as numerous as the emigrants. Most of the latter also are permanent, but it is impossible to ascertain how far they have affected the population, as the figures for 1891 are open to doubt. It may, however, be taken as certain that the advent of new comers has been on a very much smaller scale than the departure elsewhere of persons born in the district. If the proportion be taken roughly as one-third, and it was probably much less, the loss due to migration during the decade would be about 120,000. If so, the natural increase must have been at least 10 per cent.

214. The population of the Damin-i-Koh is stationary or decadent except the part that lies in the Rajmahal subdivision, where the collection of *sabai* grass (Ischaemum angusti folium) for the paper mills affords profitable employment. Elsewhere emigration has been busily at work, especially amongst the Santáls, who chafe under the restrictions imposed by the Forest Department on the indiscriminate felling of timber. Outside the Damin-i-Koh the only tracts that show a decline are Rajmahal, Sahibganj, and Poreya. In the first-mentioned tract it is due to migration across the Ganges, while in Sahibganj the plague epidemic is alone to blame; but for that an increase would probably have been recorded. Poreya is a poor and barren tract and, like the Damin-i-Koh, it has lost by emigration.

Thanas.	Population, 1901.	Percentage of variation.	
		1891—1901.	1881—1891.
DISTRICT TOTAL ...	***1,809,737***	***+ 3·2***	***+11·8***
Deoghur Subdivision ...	***297,403***	***+ 4·7***	***+12·8***
Deoghur	144,838	+ 2·9	+ 10·2
Madhupur	76,611	+ 4·2	+ 19·5
Sarath	75,954	+ 8·8	+ 12·7
Godda Subdivision	***390,323***	***+ 1·4***	***+10·2***
Godda	137,906	+ 15·3	+ 12·4
Mahagama	93,814	+ 6·2	+ 13·3
Poreya	59,354	− 2·3	+ 11·2
Portion of Damin-i-Koh ...	99,229	− 12·3	+ 5·0
Pakaur Subdivision ...	***238,648***	***+ 3·6***	***+12·2***
Pakaur	73,871	+ 8·2	+ 19·6
Mohespur	99,253	+ 2·1	+ 17·2
Portion of Damin-i-Koh ...	65,524	+ 1·1	+ 6·5
Rajmahal Subdivision ...	***276,703***	***+ ·1***	***+ 9·6***
Rajmahal	93,951	− 6·7	+ 7·5
Sahibganj	19,681	− 2·5	+ 47·5
Portion of Damin-i-Koh ...	163,071	+ 4·8	+ 7·1
Dumka Subdivision ...	***416,861***	***+ 3·1***	***+11·0***
Dumka	386,391	+ 6·5	+ 12·2
Portion of Damin-i-Koh ...	30,470	− 26·8	+ 3·3
Jamtara Subdivision ...	***189,799***	***+ 9·2***	***+19·0***
Jamtara	189,799	+ 9·2	+ 19·0

Of the most fertile and densely-populated thanas are Mahagama, Godda, and Pakaur, and these all show a marked improvement. So also do the thanas in the south of the district. The reason seems to be that there are here extensive areas covered with forest while there is no system of conservancy to prevent the people from destroying it. Although therefore there is some emigration in search of work there is none in order to find land for cultivation. The soil in these thanas is more fertile than that in the hilly tracts in the centre of the district.

Chota Nagpur States.

215. The Chota Nagpur States include seven Tributary States—Chang Bhakar, Korea, Sirguja, Udaipur, Jashpur, Gangpur, and Bonai—which are wedged in between Chota Nagpur Proper and the Central Provinces, and the two Political States of Kharsawan and Seraikela which lie within the district of Singhbhum. The earlier enumerations were here very inaccurate and no useful purpose would be served by considering the variations disclosed by them. The country is for the most part an irregular mass of hill ranges and broken plateaux covered with dense jungle, but there are occasional patches of fertile land, such as the valley of the Mand river in Udaipur and parts of the Gangpur and Jashpur table-lands. The population is exceedingly sparse and in spite of the generally inhospitable character of the country there is still ample room for expansion. The conditions of the last decade are very similar to those already described in the case of the adjoining districts of Chota Nagpur. The people are improvident aborigines and as they spend whatever they earn, they are at once affected by a deficient harvest. On the other hand

* The method of arriving at this figure has been explained in the foot-note on page 102.

they can subsist without much difficulty on jungle products eked out with a modicum of rice. Their condition since 1891 has, on the whole, been good. There was scarcity, not amounting to famine, in 1897, and 1900 and in the former year cholera raged with unusual virulence. The Bengal-Nagpur Railway, which runs through the whole length of the Gangpur State and passes near the boundary of Bonai and Udaipur, was opened ten years ago, and has done much to develop this part of the country. There has been a falling off in migration both to and from the States since 1891. In that year there were 97,240 settlers from Ranchi and Palamau, but now there are only 38,620. Allowing for deaths at the rate of 40 per 1,000 per annum, 64,647 of the settlers of 1891 would be still alive in 1901, and it would seem, therefore, that many of them must have re-crossed the boundary and returned to their old homes. The number of immigrants from places other than Ranchi and Palamau shows an increase. The decline in the number of emigrants is probably only apparent. The 1891 return shows that 15,949 natives of these States were enumerated in Jalpaiguri, 2,735 in Hooghly, and 2,038 in Shahabad, but it was surmised by Mr. O'Donnell that the great majority of these were in reality natives of Ranchi and other districts of the Chota Nagpur Division, and the figures for the present census fully confirm this conclusion. It is thus difficult to say with any pretence to accuracy to what extent the States have gained or lost by migration during the decade, but probably the ebb and flow have been fairly equal.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population	506,759	494,670	449,683	423,676
Immigrants	54,345	50,331	78,689	74,381
Emigrants	12,663	11,759	29,433	18,955
Natural population	465,017	456,089	400,427	378,250

States.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
TOTAL	1,001,429	+13·4	+30·3
Changbhak	19,648	+ 5·5	+ 33·3
Korea	35,113	— 3·1	+ 22·2
Sirguja	351,011	+ 8·1	+ 20·9
Udaipur	45,391	+ 30·9	+ 12·6
Jashpur	132,114	+ 16·3	+ 26·9
Gangpur	238,896	+ 24·8	+ 76·7
Bonai	38,277	+ 19·3	+ 33·3
Kharsawan	36,540	+ 3·02	+ 13·9
Seraikela	104,539	+ 11·4	+ 21·3

216. The States taken together show an increase of 13·4 per cent., of which some part is doubtless due to a more accurate enumeration. The greatest development is shown by Gangpur which has been opened out by the railway, and then come Udaipur and Bonai which have also been rendered more accessible. The only other State that shows a large increase is Jashpur, where the proportion of arable land is fairly high and where the people have benefited by the introduction of sugarcane and wheat cultivation, and roads have been constructed from the capital to the borders of Ranchi, Sirguja, Udaipur and Gangpur. Korea alone shows a decrease. The country is here very wild and barren, and the inhabitants are for the most part migratory aborigines.

TRIBUTARY STATES OF ORISSA.

217. The Tributary States of Orissa comprise a succession of hill ranges, the eastern outliers of the Sátpuras. The country is generally mountainous, but there are several fine valleys, notably those through which the Mahánadi, Bráhmani, Baitarani, and Burábalang rivers find an exit to the Orissa plains. There are in all seventeen states, but many of them are very small, and Mayurbhanj and Keonjhar alone contain half the total area, and nearly half the total number of inhabitants. The population is very sparse, but becomes greater on the lower levels as the plains of Orissa are approached. As in the Chota Nagpur States the earlier enumerations were very defective, and the large increase brought out by each successive census is due in a great measure to improvements in the arrangements for counting the people. At the same time there is no doubt that the population is growing rapidly under the ægis of British rule. The greater part of the country is very healthy; the inhabitants are hardy and prolific, and there is ample room for expansion. There is no registration of births and deaths in the Tributary States. It is reported that the last decade has been a healthy one generally, but the State of Baud, in the south-west, has suffered from epidemics of cholera and small-pox, and malarial affections are common there. Cholera was especially bad in 1900.

The pilgrims to Sonepur pass through Daspalla, and the Chief of that State complains that cholera is imported by them almost every year. The harvests on the whole have been good, and even in 1897 and 1900 there was no widespread famine. The crops in both years were short throughout most of the states, but with the aid of jungle products the people managed to support themselves without much suffering. In only a few instances were regular relief operations undertaken.

Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	969,973	977,829	840,450	847,990
Immigrants	66,300	76,092	78,560	83,225
Emigrants	21,205	36,194	13,708	22,689
Natural population ...	924,968	937,931	784,598	786,754

218. The census of 1881 disclosed a surprising amount of immigration, and no fewer than 199,895 persons born elsewhere were enumerated in these states, inclusive of Angul. In 1891, excluding Angul, the number of immigrants had fallen to 161,785, and there are now only 142,292. The volume of immigration is thus declining, but it is still very considerable. At the estimated death-rate of 40 per 1,000, the foreign-born population of 1891 would, in the absence of fresh settlers, have shrunk in the course of the decade to 107,560, and the excess over this number indicates an annual settlement of over 4,000 new comers. The number of emigrants has risen from 36,407 to 57,489 during the last decade, and to produce this result some 4,000 persons must have left the states yearly.* This calculation is necessarily approximate, as it assumes that the movements of the people were uniform throughout the decade, and that the whole of the migration was permanent; but it may perhaps be accepted as showing that the movement in both directions is now about equal, and that there is not at the present time any marked excess of immigration such as must have existed previous to 1891. The whole of the growth disclosed by the present census must, therefore, be ascribed to the natural growth of a prolific population and to the greater accuracy of the enumeration. To what extent the latter factor has contributed to the result it is difficult to say, as I have been unable to trace any detailed record of the procedure in 1891, but it probably does not account for more than 2 or 3 per cent., and if so, the natural growth stands at about 12 per cent.

States.	Population.	Percentage of variation.	
	1901.	1891—1901.	1881—1891.
TOTAL	*1,947,802*	*+14·8*	*+20·3*
Athgarh	43,784	+ 19·6	+ 17·8
Talcher	60,432	+ 14·7	+ 48·3
Mayurbhanj	610,383	+ 14·7	+ 37·3
Nilgiri	66,460	+ 18·3	+ 19·3
Keonjhar	285,758	+ 15·2	+ 14·2
Pal Lahara	22,351	+ 13·5	+ 33·3
Dhenkenal	273,662	+ 14·8	+ 14·7
Athmalik	40,753	+ 28·2	+ 43·3
Hindol	47,180	+ 24·2	+ 12·9
Narsinghpur	39,613	+ 17·03	+ 3·6
Baramba	38,360	+ 17·6	+ 9·4
Tigaria	23,625	+ 10·1	+ 3·4
Khandpara	62,456	+ 9·7	− 4·4
Nayagarh	140,779	+ 13·4	+ 2·3
Ranpur	46,075	+ 14·9	+ 10·0
Daspalla	51,267	+ 14·0	+ 9·5
Baud	89,250	− 1·4	+ 26·7

accuracy of the enumeration. To what extent the latter factor has contributed to the result it is difficult to say, as I have been unable to trace any detailed record of the procedure in 1891, but it probably does not account for more than 2 or 3 per cent., and if so, the natural growth stands at about 12 per cent.

219. The greatest increase is in the sparsely inhabited State of Athmallik, which has gained by immigration from Baud and the Central Provinces, and in Hindol, which has also received an accession of new settlers; in both these States, however, it is probable that the recent census was more complete than that of 1891, and that part of the increase is due to improved enumeration. With the exception of Tigaria, Khandpara, and Baud, the other States show an improvement varying from 13·5 to 19·6 per cent. The comparatively slow rate of increase in Tigaria and Khandpara is explained by the fact that the population is already much more dense than elsewhere. Tigaria has 492 inhabitants to the square mile compared with 285 in Baramba, which stands next to it in this respect, and Khandpara has 284 persons to the square mile compared with only 92 in its next door neighbour, Daspalla. With these easily explained exceptions, the growth of the population has been greatest along the borders of the Orissa Division, where the level is comparatively low, and the proportion of arable land is relatively high. The construction of the railway through Orissa, and of feeder roads in connection with it, has greatly improved the communications and raised the prices of produce in this tract. Baud alone shows a falling off, which is due, as in the case of the adjoining Khondmals subdivision of Angul, partly to the prevalence of epidemic disease and general unhealthiness and partly to the emigration of the migratory Kandhs during the scarcity of 1900.

* That is to say, 1,456 to replace death vacancies and 2,628 to produce in the course of the decade the 21,082 additional emigrants recorded at the census of 1901.

ANGUL AND KHONDMALS.

220. Angul is surrounded on all sides by the Tributary States of Orissa and the Central Provinces. It was itself a Tributary State until 1847 when it was confiscated owing to the rebellion of the Chief. The Khondmals subdivision lies between Ganjam in Madras and the State of Baud, whose Chief ceded it to Government in 1885 owing to his inability to suppress a rebellion. The two tracts were formed into a regular district in 1891. Their physical features differ in no respect from those of the surrounding States. Angul, however, is more open and better watered, and the climate is less unhealthy than that of the Khondmals where hill and jungle reign supreme. The latter tract is so unhealthly that outsiders fear to visit it, and many of the enumerators sent up from Cuttack to assist in taking the census were prostrated by fever. Cholera made its first appearance in the Khondmals, at least for many years, in 1900. It was introduced by persons fleeing from an epidemic in the adjoining States of the Central Provinces and spread with appalling rapidity. There are no statistics of births and deaths in these wild hills, but it is known that the loss of life was very great. In some villages syphilis is extraordinarily prevalent and almost all the inhabitants are suffering from the disease in one stage or another.

Subdivision.	Population in 1901.	Variation.	
		1891—1901.	1881—1891.
ANGUL	***191,911***	***+12·8***	***+5·7***
Sadar	127,697	+ 25·1	not available.
Khondmals	64,214	− 3·2	

In Angul the rents are very low. In the Khondmals no rent is charged, but the people pay a contribution of three annas a plough towards the cost of improving communications. The inhabitants of the latter tract are mostly Kandhs, but the comparatively small Hindu population has succeeded in obtaining possession of the best rice lands. The Kandhs grow oil seeds and turmeric. They are indolent, improvident, and fond of drink. The people in both tracts are usually prosperous, but their powers of endurance have been severely tried by several lean years in the latter half of the last decade. There were short crops in 1896 and again in 1899. In Angul the cultivators were able to tide over the bad times with the aid of loans, but in the Khondmals relief operations were necessary.

221. The recent census shows a great increase in Angul, but in the Khondmals there has been a loss of 3·2 per cent. This is amply accounted for by the adverse conditions of the latter part of the decade, viz., the prevalence of cholera and other diseases and the scarcity, which stimulated emigration amongst the Kandhs. There was very little immigration to the Khondmals except from Ganjam in Madras. In Angul, on the other hand, the comparatively favourable circumstances have attracted settlers from the adjoining States. In the two tracts together, the number of immigrants has risen from 9,508 in 1891 to 21,532. There were 12,753 immigrants from the Tributary States (all but 1,109 to Angul), and 3,599 from Madras, of whom all but 93 were found in the Khondmals. The emigrants number 6,482 compared with 1,939 according to the previous census, but the return of emigration in 1891 was not very accurate, as Angul was constituted a separate district just about the time of the census.

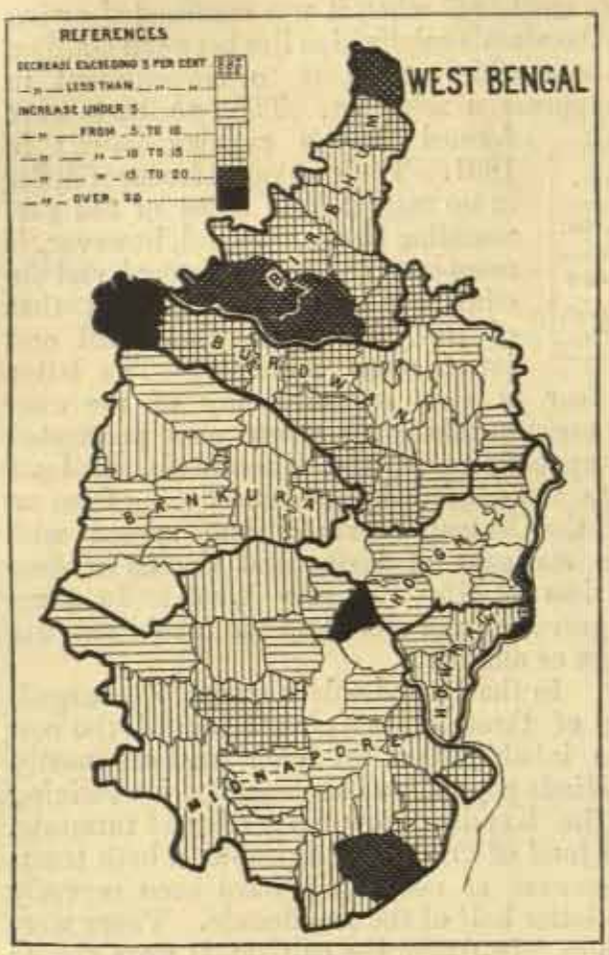
Population.	1901.		1891.	
	Male.	Female.	Male.	Female.
Actual population ...	95,035	93,976	85,768	84,290
Immigrants ...	9,363	12,169	4,939	4,569
Emigrants ...	2,294	4,184	668	1,271
Natural population ...	88,866	87,991	81,497	80,992

SUMMARY.

WEST BENGAL.

222. We are now in a position to reconstruct the figures and to glance at the variations for the groups of districts which comprise the eight Natural Divisions into which the Province has been divided. West Bengal, which is coterminous with the Burdwan Division, has grown since 1891 by 7·1 per cent. The variations in the population of districts taken as a whole have been indicated in the map of Bengal on page 39. But the circumstances of different parts of a

district often vary greatly and it is, therefore, desirable to display the changes for smaller units. The map in the margin shows the variations in the population of each thana in the districts of West Bengal. The most progressive district is Birbhum which has added 13 per cent. to its population. This result is due mainly to natural growth. The increase is most marked in the south of the district, where it represents a recovery from the unhealthiness of the previous decade, and in the extreme north, where there has been a considerable settlement of Santáls. Howrah, which comes next, owes its development to the presence of a large manufacturing town and to its proximity to the metropolis. No less than 17 per cent. of its inhabitants are immigrants attracted by the mills and other forms of remunerative employment obtainable in Calcutta and in Howrah city. Excluding the Municipalities of Howrah and Bally, the rate of increase ranges from 5 to 10 per cent., except in the north-east corner which is wedged in between decadent tracts in Midnapore and Hooghly, where it is only 2·4 per cent. Burdwan, which follows close on Howrah with a gain of 10·1 per cent., also owes a great part of its expansion to immigration, especially in the westernmost thana, Asansol, where the coal mines have created a demand for labour far in excess of the supply available locally, and nearly one-third of the inhabitants are foreign-born. The central part of the district, having escaped from the clutches of the Burdwan fever, has also grown with more than average rapidity, and the jungly and ill-drained western tract which slopes back from the banks of the Bhágirathi, is alone stationary or decadent. The increase of 5·9 per cent. in Midnapore is the resultant of a rapid growth of the population along the sea-coast and the estuary of the Hooghly, and a fair natural development in the healthy but barren and sparsely inhabited up-lands in the west of the district, combined with stagnation or decline in the ill-drained depression that intervenes between these two extremes. The district has lost slightly by migration, but not to the same extent as Bankura, where the increase of 4·3 per cent. represents less than half of that which would have been registered had none of its inhabitants sought a more hospitable home elsewhere, or gone to eke out their local earnings by working in the metropolitan districts during the cold-weather months. The number who were temporarily absent at the time of the census was greater than usual owing to the short harvests of the preceding year. The southern part of Bankura has suffered most by this exodus of the people, and in the extreme south it has been sufficient, not only to retard progress, but to actually reduce the population. The eastern part of the district resembles the adjacent part of Burdwan and, like it, has grown with fair rapidity now that it is no longer subject to the ravages of the 'Burdwan fever.'

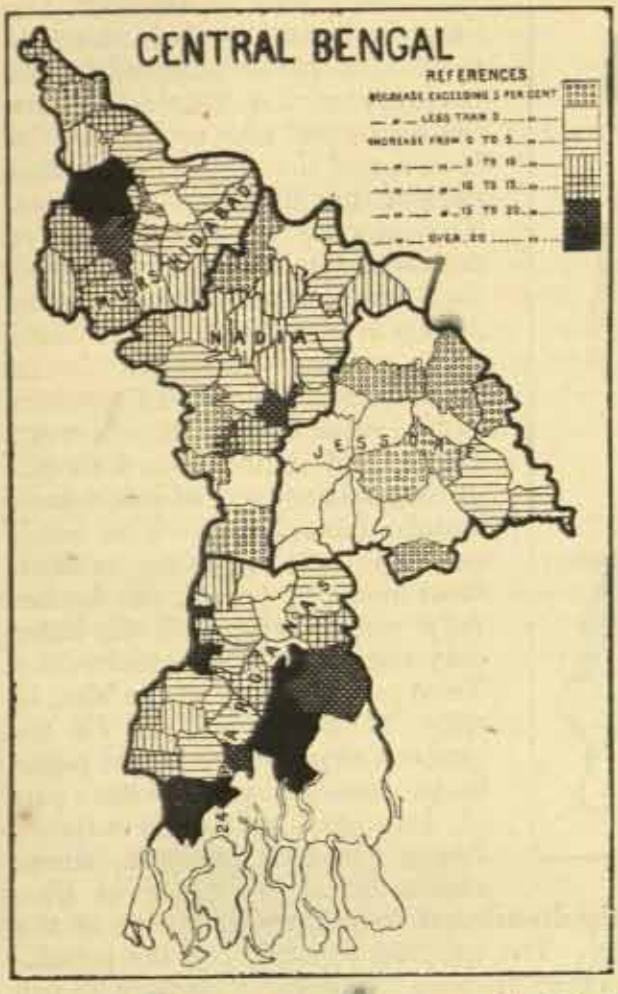


Hooghly remains to be mentioned. This district suffers from the same insanitary conditions as the west of Burdwan and the central part of Midnapore. It contains the busy and growing town of Serampur which affords employment to numerous immigrants, and if the figures for this town were excluded, the small increase of 1·4 would disappear, leaving the population almost exactly at the figure at which it stood ten years ago.

223. Central Bengal, or the Presidency Division excluding Khulna, shows an increase of 5·1 per cent. This result is due mainly to the figures for Calcutta and the 24-Parganas. Calcutta has grown by 24·2 per cent., but here the greater accuracy

CENTRAL BENGAL.

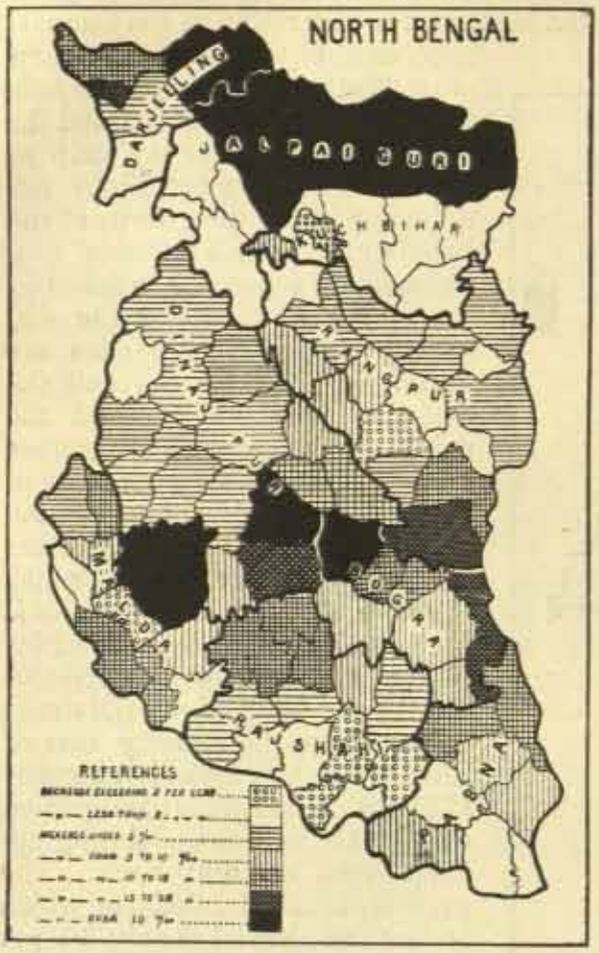
of the present enumeration has obscured the true growth. The expansion of 9·8 per cent. shown by the 24-Parganas is due mainly to the rapid progress which is being made in the reclamation of land for cultivation in the Sundarbans, and to the opening of new mills on the banks of the Hugli. Between the riparian tract and the Sundarban area there is a strip of country where the conditions are unfavourable to the growth of the population. These adverse conditions are especially marked in the north-central part of the district, where the village sites are buried in dense jungle, and the drainage is obstructed and the drinking water unwholesome. Here, as in the adjoining parts of Nadia and Jessore, the population has declined. The only other district in Central Bengal which shows a fair rate of progress is Murshidabad where the gain of 6·5 would have been slightly greater had not the emigration of Muhammadans from the country east of the Bhágirathi to the adjoining part of Malda exceeded the immigration of Santáls to the undulating uplands west of that river. The growth of East Murshidabad which is low and alluvial is only 3·1 per cent. while that of West Murshidabad which belongs to an older formation and has a healthier climate, is 12·9 per cent. The central portion of the latter tract, which is still very sparsely inhabited, has added more than a quarter to the population recorded in 1891. The other two districts of Central Bengal, Nadia and Jessore, are a region of moribund rivers and obstructed drainage. In Jessore the village sites are old and are surrounded by rank jungle. Except in the south and the extreme north-east, where the conditions resemble those of Jessore, Nadia is more open and, its soil being more sandy, there are fewer stagnant swamps. The result is that while Jessore has lost ground to the extent of 4·0 per cent., Nadia has registered a slight improvement of 1·4 per cent., the decadence in the south being more than counterbalanced by the moderate growth that has taken place in most of the other police circles of the district. The country round Bogula, on the Eastern Bengal State Railway, which showed a decrease in 1891 on account of the damage done by floods, and a small tract in the extreme west which adjoins the progressive part of Murshidabad, are the only parts where the growth of the population has been at all rapid. In Jessore the decadence is general, and the south-eastern corner is the only tract which shows even a nominal improvement. The loss of population is greatest in the country running west and south-west from the Muhammadpur thana on the eastern boundary, which possesses the evil reputation of having been the matrix both of epidemic cholera and of the 'Burdwan fever.' It will be seen, in dealing with East Bengal, that this unhealthy zone stretches eastwards and southwards beyond the Jessore boundary, and includes the north-western part of Faridpur and a small area in the north-west of Khulna.



224. North Bengal, which includes Malda and the Native States of Kuch Bihar and Sikkim, as well as the districts of the Rajshahi Commissionership, has added 5·9 per cent. to its population in the course of the decade. Excluding Sikkim, where part of the apparent expansion is attributable to the greater accuracy of the present census, the most rapid rate of increase is found in the great tea district, Jalpaiguri, where it amounts to 15·6 per cent. This district is not a healthy one, and almost the whole of the increment as compared with 1891 is due to immigration to the Duars, *i.e.*, the eastern portion of the district which was taken from Bhotan in 1865. The tea gardens are the great attraction,

NORTH BENGAL.

but there has also been a considerable extension of ordinary cultivation. The proportion of the foreign-born population is high throughout the Duars, and varies from 28 to 55 per cent. West of the Tista the little immigration that exists is more than counterbalanced by the tendency of the people to move eastwards and take up land in the Duars, and the population in this part of the district is decadent. The second place in point of development is held by Darjeeling, which is more populous by 11·5 per cent. than it was in 1891. The greatest expansion is in Kalimpong, which was annexed at the same time as the Western Duars, and which owes its rapid growth to the advent of cultivators, mainly from Nepal. The headquarters and Kurseong subdivisions have also grown, the former very considerably and the latter only slightly, while the unhealthy Terai at the foot of the hills, in spite of the growth of its tea gardens, shares in the loss of population sustained by the western part of Jalpaiguri which it adjoins. Bogra, with 11·7 per cent., shows almost the same progress as Darjeeling, but the increase is more evenly distributed and there is no part of the district where it falls below 6 per cent. The greatest expansion of the population is in the north-west corner. This tract rests on the laterite formation known as the Bárind which extends over the adjoining parts of Dinajpur, Rajshahi and Malda. It was once highly cultivated but at some period, possibly about five or six centuries ago, it relapsed into jungle, and remained desolate and uninhabited until the Santáls, who have been steadily pushing their way to the north-east, ever since the beginning of the last century, arrived on the scene, and set themselves to the task of reclaiming it. The east of Bogra which is on the bank of the Jamuna and is a great centre of jute cultivation, also shows an unusually rapid development.



225. Malda with a net growth of 8·4 per cent. has great local variations. The east-central part of the district lies in the Bárind and has shared in the rapid increase which is the present characteristic of that tract of country. The south-western portion is washed by the Ganges and, having gained considerably by alluvion, has attracted numerous settlers from other parts of the district and from Murshidabad, on the other side of the river, which has suffered a corresponding loss of territory. North and south of this progressive area, diluvion has been followed by emigration, and a slight decrease has occurred, while the centre of the district, which is low and swampy and has been very unhealthy of late years, has suffered a still greater loss of population. About half the net gain in this district seems to be due to immigration and half to natural growth. The increase in Dinajpur is 5·7 per cent. of which nearly three-quarters is due to migration. The south of the district lies in the Bárind and is very progressive. Further north the notorious unhealthiness of the climate has prevented more than a very slight growth of the population but, with the exception of one police circle, the district is free from the decadence which characterises the adjacent parts of Purnea on the west and of Jalpaiguri on the north-east.

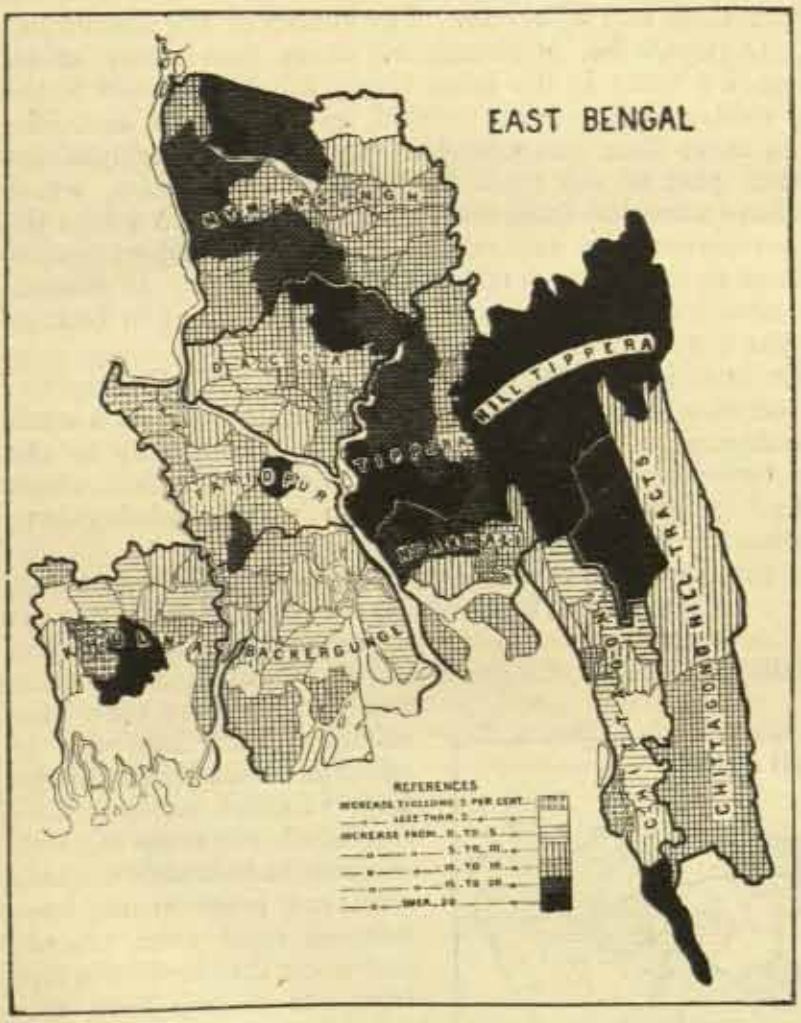
Pabna (4·3 per cent.,) and Rangpur (4·3 per cent.,) which lie respectively to the south and north of Bogra, show approximately the same rate of growth. The north of Pabna which comprises the Serajganj Subdivision and is largely given up to the cultivation of jute, is progressive, and there has been a gain of 9·4 per cent., while the south, which is water-logged and unhealthy and has suffered from floods and diluvion, has lost 2·1 per cent. Prior to 1891

Rangpur had a long history of unhealthiness and in that year, as well as in 1881, a loss of population was recorded. Its increase at the present census appears to be due largely to immigration, but apart from this there has been a slight natural increase, and it would seem that the tide has at last turned. In the estimation of the people also, the district has become more salubrious, and it may be hoped that the small increment now recorded is only a harbinger of more rapid progress during the next ten years. The greatest improvement at the present census is in the south, along the boundary of Bogra. There is a considerable area in the centre of the district where the population is still retrogressive.

The increase of 1·5 per cent. in Rajshahi is the net result of a considerable growth of population in the extreme north, in the Bárind, and a slight improvement in the rest of the northern half of the district, combined with a decline in the southern part, where the drainage is obstructed to almost as great an extent as in Jessore and malarial fevers are very prevalent. The State of Kuch Bihar forms the eastern extremity of the unhealthy belt of country which stretches westwards as far as the Kosi and shares in the loss of population which has been sustained by Purnea, the Darjeeling Terai and the west of Jalpaiguri. The decrease at the present census is 2·0 per cent. compared with 3·9 per cent. in 1891.

EAST BENGAL.

226. East Bengal with an increase of 10·4 per cent. is at present by far the most progressive part of the Province. The climate is generally healthy, the people are most prosperous and a very large proportion of them are prolific Muhammadans. The whole tract shares in the general result but the greatest apparent development is in Hill Tippera where the population is greater by 26·1 per cent. than it was in 1891. The present enumeration is possibly more accurate than its predecessor, but the statistics of birthplace show that the State has gained largely by the overflow of population from the neighbouring districts and especially from Sylhet and Plains Tippera. Natural growth accounts for less than a third of the total increment. The British district of Tippera comes next with an advance 18·7 per cent. which is attributable solely to natural growth and affords a remarkable illustration of the rate at which the human race can multiply itself when all the conditions are favourable. The soil is very fertile and there is still room for expansion, the crops have been good throughout the decade, and the district has been remarkably free from disease of all kinds. The most rapid growth is in the south where there has been a great extension of jute cultivation. In the rest of the district the improvement is most marked along the bank of the Megna; the country further east is somewhat less fertile and has suffered slightly from emigration. Excluding the Chittagong Hill Tracts which shows an increase of 16·2 per cent., attributable to a more accurate enumeration and to the natural growth of a very sparse population, Noakhali, with a growth of 13·0



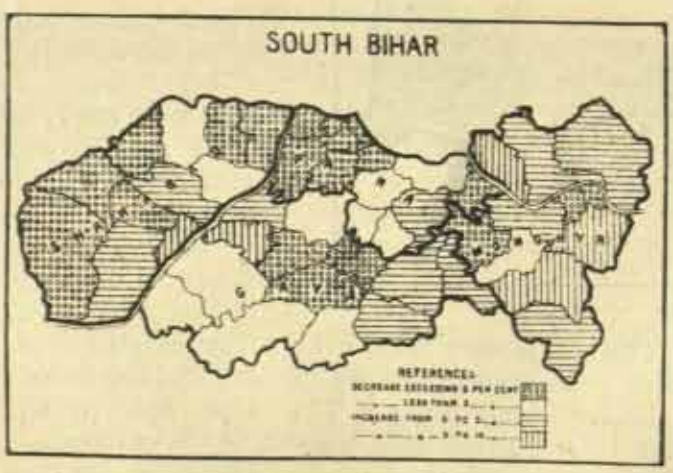
per cent. stands next to Tippera. There has been some loss by migration and the real increase is slightly greater than would appear from the census figures. The most progressive part is in the north, along the Tippera boundary, but the whole district has contributed its quota to the increment except a small tract at the mouth of the Feni river where there has been a good deal of diluvion. Mymensingh, with an increase of 12·7 per cent., follows close on Noakhali. There has been some gain by migration, but not enough to materially affect the result. The western half of the district is more progressive than the eastern. The latter is low and contains numerous large hollows which are filled with water in the rains, while to the north the climate is unhealthy and a slight loss of population has been sustained. Dacca has gained 10·6 per cent. since 1891. The increase is greatest in the north-east, where extensive clearances have been made in the Madhupur jungle, and least in the south-west, where the population is already very dense and the Padma has diluviated a considerable area. This district has lost somewhat by migration.

227. The three districts of East Bengal which lie to the west of the Padma and of the united channel of that river and the Megna, viz., Khulna, Backergunge and Faridpur, show a remarkably uniform rate of growth amounting respectively to 6·4, 6·4 and 6·2 per cent. In Khulna the result is the outcome of a very rapid extension of cultivation in the south-central and south-western portion of the district and a steady but less rapid growth in the *bil* country to the north-east, on the confines of Faridpur, combined with a decrease in the north-western corner which projects into Jessore, and in a narrow strip of country running from it first in a southerly and then in a south-easterly direction; fever is here very prevalent. Backergunge shows a rapid growth in the *bil* country to the north and in the Sundarbans to the south. The progress is least in the part between these two extremes. The centre of the district has been long settled and its population is already so dense that many of its inhabitants are fain to seek a home in the more thinly inhabited tracts to the north and south. The western part of this central tract has also sustained damage from cyclones on more than one occasion since 1891. Faridpur has gained most in the central part of the tract bordering on the Padma, where new alluvial formations have attracted fresh settlers, and in the south where the *bils* are gradually being reclaimed for cultivation. The north-western portion which borders on the most unhealthy part of Jessore is decadent. In Khulna migration accounts for nearly a quarter of the total increase, but in Backergunge and Faridpur, it has not appreciably affected the result.

The least progressive tract in the whole of Eastern Bengal is Chittagong, where there has been an increase of only 4·8 per cent. Its soil is only to a small extent alluvial, but the comparatively slow development is due mainly to the havoc wrought by the cyclone of October 1897, and to the emigration which followed it. But for this the growth of the population would probably have been twice as great as it has actually been.

228. South Bihar includes all the plague districts except Saran, and its decrease of 3·6 per cent. is mainly attributable to the direct and indirect losses caused by the epidemic, viz., a very heavy mortality, the flight of a great part of the immigrant population and, in some parts, the failure of the census staff to effect an exhaustive enumeration. Except in the west of Shahabad, the areas of greatest decadence exactly coincide with the areas which have suffered most from plague, and tracts that have been free from the disease have, as a rule, added to their population. Prior to the census the epidemic had been most virulent and most widespread in Patna, where the population has declined by 8·3 per cent. as compared with 1891. The loss is greatest in the thickly populated urban and semi-urban country on the bank of the Ganges, where the mortality due to plague was greatest. The southern part of the district which suffered least from plague has almost held its ground.

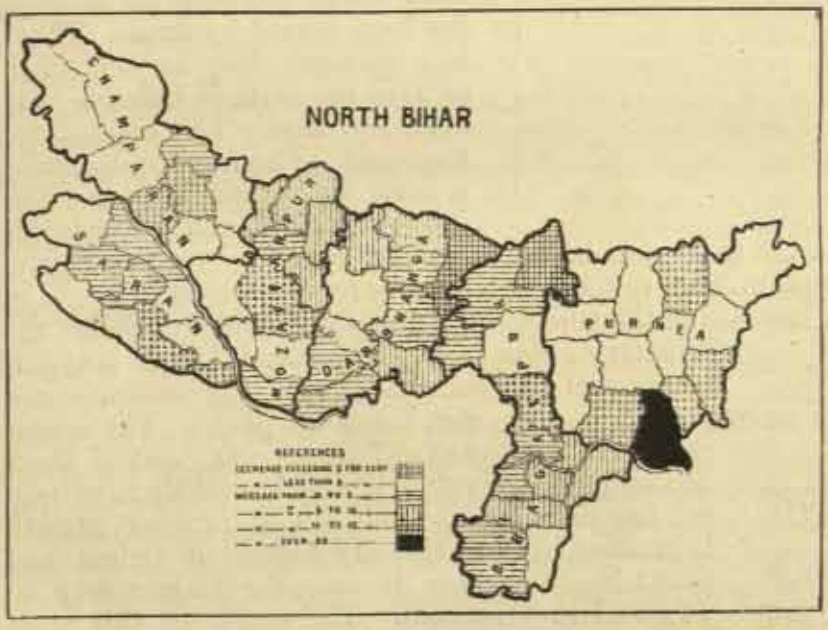
South Bihar.



Gaya, with a net decrease of 3·6 per cent., has sustained the greatest loss in the central police circles where plague has been most prevalent. There is also a slight decadence, for which plague is not to blame, in the south-west of the district, where the land is high and barren and the crops are scanty and uncertain. The Nawadah subdivision in the east, and a small tract which benefits by irrigation from the Sone in the north-west, have added to their population; both these tracts had escaped the ravages of plague up to the time of the census. Shahabad has lost 4·7 per cent. of the population recorded in 1891. The whole of the western half of the district, which has long been very unhealthy and marches with an extensive decadent area in the United Provinces, shows a comparatively heavy decrease, and so also does the north-eastern portion on the Patna border, where the result is attributable mainly to the appearance of plague shortly before the census. Three of the four police circles of the Sasaram subdivision, where the area under irrigation is greatest, alone show an increase. Monghyr, with a slight gain of 1·6 per cent., is the only district in South Bihar which has escaped a loss of population, and even here the result is due to the fact that part of the district lies on the north bank of the Ganges; the portion south of that river has sustained a small net loss an increase in four police circles having been more than obliterated by a heavy loss in four others where plague had appeared, viz., the town of Monghyr and its environs and two police circles in the west whither the epidemic spread from Mokameh in the Patna district.

North Bihar.

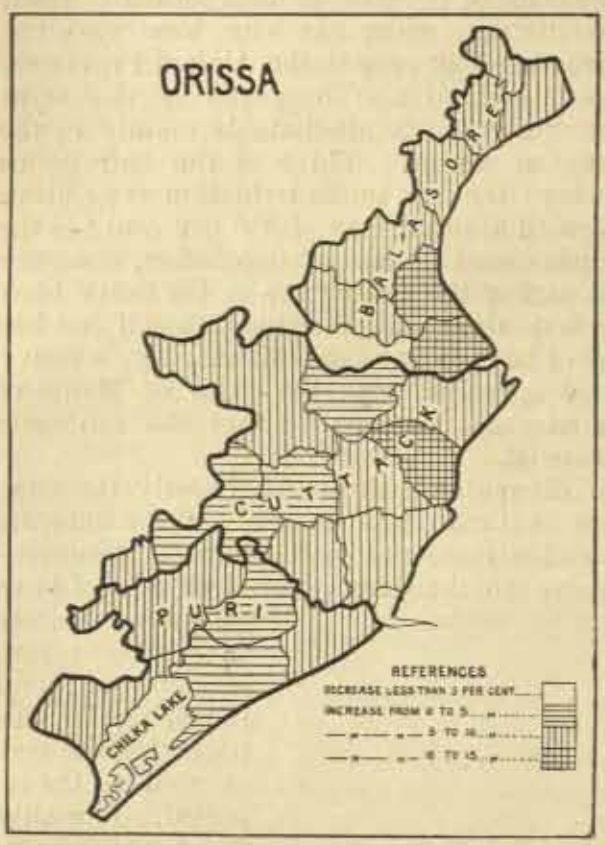
229. The population of North Bihar stands at almost exactly the same figure now as it did ten years ago, and the question that immediately occurs to one is how far the unprogressive state of this tract is attributable to the famine of 1896-97. I have shown in the notes on the Champaran, Muzaffarpur and Darbhanga districts that the real, as well as the reported, mortality was less than usual in the famine year, and that no connection can be traced between the relative severity of the famine and the variations in the population in the different parts of these districts; but it may be as well to review briefly the salient points in the argument, so far as it depends upon the census figures, from a more general stand point. The stress of famine was greatest in Darbhanga, but this district shows the largest gain of population (3·9 per cent). Purnea escaped the famine altogether, but it has sustained a loss of 3·5 per cent. or exactly the same as Champaran where the decline is greatest in the very tract that suffered least from famine. Saran, which has a decrease of 2·2 per cent. was far less severely affected than Muzaffarpur, which has gained 1·5 per cent., and its loss of population is amply accounted for by the plague epidemic which was more virulent there than in any other district except Patna. The Gopalganj subdivision, where the famine was worst has added slightly to its population. In Muzaffarpur and Darbhanga the great rice-growing tracts under the Nepal frontier, which suffered most in the famine year, show the greatest growth of population. The decadent tracts in Muzaffarpur and Bhagalpur either escaped the famine altogether, or suffered from it only in a minor degree. The true causes of the decay in parts of North Bihar must, therefore, be sought elsewhere. Champaran and Purnea are well known to be unhealthy and have suffered since 1891, not only from malarial affections but also from severe epidemics of cholera. The outbreak of this disease in Purnea in 1900 was of unparalleled severity and no fewer than 46,240 deaths were laid to its account in the annual returns of mortality. The part of Bhagalpur that has



lost population borders on Purnea and shares the unhealthiness of which that district is the victim. In Saran, as already noted, plague fully accounts for the decrease which is greatest where that disease was most prevalent.

ORISSA.

230. The Orissa Division has enjoyed a placid period of peaceful development and the rate of growth is remarkably uniform throughout. Cuttack has added 6·4 per cent. to its population, or slightly less than Balasore and Puri, both of which show an increment of 7·6 per cent. The increase is wholly due to natural growth and would have been slightly greater, but for a small adverse balance on account of migration. Throughout the greater part of the Division the rate of progress ranges betwen 5 and 10 per cent. It exceeds 10 per cent. only in two fertile but sparsely inhabited tracts, the one in the south-east of Balasore and the other in the east of Cuttack, both on the sea coast. The rate of increase is less than 5 per cent. only in the more densely populated parts of Cuttack and in a strip of country running north and south through the centre of Puri where considerable damage has been caused by floods. There is a decrease only in one small area, in the south of Balasore, and here, too, floods are blamed for the result. The loss of population is little more than nominal, and is not of a permanent nature; it is due to a large part of the male population having gone away temporarily to seek work near Calcutta, in order to recoup themselves for the loss of their crops.



CHOTA NAGPUR PLATEAU.

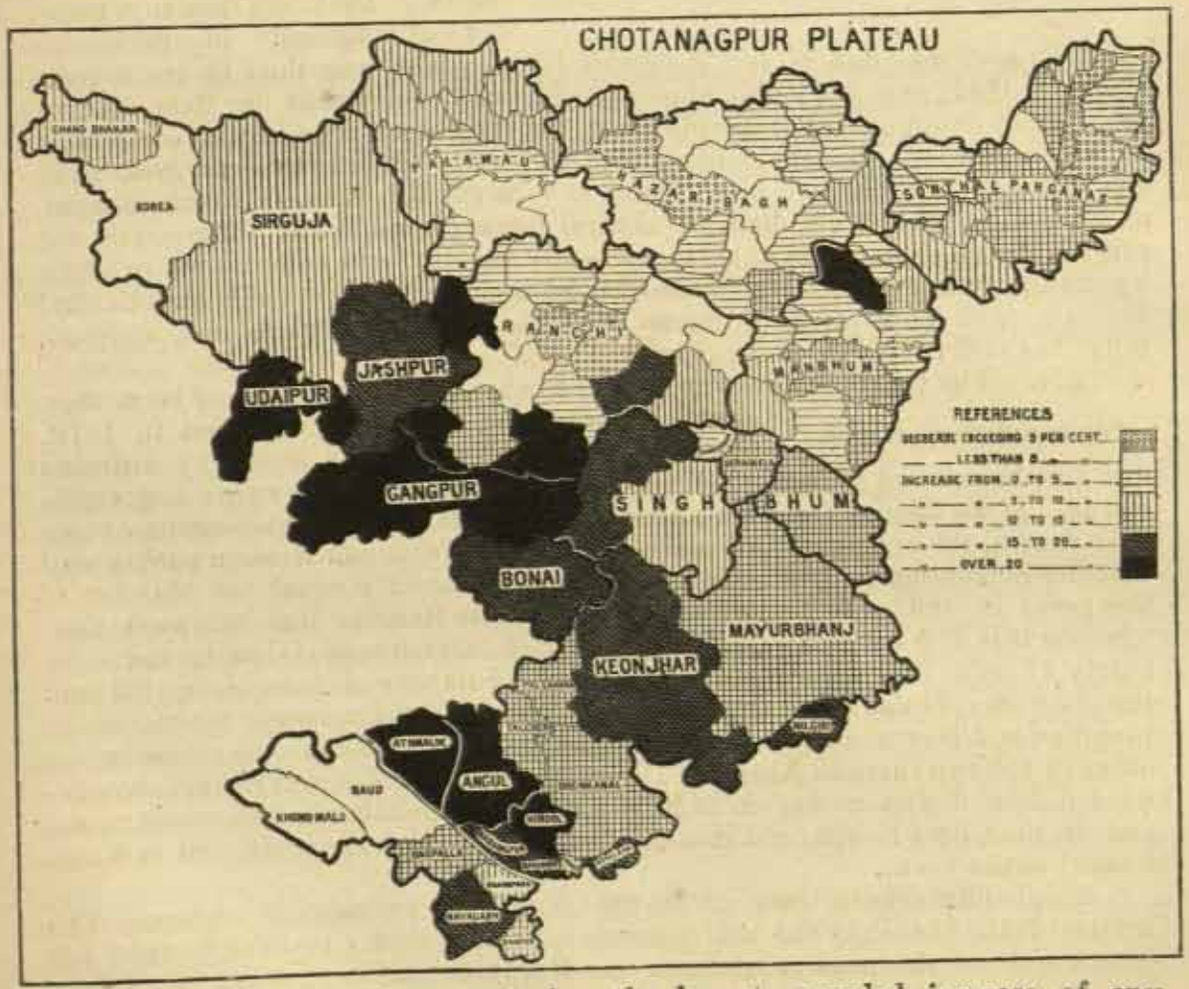
231. The true increase in the Chota Nagpur Plateau, which includes the Tributary States of Orissa and Chota Nagpur, the Sonthal Parganas and Angul, has been obscured by the extraordinary amount of emigration that has taken place. The census shows a net increase of 7·8 per cent., but if there had been no emigration it would probably have been not less than 10 per cent. The rate of growth is greatest in the Tributary States of Orissa and Chota Nagpur where it amounts respectively to 14·8 and 13·3 per cent. The whole of this tract is very sparsely inhabited, and although the proportion of unculturable land is high, there is still ample room for expansion. In a few States where the increase is very great it is probable that improved enumeration accounts for a part of it, but even so, the actual growth during the decade is probably not less than 12 per cent. The only States which have lost population are Baud in the south, where it is due to unhealthiness and emigration after the scarcity of 1900, and Korea in the north, where it is due mainly to emigration.

District.	Estimated loss by emigration.
TOTAL	*398,000*
Hazaribagh	91,000
Ranchi	128,000
Manbhum	36,000
Singhbhum	20,000
Sonthal Parganas	120,000

Angul which lies in the midst of the Orissa States shows a net increase of 12·8 per cent., but while the head-quarters subdivision has gained more than 23 per cent., the Khondmals subdivision has lost rather more than 3 per cent. The latter tract has been affected, but to a greater extent, by the adverse circumstances that have reduced the population of Baud which adjoins it and to which it formerly belonged.

232. Next to Angul and the Tributary States, the greatest progress is shown by Singhbhum which has suffered less from scarcity and has lost less by emigration than the other districts of the Chota Nagpur Division. The opening of the railway has benefited the district, and the tracts which show the greatest improvement are those through which the railway runs. The net increase is 12·4 per cent., while in Manbhum which adjoins it, it is 9·0 per cent.

The most progressive tract in the latter district is in the north-west where the rapid development of the Jheria coal field has created a great and growing demand for labour. In spite of this the district has sustained a net loss by migration, owing to the recruitment of coolies for Assam, and the true rate of increase must have exceeded the census figures by 2 or 3 per cent. Ranchi has lost by migration more than any district except the Sonthal Parganas, and its apparent increase of 5·2 per cent. represents less than half of the real growth of the population. More than 91,000 persons born in this district were enumerated in Assam. Apart from this vast emigration troubles between landlord and tenant have led to a movement of the population from the central plateau towards the west and south of the district. The migration statistics of Palamau show a slight balance in favour of the district, but the figures are not very trustworthy, and many of the ignorant emigrants probably returned Lohardaga (the old name for Ranchi) as their birth-place. The census shows an increase of 3·8 per cent., and even if we raise this to 6 per cent. to allow for a probable though unrecorded loss on account of emigrants, the rate of progress is still small for so sparsely inhabited a district. The decade has been unhealthy, and this is probably the reason why the growth of population has not been greater. The apparent increase in the Sonthal Parganas (3·1 per cent.) is rather less than in Palamau, but the real growth has been far greater. The district has sent out more than 180,000 emigrants as tea-garden coolies in Assam and Jalpaiguri and as pioneers of cultivation in the Bárind. But for this its rate of growth would have exceeded 10 per cent. There are two decadent tracts in the district. The Damin-i-Koh, which has lost by permanent emigration, and a narrow strip in the north-east of the district where plague broke out shortly before the census and caused the temporary flight of many of the settlers from other districts.

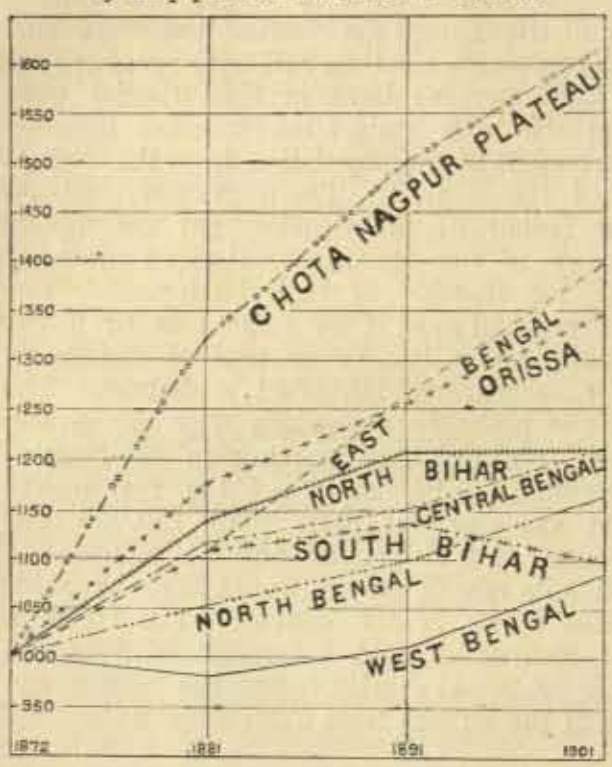


Hazaribagh with 1·1 per cent. has the lowest recorded increase of any district in the Chota Nagpur Plateau but here, as in the Sonthal Parganas and Ranchi, there has been a very heavy loss by emigration. It is probable that more than 90,000 persons left the district during the decade, and if so, the natural growth of population would be about 8 per cent. The central part of the district, where recruiting for tea-gardens was most active, shows an actual decrease. In the Giridih subdivision and the Kasmar thana the coal fields have attracted labour, and the rate of expansion is more rapid than in the rest of the district.

233. In conclusion we may briefly notice the progress in the province as a whole and in each natural division since 1872. The diagram in the margin shows the variation per 1,000 of the population of the different natural divisions at each successive enumeration. Between 1872 and 1881 the Chota Nagpur Plateau showed the greatest apparent improvement, but this was due mainly to the inaccuracy of the first census in a wild, remote and sparsely-peopled tract, where the difficulties in the way of a proper enumeration are exceptionally great. Orissa, which came second, had suffered a terrible loss of population in the great famine of 1866, and its rapid growth was the natural reaction from that calamity during a period of renewed prosperity. In North and South Bihar, as in Chota Nagpur, the census of 1872 was defective, and the increment brought to light in 1881 was thus to a great extent fictitious. The decline in West Bengal was due to the Burdwan fever, which was then at its height. Between 1881 and 1891 the apparent rate of development in East Bengal and Chota Nagpur was about the same, but the latter tract again owed part of its increase to better enumeration, and the real growth was greatest in East Bengal. Then followed Orissa and North Bihar, then North Bengal, and then in order West Bengal, Central Bengal, and South Bihar. On the present occasion East Bengal again heads the list and is followed in order by the Chota Nagpur Plateau, Orissa, West Bengal, North Bengal, and Central Bengal. The population of North Bihar is stationary, while that of South Bihar has suffered a loss of 3 per cent.

PROGRESS IN EACH NATURAL DIVISION COMPARED.

Diagram showing the variation since 1872 per 1,000 of the population in each natural division.



234. The province as a whole, is now more populous by 25 per cent. than it was at the time of the first enumeration in 1872, but the varying rate of progress shown by different districts, has caused the centre of the population to shift eastwards and southwards. If we take as the centre of population the point of intersection of two lines drawn, the one north and south and the other east and west, in such a way that the population on either side of each line is exactly equal, the position of this point in 1891 was 6·4 miles due west of the Rampur Hat railway station, whereas it is now 8·8 miles due south of it. In other words it has moved very nearly 11 miles in a south-easterly direction. This way of considering the subject, however, is defective as it fails to recognize all the movements that are in progress, and leaves out of account all changes which do not pass across one or other of the two dividing lines. If individual parts of the Province be considered it will appear that in Bihar the centre of population has moved to the east, in the Chota Nagpur Plateau and Central Bengal to the south, and in North Bengal to the north.

THE CENTRE OF POPULATION.

235. The general rate of growth shows a progressive decline. The census of 1881 brought out an increase of 11 per cent.; in 1891 it was 7 per cent., and on the present occasion it is only 5 per cent. This is due in part to the greater accuracy of each succeeding enumeration. The pioneer census of 1872 was admittedly very incomplete. A fresh count of part of the Darbhanga district taken only two and-a-half years later disclosed an excess of 25 per cent. over the population recorded in 1872. A similar revision of the figures for Madhubani in 1876 showed a gain of 20 per cent. The Nadia district suffered terribly from fever between 1872 and 1881, and in one year lost no less than 4 per cent. of its inhabitants, yet in spite of this the recorded population was greater in 1881 by more than 11 per cent. than it was in 1872. The present Deputy Commissioner of Jalpaiguri estimates

that the population of that district was under-estimated in 1872 to the extent of at least 15 per cent. In 1881 the numbering of the people was carried out with far greater accuracy, and although it is impossible to gauge, even approximately, the extent to which this affected the comparative results of the two enumerations, it is quite certain that a very large proportion of the apparent growth must have been due to this cause. It would probably be quite safe to assume that, but for this disturbing factor, the excess of the figures for 1881 over those for 1872 would have been less than the increment disclosed by the present census as compared with 1891.

But although the census of 1881 was very much more complete than that of 1872, there were still tracts where the standard of accuracy fell considerably below that attained ten years later. In 1881 there were disturbances in the Sonthal Parganas, and in Darjeeling many of the ignorant hill men fled to Nepal to escape the imaginary dangers with which the census was associated in their minds. In the remoter tracts of Chota Nagpur the difficulties encountered were very great, and the success in surmounting them was less than that attained ten years later with the aid of the experience then gained. Speaking of the apparent growth of the population of the Chota Nagpur States, as disclosed by the census of 1891, Mr. O'Donnell says that "it is no doubt, two-thirds due to improved enumeration," and mentions that in Gangpur, two hundred villages were left out of account in 1881. It would probably be safe to estimate the gain due to the greater accuracy of the count of 1891 in these remote tracts at the difference between the increase then obtained and that disclosed by the present census, or at about 420,000 in all, to which may perhaps be added another 80,000 to allow for a fictitious improvement in Saran and other districts where the earlier enumeration was less complete than the later one. The standard of accuracy in 1891 had reached a stage which left but little room for further improvement and, as compared with that census, it is probable that the general gain on this account at the present enumeration does not, at the outside, exceed 100,000, while in the districts where plague was prevalent, the census of 1901 was less complete than its predecessor. Taking the gain and loss together, it may be concluded that there has been no appreciable improvement in the matter of accuracy at the present census.

Orissa Tributary States	...	120,000
Chota Nagpur ,,	...	130,000
Sonthal Parganas	...	120,000
Darjeeling ...	...	40,000
Total	...	*420,000*

If half a million be added to the population of 1881 to allow for omissions which did not occur at the succeeding census, the net increase comes to 4,636,858, which is still about 800,000 more than that of the decade preceding the census of 1901. This must be due either to (1) a lower birth-rate, (2) a higher mortality, or (3) an adverse balance of migration. The consideration of the birth and death-rates must be left to a subsequent chapter, but it may be noted here that the appearance, for the first time, of plague has added a new factor to the causes affecting mortality, and that the recorded number of deaths from this disease between the first outbreak in 1898 and the 1st March 1901, amounting to 65,884, probably represents less than half of the loss of life that actually occurred. The cyclone of 1897 in Chittagong, which is estimated to be responsible, directly or indirectly, for some 50,000 deaths, is another calamity of a special character that cannot be left out of account.

The record of migration between Bengal and other countries and provinces is incomplete as we have no information of the number of persons who go across the Nepal boundary. The figures so far as they go, show a considerable net loss, chiefly in the direction of Assam, but it does not appear that it has been greater during the last decade than it was in the preceding ten years. So far, therefore, as the causes affecting the relative rate of growth are concerned, migration may be left out of the reckoning. If 200,000 deaths be allowed for on account of plague and cyclone, there still remains a difference of about 600,000 between the increment in 1891-1901 and that during the previous decade, which is attributable either to a lower birth-rate or a greater general mortality.*

236. Before closing this Chapter, there is one more point to which I should like to draw attention, viz., the comparative progress made by tracts of

* The conclusions arrived at in the chapter on age are that the mortality has not increased but that the birth-rate has fallen. The reasons for the decline in the birth-rate vary. In Bihar it is due to the influence of bad seasons which has led to various preventive checks on the growth of the population while in Orissa and Central and West Bengal the birth-rate prior to 1891 was abnormally high owing to the recovery, in the one case, from the famine of 1866 and, in the other, from the ravages of Burdwan fever.

varying density of population. It is generally assumed that the tendency is for people to move from densely-inhabited tracts to those that are more sparsely populated. The extent to which this is actually the case will be seen by a reference to Subsidiary Table No. III. The greatest absolute increase has occurred in thanas with a density of less than 300 to the square mile, but this is mainly owing to the figures for the Chota Nagpur Plateau where the population is unusually prolific and where most of the police circles belong to this category. The proportional rate of increase in Chota Nagpur is not so great in these sparsely inhabited police circles as it is in some that carry a much greater population. North Bengal, which also contributes largely to the total growth of population in this group of thanas, does so owing to the reclamation of the Bárind and the development of the recently acquired tracts in the east of Jalpaiguri and Darjeeling. If Chota Nagpur and North Bengal be left out of account the greatest absolute addition to the population during the last decade has occurred in thanas with a density of 500 to 600 persons per square mile. In East Bengal the greatest absolute increment has taken place in thanas with 800 to 900 persons per square mile and the greatest proportional growth in those with from 400 to 500 and then in those with from 900 to 1,000. West Bengal has gained most, both absolutely and relatively, in the thanas with a density of from 400 to 500. The proportional growth in Central Bengal is greatest in the most sparsely inhabited thanas but the actual gain is insignificant compared with that in thanas with a population of 900 to 1,000 per square mile. In North Bihar also, though the proportional increase is greatest in the police circles of very low density, the actual increment is far greater in those with a population varying from 800 to 1,000 per square mile. In Orissa the greatest increase from both standpoints has taken place in police circles with a population of from 700 to 800 per square mile. It may, therefore, be concluded that, although sparsely inhabited tracts may grow at a very rapid rate in special cases where annexation or a successful experiment in colonisation may have brought about a change of circumstances and so encouraged immigration, the general tendency is for those tracts to grow most where the general conditions of life are favourable, irrespective of the existing density of population. In other words climate is still, as a rule, a more important factor than the density of the population in determining the growth or decay of the population. In Bihar it seems doubtful if the most thickly inhabited tracts are capable of sustaining a larger population than that which they already carry, but in East Bengal there seems to be no area where the inhabitants have reached the limit which the land can support.

SUBSIDIARY TABLES REFERRED TO IN CHAPTER II.

Subsidiary Table No. I.—Showing the variations in the population since 1872.

Subsidiary Table No. II.—Comparing the census variations with those indicated by the vital statistics returns.

Subsidiary Table No. III.—Showing the variations according to density of population.

SUBSIDIARY TABLE I—SHOWING THE VARIATIONS IN THE POPULATION SINCE 1872.

NATURAL DIVISIONS AND DISTRICTS.	PERCENTAGE OF VARIATION INCREASE (+) OR DECREASE (−).			NET VARIATION IN PERIOD 1872—1901 INCREASE (+) OR DECREASE (−)	MEAN DENSITY OF POPULATION PER SQUARE MILE.			
	1891—1901.	1881—1891.	1872—1881.		1901.	1891.	1881.	1872.
1	2	3	4	5	6	7	8	9
BENGAL	***+ 5·1***	***+ 7·3***	***+11·5***	***+ 25·9***	***413***	***393***	***366***	***328***
WEST BENGAL	***+ 7·1***	***+ 3·9***	***− 2·7***	***+ 8·3***	***591***	***551***	***530***	***545***
Burdwan	+ 10·1	− ·1	− 6·2	+ 3·1	579	518	518	553
Birbhum	+ 13·0	+ ·6	− 6·9	+ 5·9	515	456	452	486
Bankura	+ 4·3	+ 2·7	+ 7·5	+ 15·2	426	408	397	389
Midnapore...	+ 5·9	+ 4·6	− 1·1	+ 9·6	533	507	485	490
Hooghly	+ 1·4	+ 6·1	− 12·9	− 6·2	881	868	818	940
Howrah	+ 11·3	+ 13·1	+ 6·2	+ 33·7	1,668	1,497	1,324	1,248
CENTRAL BENGAL ...	***+ 5·1***	***+ 3·1***	***+11·7***	***+ 21·3***	***775***	***736***	***714***	***639***
24-Parganas	+ 9·8	+ 11·8	+ 6·2	+ 31·4	986	897	802	750
Calcutta	+ 24·2	+ 11·4	− 3·2	+ 35·9	42,390	34,115	30,615	31,650
Nadia	+ 1·4	− 1·1	+ 10·8	+ 11·1	597	589	595	537
Murshidabad	+ 6·5	+ 1·9	+ 1·04	+ 9·8	622	584	572	566
Jessore	− 4·0	− 2·6	+ 33·6	+ 24·9	620	646	663	496
NORTH BENGAL	***+ 5·9***	***+ 4·4***	***+ 5·3***	***+ 16·6***	***483***	***458***	***439***	***417***
Rajshahi	+ 1·5	− ·7	+ 1·2	+ 2·7	564	555	559	549
Dinajpur	+ 3·7	+ 2·8	+ ·8	+ 9·5	397	376	365	362
Jalpaiguri	+ 15·6	+ 17·3	+ 38·9	+ 88·4	266	230	196	141
Darjeeling	+ 11·5	+ 43·4	+ 63·8	+ 162·2	214	192	134	81
Rangpur	+ 4·3	− 1·5	− 2·5	+ ·02	617	591	600	616
Bogra	+ 11·7	+ 11·2	+ 6·9	+ 33·1	699	563	505	472
Pabna	+ 4·3	+ 3·9	+ 8·2	+ 17·3	772	740	712	658
Malda	+ 8·4	+ 14·5	+ 5·9	+ 30·5	466	429	374	356
Kuch Bihar	− 2·0	− 3·9	+ 13·1	+ 6·5	434	443	461	407
Sikkim	+ 93·7	.....	.....	.....	21	11	.....	.....
EAST BENGAL	***+ 10·4***	***+ 14·1***	***+ 10·9***	***+ 39·9***	***701***	***635***	***558***	***505***
Khulna	+ 6·4	+ 9·9	+ 3·1	+ 19·6	693	587	529	504
Dacca	+ 10·6	+ 14·5	+ 14·3	+ 44·9	932	861	751	657
Mymensingh	+ 12·7	+ 13·6	+ 29·9	+ 66·5	618	548	482	371
Faridpur	+ 6·2	+ 9·9	+ 8·4	+ 26·6	849	800	727	671
Backergunge	+ 6·4	+ 13·3	+ ·7	+ 21·4	628	591	531	518
Tippera	+ 18·7	+ 17·8	+ 7·8	+ 50·8	848	713	606	562
Noakhali	+ 13·0	+ 23·0	− 2·3	+ 35·8	694	614	499	511
Chittagong	+ 4·8	+ 13·9	+ ·4	+ 20·0	583	518	454	452
Do. Hill Tracts ...	+ 16·3	+ 3·5	+ 45·9	+ 79·2	24	21	20	13
Hill Tippera	+ 26·1	+ 45·8	+ 171·2	+ 391·6	42	34	23	8
NORTH BIHAR	***+ ·1***	***+ 5·8***	***+ 14·0***	***+ 20·8***	***636***	***635***	***600***	***526***
Saran	− 2·2	+ 7·4	+ 10·5	+ 16·0	907	923	864	781
Champaran	− 3·7	+ 8·0	+ 19·4	+ 24·3	507	537	488	408
Muzaffarpur	+ 1·5	+ 5·9	+ 14·9	+ 22·6	917	903	859	747
Darbhanga	+ 3·9	+ 6·5	+ 23·1	+ 36·3	873	840	788	640
Bhagalpur	+ 2·7	+ 3·3	+ 7·7	+ 14·4	494	481	465	432
Purnea	− 3·5	+ 5·1	+ 7·8	+ 9·3	375	389	370	343
SOUTH BIHAR	***− 5·6***	***+ 2·6***	***+ 10·9***	***+ 9·7***	***511***	***531***	***517***	***466***
Patna	− 8·4	+ ·9	+ 13·6	+ 4·2	783	855	846	751
Gaya	− 3·6	+ ·6	+ 9·1	+ 5·8	437	454	459	413
Shahabad	− 4·7	+ 5·6	+ 13·9	+ 14·7	449	471	445	391
Monghyr	+ 1·6	+ 3·3	+ 8·5	+ 14·0	527	519	502	462
ORISSA	***+ 7·06***	***+ 6·8***	***+ 17·6***	***+ 34·5***	***508***	***475***	***445***	***378***
Cuttack	+ 6·4	+ 7·9	+ 16·2	+ 33·6	568	534	494	425
Balasore	+ 7·6	+ 5·2	+ 22·07	+ 39·7	520	483	459	374
Puri	+ 7·6	+ 6·3	+ 15·4	+ 32·1	411	382	359	311
CHOTA NAGPUR PLATEAU	***+ 7·8***	***+ 13·5***	***+ 32·1***	***+ 61·8***	***159***	***141***	***124***	***94***
Hazaribagh	+ 1·1	+ 5·4	+ 43·1	+ 52·6	168	166	157	110
Ranchi	+ 5·2	+ 6·7	+ 30·1	+ 46·0	167	158	148	114
Palamau	+ 3·8	+ 8·3	+ 30·9	+ 46·2	126	121	112	86
Manbhum	+ 9·0	+ 12·7	+ 28·9	+ 56·6	314	258	255	198
Singhbhum	+ 12·4	+ 20·2	+ 42·6	+ 92·8	163	145	121	85
Southal Parganas	+ 3·1	+ 11·8	+ 24·5	+ 43·7	331	321	296	230
Chota Nagpur Tributary States	+ 13·3	+ 30·3	+ 35·9	+ 100·8	62	55	43	31
Orissa Ditto ...	+ 14·8	+ 29·3	+ 27·7	+ 76·4	135	118	98	77
Angul	+ 12·8	+ 5·7	+ 105·2	+ 144·8	114	101	95	46
CITIES	***+ 10·9***	***+ 8·3***	***+ 2·9***	***+ 23·8***	***15,116***	***13,631***	***29,569***	***12,219***
Calcutta	+ 24·2	+ 11·4	− 3·2	+ 33·9	42,390	34,115	30,615	31,650
Howrah	+ 35·1	+ 28·4	+ 8·9	+ 87·4	17,810	12,956	10,090	9,541
Cossipore-Chitpur	+ 29·6	+ 19·5	+ 3·6	+ 60·6	13,583	10,474	8,764	8,456
Manicktala	+ 15·0	− 41·4	− 10·8	− 40·0	5,398	4,693	8,021	8,998
Garden Reach	+ 1·9	+ 46·7	+ 18·5	+ 75·6	9,403	9,308	6,544	5,353
Serampur	+ 13·6	+ 40·4	+ 4·5	+ 81·8	11,112	8,988	6,390	6,110
Bally	+ 11·7	+ 12·7	+ 8·9	+ 36·1	9,331	8,350	7,407	6,857
Dacca	+ 9·9	+ 4·1	+ 14·2	+ 30·8	10,060	9,147	8,756	7,690
Patna	− 7·5	− 3·2	+ 7·3	− 3·9	16,964	18,374	18,261	17,655
Darbhanga	− 9·9	+ 11·5	+ 22·7	+ 23·2	9,453	10,508	9,423	7,677
Muzaffarpur	− 7·2	+ 15·8	+ 11·0	+ 19·2	7,602	8,198	7,076	6,373
Monghyr	− 37·1	+ 3·1	− 7·2	− 39·9	4,485	7,134	6,921	7,462
Bhagalpur	+ 9·6	+ 1·2	+ 4·3	+ 15·9	9,470	8,638	8,529	8,173
Chapra	− 19·9	+ 10·9	+ 11·6	− ·8	6,537	8,193	7,381	6,612
Bihar	− 5·5	− 2·3	+ 10·5	+ 1·7	5,007	5,302	5,441	4,921
Gaya	− 11·3	+ 5·1	+ 14·3	+ 6·6	10,184	11,483	10,916	9,549

NOTE.—The density shown for North Bengal is exclusive of Sikkim and that for East Bengal is exclusive of Hill Tippera and the Chittagong Hill Tracts.

SUBSIDIARY TABLE No. II—Comparing the variation in the actual and natural population of each district with that indicated by the vital statistics returns.

Districts.	Total number of deaths reported from 1891-1900.	Number of deaths per cent. of population of 1891.	Births.			Number of births per cent. of population of 1891.	Excess or deficiency of column 6 compared with 2.	Increase or decrease in actual population at census of 1901, compared with 1891.	Increase or decrease in natural population at census of 1901, compared with 1891.
			1892—1900.	Add ⅛th for 1891.	Total 1891—1900.				
1	2	3	4	5	6	7	8	9	10
PROVINCE	***22,390,829***	***31·50***	***22,995,026***	***2,555,003***	***25,550,029***	***35·9***	***+3,159,200***	***+3,358,576***	***+3,240,973***
BURDWAN DIVISION	***2,218,060***	***28·84***	***2,300,886***	***255,654***	***2,556,540***	***33·24***	***+ 338,480***	***+550,887***	***+ 402,390***
Burdwan	416,422	29·91	409,936	44,535	445,551	33·01	+ 29,129	+ 140,595	+ 83,527
Birbhum	239,011	29·94	279,442	30,049	309,491	37·64	+ 61,480	+ 104,026	+ 97,842
Bankura	258,366	24·15	333,501	37,056	370,557	34·64	+ 112,191	+ 46,743	+ 94,342
Midnapore	730,451	27·75	800,553	88,950	889,503	33·80	+ 159,052	+ 157,648	+ 158,470
Hooghly	347,451	33·59	267,046	29,672	296,718	28·68	− 50,733	+ 14,986	− 28,134
Howrah	226,359	29·64	228,348	25,372	253,720	33·22	+ 27,361	+ 86,889	+ 56,743
PRESIDENCY DIVISION ...	***2,837,475***	***33·24***	***2,579,265***	***286,585***	***2,865,850***	***33·57***	***+ 28,375***	***+457,902***	***+ 356,724***
24-Parganas	487,162	25·75	474,117	52,680	526,797	27·85	+ 39,635	+ 187,071	+ 143,402
Calcutta	251,875	36·92	103,920	11,547	115,467	16·92	− 136,408	+ 165,491	+ 104,487
Nadia	583,490	35·49	586,119	65,124	651,243	39·61	+ 67,747	+ 23,383	+ 26,104
Murshidabad	483,053	33·81	471,000	52,333	523,333	41·83	+ 100,300	+ 82,238	+ 88,218
Jessore	717,126	37·96	569,372	63,284	632,656	33·49	− 84,490	− 75,672	− 93,958
Khulna	374,783	31·82	374,737	41,637	416,374	35·35	+ 41,591	+ 75,391	+ 88,071
RAJSHAHI DIVISION ...	***2,815,887***	***35·12***	***2,570,965***	***285,663***	***2,856,628***	***35·63***	***+ 40,741***	***+477,757***	***+ 330,851***
Rajshahi	530,354	36·83	474,149	52,683	526,832	36·58	− 3,452	+ 22,773	− 333
Dinajpur	568,923	38·37	531,720	59,080	590,800	39·80	+ 21,877	+ 84,510	+ 48,055
Jalpaiguri	249,471	36·64	207,697	23,078	230,775	33·90	− 18,696	+ 106,644	+ 63,727
Darjeeling	93,917	41·65	58,535	6,564	65,039	29·12	− 27,978	+ 25,803	+ 39,339
Rangpur	694,780	33·83	640,082	71,120	711,202	34·43	+ 16,452	+ 88,717	+ 15,957
Bogra	233,938	30·60	250,445	27,827	278,272	36·40	+ 44,334	+ 90,072	+ 110,296
Pabna	445,474	32·72	408,337	45,371	453,708	33·33	+ 8,234	+ 59,238	+ 53,819
DACCA DIVISION ...	***3,078,601***	***31·26***	***3,425,886***	***380,654***	***3,806,540***	***38·66***	***+ 727,939***	***+948,692***	***+ 960,273***
Dacca	725,674	30·29	851,140	94,571	945,711	39·48	+ 220,037	+ 254,092	+ 290,639
Mymensingh	921,389	29·53	1,110,405	123,378	1,233,783	39·53	+ 312,394	+ 442,842	+ 436,647
Faridpur	645,461	35·38	635,406	70,601	706,007	38·71	+ 60,606	+ 113,931	+ 97,071
Backergunge	786,137	36·49	828,935	92,104	921,039	42·76	+ 134,902	+ 137,787	+ 136,896
CHITTAGONG DIVISION ...	***1,185,135***	***29·02***	***1,475,429***	***163,936***	***1,639,365***	***40·2***	***+454,230***	***+530,174***	***+ 548,031***
Tippera	451,736	25·33	635,617	70,624	706,241	39·61	+ 254,505	+ 335,056	+ 345,485
Noakhali	330,972	32·77	408,052	45,339	453,391	44·93	+ 122,419	+ 132,635	+ 135,809
Chittagong	402,427	31·19	431,760	47,973	479,773	37·18	+ 77,306	+ 63,083	+ 66,757
PATNA DIVISION ...	***5,090,475***	***32·19***	***5,020,430***	***557,826***	***5,578,256***	***35·28***	***+487,781***	***−296,617***	***− 337,454***
Patna	600,432	33·85	558,751	62,083	620,834	35·00	+ 20,402	− 148,425	− 95,101
Gaya	708,209	33·11	706,330	78,481	784,811	36·70	+ 76,602	− 78,298	− 117,964
Shahabad	677,274	32·88	610,864	67,874	678,738	32·94	+ 1,464	− 97,583	− 77,295
Saran	742,143	30·10	761,177	84,575	845,752	34·31	+ 103,609	− 55,498	− 208,860
Champaran	616,031	33·12	574,106	63,790	637,896	34·30	+ 21,865	− 69,002	+ 68,021
Muzaffarpur	902,578	33·27	911,553	101,284	1,012,837	37·33	+ 110,259	+ 41,933	− 13,443
Darbhanga	843,808	30·11	897,649	99,739	997,388	35·59	+ 153,580	+ 110,656	+ 106,578
BHAGALPUR DIVISION ...	***2,682,711***	***31·25***	***2,781,853***	***309,095***	***3,090,948***	***36·01***	***+ 408,237***	***+144,949***	***+ 326,883***
Monghyr	660,809	32·45	729,234	81,026	810,260	39·78	+ 149,451	+ 32,783	+ 42,909
Bhagalpur	657,984	33·37	676,689	75,188	751,877	38·09	+ 93,893	+ 56,257	+ 102,461
Purnea	670,539	34·94	568,268	63,141	631,409	32·46	− 48,130	− 69,864	− 76,190
Malda	306,672	37·63	293,469	32,607	326,076	40·01	+ 19,404	+ 69,111	+ 57,413
Sonthal Parganas	327,707	21·53	514,193	57,133	571,326	32·57	+ 193,619	+ 85,962	+ 206,300
ORISSA DIVISION	***1,194,133***	***30·79***	***1,291,527***	***145,503***	***1,435,030***	***37·0***	***+ 240,897***	***+273,895***	***+ 266,679***
Cuttack	291,619	30·50	653,453	72,606	726,061	37·47	+ 135,042	+ 135,087	+ 126,221
Balasore	312,317	31·39	338,736	37,637	376,373	37·83	+ 64,056	+ 76,522	+ 83,173
Puri	290,797	30·77	299,336	33,260	332,596	35·19	+ 41,799	+ 72,286	+ 57,285
CHOTA NAGPUR DIVISION ...	***1,288,332***	***27·83***	***1,548,785***	***172,087***	***1,720,872***	***37·18***	***+ 432,520***	***+271,637***	***+ 386,996***
Hazaribagh	350,581	30·88	412,209	45,801	458,010	39·33	+ 98,429	+ 13,640	+ 81,433
Ranchi	337,303	29·87	396,392	44,043	440,435	39·01	+ 103,232	+ 59,049	+ 104,375
Palamau	190,540	31·89	211,439	22,493	234,932	39·37	+ 44,592	+ 22,830	.....
Manbhum	286,054	23·96	367,242	40,805	408,047	34·26	+ 122,023	+ 108,036	+ 119,077
Singhbhum	115,204	21·11	161,503	17,945	179,448	32·90	+ 64,244	+ 68,091	+ 91,109

Note.—This statement is exclusive of the figures for the Chittagong Hill Tracts, Angul and the Feudatory States; birth and death statistics are not recorded in these tracts. Still-births are not included. The number of such births were highest in 1899, when they amounted to 106,556, or about 1·2 per 1,000 of the population of 1891.

The figures in column 10 include the persons born in each district, and enumerated, not only in other parts of Bengal but also in Assam, the United Provinces, the Central Provinces, Burma (for Chittagong only) and Madras (for Puri only).

SUBSIDIARY TABLE No. III.—Showing the variation in the population according to density.

(1) (Actual variation).

Serial No.	Natural Division.	Variation in population of thanas with a density per square mile of—							
		Less than 300.		300—400.		400—500.		500—600.	
		1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.
	1	2	3	4	5	6	7	8	9
	Total ...	*+1,767,748*	*+1,003,219*	*+371,709*	*+363,041*	*+453,090*	*+364,554*	*+522,480*	*+440,963*
1	West Bengal ...	+ 63,537	+ 26,289	+ 60,348	+ 37,434	− 3,506	+ 185,953	+ 14,605	+ 68,824
2	Central Bengal ...	+ 11,203	+ 23,671	+ 2,011	+ 22,973	− 18,958	+ 7,155	− 18,836	+ 21,769
3	North do. ...	+ 223,920	+ 198,382	+ 83,583	+ 88,802	+ 20,552	+ 23,996	+ 36,122	+ 60,438
4	East do. ...	+ 280,714	+ 91,359	+ 53,189	+ 124,337	+295,029	+ 158,158	+ 300,738	+ 267,863
5	North Bihar ...	+ 113,792	+ 34,371	+ 32,121	− 32,174	+ 58,265	− 14,208	+ 131,495	− 4,508
6	South do. ...	+ 29,893	− 4,097	+ 57,529	− 17,221	+ 17,417	− 26,753	+ 11,168	− 3,819
7	Orissa ...	+ 35,865	+ 32,712	+ 46,774	+ 63,188	+ 43,038	+ 33,996	+ 30,256	+ 14,580
8	Chota Nagpur Plateau	+ 1,008,824	+ 601,032	+ 36,154	+ 95,092	+ 21,283	− 3,846	+ 16,932	+ 15,135

Serial No.	Variation in population of thanas with a density per square mile of—									
	600—700.		700—800.		800—900.		900—1,000.		Over 1,000.	
	1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.
	10	11	12	13	14	15	16	17	18	19
	+577,619	*+225,135*	*+412,963*	*+312,568*	*+429,786*	*+328,930*	*+ 172,714*	*+ 307,998*	*+ 342,275*	*+230,491*
1	+ 7,197	+ 96,215	− 2,377	+ 19,555	+ 15,681	+ 5,273	+ 29,283	+ 9,486	+ 110,517	+ 101,866
2	− 18,399	− 3,109	+ 50,822	+ 17,744	+ 56,127	+ 20,541	+ 19,891	+ 44,073	+ 74,246	+ 28,248
3	+ 44,784	+ 141,738	− 20,276	− 354	− 26,868	− 325	.....	− 128	+ 20,345	+ 21,140
4	+ 254,091	+ 46,218	+ 284,207	+ 206,679	+ 186,736	+ 288,978	+ 43,588	+ 207,125	+ 111,748	+ 200,392
5	+ 107,044	− 15,289	+ 73,603	+ 34,843	+ 149,425	+ 51,154	+ 92,670	+ 51,977	+ 10,773	− 72,742
6	+ 21,861	− 69,165	+ 39,320	− 35,762	+ 38,096	− 47,710	− 19,217	− 16,658	+ 11,516	− 68,390
7	+ 61,041	+ 15,914	+ 7,664	+ 69,863	+ 20,589	+ 11,019	.....	+ 12,123	+ 3,130	+ 20,340
8	.....	+ 11,113	.....	.....	.....	.....	+ 6,499	.....	.....	− 503

(2) (Proportional figures).

Serial No.	Natural Divisions.	Variation per cent in population of thanas with a density per square mile of—							
		Under 300.		300—400.		400—500.		500—600.	
		1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.
	1	2	3	4	5	6	7	8	9
	Total ...	*+14·4*	*+8·6*	*+ 7·3*	*+5·3*	*+ 5·0*	*+4·5*	*+ 5·7*	*+4·6*
1	West Bengal ...	+ 8·7	+ 6·3	+ 9·3	+ 4·1	− 0·3	+11·3	+ 1·0	+ 6·6
2	Central Bengal ...	+22·6	+41·8	+ 3·0	+21·9	− 3·3	+ 6·9	− 1·2	+ 1·4
3	North do. ...	+25·1	+20·7	+ 6·8	+ 7·4	+ 1·0	+ 1·3	+ 2·5	+ 3·4
4	East do. ...	+24·6	+14·0	+10·3	+11·4	+14·4	+19·9	+ 18·5	+10·5
5	North Bihar ...	+15·2	+ 5·3	+ 5·6	− 4·6	+ 4·5	− 1·1	+ 9·5	− 0·4
6	South do. ...	+ 2·9	− 0·4	+ 7·7	− 3·1	+ 3·1	− 2·4	+ 0·9	− 0·3
7	Orissa ...	+ 8·3	+ 7·0	+ 6·9	+ 8·7	+ 4·9	+ 4·8	+ 6·0	+ 2·5
8	Chota Nagpur Plateau	+13·6	+ 8·0	+ 9·6	+ 8·4	+ 9·5	− 1·7	+ 12·1	+12·3

Serial No.	Variation per cent in population of thanas with a density per square mile of—									
	600—700.		700—800.		800—900.		900—1,000.		Over 1,000.	
	1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.	1881—1891.	1891—1901.
	10	11	12	13	14	15	16	17	18	19
	+ 6·0	*+2·4*	*+ 4·2*	*+3·6*	*+ 7·1*	*+4·1*	*+ 4·2*	*+6·9*	*+ 8·5*	*+3·1*
1	+ 0·8	+ 8·5	− ·2	+ 3·5	+ 6·2	+ 2·4	+ 5·8	+ 2·2	+13·0	+ 7·4
2	− 1·5	− 0·2	+ 2·8	+ 1·5	+11·9	+ 3·6	+ 5·8	+12·4	+16·1	+ 2·8
3	+ 3·2	+ 7·7	− 1·2	.....	− 8·9	.....	.....	− 0·3	+ 9·2	+ 8·7
4	+13·2	+ 2·2	+12·3	+ 9·5	+12·3	+11·3	+ 4·3	+14·4	+14·1	+10·2
5	+ 8·2	− 1·1	+ 4·1	+ 2·1	+ 5·9	+ 1·8	+ 4·7	+ 2·8	+ 0·8	− 3·1
6	+ 1·4	− 4·8	+ 2·6	− 3·6	+ 5·9	− 3·8	− 6·8	−18·6	+ 2·5	−15·0
7	+ 9·6	+ 8·9	+ 3·0	+10·6	+ 8·2	+ 4·2	.....	+ 4·4	+12·1	+71·3
8	.....	+ 7·9	.....	.....	.....	.....	+47·4	.....	.....	− 2·4

Chapter III.

MIGRATION.

INTRODUCTORY.

237. The statistics of birthplace are contained in Imperial Table XI. The following Subsidiary Tables in which the prominent features of the statistics are pourtrayed will be found at the end of this chapter.

Subsidiary Table No. I—Showing the general distribution according to birthplace of the persons enumerated in each district.

Subsidiary Table No. II—Showing the general distribution according to the place of enumeration of the persons born in each district.

Subsidiary Table No. III—Containing proportional figures of the migration to and from each district.

Subsidiary Table No. IV—Showing the volume of migration between Commissioners' and Natural Divisions at the present census and in 1891.

Subsidiary Table No. V—Showing the gain or loss by migration between Bengal and other parts of India.

Subsidiary Table No. VI—Showing the number of immigrants from certain foreign countries.

VARIOUS KINDS OF MIGRATION.

238. The movements of the people so far as they affect the total population of each district have been dealt with in the last chapter and the present discussion will be directed mainly to the direction and character of the migration, and to the reasons that induce it.

Five different types of migration may be distinguished in this Province,* *viz.*—

I. *Casual*—Or the accidental movement backwards and forwards across the boundary line between contiguous districts. All over the province, people are constantly found moving short distances from their original home. Amongst Hindus this is especially the case with women, as men usually get their wives from villages at some little distance from their own.† The volume of casual migration is for this reason much greater in Bihar, West Bengal, and Orissa, where Hindus largely preponderate, than it is east of the Bhágirathi, where the Muhammadan element is greater. Where a man's wife is a native of another district his eldest child will usually be born there also, as it is the usual practice for the young wife to return to her parents' home for her first confinement. These minor movements pass unnoticed except along the line which happens to have been selected as the district boundary. In Subsidiary Tables I, II, and III migration between contiguous districts has been shown separately, but it does not by any means follow that the whole or even the greater part of it is due to movements of the casual nature here described. There may be a genuine permanent progression from one district to another, such as that from Hazaribagh to the Sonthal Parganas in the early half of the last century.

II. *Temporary*.—Due to a temporary demand for labour on roads or railways, journeys on business, pilgrimages and the like. This form of movement needs no illustration.

III. *Periodic*.—Due to the changing seasons. Of this nature is the annual exodus from Chittagong for the rice harvest in Akyab, or the visits paid to Bengal by Nunias and other labouring castes from upcountry during the cold weather months, when their crops are off the ground, and there is no work requiring their presence at home.

IV. *Semi-Permanent*.—Where the inhabitants of one place earn their livelihood in another, but maintain their connection with their old homes, leave their families there, return there themselves at more or less regular intervals, and look forward to the time when they may again live there permanently. The

* The classification of the different types of migration is based on that adopted by Mr Ibbetson in the Punjab in 1881 with the necessary modifications due to different local conditions.

† There appears to be no religious sentiment involved, and the reason generally assigned is that if a wife's parents' home is near she is very prone to return there whenever she quarrels with her husband, whereas if they reside at a distance this is less easy, and the husband can keep her in greater subjection. Amongst Muhammadans these considerations are outweighed by the fact that it is considered desirable to marry a woman of the same family.

settlement of Englishmen in India is generally of this nature ; so also is that of Márwáris in Bengal, or of clerks who obtain employment at a distance from their own home.

V. *Permanent.*—Where over-crowding drives people away or the superior attractions of some other locality induce them to settle there permanently with their families. The gradual development of the Sundarbans affords an example of this sort of migration.

239. The character of the migration is to a great extent reflected in the proportion of the sexes.* Where it is casual females will naturally be in excess, as almost every woman changes her residence after marriage and amongst Hindus, she usually marries at some distance from her parental home. Where it is temporary or periodic, on the other hand, except where a pilgrimage is the object, the majority of the migrants will be men. Amongst pilgrims women, in this Province, bulk much more largely than men, but they leave their homes comparatively seldom for purposes of trade or temporary work. In the case of semi-permanent migration also, males are usually in excess. A man does not ordinarily take his wife and family with him until he has decided finally to sever his connection with his old home. It is, therefore, only when a movement is of a permanent nature that the proportion of the sexes will approach equality.

CIRCUMSTANCES AFFECTING MIGRATION.

240. The census statistics refer to the distribution of the population on a particular day and the volume of temporary and periodic migration will therefore depend very largely on the date when it is taken. In respect of temporary migration it may be mentioned that in 1891 there were two large movements of a religious nature in progress on the day of the census, the *Ardhodaya Yoga* which drew many thousands of pilgrims from all parts of the country, but especially from the Eastern districts, to the banks of the Ganges, and the *Dhulut* ceremony at Nabadwip which was attended by about 15,000 persons from the neighbouring districts. At the present census the only large gathering of this nature was the *Gobind Duádasi* festival at Puri which is estimated to have been attended by about 20,000 persons. Periodic migration is mainly of two kinds. Many of the labouring classes from Bihar, Chota Nagpur, Birbhum, and Bankura, visit Bengal Proper every season as soon as their rice crop is harvested, and do not return until the commencement of the monsoon affords them employment nearer home. The census is always taken in India towards the end of the cold weather, when the number of persons thus absent from home is at its maximum. The other main form of periodic migration is in connection with harvest operations. The great harvest which attracts people from other districts is that of the winter rice. This is over before the end of January and people who had gone to assist in the operations in other districts would usually be back in their permanent homes before the end of February. Of them, therefore, the present census, like its predecessor, has for the most part taken no count. The jute season in some of the districts of East Bengal attracts a great number of labourers from Bihar, but this is during the rains and only those are included in the census returns who belong to the category of semi-permanent migrants, *i.e.*, who do not come merely for the jute season but stay on during the cold weather months as palki-bearers, earth-workers, brick-makers, etc. The *rabi* or spring harvest in Bengal Proper attracts very few labourers from other districts. It does so to a much greater extent in Bihar, but owing to the prevalence of plague it is probable that the number of foreign reapers was much smaller on this occasion in Saran and South Bihar than it was in 1891. There is one more general consideration that should be mentioned before proceeding to a discussion of the figures; the effects of permanent migration are cumulative, but those of other movements are not so. The Western Duars of Jalpaiguri were practically uninhabited when acquired from Bhutan less than forty years ago. The country is naturally fertile and much of it is well suited for tea. Under British rule, permanent settlers soon began to pour in and the waste land is rapidly being reclaimed. The

* The extent to which the varying proportion of the sexes in the different districts is due to migration will be indicated in the chapter on sex.

process will doubtless continue until the proportion of arable land is as great there as it is in the adjoining districts. The original settlers are gradually being replaced by their children born in the district, and not by new comers from outside. Consequently when the land fit for cultivation has been opened out the flow of migration will cease and a generation later there will be no indication in the census returns of the movement that has taken place. In other words permanent migration tends to exhaust itself and die out. In the case of all other kinds of migration, however, the effect is not cumulative and so long as the circumstances remain the same the volume at each succeeding census will be as great as it was before. If the demand for labour increases the number of migrants will tend to grow in the same proportion.*

241. Finally the apparent movements of the people may occasionally be fictitious, and without any existence in fact. The alteration of district boundaries leaves the actual residence of the inhabitants unchanged, but it may often happen that a man born in a tract subsequently transferred from district A to district B may describe himself as having been born in district A and so go to swell the number of persons returned as born outside the limits of district B. The number of changes in district boundaries was less in the decade preceding the present census than in the two previous ones, and it is, therefore, probable that the returns have been less affected by mistakes of this nature.

Period.	Number of district boundaries altered.
1872–1881	19
1881–1891	24
1891–1901	13

242. In the province as a whole, of every 10,000 persons 9,423 were born in the district in which enumerated, 293 in contiguous districts of the province, 167 in other parts of Bengal, 93 in other provinces, and 24 in other countries. The total amount of migration is therefore very small, and it will be still smaller if we exclude from consideration the casual migration across the borders of contiguous districts. But before considering the movements within the province we may glance briefly at those between Bengal and (1) other countries, and (2) other parts of India. The only movements of any moment between Bengal and foreign countries are—

MIGRATION BETWEEN BENGAL AND FOREIGN COUNTRIES.

(1) immigration from, and emigration to, Nepal;
(2) emigration to the Colonies;
(3) immigration from the British Islands.

The total number of persons born in Nepal but enumerated in this Province is 161,495, or slightly less than in 1891. Nearly half the total number are found in the Darjeeling district and nearly three quarters of the remainder in Sikkim, Champaran and Jalpaiguri. Most of these are permanent settlers, the great attraction in Darjeeling being the tea gardens. The expansion of the tea industry there has now practically ceased, and there is very little fresh immigration from Nepal. The original settlers are gradually being replaced by their children born in the district. The number of immigrants at the present census is therefore less than it was in 1891. There has been no change elsewhere worth noting, except in Champaran and Purnea, where it would seem that some of the Nepalese who were in British territory in 1891, must since have recrossed the frontier. We have, however, no statistics of the population of Nepal, and it is thus impossible to verify this inference, or to learn how far the influx into British territory is counterbalanced by an ebb of population in the opposite direction. It is believed that a considerable number of people go from the North Bihar districts, and especially from Champaran, Muzaffarpur, Darbhanga and Bhagalpur to the adjoining part of Nepal, but that very few go thither from Darjeeling, Jalpaiguri or Sikkim.†

* The above propositions are only intended to indicate the general tendency of each kind of migration. Periodic migration may become semi-permanent, and semi-permanent migration may become permanent. This subject will be dealt with further in paragraph 262 in connection with the movement from Bihar and the United Provinces to Bengal Proper.

† In Purnea the movements of a permanent nature across the boundary are said to show no special tendency in either direction. They are due mainly to the matrimonial arrangements of the people. Many persons living in Purnea hold land in Nepal, but they only go thither when agricultural operations require their presence. The Magistrate of Champaran thinks that the emigration to Nepal exceeds the countervailing immigration. A strip of jungle in the Nepal Terai is now being cleared for cultivation and this attracts settlers. The Magistrates of Bhagalpur and Muzaffarpur are of the same opinion. Rents, it is said, are lower in Nepal and good land is plentiful.

243. The emigration to the Colonies, though not shown in the census statistics, is known from special returns which are maintained under the orders of Government. The total number of emigrants during the decade 1891—1901 was 16,818, of whom nearly half went from Shahabad, and most of the remainder from Saran, Patna, Gaya and Muzaffarpur. On the average about 1 in 4 of these emigrants returns home. The persons born in the British Islands now number 11,886 compared with 9,544 in 1891. This is due mainly to the figures for Calcutta where there was a decrease in 1891, which was attributed by Mr. O'Donnell to some of the schedules on which Europeans were enumerated having been lost. The industrial towns near Calcutta and the coalfields of Jheria and Asansol contain more Europeans than they did 10 years previously, but there has been a falling off in Darjeeling due to the reduction of tea garden establishments. Three quarters of the British-born are males and only one quarter females. The persons who returned Afghanistan as their birthplace are chiefly itinerant traders, most of whom are only cold weather visitors; almost all of them were shown as woollen cloth dealers, but a few were returned as fruit-sellers, hide-dealers and money-lenders.

District	No. of Emigrants.
Shahabad ...	7,633
Saran	2,617
Patna	2,029
Gaya	1,423
Muzaffarpur	901
Other districts ...	2,215

MIGRATION BETWEEN BENGAL AND OTHER PARTS OF INDIA—UNITED PROVINCES.

244. The number of immigrants to Bengal from other parts of India, according to the present census, is 728,715, and the corresponding number of emigrants is 879,583. By far the greatest amount of immigration is from the United Provinces which send a continually growing supply of labourers for the mills of the metropolitan districts and the coalfields of Burdwan and Manbhum, and for earth-work, palki-bearing, etc., throughout the province. The total number of persons born in the United Provinces and States, but enumerated in Bengal, is 496,940, compared with 365,248 in 1891 and 351,933 in 1881. These figures include migration between contiguous districts along the common boundary line. If this be left out of account, the number of immigrants from the United Provinces at the present census is about 416,000. Of these nearly three-sevenths were enumerated in Calcutta, the 24-Parganas, and Howrah.* The emigrants to the United Provinces number only 128,991, of whom all but about 32,000 were found in districts contiguous to their district of birth. Of the migration between contiguous districts of Bihar and the United Provinces the most notable features are (1) the decrease in the number of settlers from Gorakhpur in Champaran and a corresponding increase in the number of these settlers in Saran; (2) the remarkable falling-off in the number of Saran-born people enumerated in Gorakhpur, which is now little more than a third of the figure (70,560), returned in 1891; and (3) the total absence of any emigration from the Chota Nagpur States to Mirzapur. The great balance of migration in favour of Bengal, in the case of migration other than to contiguous districts, is of course due to the great demand for labour in Bengal Proper, where the landless labourer is in many parts unknown and there is no local supply of field labourers, palki-bearers and earth-workers. The mills around Calcutta and the coal mines and tea gardens also need far more workers than they can procure within the limits of the province. The emigrants from Bengal to the United Provinces belong to a great extent to the educated classes, and find employment as pleaders or as clerks under Government or in the railway offices. Numerous pilgrims go from Bengal to Benares and other holy places, and of these the greater number are women. Many well-to-do widows also settle permanently at Benares for religious reasons. This explains why of the total number of Bengal-born persons in Benares (20,157) two-thirds are females. In Ballia, where the proportion of females born in Darbhanga is extraordinarily high (1,564 out of 1,574), the Census Superintendent of the United Provinces informs me that it is due to hypergamy. The low class but wealthy Bábhans

* The districts of the United Provinces from which most of the emigrants come are those in the extreme east, viz., Ballia, Azamgarh, Ghazipur, Gorakhpur, Benares, Jaunpur, Mirzapur and Allahabad. Then come the districts immediately to the west of these, viz., Fyzabad, Sultanpur, Partabgarh, Rai Bareli, Lucknow, Fatehpur and Cawnpur.

and Rájputs of Darbhanga are said to be addicted to marrying their daughters to their poorer but better bred caste-fellows of Ballia. Except for the movements between contiguous districts and the interchange of wives, most of the migration between Bengal and the United Provinces is temporary, but the extent to which this is so will be further discussed in a subsequent paragraph.

ASSAM.

245. In the case of Assam the balance of migration is heavily against Bengal. The emigrants aggregate 503,876, an increase of 85,532 as compared with 1891, while the immigrants number only 48,296, or 5,327 less than at the time of the previous census. If we exclude from the account the ebb and flow of population between contiguous districts, the loss by migration is about 457,000 and the gain only 12,000. This great exodus is, of course, mainly in connection with the tea industry; most of the coolies for the Assam tea gardens are recruited in the Chota Nagpur Plateau and the adjoining district of Gaya and West Bengal. The returns maintained under the labour law show that during the decade 1891-1900 more than a quarter of a million persons were recruited in Bengal and sent up to Assam to work on the tea gardens. Comparatively few of these return to Bengal on the expiry of their labour contracts; the majority either stay on in Assam as garden coolies or settle down there as cultivators* This enormous exodus is therefore a great drain on this Province, and constitutes one of the main reasons for the very small growth of population in the districts from which the coolies are mainly drawn.

Chota Nagpur Plateau ...	180,097
Bihar	26,268
Bengal Proper and Orissa	46,574
Total ...	252,939

OTHER PROVINCES.

246. Another noticeable feature in these statistics is the growing number of people who go from Bengal to Burma. Of the total number 79,262 are from Chittagong. Many of these are temporary emigrants in connection with the Akyab rice harvest; of the remainder, about 77,000, the majority are labourers who ultimately return home, but many are permanent emigrants who have been settled on waste land grants in Upper Burma. Most of them come from Bihar, but unfortunately the Burma census returns afford no information as to the particular districts from which they have emigrated.† The next most considerable inter-provincial movement is that with the Central Provinces, but here the interchange of population is more even; Bengal receives 62,181 and gives 44,360, so that there is a net gain of rather less than 18,000. The bulk of this migration occurs between the Tributary States of Orissa and Chota Nagpur and the contiguous districts and States of the Central Provinces. Elsewhere the immigrants are mainly general labourers, earth-workers and tea garden coolies. The volume of migration between Bengal and Madras is about a third of that with the Central Provinces, but the balance in favour of Bengal is almost as great. Three quarters of the immigrants from this Province are to be found in Orissa and the Orissa States, and more than a third of the remainder in the 24-Parganas and Calcutta. There are wings of a Madras regiment at Cuttack and Barrackpore. Elsewhere most of the Madrásis, other than those in Puri, Angul and the Orissa States, have come for employment on railways. Rajputana sends about 40,000 persons to Bengal, almost all of whom are traders, and receives barely 1,000 in exchange. Central India gives about 23,000, and receives back only 5,000. Most of this interchange of population takes place with the States of Chota Nagpur. The Panjab gives 17,000 and gets 7,000. The occupations of the Panjabis enumerated in Bengal are various.

* The persons born in non-contiguous districts of Bengal who were enumerated in Assam in 1891 numbered 368,071. At a death-rate of 40 per 1,000 an annual emigration of 14,723 persons would be needed to maintain this number, and a further annual emigration of about 11,100 persons would be required to produce the excess of 88,962 shown by the present census. If none returned home, this would mean a total emigration of about 260,000 persons in the decade. The number of ex-garden coolies who return to Bengal but little, if at all, exceeds the number of persons, other than garden coolies, who go from Bengal to non-contiguous districts of Assam, and the number of such persons is known to be very small. It consists mainly of persons connected with the railway and of a few Government officers, clerks, shopkeepers and domestic servants.

† I supplied all other provinces with full details of the birth districts of the persons born in them who were enumerated in Bengal, but the United Provinces and Assam were the only provinces from which I received similar information regarding persons who had gone to them from Bengal. On a future occasion it would be well to arrange beforehand for the preparation of a more complete return, at least for contiguous districts, of Madras, the Central Provinces and Burma.

Many are carpenters, railway mechanics, door-keepers and peons, but there are also many shop-keepers of various kinds, chiefly dealers in woollen-cloth, shawls, etc. The majority are Pathán, but there are also many Chatris and Baniyas, and many who described themselves as Sikhs by caste. The immigrants from Bombay number only 6,695, and the emigrants to that Province only 5,402. Many of the natives of Bombay who were enumerated in Bengal hold appointments in railway offices, and many are shop-keepers, contractors, masons and coolies. There are also a few sepoys of the Bombay army stationed at Alipore.

MIGRATION IN DIFFERENT PARTS OF THE PROVINCE.

247. The total number of persons born in Bengal who were enumerated outside their district of birth was 4,494,935, or 573 per 10,000 of the population. Of these, 2,279,512, or 290 per 10,000, were enumerated in contiguous districts, 1,335,840, or 170 per 10,000 in other parts of Bengal, and 879,583, or 113 per 10,000 in other parts of India. Immigration, both from contiguous and from distant districts, is most active in the neighbourhood of the metropolis and in North Bengal, while the volume of emigration is greatest in the case of the Chota Nagpur plateau and North and South Bihar. Of the number of persons enumerated in districts contiguous to those where they were born 51·9 per cent. were females. Amongst those enumerated in other parts of the Province, and in other parts of India 29·5 and 42·6 per cent., respectively, were females. The relatively high proportion of females in the case of contiguous districts is due, as already explained, to marriage, which usually means for women a move to a new home, and, in the case of Hindus, to a home in another village some distance away. The proportion is highest in Bihar and Orissa where the great bulk of the people are Hindus, and lowest in East Bengal where Muhammadans greatly preponderate.

The principal statistics of the migration to and from each district will be found in Appendix I at the end of this volume, and it is not proposed to cumber the text with details that can be more readily gathered from these figured statements. The following discussion will, therefore, be confined to a brief review of the general trend and volume of these interdistrict movements and of the reasons for the same so far as they can be ascertained.

WEST BENGAL.

248. In West Bengal, as a whole, the gains and losses due to migration almost cancel each other but there are great local variations. Burdwan attracts labourers for its coal mines, iron works, paper mills and potteries in large numbers, not only from Bankura within the division, but also from Manbhum, Hazaribagh and the Sonthal Parganas in the Chota Nagpur Plateau, from Patna, Gaya, Shahabad and Monghyr in Bihar, and from the United Provinces. On the other hand it sends out numerous clerks to Calcutta and labourers to Assam, and loses more than it gains by the casual migration across the boundary line that divides it from Birbhum and Hooghly. The loss in both cases is due mainly to the demand in these districts for wives born in Burdwan. Birbhum receives a considerable number of settlers from the Sonthal Parganas for the purpose of cultivation, and labourers, shopkeepers, zemindars' peons, etc., from Shahabad and the United Provinces. Many of its inhabitants go to Assam as tea garden coolies, but, apart from this, most of the emigration is of the casual type. Bankura sends numerous coolies to Assam, and there is also a great deal of periodic and semi-permanent emigration to other parts of Bengal, chiefly to the coal mines of Burdwan and Manbhum and the mills of Hooghly. The short crops of 1900 greatly stimulated migration in the season when the census was taken. Midnapore receives in proportion to its population a smaller number of immigrants than any district in West Bengal. If contiguous districts be excluded the foreign-born population comes mainly from the United and Central Provinces, Cuttack and Shahabad. There is a considerable permanent migration from the west of the district to Mayurbhanj and to the Assam tea gardens, and a fair amount of periodic emigration from the Contai, Tamluk and Ghatal subdivisions for employment as cultivators and field labourers in the Sundarbans, and as mill hands and coolies in the metropolitan districts. It would appear from the proportion of women amongst them that many of the migrants to the Sundarbans are beginning to settle down there permanently.